Trenitalia
ETR1000 VZI R1

Brake Control

Revision	Author	Date	Revision Description
00.1	R. Schiavo	26/04/2023	Draft Release Generated from DOORS module /Platform.-.Zefiro-Europe/01.-.Requirements Breakdown Structure.-.Zefiro-Europe/2F.-.Vehicle-Functional Integration Areas.-.Zefiro-Europe/2F_05.-.Drive and Brake Control-VF.-.Zefiro-Europe/2F_05.02.-.Brake Control-VF.-.Zefiro-Europe/Brake Control-VF.CONCEPT, current 11.0 3EGH489028-2358 revA (VZES) **reviewed for VZI-6**
00.2	G.Tarquini	13/10/2023	
00.3	G.Tarquini, R. Schiavo	16/10/2023	Modified Dragging brake, fading, speed signals diagnostic and handling, PCF and TCF diagnostic. Braking rheostat cooling req added Signal exchanged review
00	G.Tarquini, R. Schiavo	19/10/2023	Emissione
01	G.Tarquini, R. Schiavo	30/11/2023	Customer comments: 3.1.1.6 Dragging brake detection updated description with impact on service 2F_05.02.Zefiro-Europe.CONCEPT.HR53 2F_05.02.Zefiro-Europe.CONCEPT.HR109 2F_05.02.Zefiro-Europe.CONCEPT.HR56 2F_05.02.Zefiro-Europe.CONCEPT.HR59 2F_05.02.Zefiro-Europe.CONCEPT.HR62 2F_05.02.Zefiro-Europe.CONCEPT.HR65 Improved details: 2F_05.02.Zefiro-Europe.CONCEPT.HR106 2F_05.02.Zefiro-Europe.CONCEPT.HR107 2F_05.02.Zefiro-Europe.CONCEPT.HR108 2F_05.02.Zefiro-Europe.CONCEPT.HR118 Europe.CONCEPT.HR42: added reset condition 2F_05.02.Zefiro-Europe.CONCEPT.HR54 2F_05.02.Zefiro-Europe.CONCEPT.HR66 2F_05.02.Zefiro-Europe.CONCEPT.HR67 Improved details modifications (existing functionalities not tracked): 3.1.1.8.3 Failure detection 2F_05.02.Zefiro-Europe.CONCEPT.HR110 Additional functionalities:

			2F_05.02.Zefiro-Europe.CONCEPT.HR44 2F_05.02.Zefiro-Europe.CONCEPT.HR45 2F_05.02.Zefiro-Europe.CONCEPT.HR51 Additional status: 2F_05.02.Zefiro-Europe.CONCEPT.HR39 2F_05.02.Zefiro-Europe.CONCEPT.HR113 2F_05.02.Zefiro-Europe.CONCEPT.HR114 2F_05.02.Zefiro-Europe.CONCEPT.HR112 2F_05.02.Zefiro-Europe.CONCEPT.HR91 Specified signal in SIL 2 pack: 2F_05.02.Zefiro-Europe.CONCEPT.616 2F_05.02.Zefiro-Europe.CONCEPT.HR105 (new requirement) 2F_05.02.Zefiro-Europe.CONCEPT.120 2F_05.02.Zefiro-Europe.CONCEPT.1585 2F_05.02.Zefiro-Europe.CONCEPT.1587 2F_05.02.Zefiro-Europe.CONCEPT.1523 2F_05.02.Zefiro-Europe.CONCEPT.1622 3.12.1.1 VZI-6 Towed Mode 3.12.1.2 VZI-6 Towing Mode 3.4 Urgency Brake to be implemented, not optional Updated Table 1
02	G.Tarquini	14/04/2024	**Wabtec comments as per :** 2F_05.02.Zefiro-Europe.CONCEPT.HR105: listed the names of signals 2F_05.02.Zefiro-Europe.CONCEPT.HR46: not possible RS232 full redundancy for max workload CPU 2F_05.02.Zefiro-Europe.CONCEPT.HR47: not possible RS232 full redundancy for max workload CPU 2F_05.02.Zefiro-Europe.CONCEPT.HR49: not possible full redundancy for max workload CPU, in case of RS232 failure i twill shared on Ethernet the minumum for basic functionality (WSP, DNRA, braking) 2F_05.02.Zefiro-Europe.CONCEPT.HR50 – requirement allocated to TCMS 2F_05.02.Zefiro-Europe.CONCEPT.HR112 - requirement allocated to TCMS 2F_05.02.Zefiro-Europe.CONCEPT.HR91 - heavy slide SIL 0

			2F_05.02.Zefiro-Europe.CONCEPT.HR54 2F_05.02.Zefiro-Europe.CONCEPT.HR57- corrected timer 15 s 2F_05.02.Zefiro-Europe.CONCEPT.HR74 – added reset conditions 2F_05.02.Zefiro-Europe.CONCEPT.HR46 - SIL 0 2F_05.02.Zefiro-Europe.CONCEPT.HR47 - SIL 0 2F_05.02.Zefiro-Europe.CONCEPT.HR88 - Diameter correction achieved 2F_05.02.Zefiro-Europe.CONCEPT.HR89 - not used - reserved 2F_05.02.Zefiro-Europe.CONCEPT.HR44 - boolean 3.11.1.4.4.1 BCU pressure switch check (per bogie) parameter 15s 2F_05.02.Zefiro-Europe.CONCEPT.HR48 – comparison BCU only (BCU comparison with WSP is already active) 2F_05.02.Zefiro-Europe.CONCEPT.HR113 (re-used existing signal) 2F_05.02.Zefiro-Europe.CONCEPT.HR112 (re-used existing signal) 2F_05.02.Zefiro-Europe.CONCEPT.HR53 - corrected name 2F_05.02.Zefiro-Europe.CONCEPT.HR59 - corrected name 4.2.1.1.2 WSP-CU to TCMS eliminated 2F_05.02.Zefiro-Europe.CONCEPT.HR83 2F_05.02.Zefiro-Europe.CONCEPT.HR84 2F_05.02.Zefiro-Europe.CONCEPT.HR85 2F_05.02.Zefiro-Europe.CONCEPT.HR86 4.1.2.1 modified Da antivalent 2 a boolean : 2F_05.02.Zefiro-Europe.CONCEPT.HR39 2F_05.02.Zefiro-Europe.CONCEPT.HR90 2F_05.02.Zefiro-Europe.CONCEPT.3539 **Trenitalia comments:** 2F_05.02.Zefiro-Europe.CONCEPT.HR106: corrected in 72h

			Hitachi modification: **Added:** 2F_05.02.Zefiro-Europe.CONCEPT.HR111 2F_05.02.Zefiro-Europe.CONCEPT.HR115 2F_05.02.Zefiro-Europe.CONCEPT.HR116 2F_05.02.Zefiro-Europe.CONCEPT.HR117 2F_05.02.Zefiro-Europe.CONCEPT.HR118 2F_05.02.Zefiro-Europe.CONCEPT.HR119 2F_05.02.Zefiro-Europe.CONCEPT.HR120 **Removed DNRA chapter 3.8, DNRA is described in the vehicle requirement specification**
03	C. Dreoni	11/09/2024	Ed cut off and blending management 3.7 Wheel Slide Protection 3.7.1 Function Design 2F_05.02.Zefiro-Europe.CONCEPT.HR123 WSP ED brake failure management 2F_05.02.Zefiro-Europe.CONCEPT.HR122 3.11.1.4.8 WSP test 3.1.2.2 Load Compensation 2F_05.02.Zefiro-Europe.CONCEPT.1751 – modified 2F_05.02.Zefiro-Europe.CONCEPT.1752 – modified 2F_05.02.Zefiro-Europe.CONCEPT.1753 – modified 2F_05.02.Zefiro-Europe.CONCEPT.1754 – modified Re-added DNRA chapter 3.8, it reports DNRA detection from BCU Minimum friction brake demand Added 2F_05.02.Zefiro-Europe.CONCEPT.HR121 3.9 Passenger Emergency Brake 2F_05.02.Zefiro-Europe.CONCEPT.2013 –change speed ref 2F_05.02.Zefiro-Europe.CONCEPT.2014 –change speed ref 3.11.2.4.11 DNRA Test 2F_05.02.Zefiro-Europe.CONCEPT.3510 – modified 2F_05.02.Zefiro-Europe.CONCEPT.3514 – modified 2F_05.02.Zefiro-Europe.CONCEPT.3518 – modified 2F_05.02.Zefiro-Europe.CONCEPT.3519 – modified

			2F_05.02.Zefiro-Europe.CONCEPT.3520 – deleted 2F_05.02.Zefiro-Europe.CONCEPT.3888 – deleted 2F_05.02.Zefiro-Europe.CONCEPT.3522 – deleted Added chapter 3.1.2.9.1 Dragging Brake by TCMS Reset – from 3ECP413922-0502 VZES **TFS pressure shared on SIL 0 and SIL TRDP Ethernet signals** 2F_05.02.Zefiro-Europe.CONCEPT.3844 **PCF incoerence modified** 2F_05.02.Zefiro-Europe.CONCEPT.1839 **EVAPP isolation** 2F_05.02.Zefiro-Europe.CONCEPT.3937 repealed **Other modification** 2F_05.02.Zefiro-Europe.CONCEPT.HR45 - automatic drive in traction 2F_05.02.Zefiro-Europe.CONCEPT.2359 modified **Minor Modifications (08-10-2024)** Derived defined. Exported to TOD and not TCMS 2F_05.02.Zefiro-Europe.CONCEPT.HR106 2F_05.02.Zefiro-Europe.CONCEPT.HR107 2F_05.02.Zefiro-Europe.CONCEPT.HR108 2F_05.02.Zefiro-Europe.CONCEPT.HR118a SIL 0 and not SIL 2 2F_05.02.Zefiro-Europe.CONCEPT.HR118b SIL 0 and not SIL 2 2F_05.02.Zefiro-Europe.CONCEPT.HR106 splitted in 2F_05.02.Zefiro-Europe.CONCEPT.HR106 and 2F_05.02.Zefiro-Europe.CONCEPT.HR124 2F_05.02.Zefiro-Europe.CONCEPT.HR91 splitted in 2F_05.02.Zefiro-Europe.CONCEPT.HR125 Diagnostic in SIL Event in N/A reformatted in standard format 2F_05.02.Zefiro-Europe.CONCEPT.HR110 2F_05.02.Zefiro-Europe.CONCEPT.HR39 2F_05.02.Zefiro-Europe.CONCEPT.HR115 2F_05.02.Zefiro-Europe.CONCEPT.HR114 2F_05.02.Zefiro-Europe.CONCEPT.HR112

			Event referenced and splitted: 2F_05.02.Zefiro-Europe.CONCEPT.HR39 2F_05.02.Zefiro-Europe.CONCEPT.HR114 2F_05.02.Zefiro-Europe.CONCEPT.HR113 2F_05.02.Zefiro-Europe.CONCEPT.HR115 2F_05.02.Zefiro-Europe.CONCEPT.HR114 2F_05.02.Zefiro-Europe.CONCEPT.HR116 2F_05.02.Zefiro-Europe.CONCEPT.HR112 2F_05.02.Zefiro-Europe.CONCEPT.HR91

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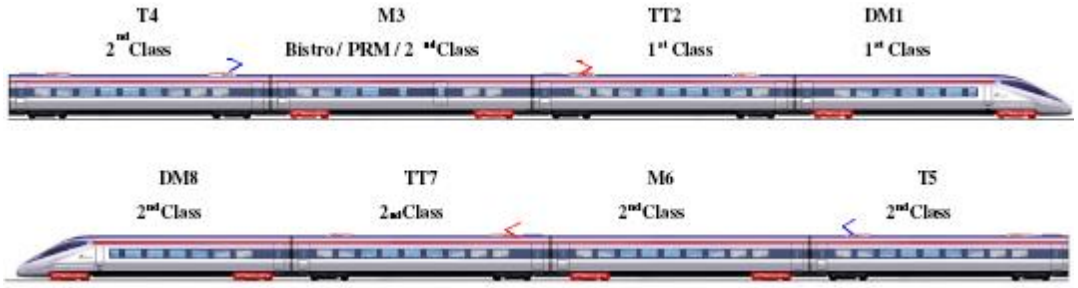
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1 Introduction

This document is a function specification for the control of the Brake system VZI-R1 for customer Trenitalia.

1.1 Train Configuration

Each VZI-R1 train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and Abbreviations

1.2.1 Definitions

Brake Mode: The type of brake which is used for the active deceleration of the train: DEP, IP or IEP.
Braking: Master Controller, or DBV controller in a brake position or ATP or TCMS speed regulation function request a deceleration

1.2.2 Abbreviations

ASFA	Anuncio de Señales y Frenado Automático
BMS	Bearing Monitoring System
BP	Brake Pipe
BCU	Brake Control Unit
CCU	Car Control Unit
C-pressure	Brake Cylinder Pressure
DBV	Driver's Brake Valve
DBV-GW	Driver's Brake Valve Gateway Control Unit
DEP	Direct Electro Pneumatic Brake
DIA	Diagnostic Event
DNRA	Detection of Non Rotating Axle
DSD	Driver's Safety Device
DVD	Driver Vigilance Device
EB	Emergency Brake
ED	Electro Dynamic Brake
EVENT	Diagnostics that will be stored in the Event database within TCMS SW
EP	Electro Pneumatic
ETCS	European Train Control System
HW HMI	Human Machine Interface, e.g. buttons, switches and indication lamps
IDU	Intelligent Display Unit
IEP	Indirect Electro Pneumatic Brake
IP	Indirect Brake
IP Bus	Ethernet bus IP protocol based
MOCU/M	Control Unit / Motor converter

PB	Parking Brake
PCU	Propulsion Control Unit
PEB	Passenger's Emergency Brake
TBU	Train Base Unit (4car train level)
TCMS	Train Control and Management System
TSI	Technical Specification for Interoperability
UIC	Union Internationale de Chemins de Fer
WSP	Wheel Slide Protection
WSP-CU	Wheel Slide Protection Control Unit

2 System Overview

2.1 System Configuration

VZI-R1 brake control system consists of an electro-pneumatic direct brake and an indirect brake. The direct brake is used for normal service and urgency (optional) brake in blending mode with electro dynamic. The indirect brake mode is used for the safe emergency brake and back-up indirect service brake in case of faulty direct brake.

The train is equipped with a main reservoir pipe and a brake pipe. The main reservoir pipe feeds the brake auxiliary reservoir and all other air consumers. The brake pipe is always active to control the indirect brake.

The driver has following control elements in the driver's cab to control the brake system:

- Master control handle (traction/brake controller for direct brake control)
- Driver's brake valve (indirect brake control)
- Two emergency stop buttons

In each car one brake frame is located in the underframe including two brake control units:

- BCU - Controls DEP brake, DNRA function, parking brake supervision, redundant WSP function
- WSP-CU - Controls WSP function, C-pressure supervision, redundant DNRA function

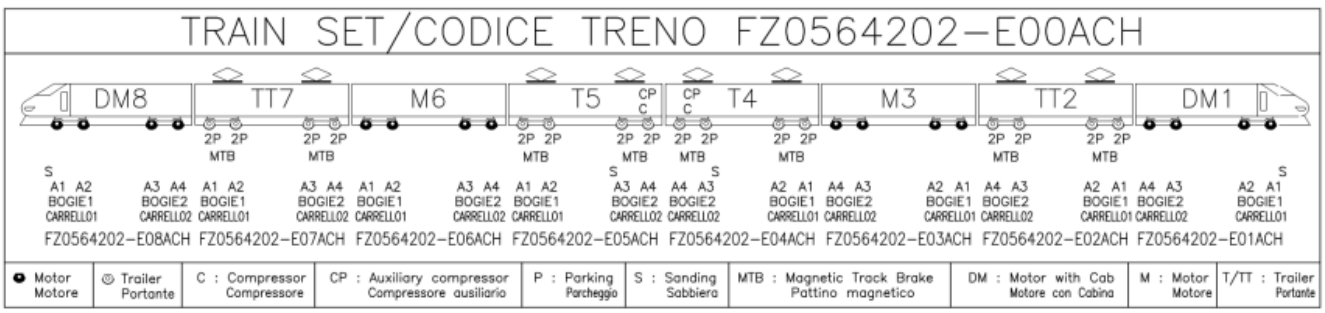
The WSP is controlled by the WSP control unit on axle level, in case of failure the control is automatically take over by the redundant WSP function in the BCU.
The DNRA is controlled by the BCU on axle level, in case of failure the control is automatically take over by the redundant DNRA function in the WSP-CU.

In the DM1/DM8 car the indirect brake control equipment is located on the cabin panel in the driver's desk. The electrical indirect brake functions are controlled by the DBV control unit (DBV-GW) on the cabin panel. The brake pipe is controlled by the Minitrol panel and the driver's brake valve.
In case of failure of the indirect electro-pneumatic brake pipe control (IEP), the brake pipe can be controlled in a pure pneumatic way operating the RIBK depannage cock in the DBV.

The bogie brake equipment is divided by the following bogie types.

- Motor bogie:
  - 4 wheel mounted brake discs
  - 4 wheel mounted brake caliper units
  - Air suspension pressure sensor for load compensation
- Trailer bogie
  - 6 axle mounted brake discs
  - 4 axle mounted brake caliper units with parking brake
  - 2 axle mounted brake caliper units without parking brake
  - Air suspension pressure sensor for load compensation

<2.1 - D1: Brake system train layout>



2.1.1 Brake Modes

The VZI-R1 brake system is able to perform following brake modes:

- **Service brake normal operation**

Direct electro-pneumatic brake mode (DEP):
The brake request is given by the master controller or TCMS service brake request. The brake blending is controlled by TCMS based on the service brake request. The friction brake demand will be transferred by TCMS via IP to the BCU. Based on the friction brake force demand the BCU is building up the cylinder pressure by the direct brake.

- **Service brake back-up indirect brake mode**

Indirect electro-pneumatic brake (IEP):
The brake request is given by the driver's brake valve (DBV), the brake request is electrically transferred from the DBV to the Minitrol panel located on the cabin panel. The brake demand is set by the Minitrol and transferred via the brake pipe by lowering the brake pipe pressure. Based on the brake pipe pressure the cylinder pressure is build-up by the UIC distributor valve and ED brake is not possible while braking with IEP.

Indirect pneumatic brake (IP):
The brake request is given by the DBV, the brake request is transferred by the DBV to a pressure level. The brake demand is transferred via the cabin panel to the brake pipe. Based on the brake pipe pressure levels the cylinder pressure is build-up by the UIC distributor valve and ED brake is not possible while braking with IEP.

The service brake request can be set by following devices:
- Master control handle (DEP brake)
- TCMS speed control (DEP brake)
- ATP (DEP brake)
- Driver's brake valve (back-up IEP or IP brake)

- **Urgency Brake Mode (Optional)**

Urgency brake is an optional brake mode function which allows to activate an emergency brake with ED-brake + direct brake.
Urgency brake is controlled by TCMS via IP IP and by the urgency brake loop. The brake cylinder pressure is build-up by the direct brake controlled by the BCU based on the friction brake force demand by TCMS. The deceleration is controlled by a speed dependent curve to lower the power input and thermal stress on the brake discs.
A monitoring device is used to supervise the brake performances (EBS), able to trigger an emergency brake by indirect brake in the case the deceleration obtain with the urgency brake is not according to the expected brake curves.

- **Emergency Brake Mode**

Emergency brake is controlled by lowering the brake pipe pressure (< 250 kPa) and the emergency brake loop. The emergency brake cylinder pressure is build-up by the UIC distributor valve. The brake pressure is controlled by 4 stages braking group to reduce the brake force at higher speeds due to adhesion and thermal input into the brake disc. At low speed the 4th stage will manage the sinter brake pad friction coefficient behaviour.

Emergency brake request can be requested by following devices:
- Master control handle EB position
- Two emergency stop buttons
- Driver's brake valve EB position
- Passenger's emergency brake function
- DNRA-BCU brake request
- TCMS brake request
- ATP brake request

2.1.2 Brake Control Layout

For the brake control on train level safety loops, train control lines and IP bus communication hardware are installed.

The safety loops are used to activate the safe brake functions as urgency (optional) and emergency brake by hardware line. Following safety brake loops are present:

- Urgency brake loop
- Emergency brake loop

For degraded mode and back-up reason train control lines are installed to transfer the service brake demand and traction demand to the brake control units BCU in degraded emergency operation mode. Therefore, following train lines are present:

- Emergency operation train line
- Traction train line
- Brake not train line

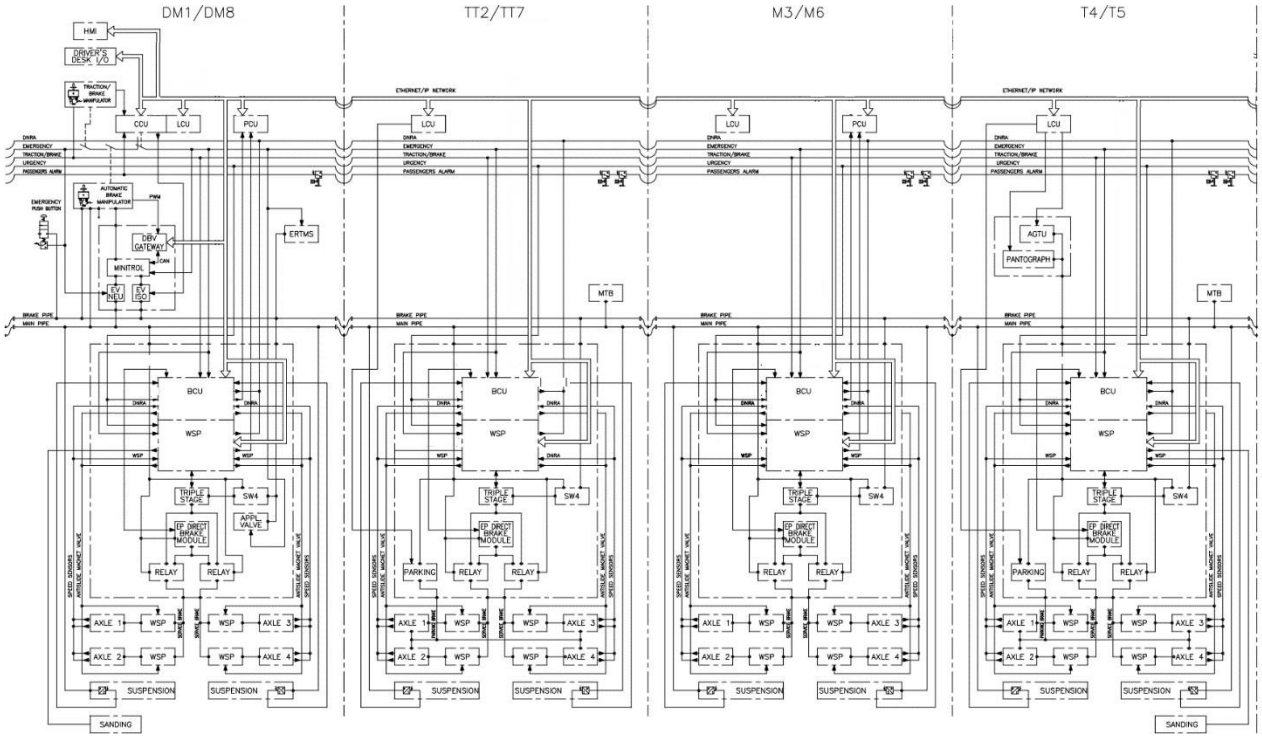
To transfer the detection of non-rotating axle signal and the passenger's emergency brake request to the active cabin and to the driver two additional train lines are installed.
Therefore, train control lines are present for following functions:

- DNRA train line
- Passenger's emergency brake train line

The DNRA train line is as well used to transfer the hot bearing monitoring signal to the active cabin in case a thermal fuse melts (above the threshold).

Urgency and emergency brake loops, emergency operation, traction, not-brake and DNRA hardware control lines are connected to the respective brake control unit BCU, WSP-CU and DBV-GW of each vehicle to control and transmit the signal for the needed function.

<2.1.2 - D1: Train control layout 4 car train base unit>

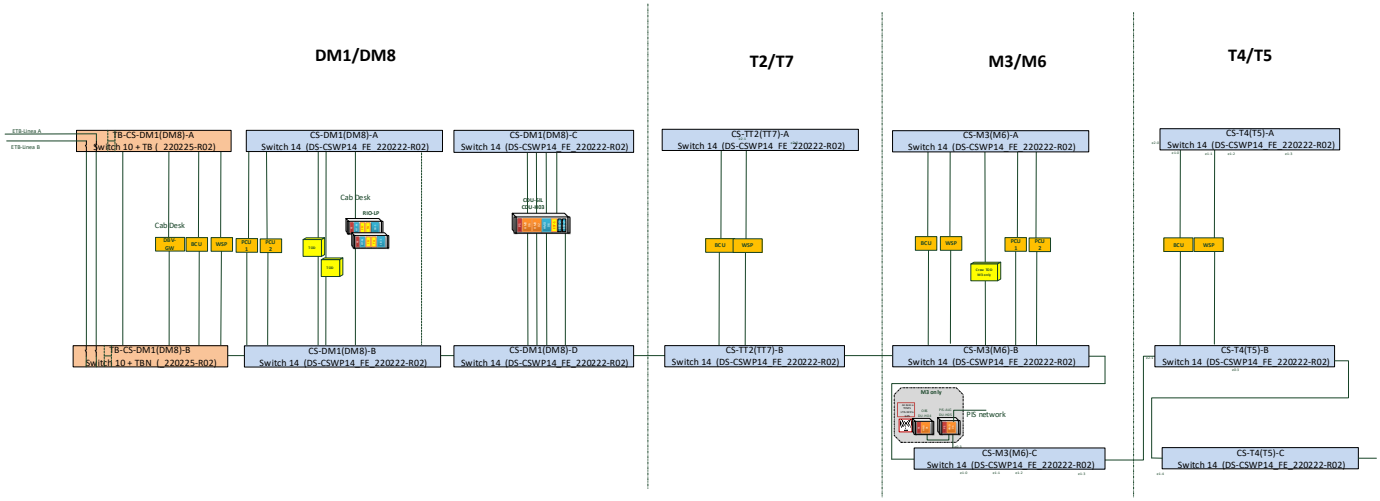


The bus communication between TCMS and brake control units is realised via the IP bus (TRDP Protocol). All software control signals and diagnostic signals are transferred by the IP bus. All brake control units BCU, WSP-CU and DBV-GW have a dual homing connection on the Ethernet network to have a redundant communication channel that makes the communication available also in case of single failure in the Device interface or for other network failure (switch, cable,..).

Optional the emergency brake supervision unit (EBS, optional device necessary only in presence of urgency brake function) is connected in dual homing on the IP bus.

The IP-bus system of the train is used also for brake software updates and terminal software remote control. Terminal software remote control can be used for online monitoring of signals and variables of the brake control units.

<2.1.2 - D2: Train bus layout>



In the train following brake control units are present and located in different cars:

- BCU: One electrical control unit in each car
- WSP-CU: One electrical control unit in each car
- DBV-GW: One electrical control unit in DM1/8 car

The BCU and WSP-CU are located on the brake frame mounted in the underframe of each car. The brake frame consists of the complete electronic control, pneumatic hardware panel and pneumatic auxiliary brake reservoir and main pipe reservoirs.
The data exchange between the BCU and WSP-CU is realised via the RS232 interface.

The DBV-GW is located in driver's cabin mounted on the cabin panel together with the pneumatic hardware of the indirect brake control.

2.1.2.1 *Brake Control Units*

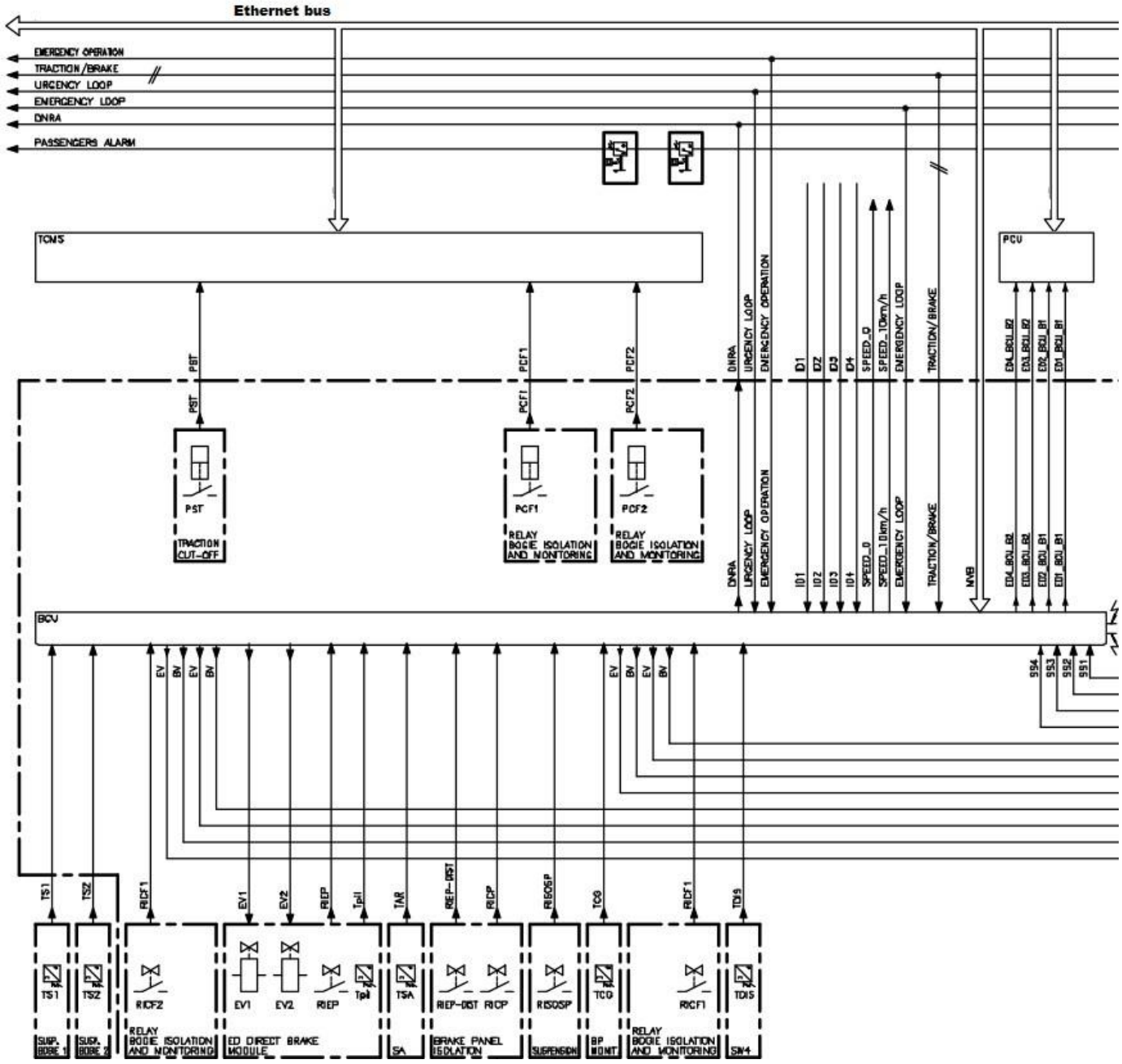
The different brake control units are responsible for the following functions. Each brake control unit is independent and has an own power supply board connected to the train battery bus nominal voltage at 110V.

BCU, connected to the IP and all hardwired train lines/safety loops:

- Direct electro-pneumatic service brake control and monitoring
- Direct electro-pneumatic urgency brake control and monitoring (optional)
- Direct electro-pneumatic emergency brake control as redundancy to indirect chain
- Indirect and direct emergency brake monitoring

- DNRA detection and signal transmission
- Parking brake supervision
- WSP redundancy in case of failure of WSP-CU control chain
- Speed signal for door control

<2.1.2.1 - D1: BCU input/output overview M3/6 car>



WSP-CU, connected to the IP Bus and hardwired safety loops:

- Brake cylinder pressure monitoring
- DNRA detection and signal transmission (redundant to BCU)
- WSP function

- Speed signal for door control

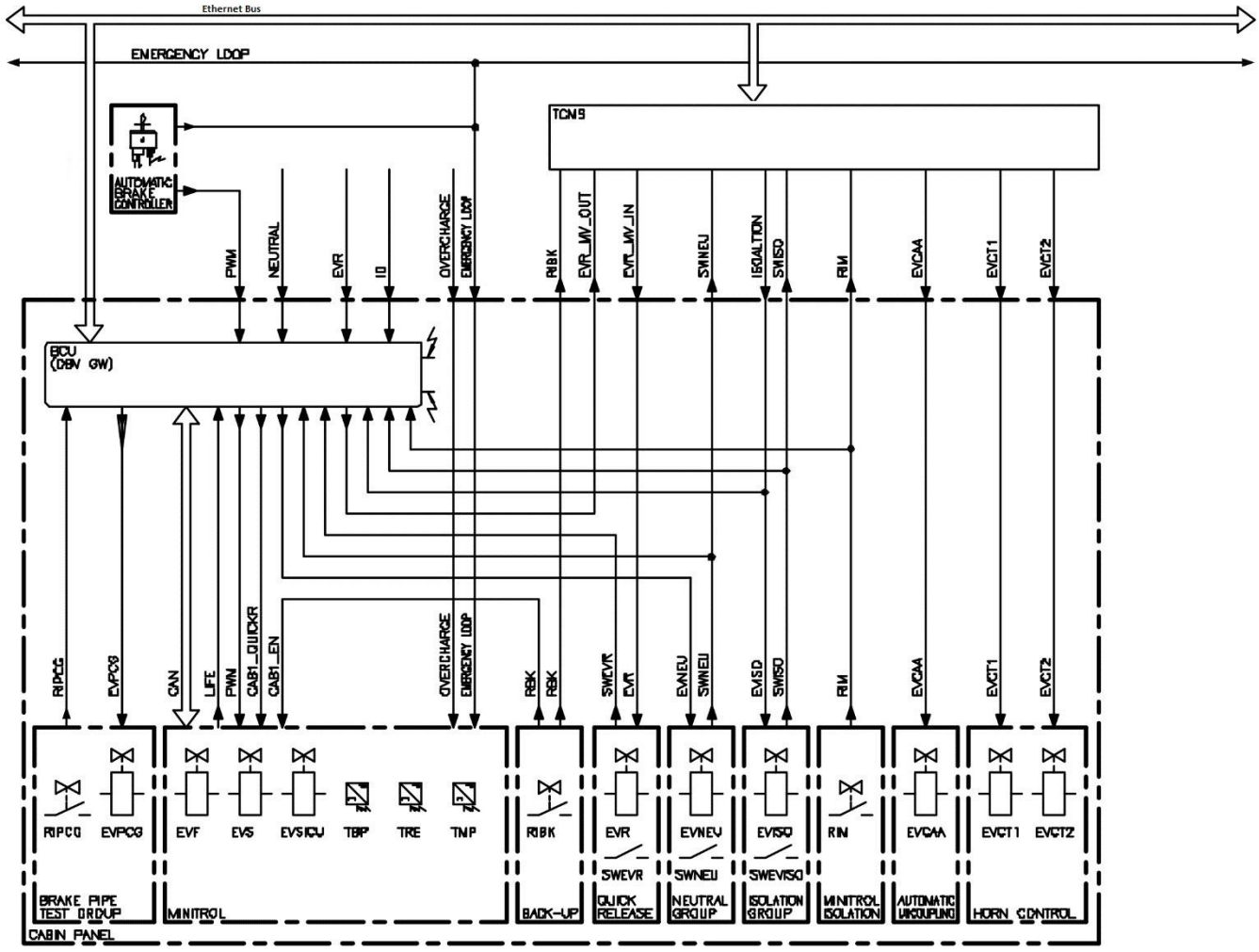
DBV-GW, connected to the IP Busand emergency brake loop:

- Brake pipe pressure monitoring
- Indirect brake pipe control monitoring
- Indirect brake test
- Connected to Minitrol

Minitrol, closed control chain connected to emergency brake loop and DBV PWM:

- Indirect electro-pneumatic brake

<2.1.2.1 - D2: Cabin panel DBV-GW/Minitrol input/output overview DM1/8 car>



2.1.2.2 *TCMS Software Brake Control*

The TCMS control layout is based on redundant control modules per 8-car train. Between two coupled 8-car trains the communication is realized by the redundant Ethernet Train Switches located in the DM1 and DM8 coaches implementing routering functionality between the 2 network layers.
The TCMS controls or monitors following brake control functions:

- Direct brake blending function on 8-car train level
- Holding brake control and monitoring
- Parking brake control and monitoring
- Passenger's emergency brake control and monitoring
- Driver's guided brake test control and monitoring
- Monitoring of brake isolation cocks
- Monitoring of urgency and emergency brake loops
- Monitoring of train lines
- Towing / towed control

In general, the TCMS is responsible for the complete diagnostic control and storage on train level.

The TCMS signal input is based on selection method for valid signals to guarantee correct system behaviour on train level.

TCMS sets the corresponding SW input signal to invalid if no current input signal is received from a reporting unit or if the reporting unit reports itself or a specific signal is set as faulty (e.g. TCMS digital input reports an input channel error). Otherwise, if current signals are received without error, TCMS sets the corresponding SW input signal to valid.

If signals are invalid a maintenance event or driver event is set based on the effect on train level.

Plenty of brake system output signals are read in redundant way by TCMS input and IP bus signal from the brake control units.

If redundant signals are valid TCMS checks whether these signals are plausible (inverse or equal) regarding their respective logic. In case of a mismatch of two valid signals, TCMS sets both signals to invalid as the failure origin cannot be identified. In this case a maintenance or driver event is set based on the effect on train level.

3 Functional requirements

3.1 Service Brake - Direct Electro-Pneumatic Brake

3.1.1 Function design

The direct service brake is used to decelerate the train in normal operation conditions. It shall ensure a high level of reliability with the maximum use of the ED brake.

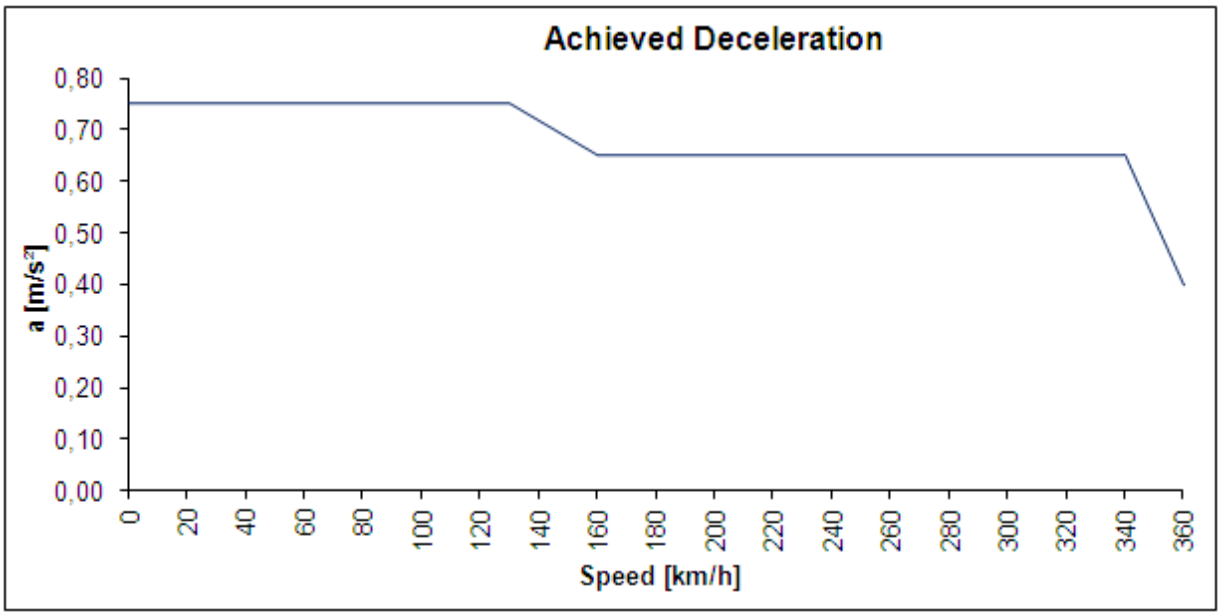
The service brake request can be ordered by the driver via master controller, TCMS train control and ATP.
The service brake is defined by 0-100% brake request which is the input parameter for the blending logic to calculate the brake force demand on car level.
The master (Traction/Brake) controller and automatic speed control is mainly used for blended brake control.

The service brake demand is communicated to BCU through a redundant IP bus but in case of a serious failure on the network communication, the service brake demand is conveyed through train lines, this is only used in emergency operation of the train (see description for degraded train control).

The 8car train has following maximum service brake deceleration characteristic:

- 0.4 m/s² increase linearly to 0.65m/s² between speeds 360 - 340 km/h
- 0.65 m/s² between speed range 340 km/h to 160 km/h
- 0.65 m/s² increase linearly to 0.75m/s² between speeds 160 - 130 km/h
- 0.75 m/s² between speed range 130 km/h to 0 km/h

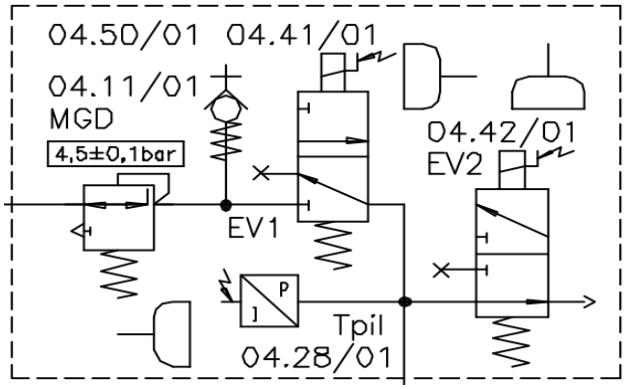
<3.1.1 - D1: Deceleration curve characteristic>



The direct brake is controlled by the BCU in each car. The direct brake module is mounted on the brake panels to control the brake actuation in service and urgency/emergency brake.

The direct brake module contains following components:

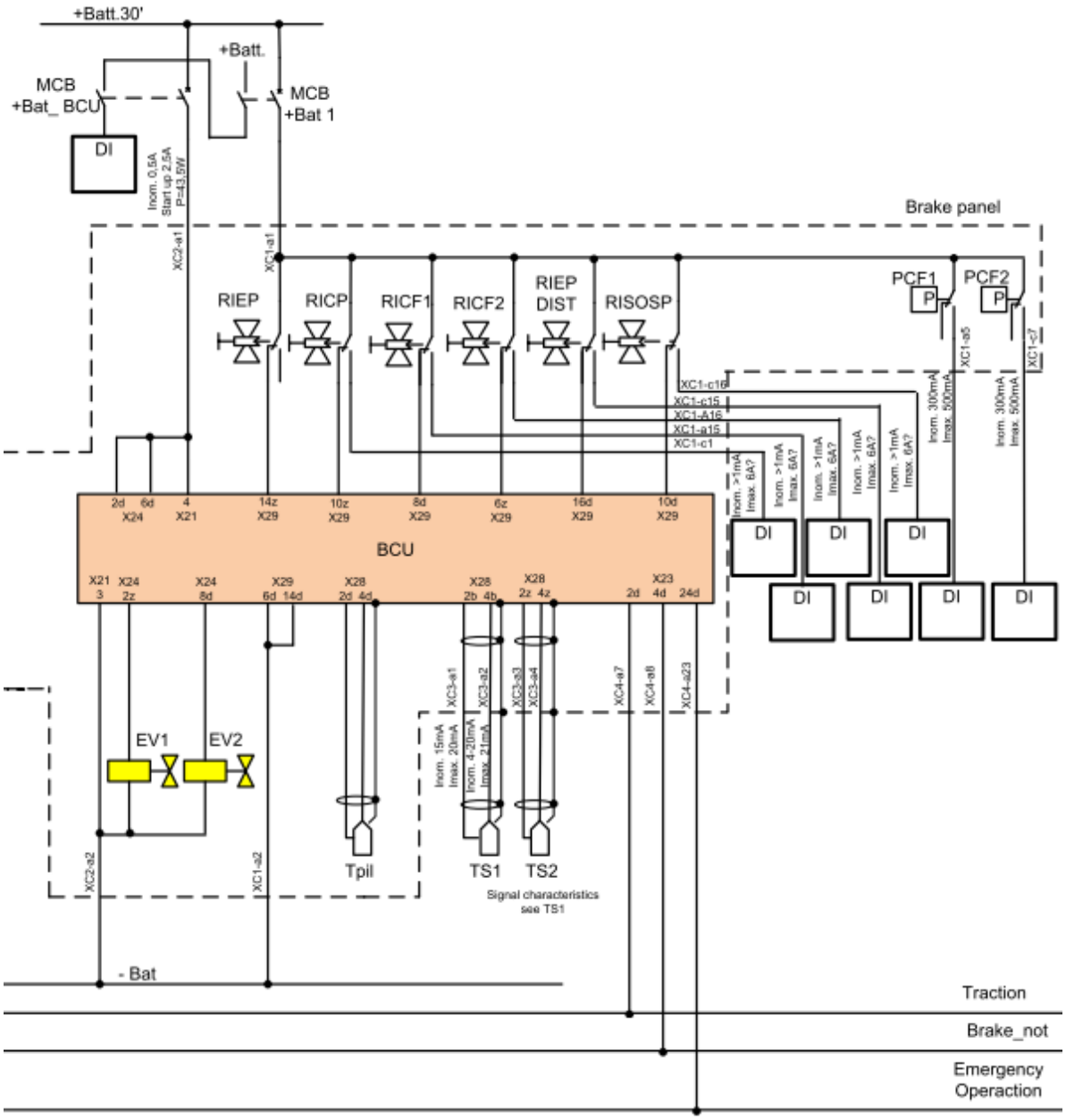
- EV1 (ref. 04.41) braking apply magnet valve;
- EV2 (ref. 04.42) braking release magnet valve;
- Tpil (ref. 04.11) pressure transducer;
- MGD (ref. 04.42) pressure reducer;
- Test point (ref. 04.50).



Electro pneumatic brake is piloted by BCU, which manages the magnet valves EV1 (to apply the brake) and EV2 (to release the brake) in a close loop by means of the pressure transducer Tpil to generate the control pressure to the relay valves of each bogie.

EV1 is setting or hold the target pressure level, EV2 is venting the air in case the target pressure is lowered.

<3.1.1 - D2: Direct brake control logic>



In case of failure in the direct electro-pneumatic brake control the missing brake force is shared and transferred to the remaining cars in the 8-car train according the defined blending logic. If more than two cars have a faulty direct brake the driver is informed to switch to indirect service brake mode defined in chapter 3.2.

The speed limitation based on available brakes is described in the operational manual depending on which track the train is operating on.

3.1.1.1 *Direct Electro-Pneumatic Brake Demand*

The direct brake demand can be required by the master controller, automatic speed regulation by TCMS and the ATP system.

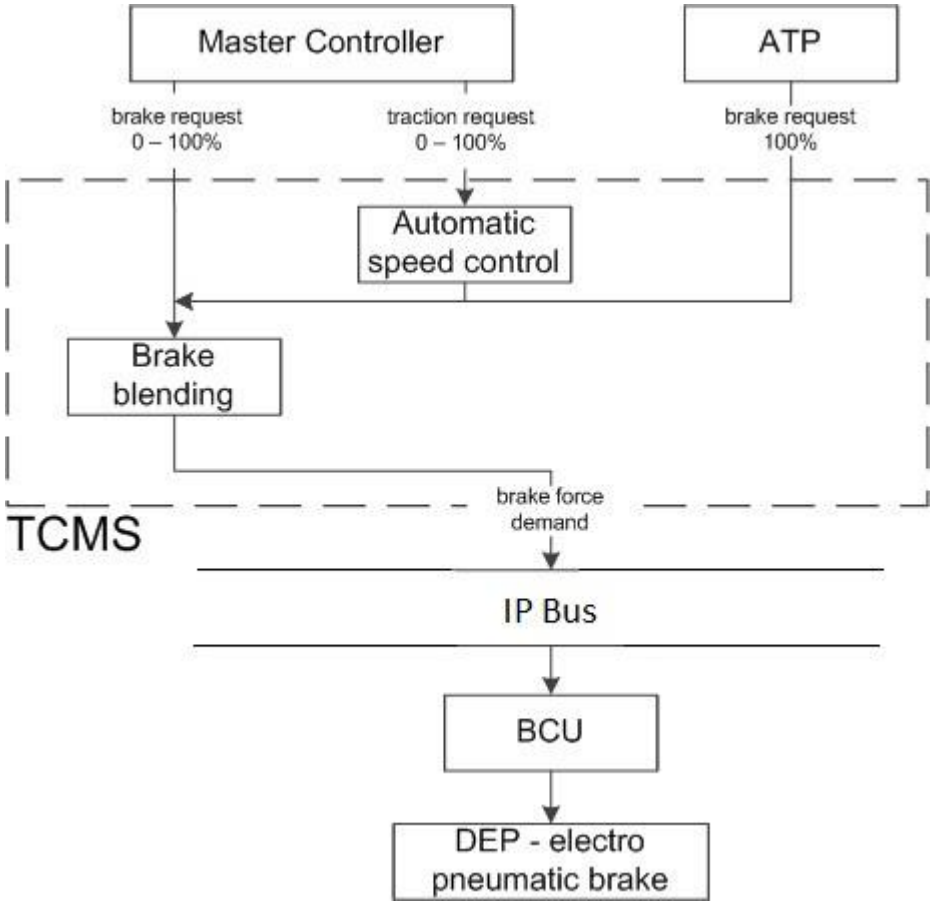
In the driver's cab one master controller is foreseen, the device has a service brake range of 0% to 100% and will increase the brake demand linearly over the full movement of the master controller brake portion.

TCMS automatic speed regulation can set a certain service brake force to regulate a pre-defined speed level or to achieve a certain speed limit selected via the speed lever in driver's cab.

In case of ATP (ETCS) service brake demand always full service brake (100%) will be applied.

Always the highest brake demand from master control handle, ATP and speed regulation (train control) shall have priority to set the brake demand to the traction and brake control units.

<3.1.1.1 - D1: Direct brake control and brake demand>



At direct service brake the traction is blocked by the TCMS software.
The traction block due to manual service brake request shall be reset if traction is requested again.

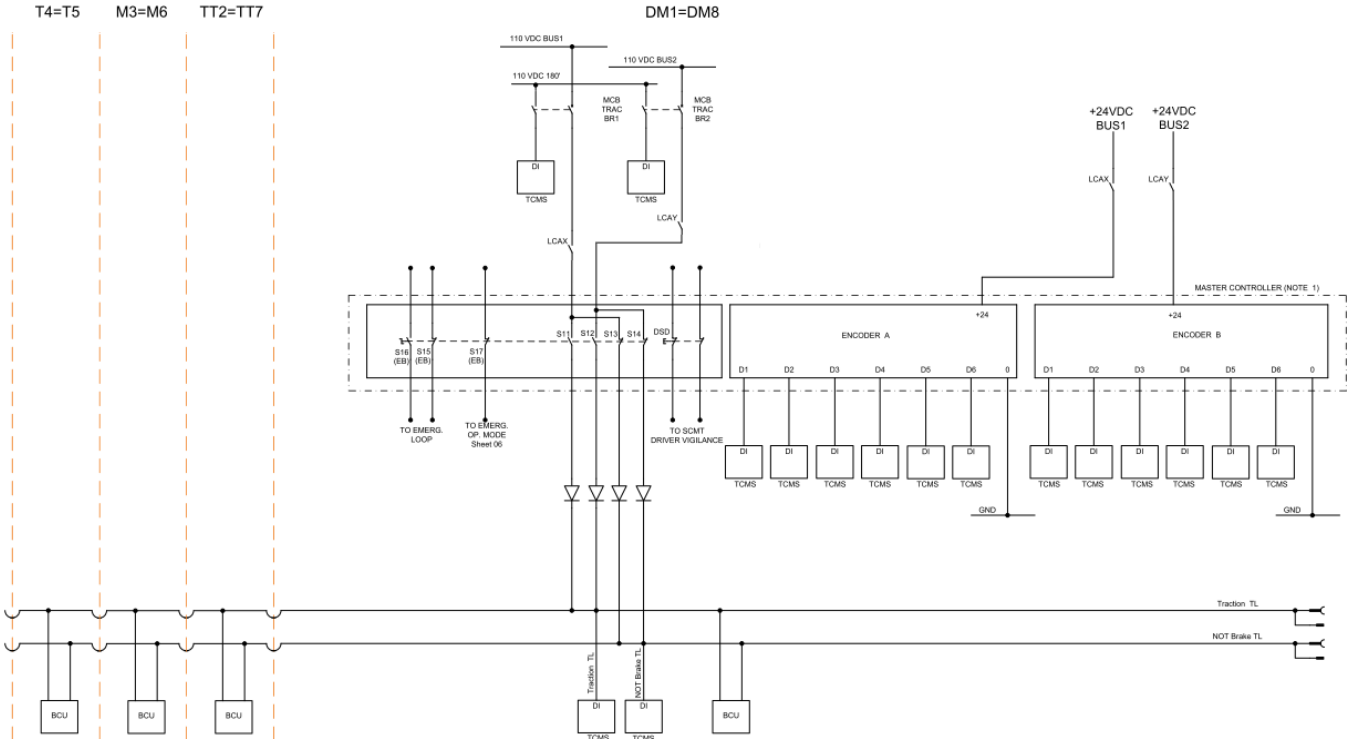
Traction safe by hardware is only foreseen via the traction safe trainline at emergency brake, refer to emergency brake function 3.5.
In automatic speed regulation function and ATP brake request the software traction block is reset automatically.

3.1.1.1.1 DEP Brake In Emergency Operation Mode

In addition to the IP bus brake demand request by TCMS hardwired train lines are installed to transfer the status of "braking" and "traction".

The "Braking NOT" train line is reset (low) if the master controller is in any brake position, "traction" train line is set (high) if the master controller is in any traction position.

<3.1.1.1 - D2: Master Controller and traction/brake train line interface>



In normal condition the train lines will be used only for diagnostic, the BCU will brake according the TCMS brake demand via IP bus . If a mismatch is detected between "Brake NOT" train line status low and no brake demand active, a maintenance event is set by TCMS.

In emergency operation mode the "emergency operation" train line will be set high, in this case the BCU applies a predefined direct service brake in case the "braking" train line is reset (low). The brake force is as well used to keep the train at standstill in a slope as the holding brake function is not active in degraded emergency operation mode.

The logic is described in diagram 3.1.1.1 - D3.

<3.1.1.1 - D3: Direct brake degraded control>

		Emergency operation train line	Traction train line	Braking train line	Train behviour
Normal mode	Traction	0 V	110 V	110 V	According TCMS brake demand
	Coasting	0 V	0 V	110 V	
	Braking	0 V	0 V	0 V	
Emergency mode	Traction	110 V	110 V	110 V	Traction with predefined traction force
	Coasting	110 V	0 V	110 V	Coasting
	Braking	110 V	0 V	0 V	Braking with predefined braking force

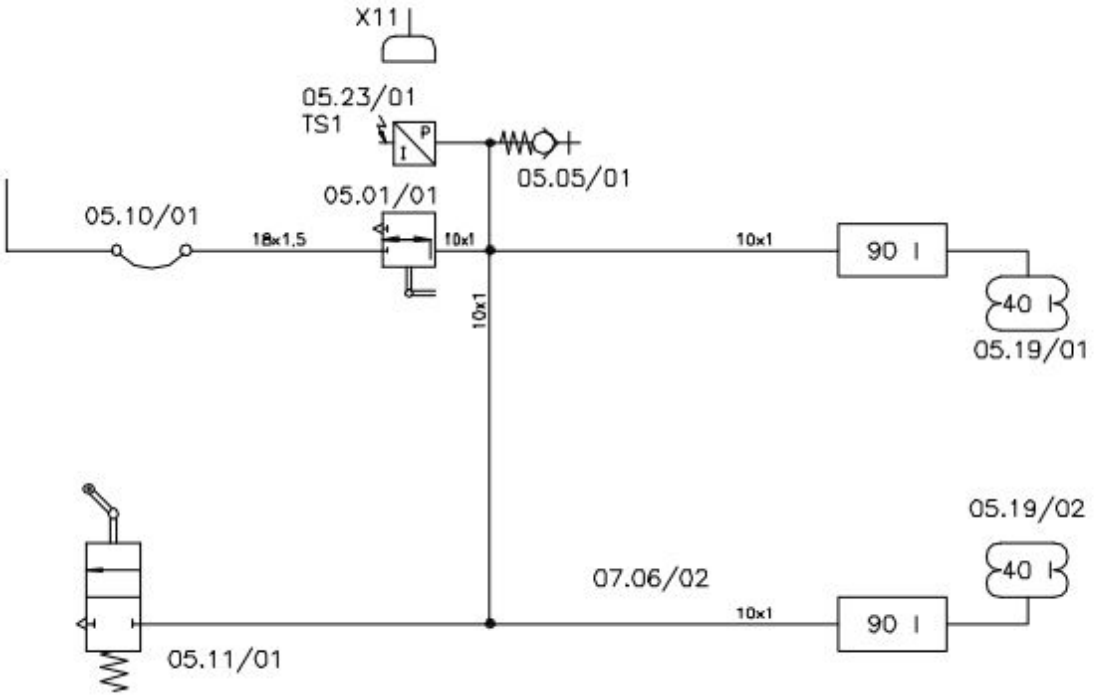
3.1.1.2 *Load Compensation*

The direct service brake force demand is based on the train load and therefore load compensated. Each bogie has one load pressure transducer TSx 05.23 located in the secondary suspension pneumatic circuit to measure the pressure after the levelling valve 05.01.

In case of defect of levelling valve 05.01 and overpressure, the safety valve 05.11 is venting the air suspension below if the carbody is moved out of the admissible section.

In case a pressure is out of the normal specified range, a diagnostic event is set by TCMS for overpressure or broken air suspension.

<3.1.1.2 - D1: Secondary suspension pneumatic diagram>



The load compensation is done by TCMS on car level based on the average pressure level given by both bogie pressure sensors TS1 and TS2.

Based on the average pressure (refer to table 3.1.1.2-T1) TCMS is calculating the actual car weight after all doors has been closed in the station.

Mass and suspension parameters according to:

- Suspension pressure measurement T59
- Mass parameters ED050025341 rev. 05

<3.1.1.2 - T1: Overview suspension pressure levels in kPa>

Mass and suspension parameters			Suspension (car average) - OMWO - Service Weight [bar]	OMWO (Service Weight) [kg]		Suspension (car average) - Default Pressure [bar]	Suspension (car average) - Suspension Broken Pressure [bar]	Suspension (car average) - Normal Weight (CN - DMNP) [bar]	DMNP (CN Weight) [kg]	Suspension (car average) - Exceptional Weight (CN) [bar]	DMNP 4P (Exceptional Weight) [kg]
DM1	**BOGIE A**	**A1**	**5,05**	15625	30950	**5,1**	**3,9**	**5,15**	31911	**5,26**	33257

		A2		15325							
	BOGIE B	**A3**		15690	31315				32561		35980
		A4		15625							
TT2	**BOGIE A**	**A1**	**4,84**	13240	26430	**5,06**	**3,68**	**5,28**	29082	**5,4**	31920
		A2		13190							
	BOGIE B	**A3**		14325	28595				30434		33366
		A4		14270							
M3	**BOGIE A**	**A1**	**4,86**	14570	29045	**4,95**	**3,81**	**5,04**	30934	**5,22**	35243
		A2		14475							
	BOGIE B	**A3**		15625	31165				32185		37086
		A4		15540							
T4	**BOGIE A**	**A1**	**4,305**	12290	24540	**4,645**	**3,11**	**4,985**	28130	**5,085**	30626
		A2		12250							

	BOGIE B	**A3**		12815	25560				28959		31602
		A4		12745							
T5	**BOGIE A**	**A1**	**4,33**	12345	24705	**4,74**	**3,13**	**5,15**	29134	**5,26**	31760
		A2		12360							
	BOGIE B	**A3**		12850	25675				28249		30722
		A4		12825							
M6	**BOGIE A**	**A1**	**4,71**	14465	28960	**5,12**	**3,54**	**5,54**	33630	**5,64**	36257
		A2		14495							
	BOGIE B	**A3**		14955	30015				32456		34930
		A4		15060							
TT7	**BOGIE A**	**A1**	**4,805**	13435	26905	**5,205**	**3,64**	**5,615**	31599	**5,725**	34225
		A2		13470							
	BOGIE B	**A3**		14030	28120				30502		32975

		A4		14090							
DM8	**BOGIE A**	**A1**	**4,91**	15480	30870	**5,21**	**3,78**	**5,51**	33846	**5,58**	36598
		A2		15390							
	BOGIE B	**A3**		14920	29975				32792		33643
		A4		15055							

Table 1 - Mass and suspension parameters

The static weight is calculated based on the car weight and both bogie weight:

w = Weight [kg] = (bogie type x + bogie type y) weight [kg] + average load pressure weight [kg]

The dynamic weight used for the brake blending calculation is considering in addition the rotational mass in the bogie:

Dw = Dynamic Weight [kg] = (bogie type x + bogie type y) weight [kg] + 2 * dynamic weight bogie + average load pressure weight [kg]

In case of isolated (RISOSP isolation cock) or broken suspension pressure TCMS is using the default pressure value to calculate the brake force in blending operation.

3.1.1.3 *Blending*

The brake blending is performed in the TCMS software. The calculation is performed on complete 8-car train level to achieve the best brake force distribution between electro-dynamic brake and friction brake.

Use of Electro-Dynamic brake minimizes the use of friction brake and reduces the wear of brake components in the bogies. The blending management with DEP brake is based on predictive blending to avoid initial fill of brake cylinders and friction material wear.

The blending logic distributes the friction brake force to all brake discs to achieve the same brake power and LCC values if the ED brake force is not sufficient. With ED brake available the brake force is distributed based on two constraints, to have equal thermal stress in all brake discs and to respect the adhesion limit.

The brake force is load compensated based on the actual weight of the complete train.

The overall brake force is calculated based on following input variables:

$$B_{ref} = \frac{B_r \cdot a \cdot D_w}{1000} - T_r$$

Br = Brake request 0-100%
a = Deceleration
Dw = Sum of dynamic weight on car x (static + rotational mass)
Tr = Train resistance in kN for 8car train

In order to define the needed friction brake effort per car, TCMS calculates the total required friction brake effort based on the ED brake available signal by the propulsion system. TCMS sends the ED brake request to the propulsion control unit depending on the dynamical brake curve.
Each propulsion control unit replies with the electro-dynamic brake achieved force related to train speed, adhesion limits, line voltage and catenary receptivity.
With this information TCMS calculates the actual ED brake demand to the propulsion and to add friction brake effort if necessary to reach the required total brake effort.

In case of degraded condition, for example the loss of brake force in one car, the missing brake force is shifted to the available cars in consideration of the maximum allowed adhesion value.

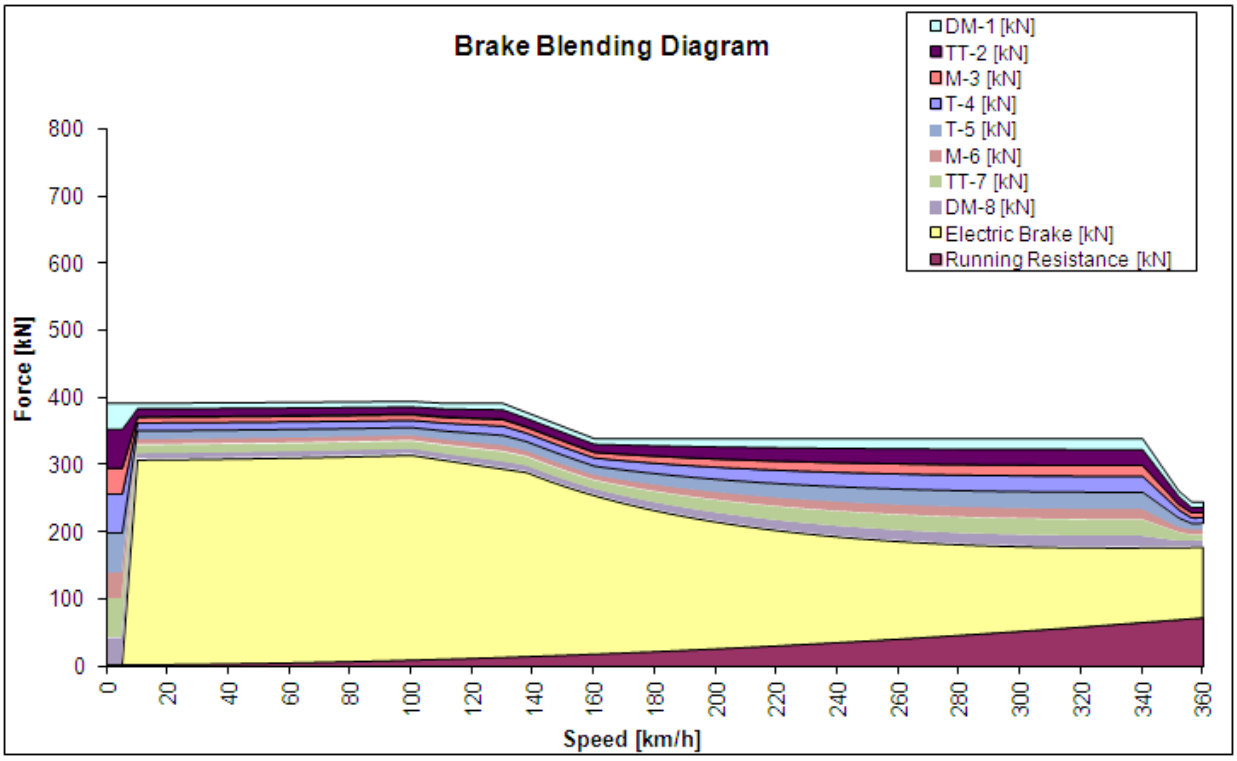
The brake force blending diagram for a 8car train is shown below:

- 25kV AC - 100% ED brake regenerative - Diagram 3.1.1.3 - D1
- 3kV DC / 25kV AC - 100% ED brake rheostatic - Diagram 3.1.1.3 - D2
- 3kV DC / 25kV AC - 0% ED brake - Diagram 3.1.1.3 - D3

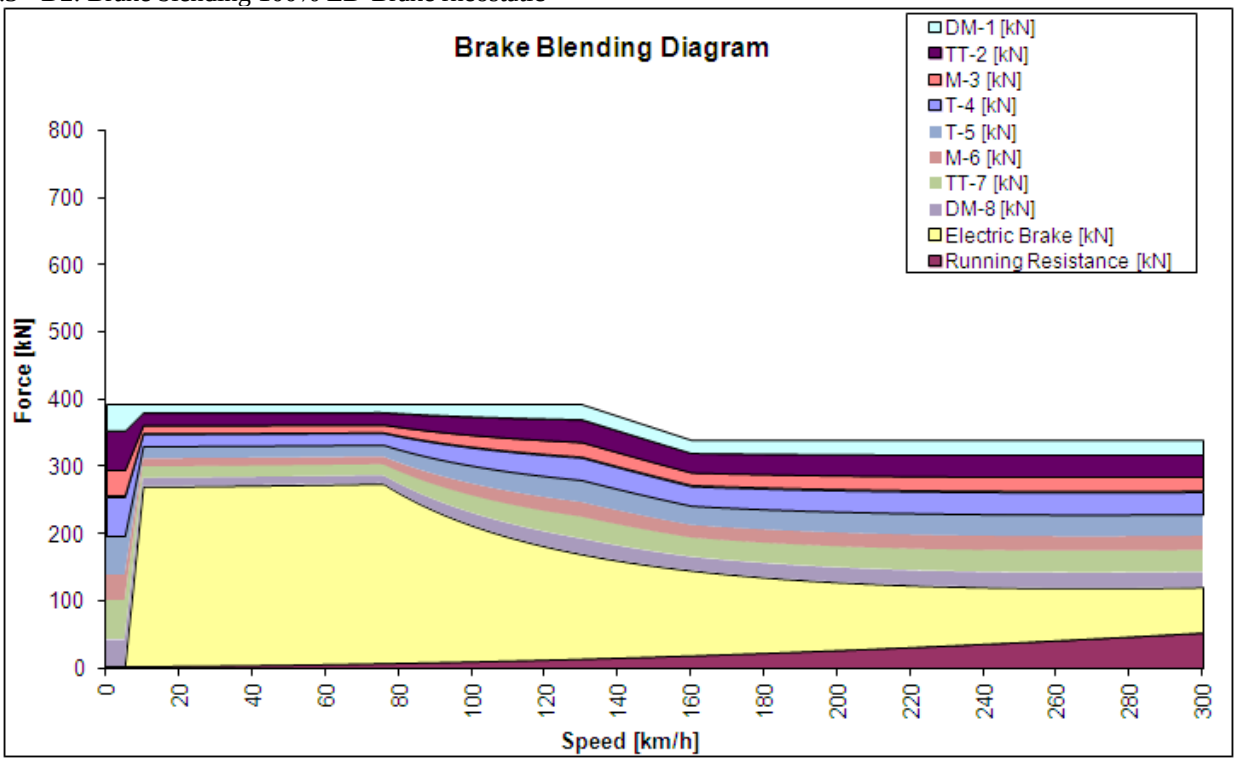
Minimum friction brake demand per car

Depending by the day number (1 to 31 respective to the month), TCMS define 3 cars per day. If the calculated friction brake request per car is lower than (8.36 kN for motor cars and 8.28kN for trailer cars), the friction brake request for the car shared to the BCU is set to zero (0 kN) and the brake force is redistributed to the other cars. This is limited to a maximum of 3 BCUs per trainset.

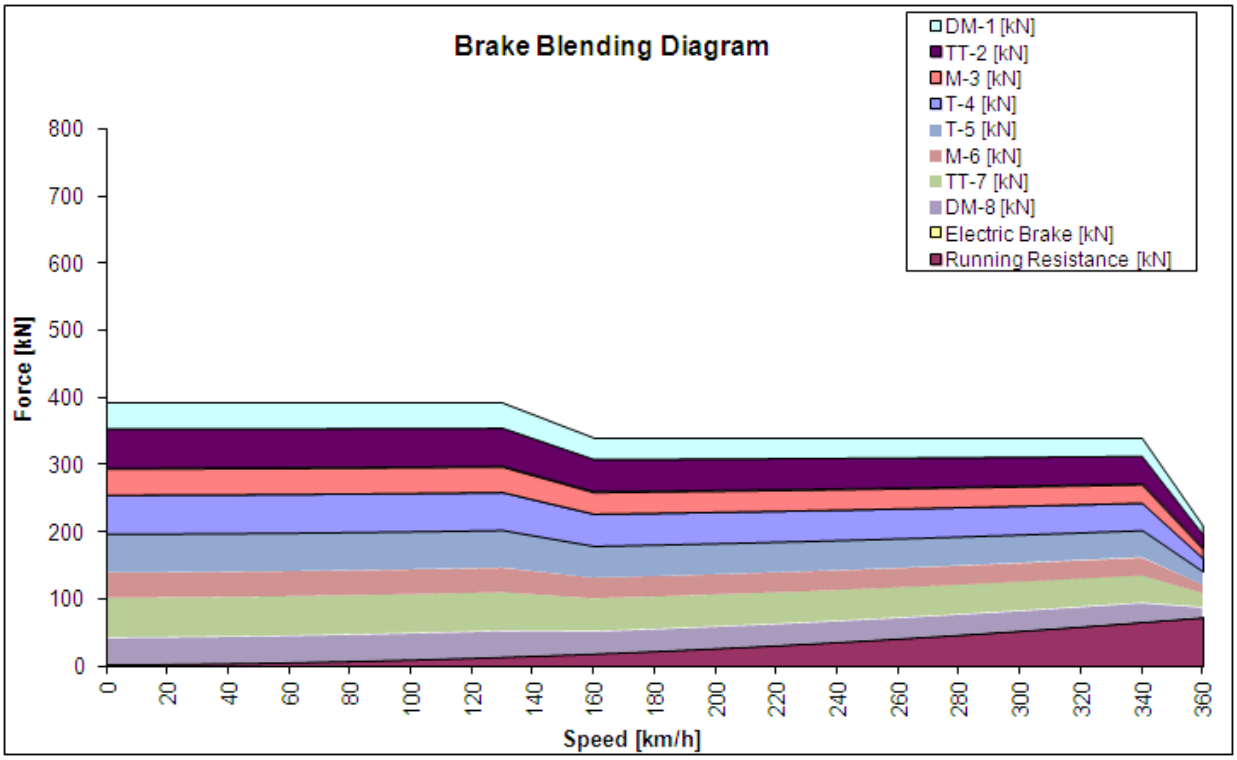
<3.1.1.3 - D1: Brake blending 100% ED-Brake regenerative>



<3.1.1.3 - D2: Brake blending 100% ED-Brake rheostatic>



<3.1.1.3 - D3: Brake blending 0% ED-Brake>



3.1.1.3.1 Degraded Blending Modes

In case of any error with electro dynamic brake or their control, ED brake is disabled in the affected motor car on axle level. During these degraded conditions the adhesion limit can be reached on axle level, in this case first the pneumatic friction brake force is limited based on available adhesion.

The loss of ED brake effort is compensated by the blending function, first with additional ED brake force on other motor cars if available or by the friction brake. The friction brake force is redistributed if a complete motor car has no ED brake available.

The following scenarios are covered by the brake force re-distribution in case of loss of DEP brake in one or more cars:
- 25kV AC - 100% ED brake regenerative + loss of 1 car DEP brake
- 25kV AC - 100% ED brake regenerative + loss of 2 car DEP brake
- 3kV DC - 100% ED brake regenerative + loss of 1 car DEP brake
- 3kV DC - 100% ED brake regenerative + loss of 2 car DEP brake
- 3kV DC / 25kV AC - 0% ED brake + loss of 1 car DEP brake
- 3kV DC / 25kV AC - 0% ED brake + loss of 2 car DEP brake

As already described, if more than two cars are without DEP brake performance the direct brake mode has to be disabled and switched to indirect service brake back-up mode as described in chapter 3.2.

ED brake failure management

If the dynamic brake is excluded (e.g. a failure event) in an axle or a bogie, the ED brake is kept active where the failure is not present. The friction brake and ED brake are blended in order to respect the adhesion limit and the missing force is compensated.

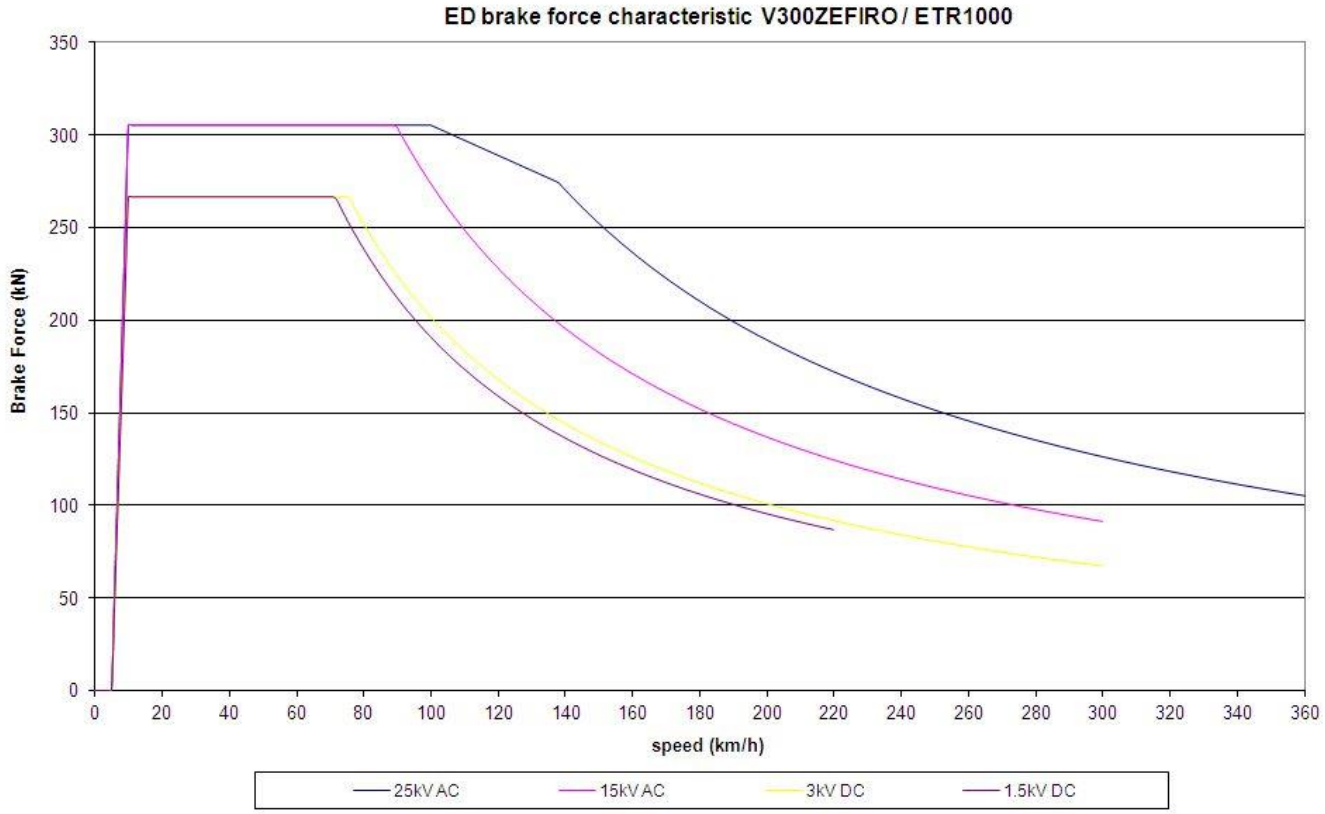
3.1.1.4 *Dynamic Brake*

The dynamic brake curve is based on the line voltage and stored in TCMS for the blending logic.
In case of service on tracks with 25kV AC and 15kV AC the dynamic brake curve is based on the regenerative brake and rheostatic brake as a backup.

In service at conventional tracks with 3kV DC and 1.5kV DC the dynamic brake curve is based on the rheostatic brake capacity by the brake resistor.

The diagram 1 is showing the ED-brake characteristic under different line voltage.

<3.1.1.4 - D1: ED-brake force characteristic under different line voltage>



3.1.1.4.1 Dynamic brake cut out

It's available a function for dynamic brake cut out. For this purpose, a softkey can be operated from the IDU and ED brake effort reference demand generated by TCMS is set to null (0 kN) for the whole train (in case of multiple operation the inhibition is valid for both consists). In this scenario, the braking effort is originated only by BCU (electro-pneumatic brake effort).

The softkey shall be available to the driver in the active cab and the cut in/off of the ED brake effort can be activated only under certain preconditions (master controller in coasting). Both states come to the driver with dedicated operating messages.

3.1.1.5 *Direct Electro-Pneumatic Monitoring and Diagnostics*

The DEP friction brake is monitored continuously by the BCU measuring and evaluating following data, refer to diagram 3.1.1.5 - D1 and 3.1.1.5 - D2:

- Direct electro-pneumatic brake (DEP) pilot pressure (transducer Tpil 04.28) of the direct brake group 04.50
- Power supply to direct brake control valve 04.41 EV1 and 04.42 EV2
- Brake applied pressure switch 04.33 per bogie (PCF1 and PCF2)

The friction brake force is monitored continuously by the WSP-CU measuring and evaluating:

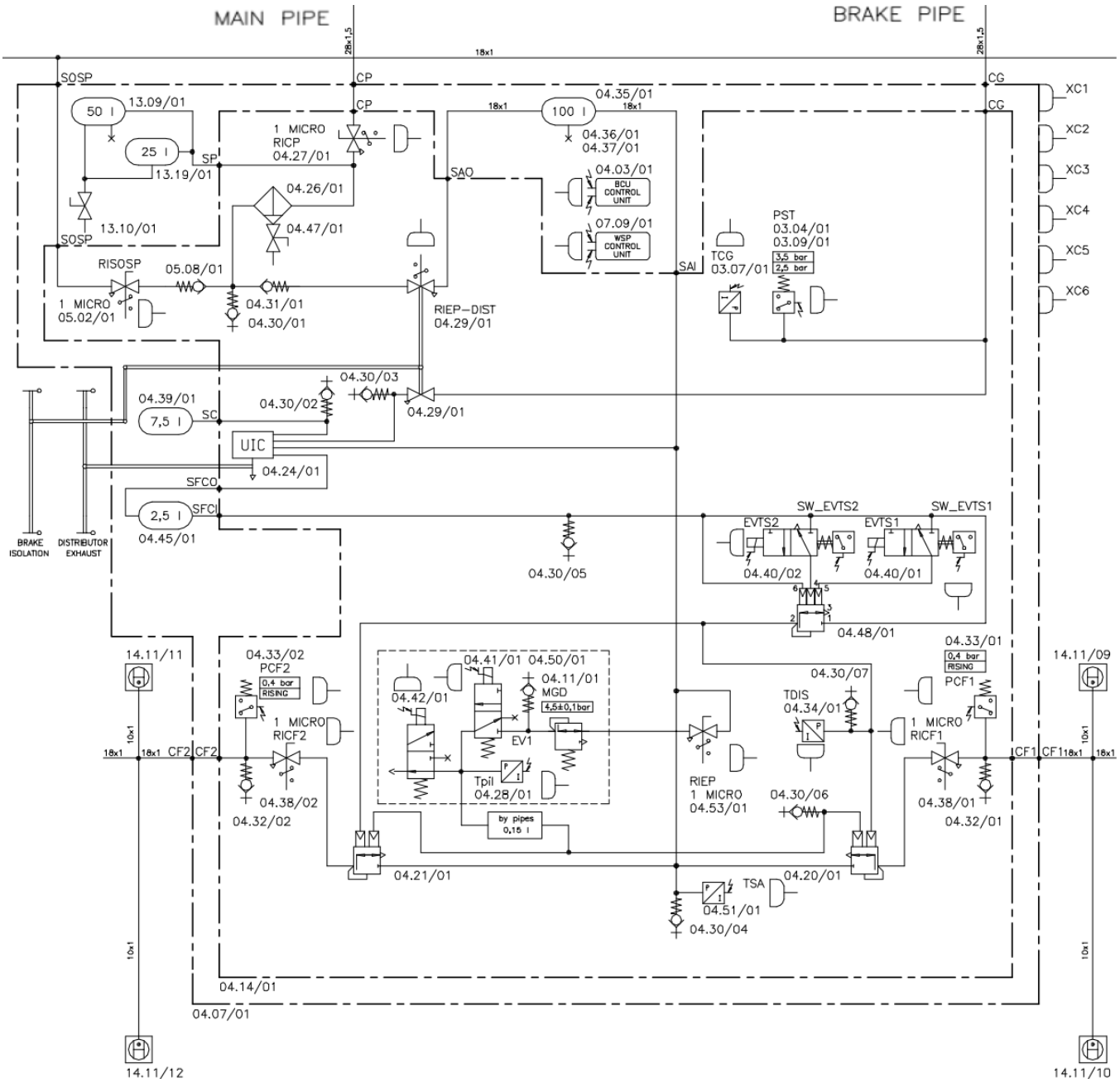
- Cylinder pressure transducer per axle TCF (as part of the WSP dump valve 07.01)

The cylinders pressure is transmitted by IP to TCMS. TCMS reads by independent digital input the pressure switches 04.33 (PCF1 and PCF2), one located per bogie to have information if the brake is applied or released.

In addition to the TCMS monitoring the brake pressure status is shown by pneumatic driven visual indicators 14.11 on bogie level. The external indicator located in the multiframe work next to the bogie has the following logic:

- Brake applied (pressure > 55 kPa +/-10 kPa) = Red flag
- Brake released (pressure < 20 kPa +/-10 kPa) = Green flag

<3.1.1.5 - D1: DEP pneumatic control (brake frame of M3/M6 car as example)>

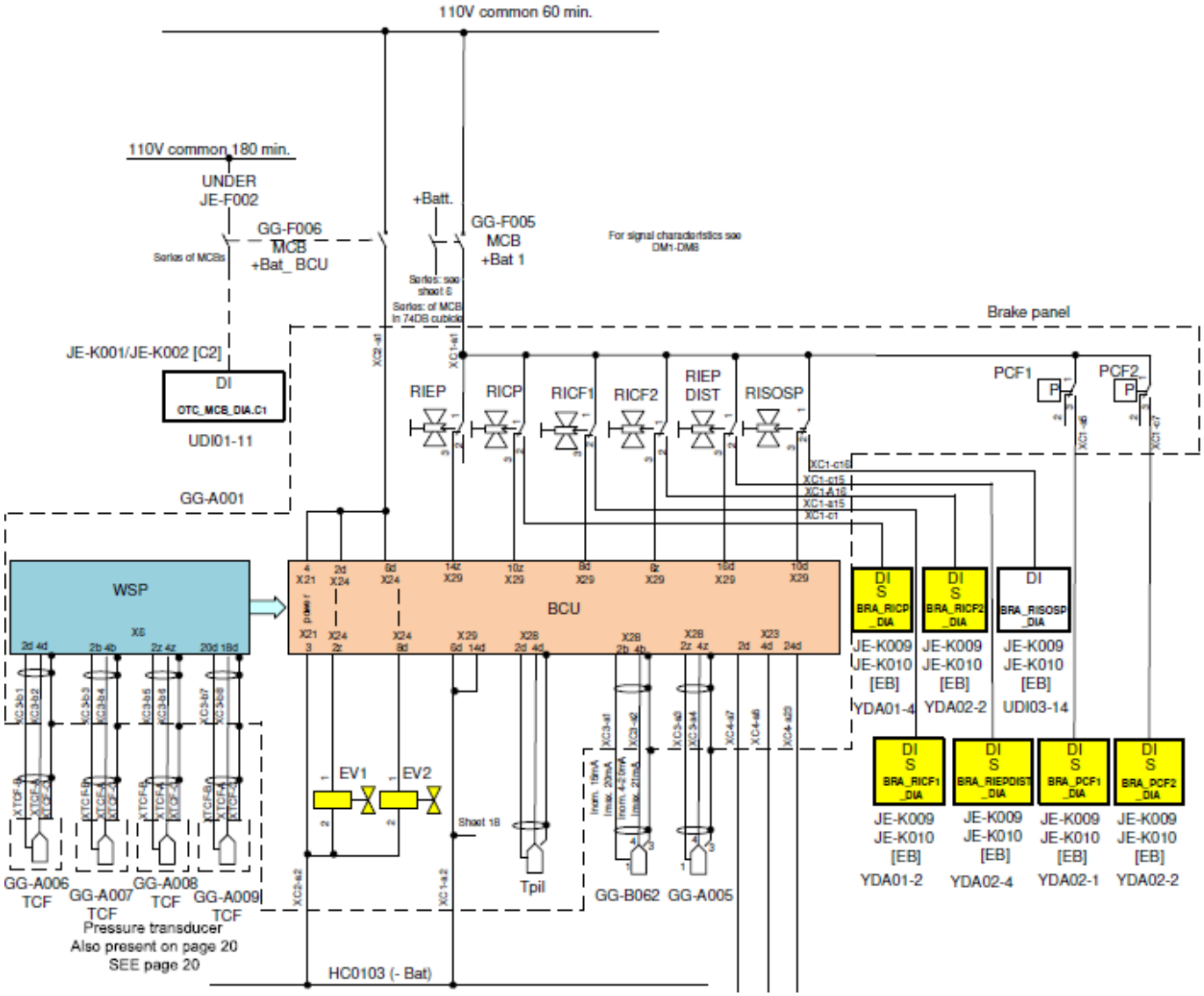


In case of any fault of the direct brake control group 04.50 a diagnostic event is set and the direct brake in this car is considered as not available in the brake blending calculation described in chapter 3.1.1.3.

The cylinder pressure per axle is supervised to set a diagnostic event when the achieved brake force is less or higher than the expected value. Based on this pressure the BCU is calculating an achieved brake force and provide this information to TCMS via the IP .

If the brake is considered as applied, the visual indication element "Brake applied" lamp on the driver's desk is illuminated by TCMS. The status of TCMS is set based on the input of the PCF1/2 or BCU IP input.

<3.1.1.5 - D2: DEP monitoring control logic>



Brake auxiliary reservoir supervision (DEP inexhaustibility)

The brake auxiliary reservoir 04.35 / 04.25 located in the brake frame per car with a capacity of 100l is monitored by the pressure sensor TSA 04.51 read by the BCU, refer to diagram 3.1.1.5 - D1.

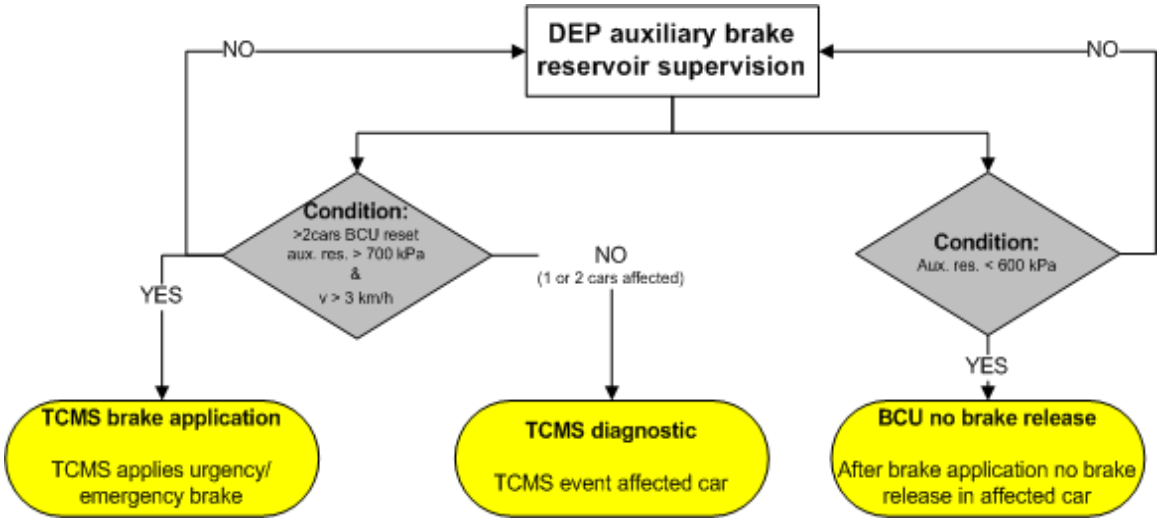
In case a low brake auxiliary pressure < 700 kPa is detected by the BCU and the dedicated IP signal is sent to TCMS, an alarm event is shown to the driver by TCMS.

If in more than two cars low brake auxiliary pressure < 700 kPa is detected by the BCU, TCMS immediately requires an urgency/emergency brake application until the affected cars has been isolated by isolation cock RIEP-DIST.
Isolation cock RIEP-DIST vents the complete brake auxiliary reservoir pressure.

After a brake demand has been ordered and the release command is given, the BCU will not release the DEP brake in the affected vehicle if the auxiliary reservoir pressure is < 600 kPa.
In this case the vehicle has to be isolated by isolation cock RIEP-DIST 04.29, the same action as for TCMS brake reservoir pressure supervision has to be performed to disable the complete brake performance of this car.

Both kind of parallel auxiliary brake reservoir supervision by BCU and TCMS are shown in diagram 3.1.1.5 - D3.

<3.1.1.5 - D3: DEP inexhaustibility/auxiliary brake reservoir supervision>



The TCMS DEP inexhaustibility/auxiliary brake reservoir supervision and BCU auxiliary reservoir supervision is disabled in case the complete brake performance of one car is isolated by RIEP-DIST isolation cock. This is not the case if only the direct brake isolation cock RIEP is in isolated position.

3.1.1.6 *Dragging brake detection*

The dragging brake detection alerts the driver about not released or unwanted applied brake for a defined time during driving.

The alarm is generated in the BCU and transmitted via IP to TCMS. In case an alarm is detected the indication lamp "Brake applied" will be illuminated on driver's desk and an alarm is shown on the IDU.

The dragging brake monitoring is performed also by TCMS using the pressure switch PCF1/PCF2 signal acquired by safe digital inputs.

Dragging brake diagnostic is managed as priority 1 by the BCU. In case of failure for TCFx of one bogie OR in case of TCFx and TCFy mismatch for pressure signal more than 0,15 bar OR in case of BCU not refreshing correctly the life signal, the dragging brake diagnostic is allocated to the TCMS if there are not failures on TCMS pressure switch.

If there are multiple failure leading to the unavailability of TCF signals with PCF declared faulty, dragging brake diagnostic is not available and a dedicated message is provided.

Failure	**Impact for the service (respective to the dragging brake diagnostic only)**
Single failure or multiple failures of TCF	None, TCMS in charge for the dragging brake diagnostics
Single failure or multiple failure of PCF	None, BCU in charge for the dragging brake diagnostics

TCF failure/failed adjustment (OR BCU life not refreshed) & PCF failure **on the same bogie**	Dragging brake not available event. It is necessary to isolate the affected bogie / car at the first convenient time. It is not necessary to isolate the bogie/car immediately because it is a scenario of 2 failures and a limited time where it is not realistic to have a dragging brake in the specific position. Anyway, in case of dragging brake protection not available it is necessary to proceed to isolate the bogie or the car or restore the functionality in not more than 72h.

[Rev.2] If there is a mismatch between BCU status and train status and train status is in coasting or traction and BCU status is in braking or emergency for 15s, then a dragging brake event is triggered.

3.1.1.7 *Direct Brake Isolation Cock Monitoring*

The brake function of one car is possible to be isolated mechanically by closing the isolation cocks for one bogie or for one complete car. In normal conditions all isolation cocks are in not isolated position. Each isolation cock has an electrical contact for monitoring and diagnostics.

The following isolation cocks are monitored by BCU and/or TCMS, refer to diagram 3.1.1.5 - D2:
- RICP 04.27 isolation cock of complete brake frame incl. air suspension for maintenance purpose
- RIEP-DIST 04.52 isolation cock for complete brake pressure (direct and indirect brake)
- RIEP 04.53 isolation cock for direct electro pneumatic brake
- RICF1 04.38/01 isolation cock cylinder pressure bogie 1
- RICF2 04.38/02 isolation cock cylinder pressure bogie 2

The isolation cock RICP 04.27 is used during maintenance to vent the pressurized air of the complete brake frame including the main pipe reservoir.

In case of faulty indirect brake the isolation cock RIEP-DIST 04.52 has to be isolated, in this case the complete car has no brake performance available and the auxiliary brake reservoir is vented.
If only the direct brake is faulty, the isolation cock RIEP 04.53 has to be isolated. The indirect brake is still working without any restrictions.

In case of problems with the bogie brake equipment, each bogie can be isolated separately by using isolation cock RICF1/2 04.38. In case one RICF isolation cock is in isolated position the direct service brake is disabled in complete car and not considered available for blending calculation, the missing force is distributed to the remaining available cars (refer to blending function 3.1.1.3).
The indirect brake and direct emergency brake is not affected.

3.1.1.8 *Failure Information*

3.1.1.8.1 Normal condition

The Master controller is used to control DEP brake mode in direct service brake mode. ED brake is used as much as possible and in case of insufficient ED brake effort additional friction force is added in trailer bogies and on motor bogies.

3.1.1.8.2 Failure cases

1. Achieved DEP service brake lower than requested in one car
2. Achieved DEP service brake higher than requested in one car
3. Achieved DEP brake force lower than requested in more than one car
4. DEP brake failure in one or two cars

5. DEP brake failure in more than two cars

3.1.1.8.3 Failure detection

1. BCU internal supervision between brake demand, Tpil and TCF pressure detects discrepancy and generates an missing brake event via TCMS. WSP supervision detects error on EWS dump valves and generates event via TCMS.

TCF monitoring (FTIT10002-E16PSP—1041)
WSP monitors BC pressure via a pressure transducer installed on each dump valve and it forward BC pressure information to TCMS via ETH.

Hardware signal monitored: From: BC pressure transducer (on dump valve) To: WSP Type: AI 4-20 mA / 0-10 bar Pins X6-4d (+15V) X6-2d (Signal)

Value:	Current [mA]	<2	Pressure [bar]	0xFFFF	Status:	TCFx pressure sensor faulty
		2= I = 19		0 = P = 10		Reset failure
		> 19		0xFFFF		TCFx pressure sensor faulty

2. BCU internal supervision between brake demand, Tpil and TCF pressure detects discrepancy and generates an overbrake event via TCMS.
3. Same as above for several cars.
4. BCU internal diagnostic, DEP brake faulty event is set to TCMS. Based on this event TCMS re-distributes brake force to all remaining cars (blending).
5. BCU internal diagnostic, DEP brake faulty event is set to TCMS in more than two cars. Based on this event and amount of faulty cars TCMS alarm is set to the driver to switch to indirect service brake mode (back-up mode).

3.1.1.8.4 Failure handling

1. DEP in local car will be disabled in case of brake force discrepancy beyond acceptable limit. Lost brake force is compensated by additional brake force on other cars. No effect on train level.
2. DEP in local car will be disabled in case of brake force discrepancy beyond acceptable limit. Loss of DEP brake in one car is compensated by additional brake force on other cars. No effect on train level.
   Loss of DEP brake in more than two cars, stopping of train and switch to IEP brake mode.
3. DEP in local car will be disabled in case of brake force discrepancy beyond acceptable limit. Lost brake force is compensated by additional brake force on other cars. No effect on train level.
4. DEP in local car will be disabled in case of direct brake failure. Lost brake force is compensated by additional brake force on other cars. No effect on train level.
5. Driver shall activate indirect service brake mode as specified in chapter 3.2 and isolate faulty DEP brake by isolation cock RIEP if necessary.

3.1.2 Requirements

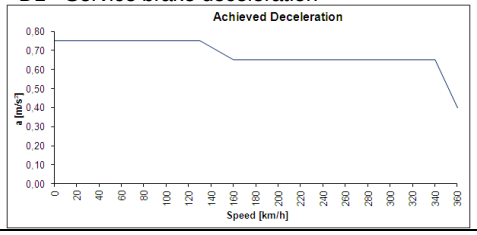
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3159	TCMS shall set the corresponding SW input signal to invalid if no input signal is received from a reporting unit or if the reporting unit reports itself or a specific signal is set as faulty (e.g. TCMS digital input reports an input channel error). Otherwise, if signals are received without error, TCMS sets the corresponding SW input signal to valid. If signals are invalid an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3158	If redundant signals are valid TCMS shall check whether these signals are plausible (inverse or equal) regarding their respective logic. In case of a mismatch, TCMS shall invalidate both signals and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.2.1 *DEP Brake Control*

ID	2F_05.02.Zefiro-Europe.TRS.16
Description	Service brakes shall be applied by: – The driver via two control handles: 1. one master controller (drive/brake control handle) for blended brake 2. one DBV brake control handle for pneumatical brake – the speed control & regulation function – The ERTMS and other national ATP systems
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1510	Direct EP brake demand shall be derived from the master control handle.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1511	Maximum service brake (100%) shall be applied by ATP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1512	Direct EP brake shall be applied by speed regulation function (defined maxmimum brake demand).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.549	The highest brake demand from master control handle, ATP and speed regulation (train control) shall have priority.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.84
Description	The service braking force demand is communicated to all BCU and PCU along the train using communication buses and logical train lines.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.617	The brake demand by TCMS shall be sent via the IP to the BCU in kN. The brake demand shall be load compensated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.60	The BCU shall generate c-pressure according to the brake force demand in kN received from TCMS via IP	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.616	The BCU shall send the achieved brake force based on the measured c-pressure (04.28 Tpil sensor) per car to TCMS via IP in a the safe class SIL 2 Ethernet signal list.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1513	Maximum deceleration (100% Service Brake) with and without ED-brake shall be based on following deceleration characteristic: – 0.4 m/s² increase linearly to 0.65m/s² between speeds 360 - 340 km/h – 0.65 m/s² between speed range 340 km/h to 160 km/h – 0.65 m/s² increase linearly to 0.75m/s² between speeds 160 - 130 km/h – 0.75 m/s² between speed range 130 km/h to 0 km/h	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1961	<D1 - Service brake deceleration>  	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.615	The brake force demand shall be jerk limited. Maximum jerk shall be 0.6 m/s³.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.1526	The service brake DEP of one car shall be considered as not available if one of the following conditions are true: – RIEP isolation cock DEP is isolated – RIEP-DIST isolation cock brake frame R-pressure is isolated – RICP isolation cock brake frame is isolated – RICF1 or RICF2 isolation cock for bogie are isolated – DEP brake faulty (Service brake) – BCU communication fault (life sign watchdog for both ethernet lines) The affected car shall not be considered in brake blending calculation.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3025	{In case the communication is lost (life sign) to more than 2 BCUs for both ethernet lines OR "DEP brake faulty" IP signal is set by more than 2 BCUs} AND trainline "braking not" is de-energised the TCMS shall open the urgency brake loop in case direct brake mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2596	If the communication from BCU to TCMS is lost the direct brake c-pressure shall be released if no urgency or emergency brake is active.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1527	The BCU shall set IP signal "DEP brake faulty" when the DEP brake is disturbed (summary event) in case of: – Service brake integrity (Target not according control pressure) or – Open/short circuit EV1/2 or – Tpil transducer faulty	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1541	If BCU reports "DEP brake faulty" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.790	The achieved brake force in kN (0...100 kN) per car shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR105	[Rev. 02] BCU and WSP shall share with TCMS in the safe class SIL 2 Ethernet signal list: - Brake cylinder pressure per axle (TCF) - Isolation cocks acquired - DEP pressure target (Tpil) - IP pressure target (TDIS) TCF (req.1041); RICF1/2 (reqq. 0057,0058) RIEP-DIST (req.0052) RICP (req. 0054) RISOSP (req. 0055) TDIS (req. 1552)			BCU	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.340	Brake cylinder pressures (sensor TCF) in kPa (0...1000 kPa) shall be indicated per axle on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.1001	The overall brake status applied/released/isolated of the friction brake shall be indicated per bogie on the IDU based on the WSP-CU IP signal "brake cylinder pressure present BG1/2" derived of TCF sensors and the brake isolation cocks RICF1/2, RIEP-DIST and RICP. In case the WSP-CU is not available (communication fault) in one car the brake status shall be inidcated per bogie based on the PCF1/2 pressure switch TCMS input.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3878	The direct brake status OK/faulty/isolated of the DEP friction brake shall be indicated per car on the IDU based on following signals: – OK if IP signal "DEP brake faulty" is not set and communication to BCU is ok – Faulty if IP signal "DEP brake faulty" is set OR communication is lost to BCU – Isolated if isolation cock RIEP (BCU signal) OR RIEP-DIST (BCU OR TCMS input) OR RICP (BCU OR TCMS input) OR both RICF1 AND RICF2 (BCU OR TCMS input) are isolated	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1963	When the ED-brake is active (each MCM per axle has ED brake achieved in one car) the IP signal "Dynamic brake activated" shall be sent to BCU and WSP-CU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR05	If the communication between TCS and a BCU is lost on both channels: - The BCU shall not apply the brake effort in that car The TCS shall create the “DEP brake Faulty event”	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR06	If the communication between TCS and a BCU is lost on both channels: The TCMS shall create the “DEP brake Faulty event”	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_05.02.Zefiro-Europe.CONCEPT.HR07	If the communication between TCS and a PCU is lost: - the ED from that PCU is considered faulty. - The friction effort calculation takes into account the ED missing in that car as reported in 2F_05.02.Zefiro-Europe.CONCEPT.612	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_05.02.Zefiro-Europe.CONCEPT.HR08	If the communication between TCS and a PCU is lost: - The PCU shall not apply the ED effort in that car	Derived Requirement	4S-05 Propulsion		SIL0
2F_05.02.Zefiro-Europe.CONCEPT.3839	When traction is active the IP signal "Traction activated" shall be set by TCMS to BCU and WSP-CU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.165	There shall be three digital train lines 0-110 V connected to all BCUs. These trainlines are: – Traction (motoring), energized if master controller in "traction" position – Not braking, de-energized if master controller in "braking" position – Emergency operation active if the switch is turned to "em operation" The train lines are used for emergency operation as backup to the IP .	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1028	The service brake demand shall be received as backup on one digital braking train line during the	Derived Requirement	4S_07.-.Braking System		SIL 2

	emergency operation mode. The trainline shall be de-energised if the master controller is in brake position. In neutral or in traction position the train line braking is energised. The BCU shall apply a defined DEP service brake if all of the following conditions are true: – The "braking not" train line is de-energised – The "emergency operation" train line is energised.				
2F_05.02.Zefiro-Europe.CONCEPT.2971	The defined DEP service brake force in case of emergency operation active as defined in concept 1028 shall be the following: DM/M car = 26 kN TT/T car = 40 kN	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2024	The logic behind the tree trainlines is the following. The trainlines shall only be evaluted if "em operation" is active. The braking trainline is inverted. Emergency operation trainline / Braking trainline / Traction trainline Braking: 1 / 0 / 0 Traction: 1 / 1 / 1 Coasting: 1 / 1 / 0 Not valid: 0 / 0 / 0	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2025	If the trainlines status reported from each available BCU is not the same status based on pattern defined in concept 2024, an event shall be set to indicate inconsistency between all available BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1861	The status of the three train lines traction, brake and emergency operation shall be sent by BCU to TCMS via IP signals: "Traction train line energised" "Braking train line energised" "Emergency operation train line energised"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2972	In case the traction and brake train line are inconsistent (brake 0 and traction 1) the BCU shall sent IP signal "DIA traction/brake train line inconsistency".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2973	If the traction/brake train line inconsistency signal is sent by the BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2886	If emergency operation mode is active ED brake shall be blocked and only friction brake force demand shall be sent to the BCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.157
Description	The BCU's and PCUs in a half train shall be connected via IP for exchange of status and brake control signals.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.168	The BCU and WSP-CU shall be connected to IP bus for data exchange with TCMS and other subsystems.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1827	The BCU and WSP-CU shall be connected to the IP network .	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_05.02.Zefiro-Europe.TRS.180
Description	If service brake is requested manually, the traction shall be blocked until traction is requested by the driver using the master controller. No automatical restore is allowed.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.171	If the master controller is put into any service brake position, traction shall be blocked by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	until the master controller is moved to coast or traction position.				
2F_05.02.Zefiro-Europe.CONCEPT.1032	If the DBV is put into full service brake position (MB = full service brake SW4), the traction block by TCMS shall be set until the DBV is restored in brake release position (RP = release position SW2) and master controller has to be moved back to coast position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.2 *Load Compensation*

ID	2F_05.02.Zefiro-Europe.TRS.150
Description	The Service brake force shall be load compensated
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.536	The pressure transducers TS1 (05.23/01) for bogie 1 and TS2 (05.23/02) for bogie 2 shall be read by the BCU. The data shall be sent to TCMS via IP as pressure values in kPa for each pressure transducer indvidually.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.352	Load weight values shall be updated after the TCMS door released status has been reset (not released).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1276	The load value in kg to be used in the brake blending function shall be calculated by TCMS based on the average air suspension pressure in kPa according following table concept 2606 linear from service weight to maximum exceptional load:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2606	Pressures are defined in Table 1 - Mass and suspension parameters	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1750	The average pressure shall be calculated out of the TS1 and TS2 pressure sensors for each car. Average pressure = (TS1 + TS2) / 2 This pressure corresponds linearly to a load in this car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1751	Mass is calculated in each car individually using the following formula: w = Mass [kg] = (bogie type x + bogie type y) mass [kg] + 4* load pressure [kPa]*K*10 / 9,81) Where K for VZI R1: KDM1 20,51 KTT2 20,34 KM3 20,32 KT4 20,1 KT5 20,1 KM6 20,32 KTT7 20,49 KDM8 20,39	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1752	The bogie masses are (parameters for VZI R1 [kg]): **DM1** | **CARRELLO A** | 10062	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	CARRELLO B	9978					
TT2	**CARRELLO A**	7445					
	CARRELLO B	7445					
M3	**CARRELLO A**	9978					
	CARRELLO B	9978					
T4	**CARRELLO A**	7445					
	CARRELLO B	7447					
T5	**CARRELLO A**	7447					
	CARRELLO B	7445					
M6	**CARRELLO A**	9978					

	CARRELLO B 9978					
TT7	**CARRELLO A** 7445					
	CARRELLO B 7445					
DM8	**CARRELLO A** 9978					
	CARRELLO B 10062					
2F_05.02.Zefiro-Europe.CONCEPT.1753	The total train mass shall be calculated as the sum of all cars (mass in all cars).		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1754	Dynamic Mass for each car is calculated using the following formula: Dw = Dynamic Mass [kg] = [(bogie type x + bogie type y) mass + 2*dynamic mass bogie] + 4*load pressure [kPa]*K*10 / 9,81) Dynamic mass trailer bogie = 1202 kg Dynamic mass motor bogie = 1378 kg K as per 2F_05.02.Zefiro-Europe.CONCEPT.1751		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1755	Total dynamical train weight is calculated per 8-car segment: Total dynamical train weight shall be calculated as the sum of rotating massed (dynamic weight in each 8-car segment)		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.605	The load sensor TS1 shall be considered faulty if sensor signal is out of range (valid range 4-20 mA). Then the IP signal "Load pressure sensor TS1 faulty" or "Load pressure sensor TS2 faulty" shall be set by the BCU.		Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1004	If load sensor TS1 is indicated faulty from BCU via IP , an event shall be set.		Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1543	If load sensor TS2 is indicated faulty from BCU via IP , an event shall be set.		Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.607	If one pressure sensors for load weighing (TS1 or TS2) in the car is defective TCMS shall consider the second valid sensor.		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1198	If both sensors (TS1 and TS2) are defective or both secondary suspension systems are isolated by isolation cock RISOSP (05.02/01) then the		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	default weight pressure value shall be used for load calculation.				
2F_05.02.Zefiro-Europe.CONCEPT.1516	Isolation cock RISOSP (05.02) to isolate the secondary air suspension of one car shall be read by the BCU. If the cock is in isolated position the IP signal "Secondary suspension isolated (RISOSP)" shall be set.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2377	RISOSP isolation cock signal shall be connected to TCMS input. The input is set (1) if the isolation cock is in isolated position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2376	An event shall be set if the IP signal "Secondary suspension isolated (RISOSP)" is set by the BCU OR the TCMS input is set that the cock is isolated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1518	An event shall be set if IP signal "Secondary suspension isolated (RISOSP)" and TCMS input of isolation cock RISOSP are not consistent.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1006	Suspension overpressure: If TCMS detects the suspension pressure is >= 2F_05.02.Zefiro-Europe.CONCEPT.2606 Maximal exceptional load pressure + 0,8 bar for 15 minutes on one bogie an event shall be set Reset if the condition is not true Filter if transducer faulty	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1520	Suspension low pressure If TCMS detects the suspension pressure is <= 2F_05.02.Zefiro-Europe.CONCEPT.2606 Suspension broken level for 15 minutes an event shall be set Reset if the condition is not true Filter if transducer faulty OR (MRP lower than 6 bar & speed < 5 km/h) OR suspension isolated OR Ethernet double failure communication	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2418	If TCMS detects a overload pressure more than maximum exceptional weight on the average pressure value the maximum exceptional weight pressure shall be used for brake calculation.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2417	If TCMS detects a pressure less than service weight on the average pressure value of the two bogies the service weight shall be used for brake calculation.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1519	CBM: If the two pressure trandsducers TS are valid and have a pressure deviation of more than 800 kPa for v < 5 km/h and 30 minutes an event shall be set for possible failure of pressure sensor or suspension system (leakage) Filter for MP < 6 bar	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.1.2.3 *Blending*

ID	2F_05.02.Zefiro-Europe.TRS.103
Description	ED braking and blending shall only be enabled if – communication ok between the entities involved (BCU,PCUs) in blending control in the effected car – DEP brake is available and without failure In other cases ED brake shall be disabled in the effected car.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.357	The direct brake blending shall be considered as available if:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	– DEP is reported as not faulty in minimum of 6 BCUs per 8-car train (IP signal DEP faulty = low (0)) – Service brake mode is DEP (service brake mode selector switch in position "Direct brake mode") – No IP communication error with more than two BCU per 8-car train				
2F_05.02.Zefiro-Europe.CONCEPT.3001	If the direct brake is considered as not available according concept 357 an event shall be set to inform the driver to switch to "Indirect brake mode".	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1024	If another brake mode than DEP is used, i.e. Indirect brake mode, blending shall be disabled in the whole train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3885	The electro-dynamic brake shall be fade-off all low speed according the ED brake force characterisation curve. On the same time the friction brake shall be increased to compensate the ED brake force. The demanded deceleration on train level shall be kept until standstill avoiding any jerks during fade-off phase.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR09	When the vehicle speed < 6 km/h the ED effort request shall be reset with a ramp of 1 sec and the friction effort request shall increase to compensate the ED reset		4S_09.03.05.-.TCMS Software		
2F_05.02.Zefiro-Europe.CONCEPT.HR010	If the vehicle speed < 10km/h the PCUs reduce the ED availble with a ramp that reset the effort at 5 km/h		4S_05 Propulsion (Propulsion Control)		

ID	2F_05.02.Zefiro-Europe.TRS.149
Description	The brake force for the whole train shall be compensated by cross blending of ED between cars and DEP braking, thus assuring that the required deceleration is obtained in any case. Loss of braking performance in a car shall be compensated by a higher braking force in other cars as high as permitted by the maximum adhesion level at train level.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.117
Description	The service brake function shall cross blend between ED and DEP over all cars in the train and make maximum use of available ED brake power on all motor axles before DEP is used for compensation.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.124
Description	Regenerative braking shall be used whenever possible to the maximum extend.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.153
Description	The total brake force per half train is calculated: $F_{TOT} = a \cdot \sum_i w_i \quad , i=1..4$ a = requested deceleration at train level, wi = weight on car i
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.362	The brake blending control for direct service brake shall be done on 8-car train level.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

1. Calculation of total brake reference per 8car train

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1782	The amount of service brake demand shall be a percentage value 0-100% originating from the master control handle, ATP system or automatic train control. The percentages for the master control handle shall be linear for the available brake positions of the master controller. For the master control handle the service brake demand shall be stepless from 0% to 100%. For ATP request the request shall be 100%	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.174	Total brake force for the 8car train shall be calculated with following formula: $B_{ref} = \frac{B_r \cdot a \cdot D_w}{1000} - T_r$ Br = brake request 0-100% a = deceleration Dw = Dynamic train weight (sum of all Dw per car) Tr = train resistance in kN for the train	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1783	Resistance for 8cars train (15°C): $T_r = \frac{3670 + 25.5 \cdot v + 0.47 \cdot v^2}{1000}$ v = train speed (km/h)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2419	Resistance for multiple 8+8car train (15°C): $T_r = \frac{7340 + 51 \cdot v + 0.84 \cdot v^2}{2000}$ v = train speed (km/h)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2. Calculation of dynamic brake (Elref) content of Bref

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1785	Service brake on train level: On a 8car train level it is expected that 100% ED-brake from 16 MCMs (8 motor bogies) shall give following brake force available (MaxAed) depending on the line voltage, train speed and propulsion internal limitations: – AC 25/15 kV shall be 305 kN of dynamic braking effort – DC 3/1.5 kV shall be 266 kN of dynamic braking effort	Derived Requirement	4S_05.-.Propulsion		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1786	The demanded dynamic brake effort reference percentage 0-100% from TCMS to the PCU is always a percentage of the available maximum dynamic brake (MaxAed) from one motor converter based on the available line voltage. The reference is constant and independent of train speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1787	The maximum available dynamic brake (MaxAvEl, kN) in the 8-car train shall be calculated by: MaxAvEl = number of available MCMs * MaxAed (from one motor converter)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	If MaxAvEl is the maximum available dynamic brake from all MCMs in the TBU, the dynamic brake reference AvEl for the TBU shall be calculated by: If Bref is ≥ MaxAvEl → AvEl = MaxAvEl If Bref is < MaxAvEl → AvEl = Bref				
2F_05.02.Zefiro-Europe.CONCEPT.HR37	TCMS shall send AvEl value for each car to the BCU	Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.371	Each of the PCUs (in a 8-car train) shall sent the available ED brake force in kN per MCM (MaxAvEl) via IP signal "Available braking effort for MCM1...4" to TCMS and BCU (it is possible to send to TCMS and then TCMS to BCU and not directly PCU to BCU).	Derived Requirement	4S_05.-.Propulsion		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.183	The dynamic brake demand AvEl shall be sent to the PCUs via the IP signal "Traction/Dynamic brake demand" to the available MCMs. AvEl at this point shall also compensate for loss of dynamic braking anywhere in the train by adjusting AvEl per MCM to compensate for non effective MCMs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.154
Description	The mechanical friction brake force during blending in a half train is calculated: $F_F = F_{TOT} - (EDBA_1 + EDBA_2)$ EDBA1 = EDBA achieved on first motor coach of half-train, EDBA2 = EDBA achieved on second motor coach of half-train.
Safety level	N/A

3. Calculation of friction brake (FBref) content of Bref

When brake is applied at t=0s it is only possible to use the dynamic brake available curve (AvEl) which is sent from each MCM to TCMS to calculate the required amount of friction brake. This is because it takes several seconds to calculate the actual dynamic brake achieved AhEl from each MCM.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1796	This means that from t = 0s to t = 2s that $FB_{ref} = B_{ref} - AvEl$	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.612	After t = 2s, the amount of achieved dynamic brake (AhEl) shall be used to calculate FBref $FB_{ref} = B_{ref} - AhEl$	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1797	The friction brake demand in kN is sent to the BCUs via the IP to the 8 BCUs in the train, refer to calculation of TrBref and MoBref.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3855	In case slide is detected on any axle of one car the ED brake achieved value AhEl shall be kept with the same value until the slide is not active any more or the brake demand has been changed to avoid compensation of friction brake force during ED brake slide regulation.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.155

Description	If term FF ¹ 0, additional friction effort is needed to obtain the required deceleration at half-train level. In this case term FF is shared between motor and trailer axles: FF is applied on trailer axle first (where the adhesion is less stressed due to ED brake priority) till the adhesion limit and the remaining portion (if any) on all axles till the adhesion limit.
Safety level	N/A

4. Calculation of friction brake (FBref) sent to each brake control unit

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1799	The friction value sent out to each BCU for service brake mode is based on factor x_tr for trailer car and x_mo for motor car multiplied with Fbref per train: TrBref shall be 0.15 of FBref x_tr = 0.15 MoBref shall be 0.1 of Fbref x_mo = 0.1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR121	**Minimum friction brake demand per car** [Rev. 03 – BL3] If the calculated friction brake demand per car (MtBrReq) is lower than (8.36 kN for motor cars and 8.28 kN for trailer cars), on less than 3 cars, then TCMS set to 0 the brake request to the BCU (MtBrRef) on a maximum of 3 cars chosen on dauly basis according to the day number respective to the month (1-31)..	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

5. Limiting brake force per car due to adhesion limits

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1808	Before the friction brake demand MoBref (kN) and TrBref (kN) are sent to the BCUs adhesion A (non-dimensional) limits shall be considered in the maximum brake force demand.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1809	If v ≤ 200 km/h A = 0.15 If v > 200 km/h $A = 0.15 + (0.10 - 0.15) \cdot \left(\frac{v - 200}{350 - 200} \right)$ Maximum force is calculated based on A = acc. TSI and g = 9.81 m/s².	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1817	The maximum brake force due to adhesion at car level = MaxAdhCar (kN) $MaxAdhCar = \frac{w_{car} \cdot A \cdot g}{1000}$ Where w_car = car weight (kg)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1818	By comparing MoBref and TrBref with the respective cars MaxAdhMo and MaxAdhTr, the amount of friction brake demand requested is limited on the maximum brake force due to adhesion.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	MaxAdhMo and MaxAdhTr shall be calculated on car level as specified in concept 1817. In a trailer car the brake force TrBref shall be less than equal to MaxAdhTr. In a motor car the brake force shall be controlled by: If MoBref + AhEl > MaxAdhMo the friction brake reference in the motor car shall be limited to: $MoB_{ref} = (MaxAdhMo - AhEl)$ AhEl = Maximum value of electro-dynamic brake achieved at car level MaxAdhMo = Maximum brake value due to adhesion				

6. Adjustment of MoBref and TrBref depending on achieved braking on train level

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1820	In case of 1 motor car with unavailable electrodynamics brake the brake force shall be shared according:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2975	For motor car with MCM available the factor xmo is: x_mo = 0.06 The new MoBref is: Mo_Bref = Fbref * 0.06	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1821	For motor car with no MCM available the factor xmo is: x_mo = 0.1 The new MoBref is: Mo_Bref = Fbref * 0.1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1822	For the trailer cars the factor xtr is: x_tr = 0.18 The new TrBref is: Tr_Bref = Fbref * 0.18	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1823	In case of 2 motor cars with unavailable electrodynamics brake in the 8car train the brake force shall be shared according:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2976	For motor car with MCM available the factor xmo is: x_mo = 0.05 The new MoBref is: Mo_Bref = Fbref * 0,05	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.1824	For motor car with no MCM available the factor xmo is: x_mo = 0.1 The new MoBref is: Mo_Bref = Fbref * 0.1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1825	For the trailer cars the factor xtr is: x_tr = 0.175 The new TrBref is: Tr_Bref = Fbref * 0.175	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2977	In case of 3 motor cars with unavailable electrodynamics brake in the 8car train the brake force shall be shared according:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2978	For motor car with MCM available the factor xmo is: x_mo = 0.04 The new MoBref is: Mo_Bref = Fbref * 0,04	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2979	For motor car with no MCM available the factor xmo is: x_mo = 0.1 The new MoBref is: Mo_Bref = Fbref * 0.1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2980	For the trailer cars the factor xtr is: x_tr = 0.165 The new TrBref is: Tr_Bref = Fbref * 0.165	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

7. Adjustment of MoBref and TrBref depending on available BCUs on train level

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2981	On a 8-car train level it shall be implemented that the friction brake during direct service brake can be adjusted to compensate unavailable brake on car level (BCU)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2994	If no car is unavailable then no compensation shall be done. If one car has failed friction brake then FBmis in kN is: $FB_{mis} = \sum_{i=1}^{4}\left(\left(1 - BCU_{moav_i}\right)\cdot MoB_{ref_i} + \left(1 - BCU_{trav_i}\right)\cdot TrB_{ref_i}\right)$ MoBrefi and TrBrefi: Are the friction brake forces calculated at section 4 of this chapter for each car. BCUmoav: Is the BCU availability of the motor car i (binary value: 1 = available, 0 = failed). BCUtrav: Is the BCU availability of the trailer car i (binary value: 1 = available, 0 = failed).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2995	The following factors shall be calculated which take into account the number of brake discs of each car and the availability of its BCU: $Disc_{av} = \sum_{i=1}^{4}\left(Disc_{mo}\cdot BCU_{moav_i} + Disc_{tr}\cdot BCU_{trav_i}\right)$ Discmo and Disctr: are the number of brake discs of each type of car (8 for motor cars and 12 for trailer cars).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.2996	The calculation of the compensation factors are: For motor cars: $x_{moav} = \frac{Disc_{mo} \cdot BCU_{moav}}{Discs_{av}}$ And the compensation force of the motor car is (kN): $Mo_{comp} = FB_{mis} \cdot x_{moav}$ For trailer cars: $x_{trav} = \frac{Disc_{tr} \cdot BCU_{trav}}{Discs_{av}}$ And the compensation force of the trailer car is (kN) $Tr_{comp} = FB_{mis} \cdot x_{trav}$	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2997	For the available BCUs the new reference brake force is: $MoB_{ref} = FB_{ref} \cdot x_{mo} + Mo_{comp}$ $TrB_{ref} = FB_{ref} \cdot x_{tr} + Tr_{comp}$ Where x_mo = the factor decided in section 4 for the motor car and x_tr = the factor decided in section 5 for the trailer car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2998	Mo_comp and Tr_comp has been added to remaining BCUs, the MaxAdhCar shall remain the maximum amount of braking that can be demanded from one car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2885	If MoBref is > 2 kN and < 8.36 kN the minimum demanded friction brake force shall be 8.36 kN. If TrBref is > 2 kN and < 8.28 kN the minimum demanded friction brake force shall be 8.28 kN.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3841	In case the amount of the friction brake calculated at train level is less than the minimum demanded friction brake force of one single car, all the amount of friction brake at train level shall be applied in only one car trailer car, (selected T car will be changed every 24h to minimise differences in pad wear between T cars). In case the selected trailer car is not available, the minimum demanded friction brake force shall be sent to the other trailer car. At the moment where the amount of demanded friction brake will be higher than the minimum demanded friction brake force of one single car, the next car shall apply the minimum brake force to have a maximum overbrake according the minimum brake force defined in concept 2885. In case the total friction brake force is more than the minimum value multipled by available cars, the normal brake effort distribution shall be used for the blending calculation as defined in 3.1.2.3.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.2.4 *Dynamic Brake*

ID	2F_05.02.Zefiro-Europe.TRS.178
Description	

Line Voltage [kV]	Maximum Power at Rail, Regenerative [%]	Maximum Power at Rail, Regenerative & Rheostatic [%]
17,5	0	
19	77,8	
22.5	100	
25	100	
27,5	100	100
29	0	53,33
31	0	0

Safety level	N/A

Electro-dynamic brake force limitation at 25kV AC and 15kV AC. ED brake curve shall use regenerative when line voltage can be used or the rheostatic brake if line voltage is limited.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.390	All dynamic brake references shall be based on theoretical dynamic brake curve according to reference document defined in [7] for 25 kV AC.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_05.02.Zefiro-Europe.TRS.95

Description	Electro-dynamic braking shall work independently of the line voltage. The required braking power according to the nominal brake curve shall be compensated by rheostatic brake force by converting power into heat in the braking resistors if the line is not receptible or power feed back not allowed.
Safety level	SIL 0

For the ED brake system, a rheostatic brake is installed to compensate for limited line receptibility. Rheostatic brake can compensate 100% of the ED power for 3 kV DC and 1,5 kV DC.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.70	Rheostatic brake shall be used to convert kinetic brake energy into thermal energy when catenary line is unable to receive it. Rheostatic shall be able to convert 5,6 MW rail power (8 car train; not continously) into heat.	Derived Requirement	4S_05.-.Propulsion		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.384	In case the regenerative brake power is suddenly switched off due to catenary conditions (that can be handled by the propulsion system without protective actions) the rheostatic brake shall be used. The available ED brake effort shall be limited to the value of rheostatic ED brake.	Derived Requirement	4S_05.-.Propulsion		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.176
Description	

Line Voltage [V]	Maximum Power at Rail Regenerative [%]	Maximum Power at Rail Rheostatic [%]
2000	0	0
2600	40	100
2700	77,8	100
3000	100	100
3400	100	100
3600	100	100
3700	50	91,7
3900	0	75
4200*	0	0
* Short term operatiion		

Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.389	All dynamic brake references shall be based on theoretical dynamic brake curve according to reference document defined in [7] for 3 kV DC.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2609	For commissioning purpose only: An IDU soft key shall be present to disable dynamic brake.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1791	For dynamic brake cut out during testing, a push button can be operated from the IDU and ED brake effort reference demand shall be set to 0 for the whole train.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3989	The push button described with requirement 1791 shall be available to the driver in the active cab and at any speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3992	When the driver operates a cab change the inhibition shall be stored and thus still applied from the new active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3993	The cut in/off of the ED can be operated only when the MCH is in coast and the request shall	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	be also transferred to the traction converter (via the dedicated IP variable).				
2F_05.02.Zefiro-Europe.CONCEPT.3995	In case the precondition for the cut in/out of the ED are not met an event shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3996	The cut in/out function can be actuated also with the LCB closed and at a speed >0.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3991	In case of Multiple operation the inhibition is valid for the whole train (both consist).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3988	In case the dynamic brake cut out push button is operated an event shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR123	If the dynamic brake is excluded (e.g. a failure event) in an axle or a bogie, the ED brake is kept active where the failure is not present. The friction brake and ED brake shall be blended in order to respect the adhesion limit and the missing force shall be compensated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.5 *Braking Rheostat Cooling*

The PCUs control the cooling of the braking resistances activating the fans at half or full speed depending on the rheostat temperature and vehicle speed or the braking power. At low speed the fan speed is not full if the temperature is not high to reduce the noise

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR17	The brake resistor fan shall be ordered to run at **half speed** IF the train speed is ≤ "High vehicle speed, upper level (57km/h)" AND the brake resistor temperature is above a predefined level "Brake resistor temp 1/2-speed on, low vehicle speed" (65°C, hysteresis 60°C) . OR the brake chopper power is above a predefined level "Brake resistor power 1/2-speed on" (8kW, hysteresis 2kW).	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR18	The brake resistor fan shall be ordered to run at **Full speed** IF the train speed is ≤ "High vehicle speed, upper level" (57km/h) AND the brake resistor temperature is above a predefined level "Brake resistor temp 1/1-speed on, low vehicle speed" (600°C, hysteresis 300°C). OR the train speed is > "High vehicle speed, upper level" (57km/h, hysteresis 55km/h). AND the brake resistor temperature is above a predefined level "Brake resistor temp 1/1-speed on, high vehicle speed" (65°C, hysteresis 60°C). OR the power to the brake resistor is higher than a predefined level "Brake resistor power 1/1-speed on" (300kW, hysteresis 250kW)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR19	The brake resistor fan shall be **turned off** IF the brake resistor temperature is below a predefined level " Brake resistor temp 1/2-speed off, low vehicle speed " (60°C) and Brake resistor power is below "Brake resistor power 1/1-speed off"" (2kW)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR20	The transition from half fan speed to full fan speed and viceversa is delayed of minimum 2 s from the previous order	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0

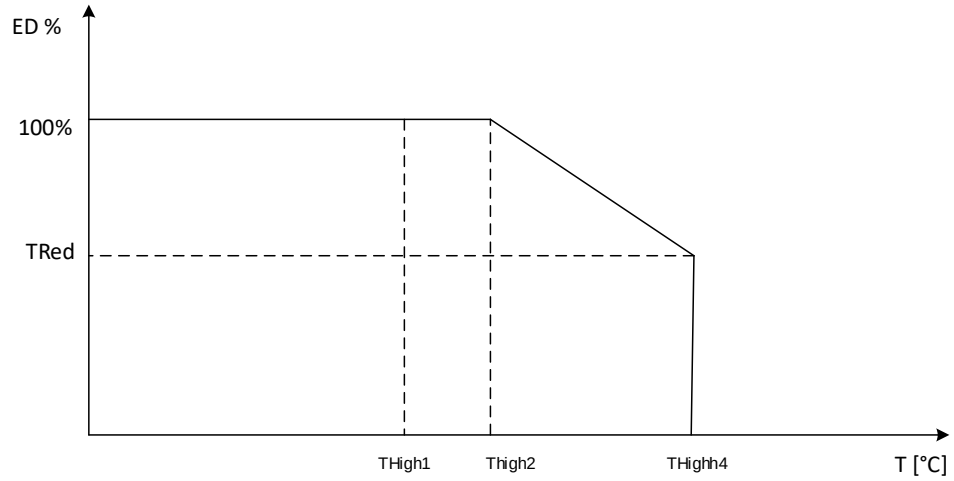
2F_05.02.Zefiro-Europe.CONCEPT.HR21	The check on fan speed (coherence between command and status) has to be verified after 1s from the order	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0

3.1.2.6 *Braking Rheostat Fan supervision*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR22	IF order half speed on and !half fan speed switched on THAN Full fan speed activation	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR23	IF order half speed off and !half fan speed switched off THAN half fan speed activation	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR24	If half speed doesn't work an event shall be created: «*Brake resistor fan half speed fault*»	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR25	IF order full speed on and !full fan speed switched on THAN half fan speed activation	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR26	IF order full speed off and !full fan speed switched off THAN full fan speed activation	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR27	If full speed doesn't work an event shall be created: «*Brake resistor fan full speed fault*»	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR28	IF Full Fan speed on fault AND Half Fan speed on fault THAN ED cut off, EDavaiable sent to TCMS shall be reset	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR29	IF Full Fan speed on fault AND Half Fan speed on fault THAN an event shall be created :"B*rake resistor fan speed fault»*	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0

3.1.2.7 *Braking Rheostat Temperature supervision*

The braking temperature measurement and high temperature alarm (Thepsys device) is acquired from PCUs but can be also estimated by a thermal model taking into account different thermal time constants depending on the status of brake resistor fans and the brake power chopper. The temperature supervision function has in input the chosen value between estimated and measured temperature, the choice criteria is a propulsion internal algorithm.
The ED effort shall be handled with the law depending on temperature reported in the following picture. The ED available shall be sent to TCMS with a limitation applied.



Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR30	IF Rhestat temperature is above THigh2 (600°C) THAN a linear ED reduction from 100% ED available to 50% ED available between Thigh2 and Thigh4 (650°C) is applied	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.HR31	IF Rheostat temperature is above Thigh4 (650°C) OR High Temperature alarm digital input is set THAN ED effort shall be reset	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR32	IF Rheostat temperature is above Thigh4 (650°C) OR High Temperature alarm digital input is set an event shall be created:"Brake resistor high temperature"	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR33	IF T<THigh1 (575°C) THAN Braking Rheostat High Temperature protection shall be reset	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR34	IF Rheostat temperature is above Thigh2 (600°C) for 600 sec THAN a ED effort shall be reset	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR35	IF Rheostat temperature is above Thigh2 (600°C) for 600 sec an event shall be created *"Brake resistor cool-down time has expired"*	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR36	iF 2 High temperatur protection events occur in 300s THAN ED shall be cut off	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0

3.1.2.8 *DEP Monitoring and Diagnostics*

ID	2F_05.02.Zefiro-Europe.TRS.182
Description	The status (applied, released, cut-out, faulty) shall be indicated for the pneumatic brake on a brake cylinder or axle basis.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.356	Functionality and performance of DEP brake shall be supervised continuously.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3925	The auxiliary brake reservoir pressure value TSA shall be send to TCMS via IP signal "Auxiliary brake reservoir pressure (TSA)".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3026	If the brake (auxiliary) reservoir pressure is ≤ 600 kPa (TSA pressure ≤ 600 kPa bar) no DEP brake release command shall be processed by BCU and brake shall remain applied.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1612	If the WSP-CU indicates brake released setting IP signal "Brake cylinder pressure released BG1", the brakes on bogie 1 shall be considered released. In case the WSP-CU is not available (communication fault) in one car the brake status shall be indicated based on the PCF1 pressure switch TCMS input for bogie 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1613	If the WSP-CU indicates brake released setting IP signal "Brake cylinder pressure released BG2", the brakes on bogie 2 shall be considered released. In case the WSP-CU is not available (communication fault) in one car the brake status shall be indicated based on the PCF2 pressure switch TCMS input for bogie 2.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.792	In case the brake cylinder pressure on axle 1/2/3/4 does not increase when requested the signal "DIA Missing brake axle 1...4" shall be sent by BCU via IP to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1279	In case the brake cylinder pressure on axle 1/2/3/4 does not increase if requested indicated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

	by signal "DIA Missing brake axle 1...4" is sent by BCU, an event shall be set per axle.				
2F_05.02.Zefiro-Europe.CONCEPT.793	In case brake cylinder pressure on axle 1/2/3/4 does not decrease when requested the signal "DIA overbrake axle 1...4" shall be sent by BCU via IP to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1280	In case brake cylinder pressure does not decrease on axle 1/2/3/4 when requested indicated by signal "DIA overbrake axle 1...4" is sent by BCU, an event shall be set per axle.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.625	In case traction and brakes (not holding brake) are applied at the same time a fault event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR110	WSP shall provide a TCF sensor runtime diagnostic as per following instruction: If i_current < 2 mA OR > 19 mA, set TCFx in failure. Else: reset failure	Derived Requirement	4S_07.-.Braking System	SIL 2	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1622	If brake cylinder pressure sensors TCF is faulty the signal "DIA TCF1...4 pressure sensor faulty" shall be sent by WSP-CU to TCMS via IP per axle.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.338	If brake cylinder pressure sensors TCF is faulty and signal "DIA TCF1...4 pressure sensor faulty" is sent by WSP-CU, an event shall be set per axle.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1535	If the pressure control valves for DEP brake EV1 has a open circuit the IP signal "DIA open circuit pressure control (EV1 - DEP MV)" shall be set by the BCU to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1536	If the pressure control valve for DEP brake (EV1) has an open circuit and signal is set by BCU, an event shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.982	If pressure control valve for DEP brake EV1 has a short circuit, the IP signal "DIA short circuit pressure control (EV1 - DEP MV)" shall be set by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1282	If pressure control valve for DEP brake (EV1) has a short circuit and signal is set by BCU, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1836	If the pressure control valves for DEP brake EV2 has a open circuit the IP signal "DIA open circuit pressure control (EV2 - DEP MV)" shall be set by the BCU to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1837	If the pressure control valve for DEP brake (EV2) has an open circuit and signal is set by BCU, an event shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1537	If pressure control valve for DEP brake EV2 has a short circuit, the IP signal "DIA short circuit pressure control (EV2 - DEP MV)" shall be set by the BCU to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1538	If pressure control valve for DEP brake (EV2) has a short circuit and signal is set by BCU, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.983	If the EP pilot pressure sensor Tpil is faulty the IP signal "Tpil - pressure sensor faulty (Short/Open)" shall be set by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1283	If the EP pilot pressure sensor Tpil is reported to be faulty an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3559	If the TSA auxiliary reservoir pressure sensor is faulty the IP signal "DIA TSA - pressure transducer faulty (Short/Open)" shall be set by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3560	If the TSA auxiliary reservoir pressure sensor is reported faulty by BCU via IP signal "DIA TSA - pressure transducer faulty (Short/Open)" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.3002	The BCU shall sent the IP signal "DIA service brake integrity failure" in case an inconsistency between service brake demand (DEP brake target pressure) and DEP brake control pressure for more than 2 s.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3003	If the signal "DIA service brake integrity failure" is set by the BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3553	After start-up BCU shall set the IP signal "BCU start-up OK" to TCMS if no hardware failure has been detected.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3554	In case the BCU resets IP signal "BCU start-up OK" and speed v > 0 km/h an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3793	If BCU detects a minor fault (summary event) the IP signal "BCU minor fault" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3789	A watchdog counter shall be sent by BCU to TCMS via IP signal "BCU life sign".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.514	The pressure switch PCF1 and PCF2 per bogie shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1839	If the bogie cylinder pressure switch PCF1/2 TCMS input is set AND WSP-CU IP signal "Brake cylinder pressure released" is set an implausibility event on bogie level shall be set. Status pcf released if pressure is higher than 0,825 bar (set error type 1) Status pcf braked if pressure is lower 0,185 bar (set error type 2) Status pcf braked if pressure is higher than 0,825 bar (reset error type 1) Status pcf released if pressure is lower than 0,185 bar (reset error type 2) Timer for set: 15 s Timer for reset: 5s	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.1.2.9 *Dragging Brake Detection*

ID	2F_05.02.Zefiro-Europe.TRS.202
Description	Dragging brake shall be detected by the brake system and by TCMS and the driver shall be alerted.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
Europe.CONCEPT.HR40	Dragging brake diagnostic "DIA dragging brake alert bogie 1" and "DIA dragging brake alert bogie 2" are in charge to the BCU or TCMS according to the Table 2 Note: for the details of the BCU check diagnostic on sensors, refers to 3.11.2.4.5, 3.11.2.4.6.	Derived Requirement	4S_07.-.Braking System 4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR106	**Dragging Brake not available** If BCU life is not refreshed & BCU pressure switch check bogie X is = check nok OR If BCU transducers faulty bogie x = faulty & BCU pressure switch check bogie X is = check nok	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR124	TOD shall provide an event, major event: Dragging brake diagnostic protection not available on bogie X. **Note**: in this case it is possible to isolate the bogie or the affected car at the first convenient time, not more than 72h, because it is related to a negligible probability event where there are at least 2 failures & dragging on a specific bogie.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR107	If the pressure switch check is not possible the pressure switch use for dragging brake is allowed. The probability to have the missing diagnostic check on a pressure switch & the pressure switch faulty in the same service is negligible.	Information already covered by table	4S_09.03.05.-.TCMS Software	TCMS	
2F_05.02.Zefiro-Europe.CONCEPT.HR108	If BCU detect at least one TCFx pressure sensor faulty on one axle, BCU shall provide Transducer bogie X faulty	Derived Requirement	4S_07.-.Braking System	BCU	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR118a	TCMS shall provide an event related to the dragging brake status, antivalent 2, 2 = BCU in charge for dragging brake diagnostics 1 = TCMS in charge for dragging brake diagnostics 0 = Dragging brake diagnostic not available	Derived Requirement	4S_09.03.05.-.TCMS Software	TCMS	SIL 0

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check ok	Check ok	Check ok	Check ok	Check ok	BCU		
Check nok	Check ok	Check not possible [-]	Check ok	Check ok	Check ok	Check ok	TCMS	BCU	BCU maintenance fault “Transducer bogie 1 setting out of range”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check nok	Check ok	Check not possible [-]	Check ok	Check ok	Check ok	BCU	TCMS	BCU maintenance fault "Transducer bogie 2 setting out of range"
Check ok	Check ok	Check nok	Check ok	Check ok	Check ok	Check ok	BCU	BCU	BCU maintenance fault "Pressure switch bogie 1 faulty"
Check ok	Check ok	Check ok	Check nok	Check ok	Check ok	Check ok	BCU	BCU	BCU maintenance fault "Pressure switch bogie 2 faulty"
Check ok	Check ok	Check ok	Check ok	Check nok	Check ok	Check ok	TCMS	BCU	BCU maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check ok	Check ok	Check ok	Check nok	Check ok	BCU	TCMS	BCU maintenance fault "Transducer bogie 2 faulty (TCF bogie X failure)"
Any	Any	Check ok	Check ok	Any	Any	Check nok	TCMS	TCMS	
Check nok	Check nok	Check not possible [-]	Check not possible [-]	Check ok	Check ok	Check ok	TCMS	TCMS	BCU maintenance fault "Transducer bogie 1 setting out of range" "Transducer bogie 2 setting out of range"

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check nok	Check ok	Check not possible [-]	Check nok	Check ok	Check ok	Check ok	TCMS	BCU	BCU maintenance fault "Transducer bogie 1 setting out of range" BCU maintenance fault "Pressure switch bogie 2 faulty"

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check nok	Check ok	Check not possible [-]	Check ok	Check nok	Check ok	Check ok	TCMS	BCU	BCU maintenance fault "Transducer bogie 1 setting out of range" BCU maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check nok	Check ok	Check not possible [-]	Check ok	Check ok	Check nok	Check ok	TCMS	TCMS	BCU maintenance fault "Transducer setting out of range" BCU maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"
Any	Any	Check not possible [-]	Check ok	Any	Any	Check nok	TCMS	TCMS	
Any	Any	Check ok	Check not possible [-]	Any	Any	Check nok	TCMS	TCMS	
Any	Any	Check not possible [-]	Check not possible [-]	Any	Any	Check nok	TCMS	TCMS	

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Any	Any	Check nok	Check ok	Any	Any	Check nok	Not available	TCMS	TCMS: Dragging brake diagnostic protection not available on bogie 1
Any	Any	Check ok	Check nok	Any	Any	Check nok	TCMS	Not available	TCMS: Dragging brake diagnostic protection not available on bogie 2

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Any	Any	Check nok	Check nok	Any	Any	Check nok	Not available	Not available	TCMS: Dragging brake diagnostic protection not available on bogie 1 Dragging brake diagnostic protection not available on bogie 2 BCU maintenance fault “Pressure switch bogie 1 faulty” BCU maintenance fault “Pressure switch bogie 2 faulty”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check nok	Check nok	Check not possible [-]	Check ok	Check ok	Check ok	BCU	TCMS	BCU maintenance fault “Transducer bogie 2 setting out of range” BCU maintenance fault “Pressure switch bogie 1 faulty”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check nok	Check ok	Check not possible [-]	Check nok	Check ok	Check ok	TCMS	TCMS	BCU maintenance fault “Transducer bogie 2 setting out of range” BCU maintenance fault “Transducer bogie 1 faulty”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check nok	Check ok	Check not possible [-]	Check ok	Check nok	Check ok	BCU	TCMS	BCU maintenance fault "Transducer bogie 2 setting out of range" BCU maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"
Check ok	Check ok	Check nok	Check nok	Check ok	Check ok	Check ok	BCU	BCU	BCU maintenance fault "Pressure switch bogie 1 faulty" BCU maintenance fault "Pressure switch bogie 2 faulty"

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check nok	Check ok	Check nok	Check ok	Check ok	Not available	BCU	TCMS: Dragging brake diagnostic protection not available on bogie 1 BCU maintenance fault “Pressure switch bogie 1 faulty” BCU maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check nok	Check ok	Check ok	Check nok	Check ok	BCU	TCMS	BCU maintenance fault “Pressure switch bogie 1 faulty” BCU maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check ok	Check nok	Check nok	Check ok	Check ok	TCMS	BCU	BCU maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)” BCU maintenance fault “Pressure switch bogie 2 faulty”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check ok	Check nok	Check ok	Check nok	Check ok	BCU	Not Available	TCMS: Dragging brake not available on bogie 2 BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)” BCU: maintenance fault “Pressure switch bogie 2 faulty”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check nok	Check ok	Check nok	Check ok	Check ok	Not available	BCU	TCMS: Dragging brake not available on bogie 1 BCU maintenance fault “Pressure switch bogie 1 faulty” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check ok	Check nok	Check ok	Check ok	Check nok	Check ok	BCU	TCMS	BCU maintenance fault “Pressure switch bogie 1 faulty” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check nok	Check nok	Check not possible [-]	Check not possible [-]	Check nok	Check ok	Check ok	TCMS	TCMS	BCU maintenance fault “Transducer bogie 1 setting out of range” BCU maintenance fault “Transducer bogie 2 setting out of range” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check nok	Check nok	Check not possible [-]	Check not possible [-]	Check ok	Check nok	Check ok	TCMS	TCMS	BCU maintenance fault “Transducer bogie 1 setting out of range]” BCU maintenance fault “Transducer bogie 2 setting out of range]” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)] BCU:

Check nok	Check nok	Check not possible [-]	Check not possible [-]	Check nok	Check nok	Check ok	TCMS	TCMS	BCU maintenance fault “Transducer bogie 1 setting out of range” BCU maintenance fault “Transducer bogie 2 setting out of range” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”
Check ok	Check nok	Check nok	Check not possible [-]	Check nok	Check ok	Check ok	Not available	TCMS	TCMS: Dragging brake not available on bogie 1

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
									BCU maintenance fault "Pressure switch bogie 1 faulty BCU maintenance fault "Transducer bogie 2 setting out of range" BCU: maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check ok	Check nok	Check nok	Check not possible [-]	Check ok	Check nok	Check ok	BCU	TCMS	BCU maintenance fault "Pressure switch bogie 1 faulty" BCU maintenance fault "Transducer bogie 2 setting out of range" BCU: maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"

Check ok	Check nok	Check nok	Check not possible [-]	Check nok	Check nok	Check ok	Not available	TCMS	TCMS: Dragging brake not available on bogie 1 BCU maintenance fault “Pressure switch bogie 1 faulty” BCU maintenance fault “Transducer bogie 2 setting out of range” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
Check nok	Check ok	Check not possible [-]	Check nok	Check nok	Check ok	Check ok	TCMS	BCU	BCU maintenance fault "Pressure switch bogie 2 faulty" BCU maintenance fault "Transducer bogie 1 setting out of range" BCU: maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"

Check nok	Check ok	Check not possible [-]	Check nok	Check ok	Check nok	Check ok	TCMS	Not available	TCMS: Dragging brake not available on bogie 2 BCU maintenance fault “Pressure switch bogie 2 faulty] BCU maintenance fault “Transducer bogie 1 setting out of range” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”
Check nok	Check ok	Check not possible [-]	Check nok	Check nok	Check nok	Check ok	TCMS	Not available	TCMS: Dragging brake not available on bogie 2,

									BCU maintenance fault “Pressure switch bogie 2 faulty” BCU maintenance fault “Transducer bogie 1 setting out of range” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)” BCU: maintenance fault “Transducer bogie 1 faulty (TCF bogie X failure)”
Check nok	Check nok	Check not possible [-]	Check not possible [-]	Check nok	Check nok	Check ok	TCMS	TCMS	BCU maintenance fault “Transducer bogie 1 setting out of range”

BCU transducers check at the brake test bogie 1 TC_BI_#BCU_S_Trans_outranb_1	BCU transducers check at the brake test bogie 2 TC_BI_#BCU_S_Trans_outranb_2	BCU pressure switch check at the brake test bogie 1 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_1	BCU pressure switch check at the brake test bogie 2 (type 1 or 2) TC_BI_#BCU_S_PCFbtcheckb_2	BCU transducers runtime diagnostic bogie 1 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf1Flt	BCU transducers runtime diagnostic bogie 2 (TCFx pressure sensor faulty on one axle) TC_BI_8WSP_S_PrsSnsrTcf2Flt	BCU life active	Dragging brake diagnostic in charge to		Other reactions (SIL 2)
							Bogie 1	Bogie 2	
									BCU maintenance fault "Transducer bogie 2 setting out of range" BCU: maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)" BCU: maintenance fault "Transducer bogie 1 faulty (TCF bogie X failure)"

Table 2 - Dragging Brake Surveillance System

Nr	Description	Applicability Details	Derived to	User Interface	SIL
Europe.CONCEPT.HR41	**BCU dragging Brake Diagnostic** BCU diagnostic for dragging brake is the following: If brake demand DEP = 0 & BP > 4,8 bar (checking TCG (BP transducer) or the TDIS > 0,5 bar if TCG is faulty) Max (TCF1; TCF2) > 0,25 bar for 10 secondi Then trigger EVENT dragging brake Filter if: TCF1 & TCF 2 faulty; RS232 faulty;	Derived Requirement	4S_07.-.Braking System	EVENT	SIL 2
Europe.CONCEPT.HR42	**TCMS dragging Brake Diagnostic** If BP > 4.8 bar & status = not braking PCF must trip within 20 seconds. Reset condition: If BP > 4.8 bar & status = not braking & PCF in braking status for at least 3 s Filter if: (RIEP DIS = 1)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2

Dragging brake for BCU status disaligned with train status

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR117	Each BCU shall provide the braking status as follows (U8): BCU_X_STATUS Bit 0 = coasting Bit 1 = braking Bit 2 = traction Bit 3 = emergency braking Bit4-7= not valid	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR118b	Each WSPCU shall provide the braking status as follows (U8): WSP_X_STATUS Bit 0 = coasting Bit 1 = braking Bit 2 = traction Bit 3 = emergency braking Bit4-7= not valid	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR119	If TCMS detect continously for 15s: Train status is in coasting OR traction & One or more BCU (less than 5) are in braking status OR emergency braking status Then TCMS shall trigger a dragging brake EVENT (2F_05.02.Zefiro-Europe.CONCEPT.2052 2F_05.02.Zefiro-Europe.CONCEPT.2051 2F_05.02.Zefiro-Europe.CONCEPT.4214)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL2

2F_05.02.Zefiro-Europe.CONCEPT.HR120	The specific EVENT defined by is recorded as (BCU X dragging for brake status problem) and directed for maintenance only BCU_X_drag_status	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.512	Not used	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.513	Not used	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3004	Not used	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3005	Not used	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2052	If dragging brake is triggered by TCMS or BCU, the indication lamp "Brake applied" shall be connected to a TCMS output.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2051	If the brakes are considered as applied according concept 1612 and 1613, the lamp "Brake applied" shall be illuminated by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.515		Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.4214	GIVEN dragging brake is detected by BCU and sent to TCMS with the IP signal "*DIA dragging brake alert bogie 1*" OR dragging brake is detected by BCU and sent to TCMS with the IP signal "*DIA dragging brake alert bogie 2*" THEN a serious fault alarm shall be set by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.1.2.9.1 Dragging Brake by TCMS Reset

ID	2F_05.02.Zefiro-Europe.TRS.346
Description	In case the driver identifies an undue dragging brake event generated by a faulty brake pressure switch on the bogie, he shall be able to reset the fault.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.4279	**GIVEN** (At least one "Dragging Brake" valid event is active in the consist) **WHEN** ((Vehicle is at "Standstill") **AND** (All the bogies in the cars in which "Dragging Brake" events have been activated are pneumatically isolated)) **THEN** ("Bypass PS" "Soft Key" shall be represented in the "Brake Page" of IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.4280	**GIVEN** ("Bypass PS" "Soft Key" on IDU pushed) **WHEN** ("Return" button on IDU pushed) **THEN** (IDU shall go back in the previous page and TCMS shall not perform any action)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4281	**GIVEN** ("Bypass PS" "Soft Key" on IDU pushed) **WHEN** ("Hard Key" of IDU acknowledged and accepted by TCMS) **THEN** ((Monitoring Function of "Dragging Brake" by TCMS shall be maintained active only in the cars of the consist in which the "Dragging Brake" events are not active) **AND** (Car is in "Dragging Brake Bypassed" status))	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.4282	**GIVEN** (Car is in "*Dragging Brake Bypassed*" status) **THEN** ((All the "*Dragging Brake*" active events shall be reset))	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.4283	**GIVEN** (Exist at least one car is in the consist in "*Dragging Brake Bypassed*" status) **THEN** (Event "*Speed limit pressure switch brake cylinder faulty removed*" shall be set by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4284	**GIVEN** (Car is in "*Dragging Brake Bypassed*" status) **THEN** (Event "*Dragging brake not monitored anymore on the car by TCMS*" shall be set by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4285	**GIVEN** ((Car is in "*Dragging Brake Bypassed*" status) **AND** (Vehicle is not in "*Towing Mode*") **AND** (NO "*Dragging Brake*" events active)) **THEN** ("*PS OK*" icon shall be represented by TCMS to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4286	**GIVEN** ("*PS OK*" icon is represented on IDU) **WHEN** (At least one "*Dragging Brake*" event is active) **THEN** ("*PS OK*" icon shall not be represented anymore by TCMS to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4287	**GIVEN** ("*PS OK*" icon is represented on IDU) **WHEN** (Vehicle is in "*Towing Mode*") **THEN** (Icon "*Towing Mode*" is represented on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.1.2.10 *Isolation Cock Monitoring*

ID	2F_05.02.Zefiro-Europe.TRS.57
Description	There shall be a possibility to cut out the pneumatical brakes on a bogie basis. For each car two cut out switches for the pneumatic brake shall be provided.
Safety level	N/A

ID	2F_05.02.Zefiro-Europe.TRS.144
Description	The status of the cut out switches shall be monitored by the diagnostic system and the driver shall be alerted in case of any cut out switch in "cut out" position.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.120	There shall be a isolation cock RICF1/2 with position switch for each bogie isolate brake cylinder pressure in the bogie. Isolation cock shall be located on brake frame panel. Isolation cocks RICF1/2 04.38/1...2 shall be read by the BCU. The status of the cocks RICF1 04.38/1 and RICF2 04.38/2 for each bogie shall be sent to TCMS via IP signal "Brake bogie 1/2 isolated (04.38/1...2 RICF)". This signal shall be allocated to the SIL 2 safe Ethernet group.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1971	The status of the isolation cocks RICF1 04.38/1 and RICF2 04.38/2 shall be read by TCMS input.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1972	The isolation cocks RICF1 04.38/1 and RICF2 04.38/2 shall be connected to TCMS digital input (MIO). If the isolation cock is in isolated position the input shall be set (1).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1583	If the isolation cock RICF1 04.38/1 is in isolated position (reported by TCMS input OR BCU IP signal) an event "Brake Bogie 1 is isolated" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1584	If the isolation cock RICF2 04.38/2 is in isolated position (reported by TCMS input OR BCU) an event "Brake Bogie 2 is isolated" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2379	If RICF1 status received by BCU and RICF1 status read by TCMS are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2380	If RICF2 status received by BCU and RICF2 status read by TCMS are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1585	There shall be a isolation cock RCIP (04.27/01) for isolation of complete brake frame located on brake frame panel in each car. Isolation cock RICP 04.27/01 shall be read by the BCU.The status of the isolation cock RICP 04.27/01 shall be sent to TCMS via IP signal "Isolation cock RICP (04.27/01) supply brake frame isolated". This signal shall be allocated to the SIL 2 safe Ethernet group.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1965	The status of the isolation cock RICP 04.27/01 shall be read by TCMS input.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1966	The isolation cock RICP 04.27/01 shall be connected to TCMS digital input (MIO). If the isolation cock is in isolated position the input shall be set (1).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1586	If the isolation cock RICP is in isolated position (reported by TCMS input OR BCU) an event "Brake frame isolated in car x" shall be set. The driver shall be advised to isolate brake isolation cock RIEP-DIST in addition to RICP isolation cock.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	The status isolated shall only be set based on valid signals.				
2F_05.02.Zefiro-Europe.CONCEPT.2378	If RICP status received by BCU and RICP status read by TCMS are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1587	There shall be a isolation cock RIEP-DIST (04.29/01) for isolation of brake reservoir located on brake frame panel in each car. Isolation cock RIEP-DIST 04.29/01 shall be read by the BCU. The status of the isolation cock RIEP-DIST 04.29/01 shall be sent to TCMS via IP signal "Brake isolated (04.29/01 RIEP-DIST)". This signal shall be allocated to the SIL 2 safe Ethernet group.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1967	The status of the isolation cock RIEP-DIST 04.29/01 shall be read by TCMS input.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1968	The isolation cock RIEP-DIST 04.29/01 shall be connected to TCMS digital input (MIO). If the isolation cock is in isolated position the input shall be set (1).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1588	If the isolation cock RIEP-DIST is in isolated position (reported either by TCMS input OR BCU) an event "Brake is isolated in car x" shall be set. The status isolated shall only be set based on valid signals.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2270	If RIEP-DIST status received by BCU and RIEP-DIST status read by TCMS are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1523	There shall be a isolation cock RIEP 04.53/01 for isolation of DEP brake located on brake frame panel in each car. Isolation cock RIEP (04.53) shall be read by the BCU. The status of the isolation cock RIEP 04.53/01 shall be sent to TCMS via IP signal "Direct brake isolated (04.53/01 RIEP)". This signal shall be allocated to the SIL 2 safe Ethernet group.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1524	If the isolation cock RIEP 04.53/01 is in isolated position reported by BCU IP signal "Direct brake isolated (04.53/01 RIEP)" an event "DEP brake is isolated in car x" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2 Service Brake - Indirect Brake

3.2.1 Function Design

The indirect service brake is used as back-up mode of direct electro-pneumatic brake.
When one or two cars have faulty DEP brake the missing brake force can be shifted to other cars with DEP brake available by the brake blending logic.

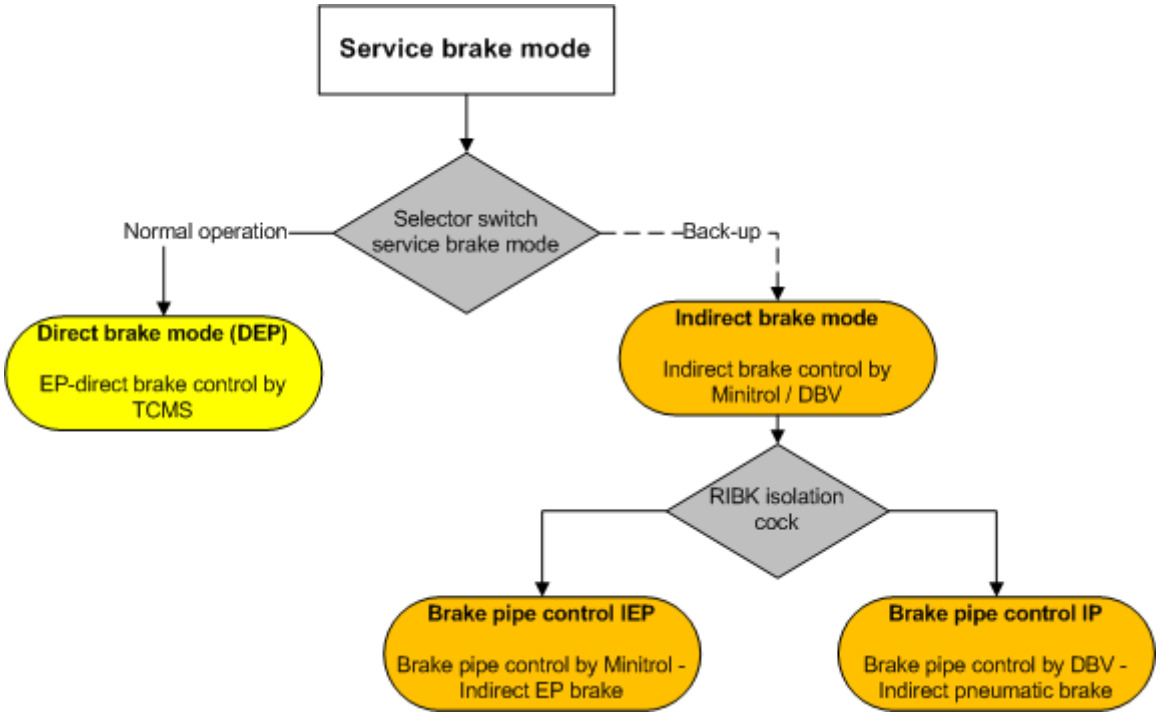
In case of direct brake failure in more than two cars the back-up indirect brake has to be activated by the driver.

The driver will be informed by TCMS via an IDU alarm if more than two cars with DEP brake are faulty and the driver has to switch manually to indirect brake mode.
The indirect brake mode is activated by the rotary selector switch "Service brake mode" in the driver's cab.

The indirect brake mode consists of two possibilities to control the indirect service brake demand via the brake pipe.
- Electro-pneumatic indirect brake pipe control (IEP)
- Pneumatic indirect brake pipe control (IP)

In normal condition the brake pipe control mode is IEP, in case of failure the second brake pipe control mode IP can be activated. IP brake pipe control is without involvement of any electronics. The IP brake pipe control mode is activated by turning RIBK isolation cock in IP position.

<3.2.1 - D1: Indirect brake function control flow chart>



The indirect brake is controlled by the driver's brake valve located on driver's desk described in chapter 3.2.1.1.

Indirect brake mode according UIC is performed with four stage deceleration with three different control pressures based on the following speed threshold.

On motor car:
- 360-300 km/h low pressure
- 300-230 km/h medium pressure

- 230-50 km/h high pressure
- 30-0 km/h medium pressure

On trailer car:
- 360-300 km/h low pressure
- 300-230 km/h medium pressure
- 230-30 km/h high pressure
- 50-0 km/h low pressure

During indirect brake mode no ED brake is available, the indirect brake mode is a pure pneumatically brake.

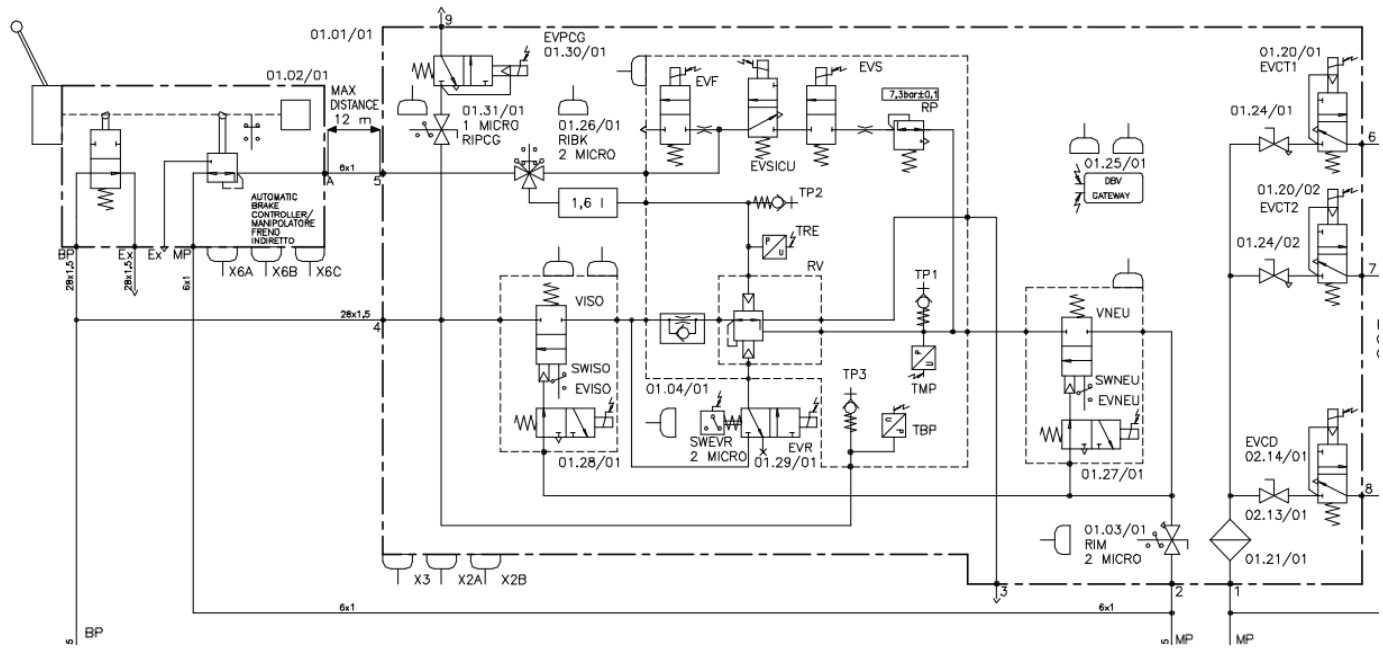
Indirect brake functions are possible to be cut-out mechanically by turning the isolation cock for one bogie RICF or for a complete car RIEP-DIST. The brake isolation follows the same logic as described in chapter 3.1.

The following isolation cocks are relevant for the indirect brake mode, they are monitored by TCMS and BCU and located on the brake panel in each car:
- RICP: Isolation of air supply to brake frame from main pipe in one car, supervision covered in section DEP brake chapter 3.1
- RIEP-DIST: Isolation of all brakes in one car (direct and indirect), supervision covered in section DEP brake chapter 3.1

On the cabin panel (Minitrol) located in the DM car following isolation cocks are present and monitored by TCMS and DBW-GW:
- RIM 01.03: Isolation cock of the complete cabin panel for maintenance purpose
- RIBK 01.26: Three way isolation cock to select the indirect brake pipe control mode (to activate IP brake)
- RIPCG 01.31: Isolation cock of the brake pipe control test venting valve EVPCG 01.30, this isolation cock is only monitored by DBW-GW

In normal conditions the RIBK isolation cock is in IEP brake pipe control position, the pneumatic input of the DBV (pneumatic input 5 to cabin panel, refer to diagram 3.2.1 - D2) is disabled.

<3.2.1 - D2: Indirect brake pneumatic control logic>

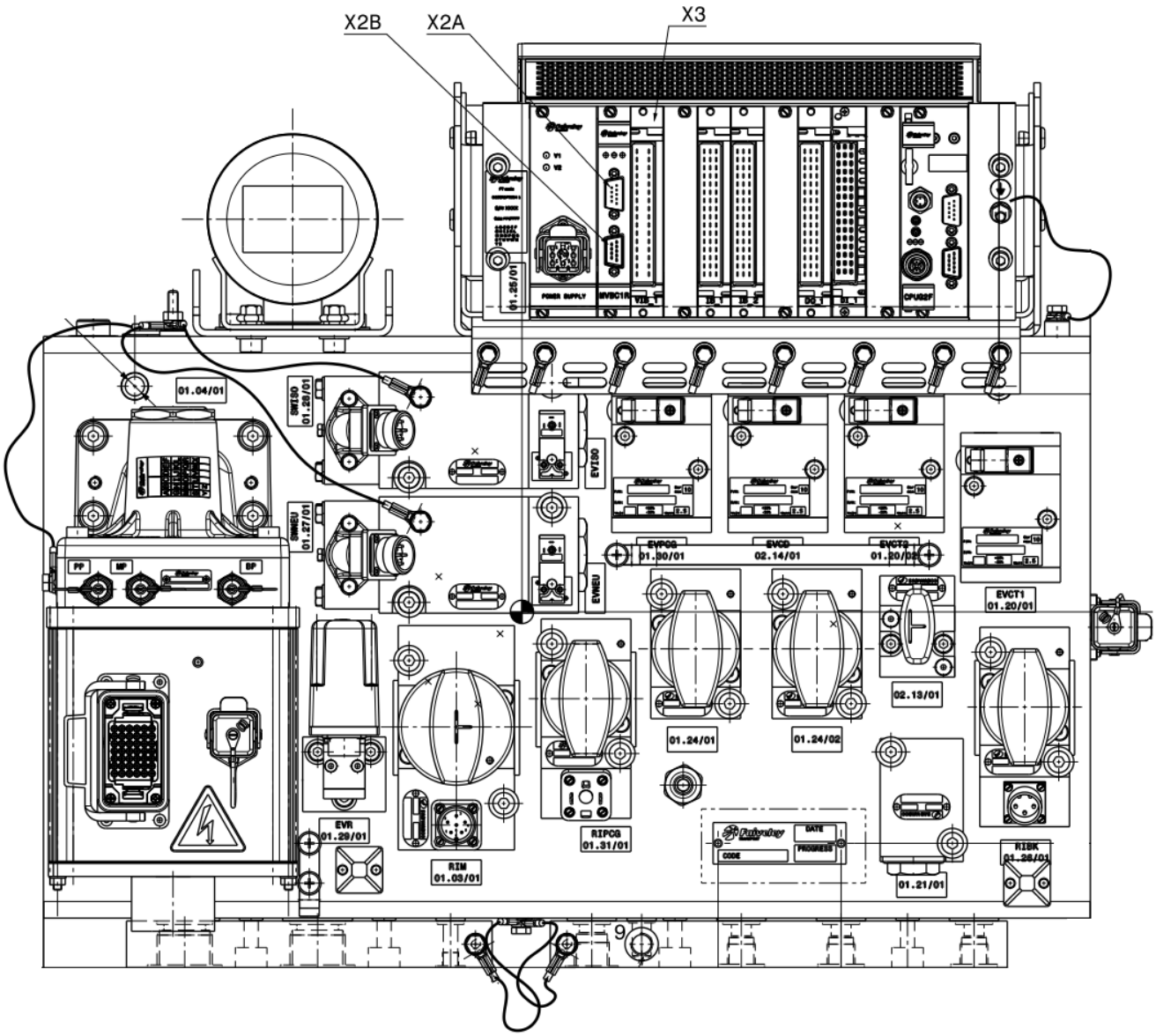


The indirect brake control consists of the following pneumatic hardware, the components are described in the related chapters:
- Driver's brake valve 01.02
- Cabin panel 01.01

Components located on cabin panel 01.01:

- Minitrol IEP control group 01.04
- Brake pipe test control valve EVPCG 01.30
- Brake pipe isolation valve VISO 01.28 and neutral valve VNEU 01.27
- Quick release magnet valve EVR 01.29
- Horn control magnet valves EVCT1/2 01.20/01-02
- Coupler control valve EVCD 02.14
- Isolation cocks RIM 01.03, RIBK 01.26 and RIPCG 01.31 with electrical supervision
- Isolation cocks 01.24/01-02 and 02.13 for horn and coupler pneumatic isolation

<3.2.1 - D3: Cabin panel>

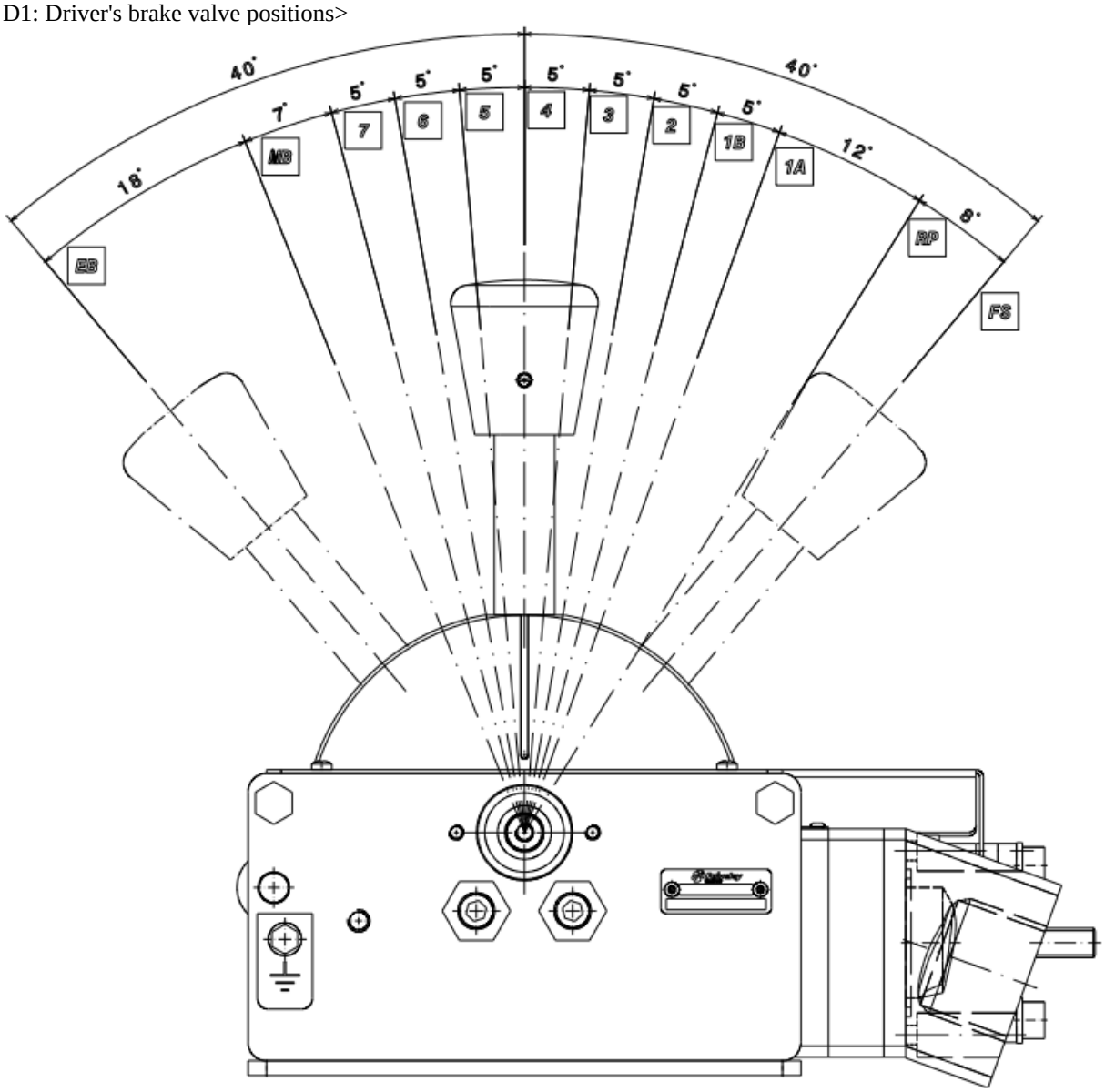


3.2.1.1 *Driver's Brake Valve*

The driver's brake valve is located on the driver's desk, the driver's brake valve is position dependent and has the following steps:

- Filling stroke FS
- Release position RP
- Last release position 1A
- First brake position 1B
- Brake steps 2 - 7
- Full service brake MB
- Emergency brake EB

<3.2.1.1 - D1: Driver's brake valve positions>



The driver's brake valve is designed with an electronic portion and a pneumatic portion.

Electronic portion:
The driver's brake valve is equipped with a PWM sensor to transfer the different positions to the Minitrol panel to build-up the IEP brake pipe demand pressure.

In addition, the DBV has 7 electric position switches used for service brake diagnostic and emergency brake loop interface.

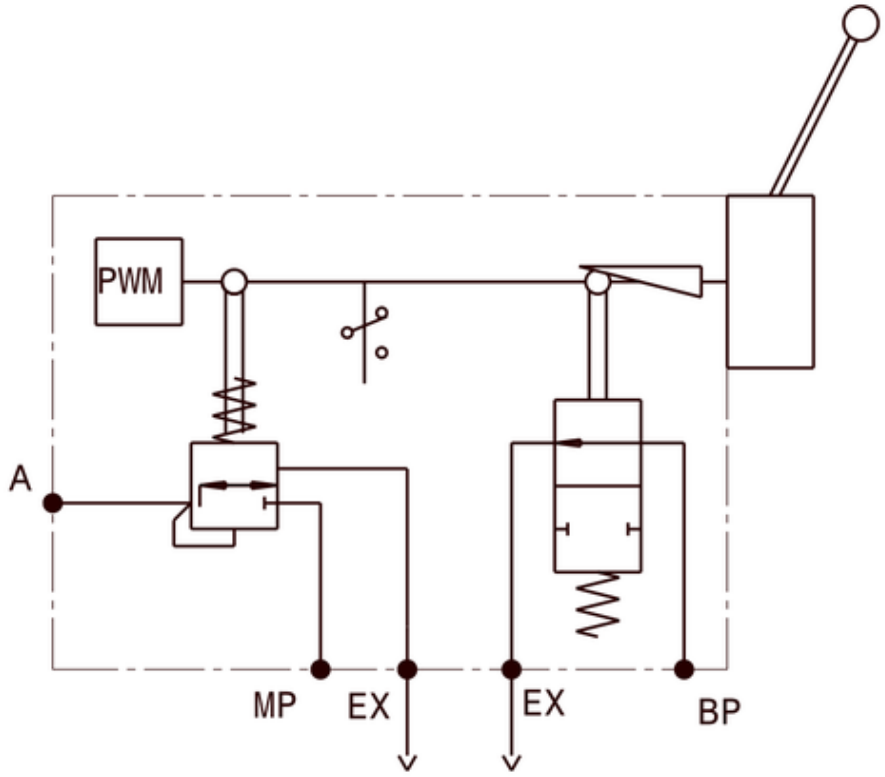
<3.2.1.1 - D2: DBV IEP brake pipe control steps>

DBV Position			Pressure Target Minitrol	Angle	PWM Duty
FS	Filling Stroke	Unstable	5,00 ± 0.05	+ 40°	34,2-36,2
RP	Release Position	Stable	5,00 ± 0.05	+ 32°	37,3-38,9
1A	Last brake release pos.	Continuous	< 4,75	+ 20°	41,8-43,4
1B	First brake apply pos.	Stable	4,55 ± 0.05	+ 15°	43,6-45,2
2	Second pos.	Stable	4,40 ± 0.1	+ 10°	45,5-47,1
3	Third pos.	Stable	4,30 ± 0.1	+ 5°	47,3-48,9
4	Fourth pos.	Stable	4,15 ± 0.1	0°	49,2-50,8
5	Fifth pos.	Stable	4,00 ± 0.1	- 5°	51,0-52,6
6	Sixth pos.	Stable	3,90 ± 0.1	- 10°	52,9-54,5
7	Seventh pos.	Stable	3,75 ± 0.1	- 15°	54,8-56,4
MB	Full service brake	Stable	3,30 ± 0.1	- 22°	57,7-59,3
EB	Emergency brake	Stable	0	- 40°	63,8-65,8

Pneumatic portion:
Back-up pressure regulator by self-lapping valve to control the brake pipe pressure in IP brake pipe control mode.

The driver's brake valve pneumatic connections are (refer to diagram below):
BP = Brake Pipe - up to maximum 600 kPa
MP = Main Pipe - up to maximum 1000 kPa
A = Pilot Pressure - up to maximum 500 kPa
EX = Exhaust port - venting port of brake pipe

<3.2.1.1 - D3: DBV pneumatic diagram>



The brake pipe is filled or maintained up to 500 kPa via the main pipe pressure port MP if the IP brake pipe control mode is active. The brake pipe pilot pressure is setting the brake pipe pressure based on the position of the driver's brake valve handle via port A. In case the brake pipe pressure is lowered by the service brake steps the pressure is vented by the exhaust port EX of the self lapping valve portion.

In case of emergency brake position of the DBV the brake pipe port BP is directly vented via the exhaust port EX with a diameter of 1" of the emergency valve to allow a fast drop of the brake pipe pressure.

The DBV self lapping valve to regulate the brake pipe pressure in IP brake pipe mode has the back-up service brake pressure steps as indicated in diagram 3.2.1.1 - D3. The "A" pressure port is the pressure to the relay valve which set up the final brake pipe pressure.

The ration between the relay pilot pressure and the brake pipe output pressure is 1:1.

<3.2.1.1 - D4: DBV pressure steps for IP brake pipe control>

REF.	POSITION	COMMENT	"A" PRESSURE [1]
FS	Filling stroke	unstable spring return position	5.00 ± 0.05 [2]
RP	Release position	notch, hard	5.00 ± 0.05
1A	Last brake	notch, soft	$4.75^{+0.10}_{-0,15}$
1B	First brake application	notch, soft	$4.55^{+0.10}_{-0,15}$
2 to 7	Intermediate positions	notch, soft	-
MB	Maximum brake application	notch, soft	$3.30^{+0.10}_{-0,15}$
EB	Emergency brake	notch, hard+end-stop	$3.30^{+0.10}_{-0,15}$

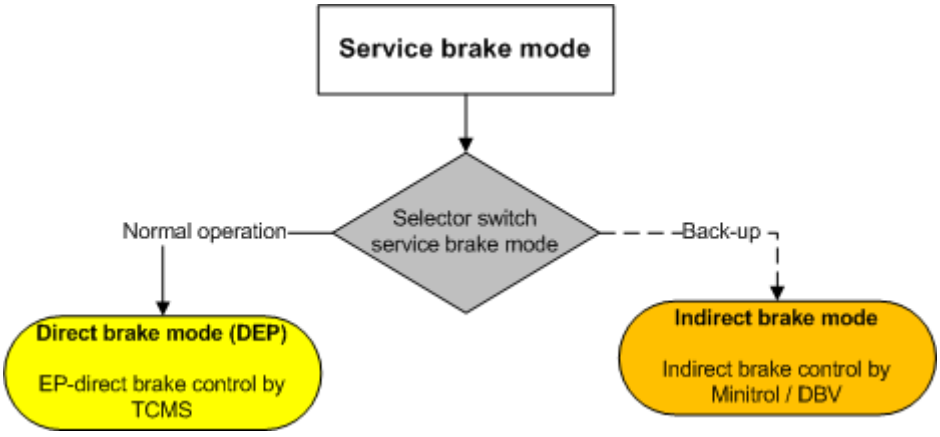
3.2.1.2 *Service Brake Mode Selection*

The service brake mode selection between direct brake (DEP) and indirect brake mode is done by the rotary selector switch "Service brake mode" located in the electrical cabinet of the driver's cab.

In case the rotary selector switch is turned to position indirect brake the driver is informed by the indication lamp "Indirect brake active" on driver's desk that the indirect brake mode is activated.

In case the direct brake (DEP) has a failure and the direct brake performance is lost in more than two cars a warning is presented on the IDU to the driver to switch to the indirect brake mode by turning the selector switch. The switching to the indirect brake mode can only be done at standstill and is not allowed during driving.

<3.2.1.2 - D1: Service brake mode selection flow chart>



In case the indirect brake mode is selected the driver's brake valve gateway control unit (DBV-GW) will activate the IEP back-up service brake if the RIBK isolation cock is in position IEP. The driver's brake valve service brake positions are now active and the DBV is controlling the IEP service brake via the PWM signals to the Minitrol.

The switching between IEP and IP brake pipe control mode is described in next chapter 3.2.1.3.

3.2.1.3 *IEP and IP Brake Pipe Control Mode*

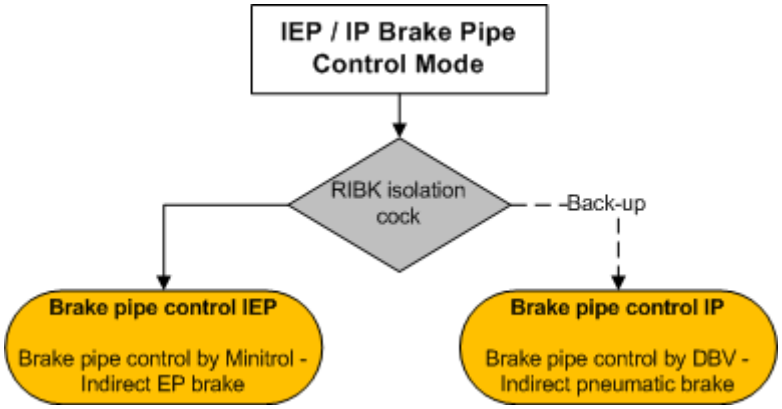
The brake pipe control mode is in default the indirect electro-pneumatic brake pipe control (IEP). In case of any failure of the IEP brake the switching of the brake pipe control mode to indirect brake pipe control (IP) can be done by turning the RIBK 01.26 cock in isolated position.

The RIBK cock is located on the cabin panel 01.01 in the driver's cab and supervised by electronic contacts of TCMS and DBV-GW control unit, refer to diagram in 3.2.1.3 - D2.

In case the RIBK cock is turned in IP position the driver is informed by the indication lamp "IP mode active" on driver's desk that the IP brake pipe control mode is activated.

In IEP brake pipe control mode and service brake control mode DEP active the driver's brake valve service brake positions are blocked by the DBV-GW and Minitrol panel, only emergency brake position is active.
In IP brake pipe control mode and service brake control mode DEP active the driver's brake valve is active in all brake position directly controlling the brake pipe without the DBV-GW and Minitrol electronics.

<3.2.1.3 - D1: Brake pipe control mode flow chart>



IEP brake pipe control mode (Minitrol)

In IEP brake pipe control mode, the brake pipe pressure is regulated by the Minitrol panel based on the PWM signal derived by the driver's brake valve.

The electro-pneumatic regulation of the brake pipe is performed by the Minitrol 01.04 hardware located on the cabin panel 01.01 in driver's cab:

- RP: Pressure reducing valve 730 kPa
- EVS: Brake pipe refill (brake applied) magnet valve
- EVF: Brake pipe venting (brake release) magnet valve
- EVSICU: Safety magnet valve to inhibit refill in case of emergency brake
- TRE: Pilot brake pipe pressure transducer
- TMP: Main pipe pressure transducer
- TBP: Brake pipe pressure transducer located after the neutral VNEU and isolation VISO valve
- RV: Brake pipe control relay valve

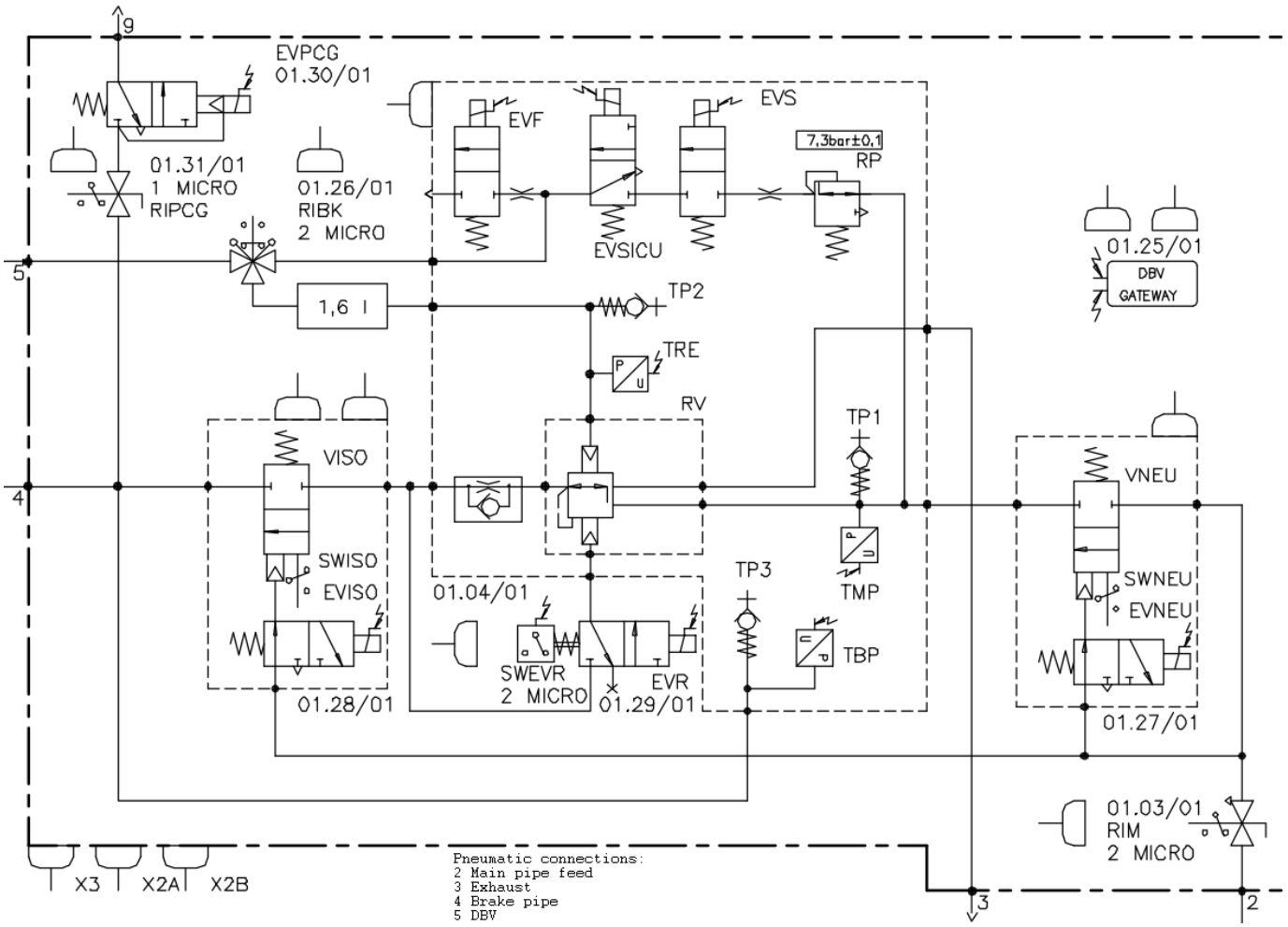
The brake pipe in DEP service brake mode is regulated and maintained at 500 kPa. Only in case of emergency brake the safety valve EVSICU is preventing the refill of the pilot pressure to the relay valve RV.

In case the Indirect service brake mode is active and the RIBK isolation cock is in IEP brake pipe control position, the brake pipe pressure is regulated by the Minitrol controlling the magnet valves EVS and EVF to set the brake pipe pilot pressure based on the PWM input signal.

The PWM input signal is based on the service brake position of the driver's brake valve. The correct setting of the brake pipe pilot pressure is supervised by the pressure transducer TRE as a circle loop function.

The complete IEP brake pipe control function is supervised by the DBV-GW. In case of any failure of the IEP control chain PWM or a failure of the involved hardware the driver is informed to switch to IP brake pipe control mode and a dedicated maintenance event is set by TCMS.

<3.2.1.3 - D2: Brake pipe pneumatic control layout IEP>



IP brake pipe control mode

In case of failure of the IEP brake pipe control mode the IP brake pipe control mode has to be activated by turning the RIBK isolation cock in IP position.

In this position the pneumatic chain coming from the Minitrol panel is disabled and the pure pneumatic control chain coming from the DBV (inlet port 5) is active. In this IP brake pipe control chain no electronics are involved, the brake pipe pilot pressure is regulated based on the pneumatic proportional self-lapping valve inside the driver's brake valve. The output pressure of the pneumatic portion of the DBV is based on the service brake position.

The service brake pressure is set in the following range at a temperature of +20°C:
- First service brake position: 455 +10/-15 kPa
- Maximum service brake position: 330 +10/-15 kPa

If IP brake pipe control is active, the overcharge function is not available.
In Direct service brake mode and IP brake pipe control active, both brake control elements master controller and DBV valve are active, both DEP brake request (by TCMS via IP) and brake request (by brake pipe) will be forwarded to brake units in each vehicle. (ED brake will be automatically inhibited as soon as brake pipe is lowered due to DBV movement, see ED brake cut out requirement).

3.2.2 Brake Pipe Control Functions

The brake pipe control functions are used to isolate the brake pipe from main pipe, to inhibit the refill and to allow the brake pipe pressure to be overcharged or quick released. The brake pipe pressure is shown to the driver via the IDU and the double manometer 01.15/01 for brake pipe (white indicator) and main pipe (red indicator) located on driver's desk

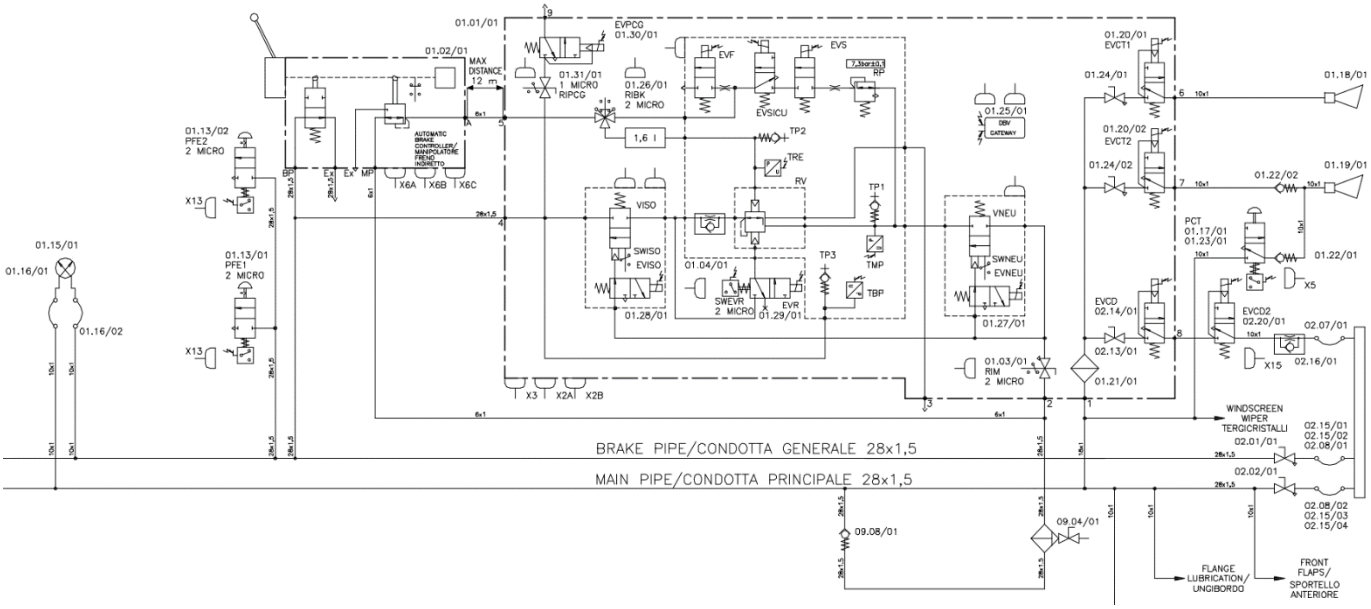
The brake pipe refill is guaranteed by the pneumatic feed of the main reservoir pressure, the train is not equipped with extra brake main pipe reservoirs. For maintaining and refill function of the brake pipe, as well in degraded condition like a certain drop of main pipe pressure, no additional main brake pipe reservoirs are necessary.

The brake pipe is protected by a check valve 09.08 and a filter element with drainage cock 09.04 against the main pipe pressure, in case of local abrupt pressure drop in the main pipe of DM car the brake pipe pressure will maintain the nominal pressure of 500 kPa, unless leakages. Pneumatic diagram of brake pipe control is shown in diagram 3.2.1.4 - D1.

The needed air to fill the brake pipe and release the brake is taken from all main pipe reservoirs located in the train. The complete main reservoir of the train has a capacity of about 1000 Nl to fulfil the Disp. RFI 30/2007 1.23 a). The disposition 1.23 b) is as well fulfilled by the given architecture as the auxiliary reservoirs are protected by check valve against the main reservoir.

In case of degraded mode, without line voltage and auxiliary power supply for air supply modules a main pipe pressure > 700 kPa is enough to fill the brake pipe to 500 kPa and release the indirect brake.

<3.2.1.4 - D1: Brake pipe pneumatic control/cabin equipment>



Brake pipe isolation

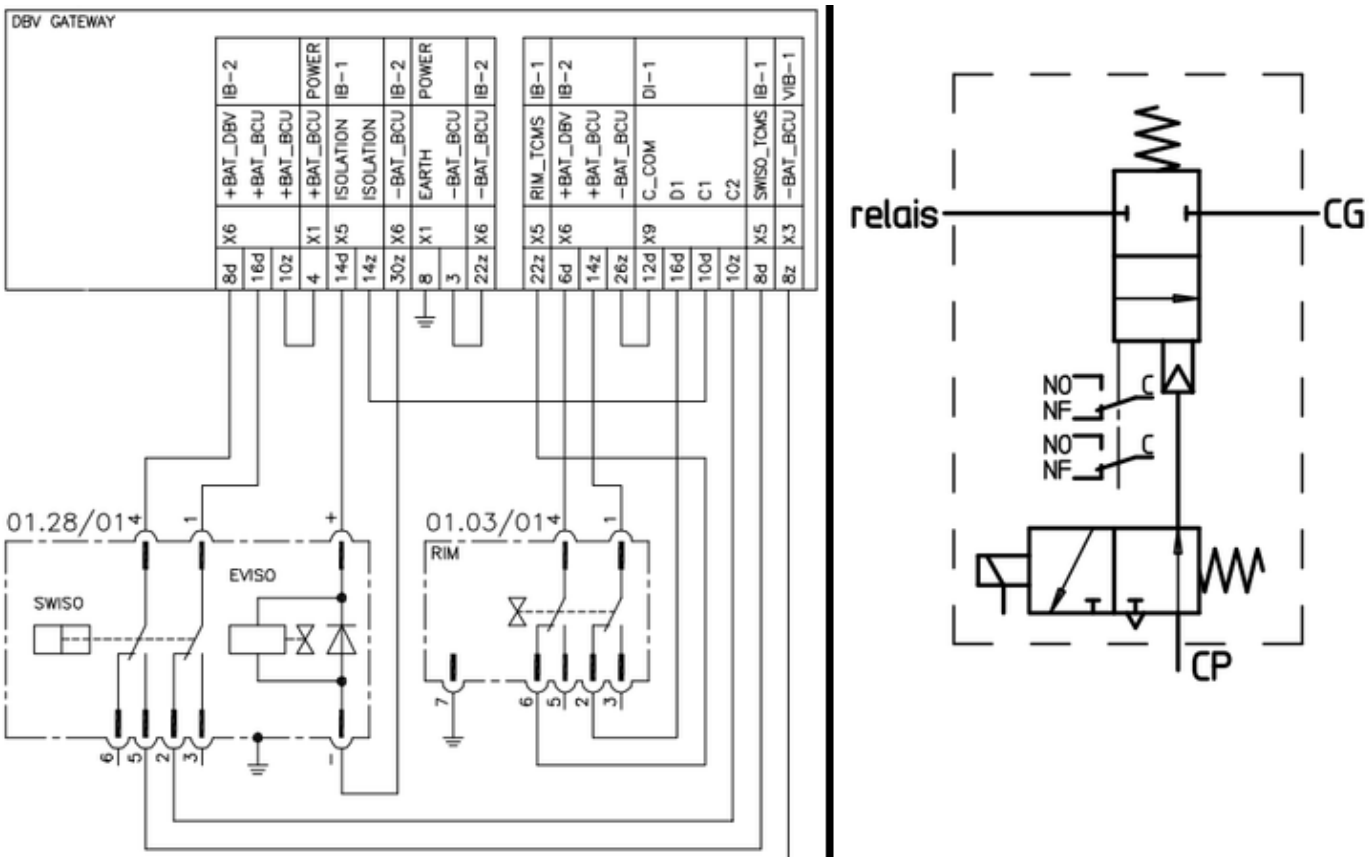
In case the cab is not active the brake pipe pressure is isolated by isolation valve VISO 01.28. Therefore, the isolation valve pre-control pressure electric magnet valve EVISO is directly hardwired connected to the cab active relay condition. The VISO valve is located after the brake pipe control relay valve RV.
To isolate the brake pipe the EVISO shall be energized (high active command).

The train control logic will energise the EVISO valve to isolate the brake pipe and disable the brake pipe control and refill in following conditions:

- Cab not active
- Train in towed mode
- Semi-automatic brake test by TCMS (BP continuity and tightness test)
- Manual brake test switch in position BP isolation (BP continuity and tightness test)

The EVISO valve can not be controlled directly by DBV-GW. In case no main pipe pressure is present the brake pipe isolation valve VISO will remain in isolated position.

<3.2.1.4 - D2: Isolation valve electric and pneumatic scheme>



The neutral valve VNEU 01.27 isolates the brake pipe control pressure before the brake pipe relay valve RV. The VNEU is driven in the same way as the VISO valve by a pre-control pressure electric magnet valve EVNEU.

The EVNEU is controlled by hardware, TCMS and DBV-GW. To isolate the brake pipe the EVNEU valve has to be energised. The valve is isolated in following conditions (energised):

- Hardwired in case EB loop is de-energised, refer to chapter 3.5.1
- TCMS output active in case emergency brake active and during brake test

Both isolation valve VISO and VNEU are supervised by mechanical position switch SW_ISO and SW_NEU read by TCMS and DBV-GW. In case the command to EVISO and EVNEU are not align to the feedback of the switches a diagnostic event is set.

Overcharge function

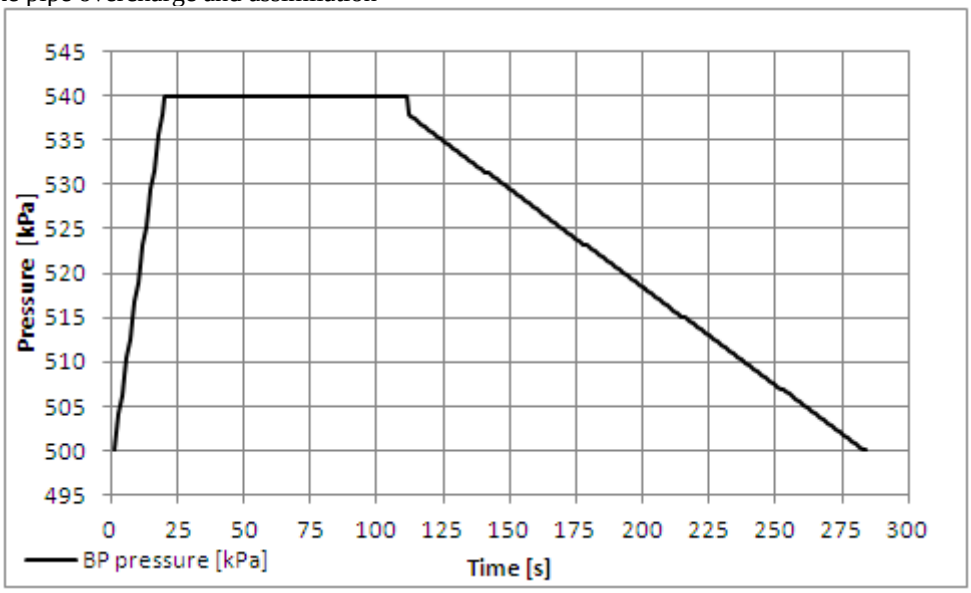
The brake pipe overcharge function is used to allow a controlled increase of the brake pipe pressure up to 540 kPa following a time based assimilation procedure of the distributors according UIC 541-03.

The overcharge function is activated pressing the push button “Overcharge” on the driver desk. The push button is connected to DBV-GW and if the signal is set the Minitrol starts the overcharge procedure. The brake pipe pressure increase is interrupted if the DBV is moved or if 540 kPa is reached in the BP.

The pressure increase and controlled decrease is realised by the normal control logic of the IEP brake pipe control magnet valves EVS and EVF.

Attention: The overcharge function is not available in IP brake pipe control mode.

<3.2.1.4 - D3: Brake pipe overcharge and assimilation>



Quick release function

The brake pipe pressure quick release function allows a fast release of the brake pressure by setting a higher filling volume to make a quick increase of the brake pipe pressure. This is realised by the EVR 01.29 magnet valve setting a pilot pressure to the relay valve RV to increase the filling portion.

The quick release function can be activated by moving the DBV in FS position. In that position an electrical signal is sent to Minitrol, which based on the preconditions, set the pilot chamber pressure to 500 kPa without ramp and ask DBV-GW to activate for 60 s the valve EVR.

The quick released is interrupted if any of the following preconditions are true:
- Big leakage failure is present
- Isolation valve active (isolated position)
- Neutral valve active (isolated position)
- Emergency brake active, command of EVR is interrupted by emergency brake loop relay Rble contact
- Service brake (brake handle is moved from FS position to any braking position)

The correct function of the EVR quick release magnet valve is supervised by the DBV-GW, therefore the mechanical switch SW_EVR located on the EVR magnet valve is read by the DBV-GW. In case of failure a maintenance event is set by TCMS and the driver is informed that the quick release function is not available.

3.2.2.1 *Brake Pipe Diagnostic*

The brake pipe is continuous monitored by the DBV-GW, Minitrol and TCMS.

Following functions are continuously supervised, in case of failure an event is set in the train diagnostic database:
- Brake pipe leakage detection by DBV-GW

Brake pipe refill/isolation monitoring:
- Supervision of VNEU 01.27 position by TCMS and DBV-GW, incoherence event not opening on order/open without order
- Supervision of VISO 01.28 position by TCMS and DBV-GW, incoherence event not opening on order/open without order
- Supervision of EVR 01.29 position by DBV-GW, incoherence event missing/undue quick release

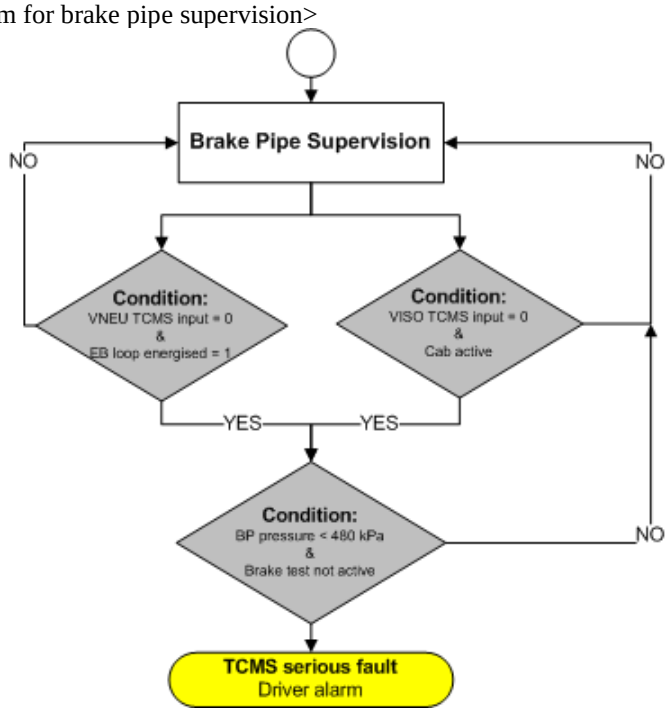
Maintenance diagnostic for components:
- Supervision of pressure sensors for brake pipe control TRE, TMP and TBP
- Supervision of quick release valve EVR
- Supervision of Minitrol control valves EVF, EVSICU and EVS

Serious fault alarm of unwanted BP re-charge during operation is presented to the driver in following cases:
- the TCMS input of SW_NEU is set (1) AND the emergency brake loop is de-energised

OR
- the TCMS input of SW_ISO is set (1) and local cab is NOT active

In case serious fault due to SW_NEU is triggered, fault shall be kept till when a new Emergency brake is commanded AND trigged condition are not true anymore.

<3.2.1.4 D3 - Serious driver alarm for brake pipe supervision>



A failure of an isolation valve VISO or neutral valve VNEU has following impact on train level:
- VISO or VNEU in undue isolated position -> Stopping failure (no refill of brake pipe possible in local cab)
- VISO or VNEU commanded but not in isolated position -> Failure indicated to driver, further operation possible

In the case the VISO valve is open in not active cab, the driver is requested to isolate the cabin panel of the not active cab to continue the service.

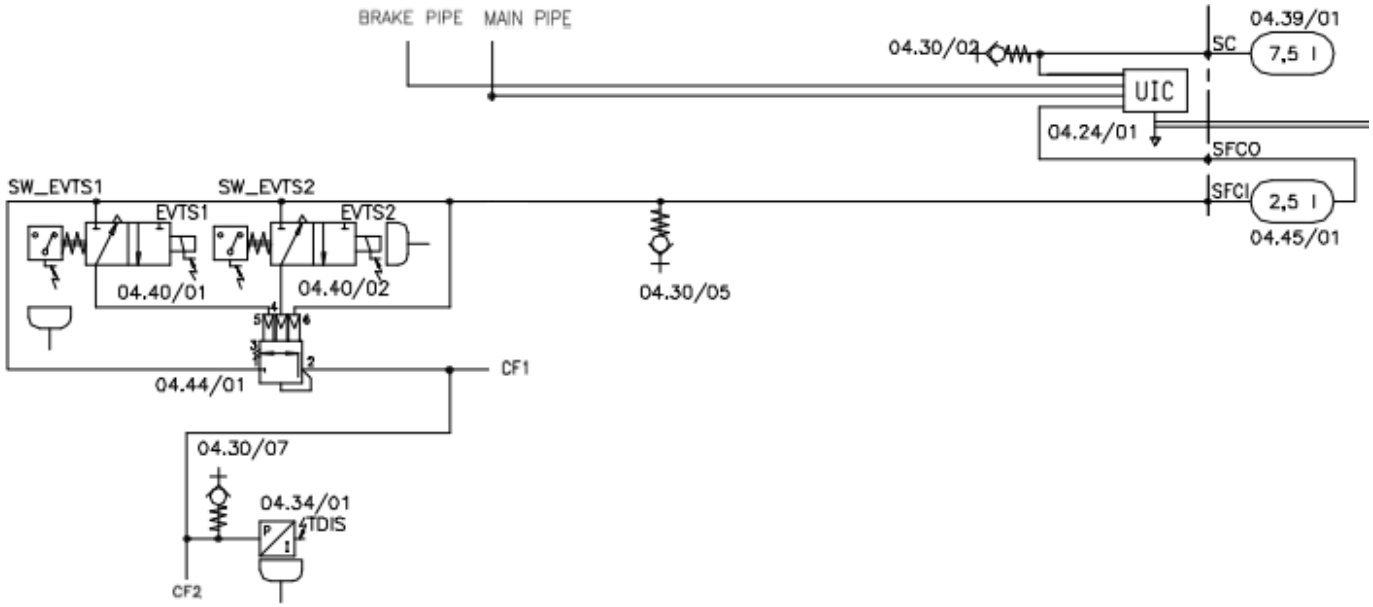
3.2.3 Local Indirect Brake Demand Control

The brake pipe pressure control is performed by the IEP or IP mode, the indirect brake demand is transferred via the brake pipe pressure.
Lowering the brake pipe pressure is controlling the distributor valve 04.24 (UIC) which build-up the brake cylinder pressure in proportion to the brake pipe depression on car level.

The brake pipe is controlled at 500 kPa in brake release condition, the maximum brake cylinder pressure is build-up at 380 kPa. In the range between 500 kPa - 380 kPa the cylinder pressure is build-up proportional to the brake pipe pressure.

After the distributor valve 04.24 the triple stage group is located to lower the cylinder pressure at higher speed to achieve the adhesion and thermal limitations. The triple stage group is always active during indirect brake mode and controlled by the WSP-CU.

<3.2.1.5 - D1: Indirect brake demand control on car level>



The triple stage group consist of the following components:
- EVTS1 04.40/01 magnet valve second step
- EVTS2 04.40/02 magnet valve third step
- Relay valve 04.44 for three step pressure control
- TDIS 04.34 pressure transducer

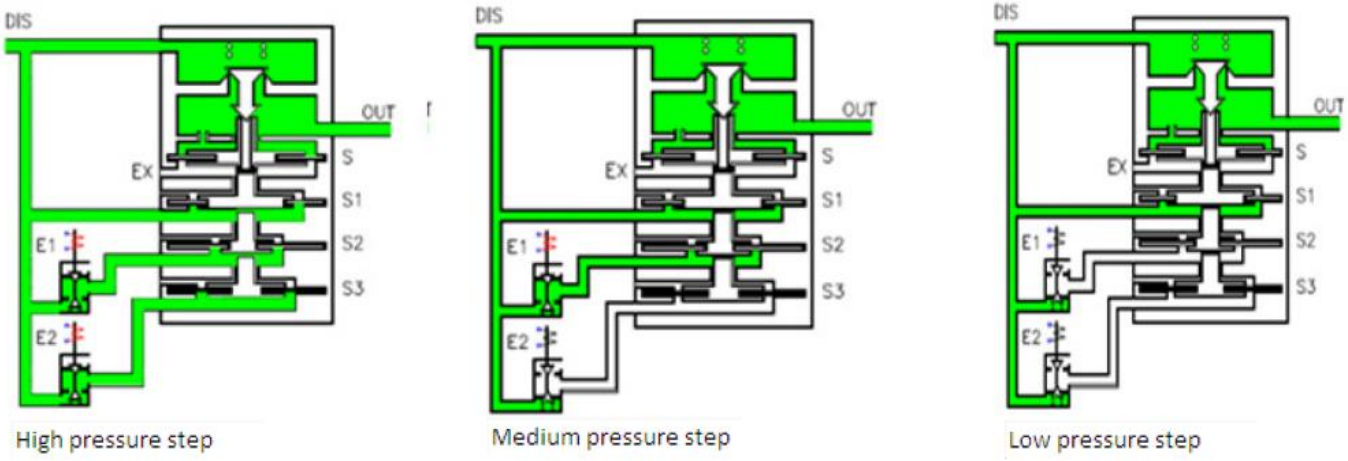
The lowest pressure step is set if both control valves EVTS1 and EVTS2 are de-energised and no pressure is present on the two ports of the triple stage relay valve 04.44.

The lowest pressure step is set if the control valve EVTS1 and EVTS2 are not energised by the WSP-CU at a speed v > 300 km/h or by the BCU 4 stage deceleration control at low speed ≤ 30 km/h on trailer cars.

The medium pressure step is set if the control valve EVTS1 is energised by the WSP-CU at a speed v ≤ 300 km/h or by the BCU 4 stage deceleration control at low speed ≤ 50 km/h on motor cars.

The highest pressure step is set if the control valve EVTS2 is energised by the WSP-CU at a speed v ≤ 230 km/h.

<3.2.1.5 - D2: Triple stage relay valve pneumatic logic>



The WSP-CU output to control the two control magnet valves EVTS1 and EVTS2 is commanded by a DIA-5 hardware board and a software controlled output with defined speed thresholds. The accuracy of the speed switching point is given by the software controlled output board, the hardware control board is switching the contacts +/- 10 km/h in regards to the reference speed.

In table 3.2.1.5 - T1 the EVTS1 and EVTS2 outputs are shown in regard to the pressure step control by the WSP-CU.
– EVTS1/2 energised = 1

<3.2.1.5 - T1: Table of EVTS1 and EVTS2 control logic>

Pressure level	EVTS1	EVTS2
High pressure step	1	1
Medium pressure step	1	0
Low pressure step	0	0

3.2.3.1 *Local Indirect 4 Stage Control*

The introduction of a 4 stage deceleration control is needed to reflect the friction coefficient behaviour of high energy sintered pads at low speeds. A significant characteristic of this kind of pads is the increase of friction value starting at a low speed < 70 km/h, to reflect this behaviour in the pure indirect pneumatic hardware control logic a 4 stage is introduced.

<3.2.1.5.1 - T1: Motor car 4 stage indirect brake control>

Motor Car			
Pressure level	Speed range	EVTS1	EVTS2
Medium step	v < 30 Km/h	1	0
High step	30 < v < 230 km/h	1	1
Medium step	230 < v < 300 km/h	1	0
Low step	300 < v < 360 km/h	0	0

<3.2.1.5.1 - T1: Trailer car 4 stage indirect brake control>

Trailer Car			
Pressure level	**Speed range**	**EVTS1**	**EVTS2**
Low step	v < 50 Km/h	0	0
High step	50 < v < 230 km/h	1	1
Medium step	230 < v < 300 km/h	1	0
Low step	300 < v < 360 km/h	0	0

The activation of 4 stage deceleration step is performed on two car level by one BCU output board DIA-5 located in trailer cars to command the 4 stage relay per car. The 4 stage is set by a high active command, the BCU output board DIA-5 has two different output channels activated at different speed levels.

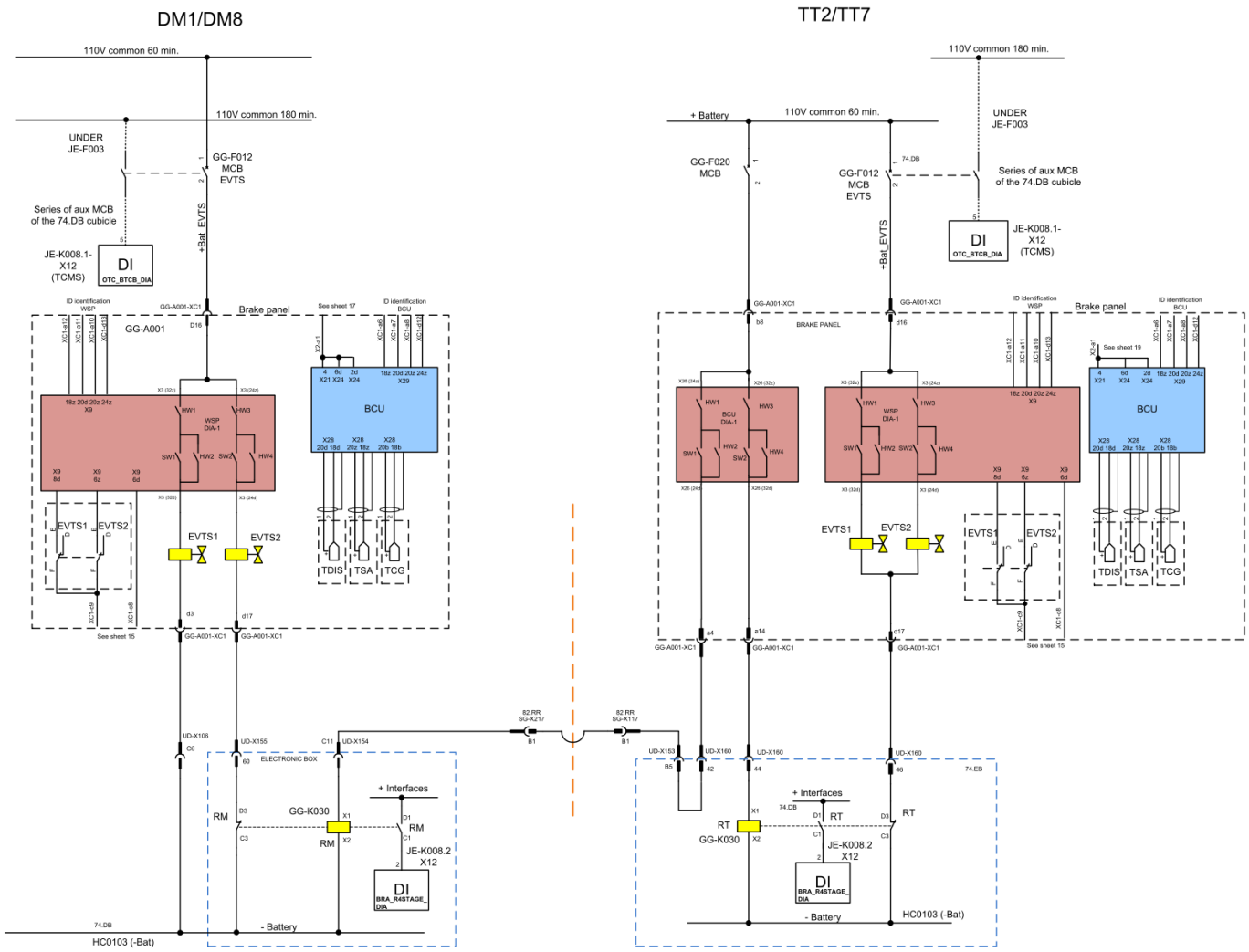
The two DIA-5 output channels are used to command independently trailer and motor car 4 stage relay.
As one BCU output board commands both motor and trailer cars a local train line is introduced on each two car train segment.

In energised status the relay RM is cutting the electrical minus of EVTS2 on motor cars to achieve medium pressure step.
In trailer cars the electrical minus of both EVTS1 and EVTS2 is cut by relay RT to achieve the low pressure step. The relay status is supervised by TCMS diagnostic input.

The BCU output board has the same architecture as the output board of WSP-CU commanding the EVTS1 and EVTS2 valve as described above (see diagram 3.2.1.5 – T1). Two output channels are used to energise the electrical 4 stage relay RM in motor and RT in trailer car at different speed threshold (see diagram 3.2.1.5.1 – D1).

The 4 stage relay RM in motor car is energised by BCU output board at a speed of 30 km/h, on trailer car the output board energise the RT relay at 50 km/h.

<3.2.1.5.1 - D1: Electrical 4 stage indirect brake control as example on DM car and TT car>



3.2.3.2 *Local Indirect Brake Demand Diagnostic*

The three stage pressure control magnet valves EVTS1 and EVTS2 are supervised by a mechanical switch SW_EVTS1 and SW_EVTS2 from the WSP-CU. In case the command to the related EVTS1 or EVTS2 valve does not correspond to the mechanical switch status a fault message is indicated to the driver.

The possible failures of the 3 stage pressure control group, the EVTS1/2 command or 4 stage control is shown in 3.2.1.5.2 - T1. The failure cases are detected by WSP-CU and transmitted to TCMS for diagnostic purpose.

The failures related to 4 stage relays RM and RT are detected by TCMS. Two possible failure cases exist in 4 stage control, unwanted 4 stage at higher speed and 4 stage missing at low speed.
The unwanted 4 stage failure can only happen on two car segment, similar to the loss of brake force of one car.

In case of fatal error on the train and complete loss of the battery bus each car will brake with the lowest pressure step to avoid damage on the brake disc at high speeds. Both EVTS1 and EVTS2 valve are not energised and DEP brake redundant path is not working as the BCU has no power supply.

If only the WSP-CU is not available, the missing triple stage pressure is applied by the DEP brake redundant path by the BCU. If a certain pressure step is set undue by the WSP-CU control logic or hardware failure, the respective car has to be isolated to avoid overbraking at high speeds.

<3.2.1.5.2 - T1: Table of SW_EVTS1, SW_EVTS2 and TCMS DIA supervision logic>

		Command			Supervision (Failures OR conditions)			
Set Condition	Speed [km/h]	EVTS1	EVTS2	4 stage	SW_EVTS1	SW_EVTS2	4st TCMS DIA	Failure Case
Medium pressure step	v < 300	1	0	0	0	0	0	Missing 2 stage
Low pressure step	v > 300	0	0	0	1	0	0	Undue 2 stage
High pressure step	v < 230	1	1	0	1	0	0	Missing 3 stage
Medium pressure step	v > 230	1	0	0	1	1	0	Undue 3 stage
Medium pressure step 4 stage M-car	v < 30	1	0	1	1	1	0	Missing 4 stage motor car
Low pressure step 4 stage T-car	v < 50	0	0	1	1	1	0	Missing 4 stage trailer car
High pressure step M-car	v > 30	1	1	0	NA	NA	1	Undue 4 stage motor car
Medium/high pressure step T-car	v > 50	1	1	0	NA	NA	1	Undue 4 stage trailer car
Low pressure step	-	0	0	0	0	0	0	Power Supply loss

The pressure transducer TDIS 04.34 is used for diagnostic between the brake pipe pressure level and the achieved pilot cylinder pressure after the distributor valve and the triple stage group.

In case of any failure of the 4 stage braking group components the WSP-CU sends a summary event to the TCMS via the IP . Following components are continuously monitored:
- EVTS1 and EVTS2 control magnet valve
- TDIS pressure sensor
- 4 stage relay RM and RT

3.2.4 Failure Information

3.2.4.1 *Normal condition*

Normal condition as a service brake back-up to DEP is the IEP brake mode. IEP service brake is controlled by the driver's brake valve and the DBV-GW control unit.

3.2.4.2 *Failure cases*

1. Failure of brake pipe control by DBV and DBV-GW in IEP brake mode
2. Neutral and isolation valve control failure
3. Incoherence between DBV-GW and TCMS input selection of brake mode (DEP/Indirect brake selector and RIBK isolation cock for IEP/IP brake pipe control mode)
4. Failure of brake pipe control by DBV in IP brake pipe control
5. Failure of brake pipe pressure sensor on cabin panel in indirect brake mode
6. Quick release failure (permanent air supply to brake pipe)
7. Failure of distributor valve on brake frame
8. Brake pipe leakage

3.2.4.3 *Failure detection*

1. Failure detection by DBV-GW, failure in IEP brake mode event sent to TCMS.
2. Electrical switch on valve read by DBV-GW, failure of valve sent to TCMS.
3. Incoherence event by TCMS for brake mode and brake pipe control selection on IDU.
4. Brake pipe pressure supervised by TCMS, DBV failure event on IDU.
5. Failure detection by DBV-GW, failure in IEP brake mode event sent to TCMS.
6. Electrical switch on quick release valve read by DBV-GW, failure of valve sent to TCMS.
7. Brake pipe pressure and pilot pressure not corresponding, event on IDU.
8. Failure detection by DBV-GW, leakage detection event on IDU.

3.2.4.4 *Failure handling*

1. Activation of IP brake mode by driver. Stopping of train to switch from IEP to IP brake mode.
2. No failure handling possible. Stopping fault (major failure) if brake pipe refill not possible.
3. Depends on brake pipe control mode selection:
   a) IEP active, TCMS input shall be ignored
   b) IP active, DBV-GW IP signal shall be ignored
4. No failure handling possible, stopping fault (major failure).
5. Activation of IP brake mode by driver. Stopping of train to switch from IEP to IP brake mode.
6. Failure handling not possible under each case, deactivation of DBV-GW. Stopping fault (major failure) or stopping of train to switch from IEP to IP brake mode.
7. Isolate affected distributor valve by isolation cock RIEP-DIST (isolation of complete brake in one car, major failure). Speed limit to be defined based on infrastructure and thermal limit of brake design.
8. No failure handling possible, could lead to stopping fault (major failure).

3.2.5 Requirements Indirect Brake

3.2.5.1 *Driver's brake valve*

ID	2F_05.02.Zefiro-Europe.TRS.105
Description	A driver's brake valve (DBV) controlling the IP brake shall be provided in the cabs.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.75	A driver's brake valve with IP brake control and emergency brake control shall be installed on the drivers' desk in the cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1075	The driver's brake valve with IP brake control and emergency brake control shall have the following brake and brake pipe control positions: – Quick release – Release position – 7 brake steps – Maximum service brake step – Emergency position	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2127	The DBV shall be able to control the BP pressure according the handle position by the DBV pneumatic control panel installed under the desk.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2137	The position of the DBV read by Minitrol shall be sent to TCMS via the following IP signals by DBV-GW: – Filling position "DBV filling position/filling stroke" – Release postition "DBV release position" – Last release position "DBV last release position 1A" – Apply position 1B "DBV last apply position 1B" – Maximum service brake "DBV max service brake position MB" – Emergency brake "DBV emergency brake position EB"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2139	The following DBV position shall be read directly by TCMS using microswitches on DBV: – Released position RP SW2 – Brake apply position 1B SW3 – Full service brake MB SW4	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2349	The following DBV position shall be connected to TCMS digital input: – Released position RP SW2 – Brake apply position 1B SW3 – Full service brake MB SW4	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.2440	If the DBV position read by TCMS are not align to the positions send by DBV-GW via IP an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2140	DBV Emergency brake position shall be connected to TCMS input using a relay contact driven by the microswitch SW7 on DBV.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2124	The DBV pneumatic panel shall be enabled (connected to the brake pipe) by energizing the Isolation valve (VISO) when the driver's desk is activated or parking mode is active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2129	The driver's desk enabled signal shall be sent to DBV-GW by TCMS via IP signal "Local cab active".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2130	The DBV pneumatic cabin control panel can be isolated by RIM isolation cock.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2131	RIM isolation cock microswitch status shall be read by DBV-GW	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2132	RIM isolation cock status shall be sent to TCMS via IP signal "Cabin panel isolation cock in isolated position (RIM)" by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2125	RIM isolation cock status shall be read by TCMS, the input shall be set (1) if the RIM isolation cock is in not isolated position (IP brake not active).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2353	RIM isolation cock shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2268	If RIM isolation cock TCMS input is not set (=0) OR DBV-GW IP signal "Cabin panel isolation cock in isolated position (RIM)" is set (=1) and event shall be generated by TCMS that RIM brake pipe isolation cock is in isolated position.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2269	If the status of RIM received by DBV-GW IP signal "Cabin panel isolation cock in isolated position (RIM)" and TCMS input is not coherent, an event shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1027	In Direct service brake mode and IEP brake pipe control active, the movement out of release position of the DBV by the driver shall generate an event.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.5.2 *Service brake mode selection*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2145	A selector rotary switch in the driver's cab shall select the active service brake mode: Direct brake mode (DEP) or indirect brake mode (IEP or IP).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2550	A selector rotary switch for service brake mode selection shall be installed on a cabinet in driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2147	The status of the selector shall be connected in a redundant way to two TCMS inputs, in nominal direct brake mode position the TCMS inputs shall be not set (0). The redundant channels shall have separate power supply to avoid single point of failure.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3917	The change of service brake mode direct/indirect shall be only allowed if following conditions are true: – Train at standstill AND – Parking brake is defined applied OR if the brake pipe < 200 kPa	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3930	If the direct/indirect service brake selector switch is turned and preconditions in concept 3917 are not set an event shall be indicated to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2359	In the transition between IEP and DEP with the switch, **if it is NOT respected the condition**	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	2F_05.02.Zefiro-Europe.CONCEPT.3917 and the indirect braker manipulator is in the braking position, it shall be kept the BP pressure at the moment of the IEP to DEP change (shall be kept in braking) and the DEP pressure is not applied (holding brake is not applied). The complete IEP to DEP status swap shall be completed condition of 3917 is satified (e.g. parking brake applied OR emergency brake applied with the train in standstill). After this action (3917) the indirect brake manipulator movement will not lead to BP change and will trigger the event as per 2F_05.02.Zefiro-Europe.CONCEPT.1027				
2F_05.02.Zefiro-Europe.CONCEPT.3931	In case the two TCMS input of service brake mode selector switch are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2148	The status of the service brake mode shall be reported via IP signal "DEP active" to DBV-GW. The signal shall be set in case TCMS status is defined direct service brake mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2361	In case the DEP brake active signal is reset (0) the DBV-GW shall activate Indirect brake mode, the IEP brake mode shall be activated in case the RIBK cock is in position IEP.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2149	DBV-GW shall send the IP signal "Indirect brake activate" to TCMS if the service brake mode Indirect Brake is active.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2179	The TCMS shall verify the "Indirect brake active" IP signal by DBV-GW before setting indirect brake mode as active. Following conditions shall be fulfiled: – Indirect brake mode has been selected by service brake mode selector switch TCMS input AND – "Indirect brake active" IP signal by DBV-GW is set (1) AND – No major failure of DBV_GW is set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2180	There shall be a indirect service brake mode active lamp (pictogram) located on driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3022	The indirect service brake mode active lamp (pictogram) shall be connected to a TCMS output.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2190	In case the indirect service brake mode is defined as active the indirect brake mode lamp (pictogram) shall be illuminated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2176	If TCMS detects a implausibilty of service mode brake mode sent by DBV-GW IP signal "Indirect brake active" different from TCMS selector status input an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2177	If TCMS detects Direct Service brake mode by selector switch and DBV-GW transmit IP signal "Indirect brake active" the Direct Service Brake mode shall be remain active. In case of failure of TCMS inputs the train shall remain in selected service brake mode. If at least one of the valid inputs indicates IEP mode then the Direct Service brake mode selector switch shall be considered in IEP mode position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2178	If the service brake mode selector switch is in indirect brake position and DBV-GW transmit Indirect brake not activated by not setting IP signal "Indirect brake active" = 0, an event shall be set to the driver to switch RIBK isolation cock to activate IP mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3024	If "DEP active" signal is set (1) via IP by TCMS and DBV position received from Minitrol is not in	Derived Requirement	4S_07.-.Braking System		SIL 0

	release position the IP signal "DBV or IP brake used in DEP mode" shall be set.				
2F_05.02.Zefiro-Europe.CONCEPT.2206	If TCMS receive the IP signal "DBV or IP brake used in DEP mode" by the DBV-GW an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.5.3 *IEP and IP brake pipe control selection*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2141	The indirect brake pipe control shall have two operative condition – IEP: indirect brake electro pneumatic control – IP: indirect brake pneumatic control The operative condition will be set by the status of two position isolation cock RIBK	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2142	The status of RIBK cock shall be read by DBV-GW.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2143	The status of RIBK cock shall be read by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2357	The status of RIBK cock shall be connected to TCMS input. If the isolation cock is in IP position the input shall be set (1).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1590	If the RIBK cock in the cab panel is turned to IP mode, the IP brake mode shall be activated. The RIBK status shall be reported to TCMS by reset (0) the IP signal "IP brake not active (RIBK isolation cock)" from DBV-GW.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2358	If major failure is set by the DBV-GW or communication is lost the brake pipe control mode shall be based on TCMS input of RIBK isolation cock.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2207	If TCMS detect RIBK cock in IP position AND the IP signal "IP brake not active" is reset by DBV-GW the IP brake pipe control status shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2211	If IP communication is lost with DBV-GW regardless of the service brake mode, an event to the driver to move RIBK to IP position shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2209	TCMS shall check the plausibilty of RIBK cock TCMS input AND DBV-GW signal "IP brake not active", in case of inconsitency an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2146	If RIBK cock is switched to IP position a permanent visual indication lamp "IP brake pipe control mode active" shall be present on driver's desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1591	If the IP brake has been activated an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.6 Brake pipe control

ID	2F_05.02.Zefiro-Europe.TRS.152
Description	During service braking the brake pipe shall be fixed at 5 bar by the DBV. The distributor valve shall remain released, ie. the IP brake shall not be used.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.204	Brake pipe pressure shall be read by the BCU (pressure transducer TCG) on each car and by DBV-GW on cabin panel (pressure transducer TBP) located in DM1 / DM8.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.602	The brake pipe pressure in all cars (transducer TCG on brake frame) shall be sent via IP signal "Brake pipe pressure TCG" to TCMS by BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3361	The brake pipe pressure in DM1 and DM8 car (transducer TBP on cabin panel) shall be sent via IP signal "Brake pipe pressure TBP" to TCMS by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2151	The brake pipe pressure from sensor TCG received by BCU and sensor TBP by DBV-GW shall be used by TCMS to define the brake pipe pressure of the train in both ends according following logic: – If DMx BCU TCG sensor is OK, the value will be used for the brake pipe pressure in vehicle DMx – If DMx TCG sensor is not OK, DBV-GW TBP senor (Minitrol) value will be used for the brake pipe pressure in vehicle DMx	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3662	The brake pipe pilot pressure in DM1 and DM8 car (transducer TRE) shall be sent via IP signal "Brake pipe pre-control pressure TBP" to TCMS by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2126	The isolation valve VISO status shall be monitored by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2227	If the VISO is not opening on order the IP signal "DIA Isolation valve (VISO) not opening on order" shall be sent by DBV-GW to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2229	If the driver's desk is enabled and IP signal "DIA Isolation valve (VISO) not opening on order" sent by the DBV-GW, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3020	The isolation valve VISO position switch SW_ISO shall be connected to a safe TCMS digital input. The TCMS input shall be set (1) in case the VISO valve is in brake pipe fill position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3017	If the TCMS input of isolation valve SW_ISO and the IP signal "DIA Isolation valve (VISO) not opening on order" by DBV-GW are inconsistent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2228	If the driver's desk is disabled command is active OR EVISO BP isolation command by TCMS is active and VISO is open without order the IP signal "DIA Isolation valve (VISO) opening without order" shall be sent to TCMS by DBV-GW.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2230	If the DBV-GW reports VISO valve is open without order and the IP signal "DIA Isolation valve (VISO) opening without order" is sent to TCMS, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2136	The EVNEU status shall be monitored by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3021	The neutral valve (EVNEU) position switch SWNEU shall be connected to a safe TCMS digital input. The TCMS input shall be set (1) in case the EVNEU is in brake pipe fill position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2231	If emergency brake loop is energised and driver's desk is active and EVNEU is closed (no refill) the DBV-GW shall sent IP signal "DIA Neutral valve (EVNEU) not opening on order" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2232	If emergency brake is not active and driver's desk is active and DBV-GW sends IP signal "DIA Neutral valve (EVNEU) not opening on order" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2233	If emergency brake is active and driver's desk is active and EVNEU is opening without order the IP signal "DIA Neutral valve (EVNEU) opening without order" (refill active) shall be sent by DBV-GW to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2234	If emergency brake is active and driver's desk is active and EVNEU is opening without order set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	by DBV-GW IP signal "DIA Neutral valve (EVNEU) opening without order" an event shall be set.				
2F_05.02.Zefiro-Europe.CONCEPT.3959	Undue brake pipe isolation event shall be indicated to the driver if the: {- the TCMS input of SW_NEU is reset (0) AND the emergency brake loop is energised AND local cab is active; OR - the TCMS input of SW_ISO is reset (0) and local cab is active}; AND Brake tightness test not active. In case of event due to SW_NEU is triggered, fault shall be kept till a new Emergency brake is released AND trigged condition are not true anymore	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3204	A serious fault undue brake pipe re-charge shall be indicated if the: – the TCMS input of SW_NEU is set (1) AND the emergency brake loop is de-energised; OR – the TCMS input of SW_ISO is set (1) and local cab is NOT active; In case serious fault due to SW_NEU is triggered, fault shall be kept till when a new Emergency brake is commanded AND trigged condition are not true anymore	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.2.6.1 *IEP brake pipe control*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2188	If IEP brake pipe control is active brake pipe control shall be managed by Minitrol panel located on cabin panel.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2205	If the driver's brake valve is moved to emergency position the control pressure (TRE) of the pneumatic path to relay valve RV in the minitrol panel shall be vented to 0 kPa and the EVSICU shall be de-energised by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2261	A major failure in brake pipe control shall be set by Minitrol panel in case of following conditions: – TRE sensor is faulty – EVS (feeding valve) / EVF (venting valve) open/short circuit – EVSICU faulty (open/short circuit) – PWM duty cycle out of range – IEP target pressure not achieved.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2431	If a major failure in IEP brake pipe control is detected the IP signal "DBV-GW major fault" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2262	If a major fault of DBV-GW / Minitrol is reported by DBV-GW setting IP signal "DBV-GW major fault" an event shall be set. The driver shall be informed to switch to IP brake pipe control regardless of the service brake mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Overcharge function

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1356	A non-latching push button for brake pipe over charge function (booster) shall be installed at driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.1357	If the non-locking push button for brake pipe over charge function has been pressed the minitrol brake system shall start the overcharge procedure according UIC541-03.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3345	As long the overcharge is activated and ongoing by Minitrol the DBV-GW shall send the IP signal "Overcharge active" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3347	If the DBV-GW sends the IP signal "Overcharge active" TCMS shall display the overcharge active on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3346	If the DBV-GW sends the IP signal "Overcharge active" to TCMS AND the brake pipe pressure will not increase > 520 kPa in 20s an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3348	As long the overcharge assimilation is activated and ongoing by Minitrol the DBV-GW shall send the IP signal "Assimilation active" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3349	If the DBV-GW sends the IP signal "Assimilation active" TCMS shall display the overcharge assimilation active on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3659	The DBV-GW shall set the IP signal "DIA overcharge command permanently active" if the overcharge button input has been continously high > 15s	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3660	An event shall be set if DBV-GW sets IP signal "DIA overcharge command permanently active" to TCMS that overcharge push button has been continously pressed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Brake pipe leakage detection

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2245	DBV-GW shall perform a monitoring of brake pipe pressure read in each car of the train by the BCU and the local brake pipe pressure transducer TBP to diagnose any leakage higher that admitted by UIC541-03.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3943	TCMS shall sent to DBV-GW the actual train configuration (single or multiple traction) via IP signal "Coupled composition active". In case the train is in multiple traction the signal shall be set (=1).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3944	The DBV-GW shall change the time criteria for brake pipe leakage detection based on actual train configuration sent by TCMS to DBV-GW via IP signal "Coupled composition active".	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2246	If any leakage of the brake pipe pressure is detected DBV-GW shall sent the IP signal "BP leakage detected" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2433	If any leakage of the brake pipe pressure is detected an alarm event shall be set by TCMS if the DBV-GW sent IP signal "BP leakage detected".	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Direct service brake mode active

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1298	If Direct service brake is active, IEP brake pipe control is active and no critical failure on IEP control by DBV-GW and driver brake valve is in any position except emergency and no overcharge and assimilation on going, the Minitrol shall maintain the brake pipe at 500 +/- 5 kPa.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2203	In Direct service brake mode and IEP brake pipe control active, if an overcharge command is	Derived Requirement	4S_07.-.Braking System		N/A

	received when the Direct service brake mode is active, the overcharge function shall be performed and consequent assimilation too. During this phase the brake pipe can be > 500 kPa.				
2F_05.02.Zefiro-Europe.CONCEPT.364	In case direct brake mode and IEP brake pipe control are selected, if the brake pipe pressure decreases below 450 kPa and emergency brake is not active the event: "brake pipe vented during DEP" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Indirect service brake mode active

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2220	If Indirect service brake mode is active and IEP brake pipe control is active the DBV position shall control the brake pipe pressure according predefined deceleration levels (electronically controlled by Minitrol).	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2221	If Indirect service brake mode is active and IEP brake pipe control is active the highest brake request of (Optional: master controller), EB loop and DBV shall be used to generare the brake demand by lowering the brake pipe pressure.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2223	If neutral valve is activated (valve closed) any release command shall be ignored by Minitrol.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2224	If neutral valve is activated (valve closed) any braking command shall be performed by Minitrol.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2225	If isolation valve is energized (valve closed) any release or braking command shall be ignored by Minitrol.	Derived Requirement	4S_07.-.Braking System		SIL 0

Quick release function

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2240	When the DBV is moved forward to the filling stroke (quick release) position a quick release command shall be sent to minitrol by DBV-GW.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2241	If Minitrol will receive the quick release command shall initiate the quick release procedure if all precondition are fullfilled (MP pressure > 6 bar, cab active, no isolation active, BP pressure < actual target - x kPa)	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2242	When quick release command is accepted, Minitrol shall directly set the pilot target pressure to 5 bar without any pressure ramp. Once accepted the command, minitrol shall also send the quick release permission to DBV GW in order to let DBV-GW activate the quick release valve EVR.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2243	The quick release procedure will be interrupted in following condition: – Big leakage – Neutral valve closed – Isolation valve closed – Emergency brake active – Indirect braking request – Main pipe pressure > 600 kPa	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2673	The quick release valve EVR shall be blocked if the emergency brake loop is de-energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3354	The quick release magnet valve EVR shall be supervised, therefore the EVR valve position switch SW_EVR shall be read by DBV-GW.	Derived Requirement	4S_07.-.Braking System		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.3352	If the DBV-GW detects a incoherence when the quick release command is active AND EVR valve status is not open the DBV-GW shall send IP signal "DIA Quick release valve (EVR) not opening on order" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3353	If the DBV-GW sends the IP signal "DIA Quick release valve (EVR) not opening on order" to TCMS an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3355	If the DBV-GW detects a incoherence when the quick release command is not active AND EVR valve status is open the DBV-GW shall send IP signal "DIA Quick release valve (EVR) opening without order" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3356	If the DBV-GW sends the IP signal "DIA Quick release valve (EVR) opening without order" to TCMS an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3357	If the DBV-GW detects a open circuit on EVR valve the DBV-GW shall send IP signal "DIA Quick release valve open circuit (EVR)" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3359	If the DBV-GW sends the IP signal "DIA Quick release valve open circuit (EVR)" to TCMS an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3358	If the DBV-GW detects a short circuit on EVR valve the DBV-GW shall send IP signal "DIA Quick release valve short circuit (EVR)" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3360	If the DBV-GW sends the IP signal "DIA Quick release valve short circuit (EVR)" to TCMS an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.6.2 *IP brake pipe control*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2250	In IP brake pipe control active brake pipe control shall be managed by pressure regulator on DBV, via DBV pneumatic panel.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2252	If IP mode is selected the diagnostic of Minitrol shall be inhibited.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2251	If IP brake pipe control is active and the driver's brake valve is in released position the brake pipe shall be regulated at 500 +/- 5 kPa.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2253	If IP brake pipe control is active, moving the DBV in braking position the brake pipe shall be regulated according the DBV regulator characteristic.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2257	If IP brake pipe control is active and brake pipe pressure on any end vehicle < 480 kPa an event of brake pipe big leakage shall be set on IDU if following conditions are true: – DBV in release position for 30s and – no emergency brake active and – driver's desk active The event shall be reset if the brake pipe pressure > 490 kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Direct service brake mode active

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2438	Direct brake request shall be working as specified in chapter 3.1 during IP brake pipe control mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.2.6.3 *Indirect brake demand control*

ID	2F_05.02.Zefiro-Europe.TRS.121
Description	A brake control pipe connecting all brake control panels distributor valves in the train according to UIC shall be provided.
Safety level	N/A

ID	2F_05.02.Zefiro-Europe.TRS.127
Description	The distributor valve on each brake panel shall apply the braking pressure in the brake cylinders following the brake pipe depression.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2152	Brake pipe pressure shall be converted to pilot control pressure by the UIC distributor valves.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2161	Indirect brake of each car shall be considered isolated if RIEP-DIST is isolated OR RICP is isolated. Note: Definition of isolation is described in chapter 3.1.2.7.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2159	The status condition of the indirect brake OK/faulty/isolated of each car shall be displayed on the IDU brake menu: – OK if IP signal "Triple stage brake faulty" is not set by WSP-CU and communication to WSP-CU is ok – Faulty if IP signal "Triple stage brake faulty" is set by WSP-CU OR communication is lost to WSP-CU – Isolated if isolation cock RIEP-DIST (BCU OR TCMS input) OR RICP (BCU OR TCMS input) OR both RICF1 and RICF2 (BCU OR TCMS input) are isolated	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2164	The auxiliary brake reservoir pressure (pressure transducer TSA) shall be read and monitored by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2165	If the TSA pressure value is > 700 kPa the BCU shall be sent the IP signal "Brake reservoir pressure avaiable (TSA>700 kPa)" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2166	If the BCU reset the IP signal "Brake reservoir pressure available (TSA>700 kPa)" AND main reservoir pressure > 600 kPa AND isolation cock RIEP-DIST is not in isolated position in the same car an event shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3027	In case more than two BCUs reset the IP signal "Brake reservoir pressure available (TSA>700 kPa)" AND speed v > 3 km/h TCMS shall open the urgency brake loop. The reset (=0) of IP signal "Brake reservoir pressure available (TSA>700 kPa)" shall not be considered if isolation cock RIEP-DIST is in isolated position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.2167	In order to fulfill the adhesion requirements and to limit the brake power on the brake disc, the control pressure output of UIC distributor shall be converted by relay valve 04.44, in a pilot pressure to relay valves 04.18 and 04.19 depending from the speed according the following ratio: – 360-300 km/h: 50% of emergency pilot pressure – 300-230 km/h: 75% of emergency pilot pressure – 230-0 km/h: 100% of emergency pilot pressure	Derived Requirement	4S_07.-.Braking System		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.2168	The control of the 3 stages of relay valve 04.44 shall be done by WSP control unit commanding the valves EVTS1 and EVTS2 per car according to the following logic: EB brake step / Speed range / EVTS1 / EVTS2 High step / 0 < speed < 230 km/h / 1 / 1 Medium step / 230 < speed < 300 km/h / 1 / 0 Low step / 300 < speed < 360 km/h / 0 / 0 1 = energized and 0 = not energized	Derived Requirement	4S_07.-.Braking System		SIL 3

3.2.6.3.1 *Indirect brake demand diagnostic WSP-CU*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2169	The valves EVTS1 and EVTS2 shall be monitored by WSP CU reading the mechanical switches EVTS1_SW and EVTS2_SW.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2271	If the command to EVTS1 is active and the status of the switch EVTS1_SW is NOT set the IP signal "DIA Missing second stage braking" shall be sent to TCMS by the WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2272	If TCMS receives the IP signal "DIA Missing second stage braking" on EVTS1 by WSP-CU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3028	If the command to EVTS1 is NOT active and the status of the switch EVTS1_SW is set the IP signal "DIA Undue second stage braking" shall be sent to TCMS by the WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3029	If TCMS will receive the IP signal "DIA Undue second stage braking" on EVTS1 by WSP-CU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2274	If the command to EVTS2 is active and the status of the switch EVTS2_SW is NOT set the IP signal "DIA Missing third stage braking" shall be sent to TCMS by the WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2273	If TCMS will receive the IP signal "DIA Missing third stage braking" on EVTS2 by WSP-CU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3030	If the command to EVTS2 is NOT active and the status of the switch EVTS2_SW is set the IP signal "DIA Undue third stage braking" shall be sent to TCMS by the WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3031	If TCMS will receive the IP signal "DIA Undue third stage braking" on EVTS2 by WSP-CU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2170	If WSP-CU has a major failure on the triple stage group (EVTS1 and EVTS2) permanently de-energised or energised and not corresponding to speed switching point (summary event) the IP signal "DIA Triple stage control faulty" shall be transmitted to TCMS by WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.2367	If triple stage control is faulty (summary event) send by WSP-CU IP signal "DIA Triple stage control faulty" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3575	If the WSP-CU detects a short circuit on triple stage EVTS1 valve the IP signal "DIA Triple stage valve EVTS1 short circuit" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3576	If the WSP-CU reports a short circuit on triple stage EVTS1 valve via IP signal "DIA Triple stage valve EVTS1 short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3577	If the WSP-CU detects a short circuit on triple stage EVTS2 valve the IP signal "DIA Triple stage valve EVTS2 short circuit" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3578	If the WSP-CU reports a short circuit on triple stage EVTS2 valve via IP signal "DIA Triple stage valve EVTS2 short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3581	If the WSP-CU detects a triple stage hardware board control failure EVTS1 the IP signal "DIA Triple Stage DIA1 HW2 failure" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3582	If the WSP-CU reports a triple stage HW control failure via IP signal "DIA Triple Stage DIA1 HW2 failure" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3583	If the WSP-CU detects a triple stage hardware board control failure EVTS2 the IP signal "DIA Triple Stage DIA2 HW2 failure" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3584	If the WSP-CU reports a triple stage HW control failure via IP signal "DIA Triple Stage DIA2 HW2 failure" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3913	If status of switch EVTS1_SW and EVTS2_SW is set at v < 30 km/h on motor and v < 50 km/h on trailer the IP signal "DIA Missing fourth stage braking" shall be sent to TCMS by the WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3914	If TCMS will receive the IP signal "DIA Missing fourth stage braking" by WSP-CU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3915	If above speed v > 35 km/h on motor and v > 55 km/h on trailer the 4 stage relay RM/RP supervision input is set (1) TCMS shall set an event undue fourth stage.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.6.3.2 *Indirect brake demand diagnostic BCU*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2172	The pilot pressure to relay valves 04.18 and 04.19 shall be monitored by BCU reading the pressure sensor TDIS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1639	The BCU shall sent the indirect pilote pressure from sensor TDIS via IP signal "Indirect brake pilote pressure TDIS" to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.162	The status of indirect brake demand by distributor valve (04.24/01) shall be checked continuously and status shall be reported to TCMS. In case of deviation between brake pipe pressure TCG and distributor valve control pressure pressure TDIS a failure shall be reported to TCMS via IP signal "DIA indirect brake integrity failure".	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1080	If BCU reports deviation between brake pipe pressure TCG and distributor valve control pressure pressure TDIS via IP signal "DIA	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	indirect brake integrity failure", an event "Indirect brake integrity failure" shall be set.				
2F_05.02.Zefiro-Europe.CONCEPT.2277	If BCU detects a TDIS sensor failure a message shall be sent to TCMS via IP signal "DIA TDIS - pressure transducer faulty".	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2278	If TDIS pressure sensor is reported as faulty and BCU sends IP signal "DIA TDIS - pressure transducer faulty" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3858	The triple stage group shall be monitored in redundancy by BCU comparing the TDIS indirect pre-control pressure with the TCG brake pipe pressure.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3859	If the second stage speed range (300 - 230 km/h) is active and the indirect pre-control pressure is less than expected pressure based on brake pipe pressure the IP signal "DIA Missing second stage braking BCU" shall be sent to TCMS by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3860	If TCMS receives the IP signal "DIA Missing second stage braking BCU" by BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3864	If the second stage speed range (300 - 230 km/h) is not active and the indirect pre-control pressure is according expected pressure based on brake pipe pressure the IP signal "DIA Undue second stage braking BCU" shall be sent to TCMS by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3865	If TCMS receives the IP signal "DIA Undue second stage braking BCU" by BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3866	If the third stage speed range (230 - 0 km/h) is active and the indirect pre-control pressure is less than expected pressure based on brake pipe pressure the IP signal "DIA Missing third stage braking BCU" shall be sent to TCMS by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3867	If TCMS receives the IP signal "DIA Missing third stage braking BCU" by BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3868	If the third stage speed range (230 - 0 km/h) is not active and the indirect pre-control pressure is according expected pressure based on brake pipe pressure the IP signal "DIA Undue third stage braking" shall be sent to TCMS by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3869	If TCMS receives the IP signal "DIA Undue third stage braking BCU" by BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3870	If BCU has detected a failure on the triple stage group (TDIS and TCG pressure not consistent in speed range) the IP signal "DIA Triple stage control faulty BCU" (summary event) shall be transmitted to TCMS by BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3871	If triple stage control is faulty (summary event) send by BCU IP signal "DIA Triple stage control faulty BCU" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3927	If the high pressure stage is set at v < 30 km/h on motor and v < 50 km/h on trailer the IP signal "DIA Missing fourth stage braking" shall be sent to TCMS by the WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3928	If TCMS will receive the IP signal "DIA Missing fourth stage braking" by BCU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2171	If the IP signal "DIA Undue second stage braking" OR "DIA Undue third stage braking" is set by the WSP-CU OR BCU a serious fault shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 2

	In case both valid signals from BCU and WSP-CU are not plausible for > 2s TCMS shall set both signals to invalid.				
2F_05.02.Zefiro-Europe.CONCEPT.3317	If the IP signal "DIA Undue second stage braking" OR "DIA Undue third stage braking" is set by the WSP-CU OR BCU an event shall be set informing the driver to isolate the affected car by brake isolation cock RIEP-DIST.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3032	If the IP signal "DIA Missing second stage braking" OR "DIA Missing third stage braking" is set by the WSP-CU OR BCU a serious fault shall be set. In case both valid signals from BCU and WSP-CU are not plausible for > 2s TCMS shall set both signals to invalid.	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3225	If the IP signal "DIA Missing second stage braking" OR "DIA Missing third stage braking" is set by the WSP-CU OR BCU and in the same car the Direct Brake is faulty, the driver shall be informed to isolate the affected car by brake isolation cock RIEP-DIST and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2.6.4 *Traction block / ED brake block*

ID	2F_05.02.Zefiro-Europe.TRS.167
Description	If service brake is applied by brake pipe, traction shall be blocked in the whole train if the brake pipe pressure is less than 4,1 +/-0,2 bar (equivalent 50% brake force). Traction shall be enabled again if the brake pipe pressure exceeds 4,6 +/- 0,2 bar.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2174	Traction block shall be set if brake pipe pressure in both end cars of the train is < 460 kPa AND speed v > 5 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1503	Traction block shall be reset if brake pipe pressure in both end cars of the train is > 480 kPa OR speed v ≤ 5 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2199	Traction shall be blocked in direct brake mode if driver's brake valve is moved from release position (SW2 is reset (0)) derived from TCMS input AND if IP signal "DBV release position" is reset (0) by DBV-GW.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.120
Description	In IP brake mode the ED brake shall be disabled in the whole train.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2427	ED brake block shall be set if brake pipe pressure in any end car of the train is < 460 kPa, whatever service mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2428	ED brake block shall be reset if brake pipe pressure in any end car of the train is > 480 kPa, whatever service mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2429	ED brake block shall be set if Indirect brake service mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.2.6.5 *Brake Pipe Control Diagnostic*

ID	2F_05.02.Zefiro-Europe.TRS.317
Description	IP Brake and brake pipe supervision: Diagnostics from the brake computer shall be stored in the train diagnostic system.

Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.985	TCMS shall make a comparison between brake pipe pressure values received by DBV-GW and BCU on the same DM car in steady state condition: – If DBV-GW TBP pressure doesn't change its value of more than ±0,05 bar in 30s, it shall be compared with the BCU TCG sensor. If the difference is higher than ±0,15 bar an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3663	Main pipe pressure transducer TMP shall be read by Minitrol, DBV-GW shall send the TMP pressure via IP signal "Main pipe pressure TMP" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.986	Pressure sensor TMP for MR-pipe located on Minitrol panel shall be supervised by Minitrol, in case of failure on sensor the DBV-GW shall sent IP signal "DIA TMP - Main pipe pressure sensor fault" to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1078	If DBV-GW detects a fault on main pipe pressure sensor TMP on Mintirol pneumatic panel (in DM cars) and sets IP signal "DIA TMP - Main pipe pressure sensor fault", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.987	Pressure sensor TRE for pilot brake pipe control pressure shall be supervised by Minitrol, in case of failure on sensor the DBV-GW shall sent IP signal "DIA TRE - Brake pipe pilot pressure sensor fault" to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1079	If DBV-GW detects a fault on pilot brake pipe pressure sensor TRE (in DM cars) and sets IP signal "DIA TRE - Brake pipe pilot pressure sensor fault", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.783	A short circuit on filling valve for the brake pipe EVS in DM car shall be reported to TCMS via IP signal "DIA Brake pipe filling valve short circuit (EVS)" by DBV-GW.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1081	If DBV-GW detects a short circuit on filling valve for brake pipe EVS (in DM cars) and sets IP signal "DIA Brake pipe filling valve short circuit (EVS)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.784	A open circuit on filling/venting valve for the brake pipe EVS and EVF in DM car shall be reported to TCMS via IP signal "DIA Brake pipe filling/venting valve open circuit (EVS/EVF)" by DBV-GW.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1082	If DBV-GW detects a open circuit on filling/venting valve for brake pipe EVS and EVF (in DM cars) and sets IP signal "DIA Brake pipe filling/venting valve open circuit (EVS/EVF)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2280	If DBV-GW detects a short circuit on EVF valve for brake pipe venting a failure shall be reported via IP signal "DIA Brake valve short circuit (EVF)" to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2282	If DBV-GW reports a short circuit on EVF valve for brake pipe venting and sets IP signal "DIA Brake valve short circuit (EVF)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2281	If DBV-GW detects a short/open circuit on EVSICU valve for the brake pipe control a failure shall be reported to TCMS via IP signal "DIA EVSICU faulty (open/short circuit)".	Derived Requirement	4S_07.-.Braking System		N/A

2F_05.02.Zefiro-Europe.CONCEPT.2283	If DBV-GW reports via IP signal "DIA EVSICU faulty (open/short circuit)" a faulty EVSICU valve for brake pipe control, an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1083	If BCU detects a faulty brake pipe pressure sensor TCG a failure shall be reported via IP signal "DIA TCG - Brake pipe pressure sensor fault" to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3651	If DBV-GW detects a faulty brake pipe pressure sensor TBP a failure shall be reported via IP signal "DIA TBP - Brake pipe pressure sensor fault" to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.785	If BCU or DBV-GW reports a faulty brake pipe pressure sensor TCG (BCU TCG in each car on brake frame, DBV-GW TBP on Minitrol panel in DM cars) via IP signal "DIA TCG - Brake pipe pressure sensor fault", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.533	The brake pipe pressure derived from sensor TCG (BCU) OR TBP (DBV-GW) send via IP signal "Brake pipe pressure TCG" OR "Brake pipe pressure TBP" shall be indicated in the cab on the IDU brake menu in a range 0...550 kPa. The BP pressure shall be derived from the local TCG (BCU) for local cab car on the IDU. As back-up from the TBP (DBV-GW).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.02.Zefiro-Europe.CONCEPT.348	If DBV-GW detects a failure on PWM input from DBV (automatic brake controller) the IP signal "DIA DBV PWM error" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1099	If DBV-GW reports a failure on PWM communication via IP signal "DIA DBV PWM error", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3329	If DBV-GW detects a failure on CAN communication to Minitrol the IP signal "DIA DBV GW Minitrol CAN communication fault" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3330	If DBV-GW reports a failure on CAN communication to Minitrol via IP signal "DIA DBV GW Minitrol CAN communication fault", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3338	If DBV-GW detects an open circuit of EVNEU valve the IP signal "DIA Neutral valve open circuit (EVNEU)" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3339	If DBV-GW reports an open circuit of EVNEU valve via IP signal "DIA Neutral valve open circuit (EVNEU)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3340	If DBV-GW detects a short circuit of EVNEU valve the IP signal "DIA Neutral valve short circuit (EVNEU)" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3341	If DBV-GW reports a short circuit of EVNEU valve via IP signal "DIA Neutral valve short circuit (EVNEU)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3362	If DBV-GW detects a minor fault on communication with Minitrol (summary event) the IP signal "DBV-GW minor fault" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3363	If DBV-GW reports a minor fault on communication with Minitrol (summary event) via IP signal "DBV-GW minor fault", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3364	If DBV-GW detects a difference for RE target pressure and pressure monitored by TRE sensor the IP signal "DIA RE Target not achieved" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A

2F_05.02.Zefiro-Europe.CONCEPT.3365	If DBV-GW reports a difference for RE target pressure and pressure monitored by TRE sensor via IP signal "DIA RE Target not achieved", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3366	If DBV-GW detects an open circuit of EVPCG brake pipe test valve the IP signal "DIA BP exhaust valve (continuity test) open circuit (EVPCG)" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3367	If DBV-GW reports an open circuit of EVPCG brake pipe test valve via IP signal "DIA BP exhaust valve (continuity test) open circuit (EVPCG)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3368	If DBV-GW detects an short circuit of EVPCG brake pipe test valve the IP signal "DIA BP exhaust valve (continuity test) short circuit (EVPCG)" shall be reported to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3369	If DBV-GW reports an short circuit of EVPCG brake pipe test valve via IP signal "DIA BP exhaust valve (continuity test) short circuit (EVPCG)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3370	The DBV-GW shall send IP signal "DBV-GW start up OK" to TCMS if no HW problems are detected in brake control unit (board failure, etc.).	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3371	If the DBV-GW is not sending the IP signal "DBV-GW start up OK" after train start-up OR the DBV-GW is reseting the signal during running an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3664	The DBV-GW shall send the life sign via IP signal "DBV-GW life sign" to TCMS if the communication is working without failures.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3665	If the DBV-GW is not sending the life sign counter via IP signal "DBV-GW life sign" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2027	The active driver's cab brake pipe pressure shall be sent back to all WSP-CUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3813	The DBV-GW (Minitrol) status isolated (by RIM cock) / faulty (DBV-GW major fault) shall be shown on IDU brake menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.3 Holding Brake

3.3.1 Function Design

Holding brake is applied as a defined direct brake force by TCMS. The holding brake keeps the train standing still and holds the train in gradient to prevent rolling.

The holding brake keeps the train at standstill in 40 permille slope with maximum exceptional load. Therefore, the holding brake force is not load compensated as the specified force is always the maximum needed to fulfil the requirements.

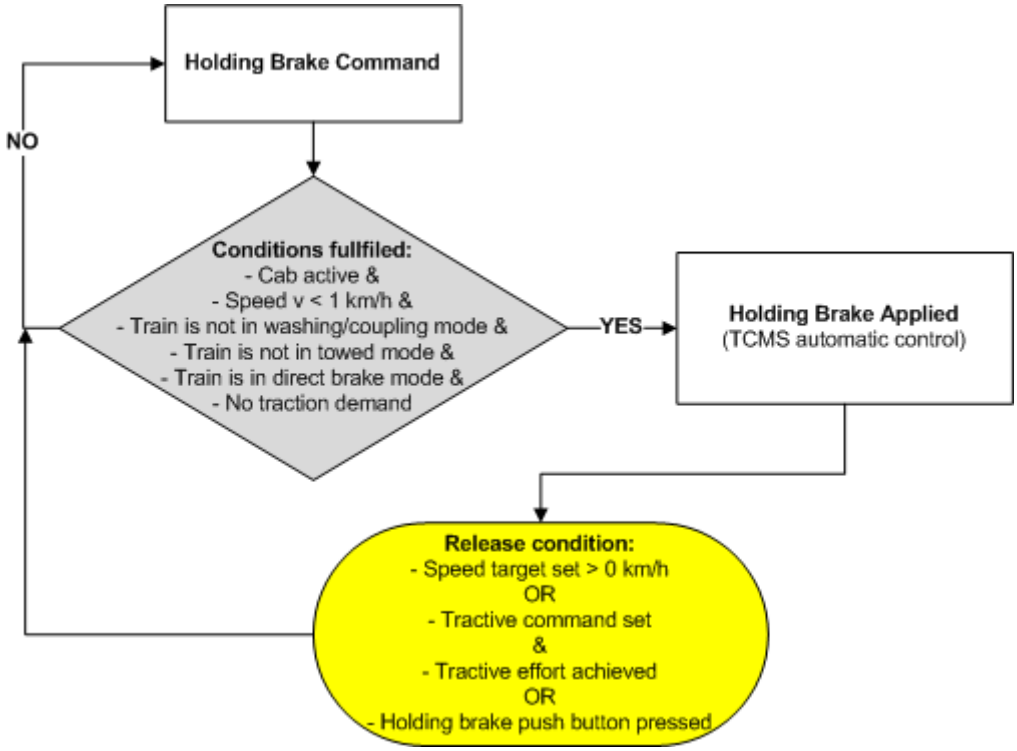
The holding brake is automatically activated when the train comes to stand still. In case a higher brake demand is set by the direct service brake, always the highest brake demand is defined.

The holding brake is automatically released when the train is accelerating from standstill and if the traction force exceeds a specific value. The driver is able to start the train without rolling backwards in a slope with 35 permille.

The holding brake can be manually released by the driver by the "Holding brake release" push button located on driver's desk.

If the indirect service brake mode is active (refer to chapter 3.2.1.2) or towed mode is active the holding brake is automatically permanent released by TCMS.

<3.3.1 - D1: Holding brake function flow chart>



The holding brake release function is depending of the parking brake applied (minimum amount of parking brake, refer to chapter 3.7) and doors released status.

1. Train at standstill, no parking brake applied and doors are not released
   – The holding brake is released as long as the holding brake release push button is pressed.

2. Train at standstill, no parking brake applied and doors are released
   – The holding brake is released as long as the holding brake release push button is pressed for more than 3s.

3. Train at standstill, parking brake applied
   – The holding brake is permanent released in case the holding brake release push button is pressed once. The holding brake is applied if the holding brake push button is pressed again once or the parking brake status is changed (released or applied and minimum amount not available).

The illuminated push button "Holding brake release" has the following visual status:
– Illuminated = holding brake command is active and applied
– Not illuminated = no holding brake active

Holding brake supervision

The holding brake force is supervised in the same logic as the direct service brake force by the BCU. In case one car has no brake force available, the needed force is shifted to all remaining available cars.

In case the train starts to move unintended and holding brake command still active the urgency/emergency brake is applied by TCMS. This supervision of the correct function of the holding brake performance is always active by TCMS in case the holding brake command is set.

The driver can reset the urgency/emergency brake TCMS demand by pressing the holding brake release push button. The emergency brake demand is reset as long the button is pressed.

During train parking mode the holding brake is active if no other higher brake request is set and the parking brake is applied.

3.3.1.1 *Failure Information*

3.3.1.1.1 *Normal condition*

In normal conditions holding brake applies automatically when the train comes to standstill and no traction demand is present.

3.3.1.1.2 *Failure cases*

1a. Holding brake not applied at standstill in direct brake mode
1b. Holding brake force not applied by driver in indirect brake mode
2. Holding brake applied at speed
3. Holding brake (DEP brake) faulty in one or more cars

3.3.1.1.3 *Failure detection*

1a. Holding brake not applied if requested and train starts to move, standstill supervision is active by TCMS and urgency/emergency brake is applied.
1b. Holding brake force not applied and train starts to move, only rollback protection active by ATP system. No roll forward supervision as in direct brake mode.
2. Visual indication of brake applied to driver and dragging brake alarm by TCMS.
3. BCU internal diagnostic and DEP brake faulty event presented to the driver.

3.3.1.1.4 *Failure handling*

1a. Urgency/emergency brake is applied while holding brake is requested. No impact on train operation.
1b. Urgency/emergency brake is applied in case of rollback. No automatic action in case of roll forward, driver is responsible to supervise train movement.
2. Driver to stop the train and isolate the car with affected direct brake active by RIEP isolation cock or switch to indirect brake mode.
3. Holding brake force re-allocation to all remaining available cars with DEP brake performance. No effect on train level.

3.3.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.20
Description	The holding brake shall apply a certain amount of brake cylinder pressure needed to keep the train stationary at 40 promille. It shall automatically be applied when the train comes to a stop.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.539	Holding brake shall be activated automatically if all the following conditions are fulfilled: – Train speed < 1 km/h AND – Train is not in washing speed mode AND – Train is not in coupling speed mode AND – Train is not in towed mode AND – No traction demand OR master controller in neutral OR brake position AND – Direct service brake mode active	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.427	Holding brake force shall be distributed over all available braked cars in the following way: Total holding brake force shall be = 8 / number of available BCUs (DEP brake not faulty) * 230 kN	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.542	The urgency brake shall be activated (urgency brake loop de-energised) in case all following conditions apply: – Train was at standstill for 3s AND – Holding brake push button not pressed AND	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

	– Train speed > 0 km/h for 0.5s (value and time to be specified during comissioning) AND – Master Controller not in traction position AND – Train is not in towed mode AND – Train is in direct service brake mode (Indirect service brake mode not active)				
2F_05.02.Zefiro-Europe.CONCEPT.1973	An event "Holding brake failure - Emergency brake applied" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.132
Description	The holding brake shall be automatically released when the applied traction is enough to start the train at the current slope in order to avoid rolling backward. The holding brake cylinder pressure shall be ramped down against the achieved traction force.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.219	Holding brake force shall be ramped down against the amount of achieved traction force.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.541	Holding brake shall start to release automatically if – v set >0 or tractive demand >0 AND – Tractive effort achieved > 150 kN per train set. OR – tractive demand / v set > 0 AND train speed v > 0	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.129
Description	One push button shall be provided in the cab for holding brake release. The driver shall be able to manually release the holding brake by pushing and holding the button down (spring loaded button).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.429	There shall be one illuminated push button for holding brake. The button shall be spring loaded and non-resting and release the holding brake if pressed.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3938	The illuminated holding brake push button shall be connected to TCMS input. The TCMS input shall be set (=1) if the holding brake push button is pressed.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3939	The lamp of illuminated holding brake push button shall be connected to TCMS output. The push button shall be illuminated (=1) if the TCMS output is set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.222	Holding brake shall be released based on parking brake condition: 1. Parking brake not applied: – Holding brake shall be released as long as the Holding brake suspend push button is pressed down for 1s or more except in door released status. – In door released status active the holding brake button shall be pressed > 3s before holding brake shall be released. – Holding brake shall be applied again if the button is released. 2. Parking brake applied (minimum amount required): – Holding brake shall be released once the holding brake push button has been pressed (positive edge for 1s) – Holding brake shall be applied once the holding brake push button has been pressed again (positive edge for 1s) OR the parking brake applied condition is reset OR cab is deactivated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.545	When holding brake is applied the holding brake push button shall be illuminated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.544	If the urgency brakes are applied due to faulty holding brake (see requirement 542) the urgency brake shall be released if the master controller has been moved to emergency brake position OR indirect service brake mode is defined.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3653	An event shall be set if the holding brake release push button has been pressed > 30s AND train speed v > 10 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.4 Urgency Brake (Included, not used)

3.4.1 Function Design

The urgency brake mode is a provision to use the electro-dynamic brake in emergency situation to achieve the shortest stopping distance. The urgency brake mode can be fully implemented by installing the emergency brake supervision (EBS) device which supervise the brake performance in blending operation.

The urgency brake is a direct electro-pneumatic brake in combination with the electro-dynamic brake in blending operation. The urgency brake is triggered by the electrical urgency brake loop and controlled by the TCMS.

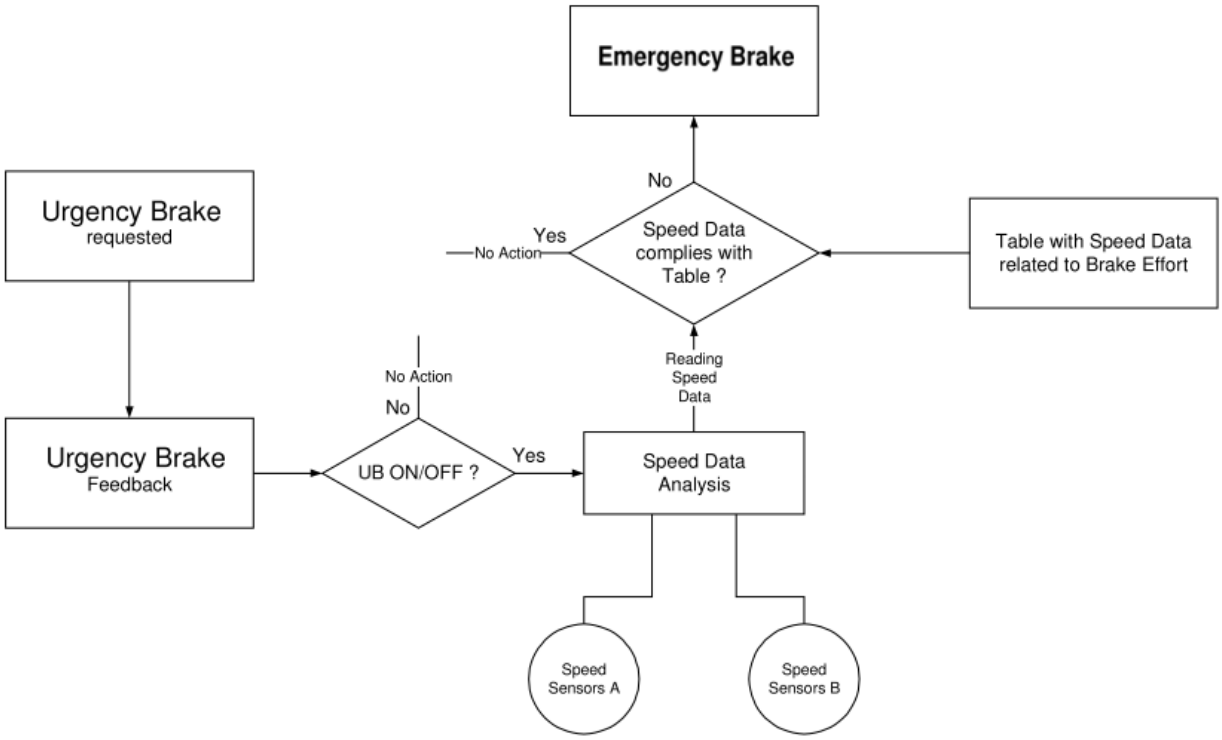
Urgency brake is done by a controlled deceleration curve to reduce the thermal load on the brake discs during the brake application. The brake curve characteristic for the friction brake is based on the availability of the ED brake.

The urgency brake is managed by TCMS like in direct service brake mode, the brake force demand is sent to the BCU via IP . In addition, the urgency brake loop is read by the BCU and WSP-CU, if the brake demand by TCMS is not send due to any reason the BCU applies a predefined pressure if the loop is de-energised to compensate the loss of friction brake force demand.

The urgency brake is load compensated, to calculate the needed brake force the same blending logic is used as described in 3.1.2.3, the only difference is that the brake force is calculated on car level.

Note:
As long the urgency brake mode with ED brake is only a provision, the urgency brake has the same performance as the emergency brake with 4-step deceleration (with 3 pressure steps, refer to 3.2.1.5) control.
In this case the BCU is following directly the input of the urgency brake loop or the safe IP UB demand to apply the 4-step deceleration by the direct braking group (same pressures as in indirect emergency brake control). As a third redundancy TCMS sends a full direct service brake force request via the IP to the BCU.

<3.4.1 - D1: Emergency brake supervision concept>



The urgency brake performance is corresponding to the emergency brake performance but with different deceleration curve characteristic.

The urgency brake will be activated by:

- Master controller (emergency position)
- Automatic train protection (ATP)
- Emergency push button
- TCMS
- Driver's brake valve
- Power supply to the urgency brake loop is cut
- Train part is lost (unintentionally de-coupled)

As the urgency brake function is only a provision, the urgency brake loop is directly connected to the emergency brake loop. Each urgency brake loop intervention is activating the emergency brake loop by de-energising the emergency brake loop.

In case the urgency brake function will be used in the future, the urgency brake loop will be disconnected from the emergency brake loop. In between the safety device emergency brake supervision (EBS) will be placed to monitor the urgency brake performance to activate the emergency brake in case the brake performance is not sufficient. The supervision is performed based on speed set points and corresponding deceleration level.

3.4.1.1 *Urgency Brake Loop*

The urgency brake loop design is in conjunction with emergency brake loop design. The urgency brake loop is feed in the active cab. The urgency loop has an intervention line and actuator line as the return of the train end.

The urgency brake loop has following intervention in the active cab feeding chain, these contacts are only working in the active cabin:

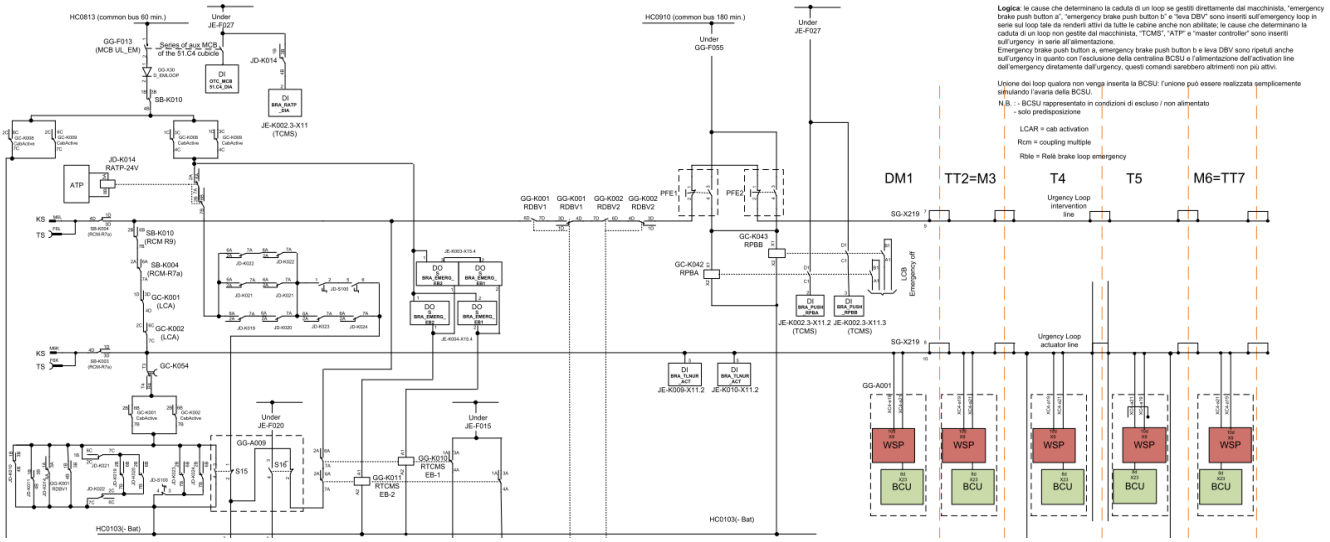
- Master controller (emergency position)
- Automatic train protection (ATP)

- TCMS

In the urgency brake intervention line contacts of following devices are implemented, these commands are always active as the devices are venting the brake pipe in parallel:

- Emergency push button
- Driver's brake valve

<3.4.1.1 - D1: Urgency brake loop design>



The urgency brake actuator line is connected to the BCU and WSP-CU for back-up function and diagnostic.
All interventions of the active cab feeding chain have a second contact used for the loop shorting for safety reason to guarantee that the loop will be de-energised.

The urgency brake loop actuator line is supervised by TCMS inputs in each DM car. In case of failure or incoherence of urgency brake loop an alarm will be presented to the driver.

In addition, each BCU and WSP-CU will compare the status of the urgency brake loop input to the IP signal sent by TCMS for diagnostic purpose.

3.4.1.2 *Failure Information*

3.4.1.2.1 *Normal condition*

In normal condition the urgency brake loop is closed and opened only if the driver is requesting a urgency brake via the master controller or the TCMS is ordering an urgency brake via the TCMS emergency relay.

3.4.1.2.2 *Failure cases*

1. Failure on urgency brake loop
2. Failure of DEP brake
3. Failure of urgency brake blending management by TCMS

3.4.1.2.3 *Failure detection*

1. Urgency brake loop supervised by two control points, event on IDU
2. Failure detected by BCU, fault message send via IP . Event on IDU
3. No failure detection possible

3.4.1.2.4 *Failure handling*

1. Isolation of direct brake in car with unwanted de-energised urgency brake loop by isolation cock RIEP. Train to be stopped to perform isolation.
2. In case of loss more than two cars switch to indirect brake mode. Train to be stopped to perform switching between direct brake to indirect brake mode.
3. Bypass of TCMS urgency brake function by switching to emergency operation mode

3.4.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.80
Description	An Emergency brake position of the master controller shall be provided. It shall be equivalent to the ultimate position of the braking position. In this position, the emergency or urgency brake shall be applied.
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.228	There shall be an emergency brake position in the master controller (traction/brake controller) connected to (intervention) the urgency brake loop feeding line.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_05.02.Zefiro-Europe.CONCEPT.431	When master controller is in emergency brake position both EB contacts of switch 1 and 2 (normaly open) shall de-energise the urgency brake loop feeding line in series. Normaly closed contact of switch 1 shall be connected to the shorting and second normaly closed contact of switch 2 shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.1055	If the master controller grey code TCMS input is in emergency brake position the IP signal "Urgency brake demand" shall be sent by TCMS to BCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 1

3.5 Emergency Brake

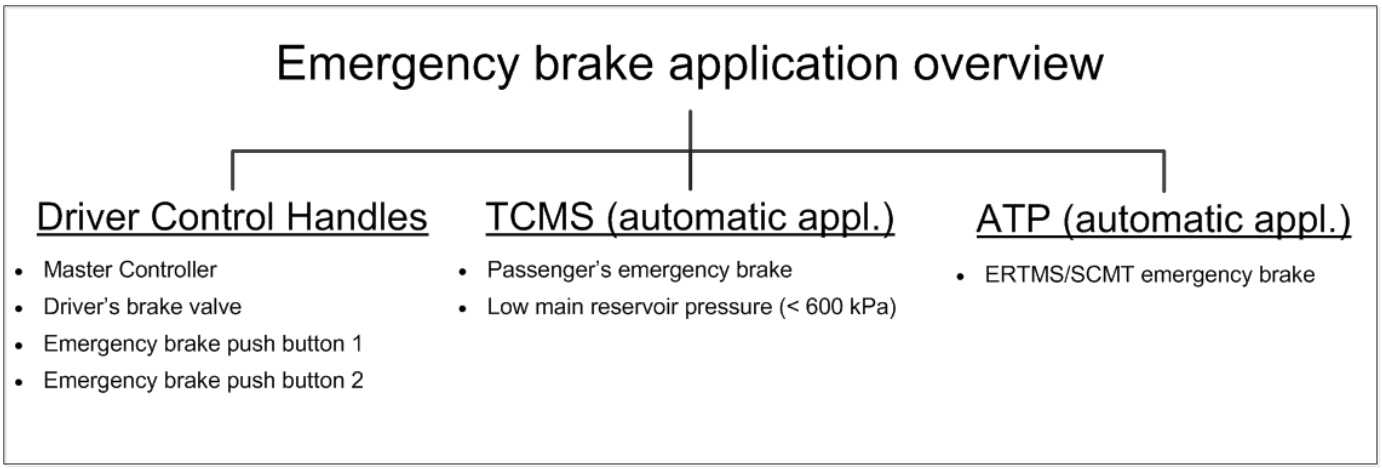
3.5.1 Function Design

The emergency brake mode is used to stop the train in the shortest braking distance and with highest level of integrity. The emergency brake is not load compensated as the service brake. The emergency brake performance is based on TSI 2014 par. 4.2.4.5.2 and the contractual requirements.

Indirect brake pipe controlled pneumatic brake application in combination with electrical emergency brake loop and second path via IP bus for direct brake command are used for the emergency brake function.

Emergency braking is done in 4 stages to reduce thermal load on the brake discs at high speed and reflect pad friction behaviour at low speed as described in chapter 3.2.1.5.

<3.5.1 - D1: Emergency brake overview>



The high level of integrity and fail safe functionality is required for emergency brake and guaranteed by a redundant concept.

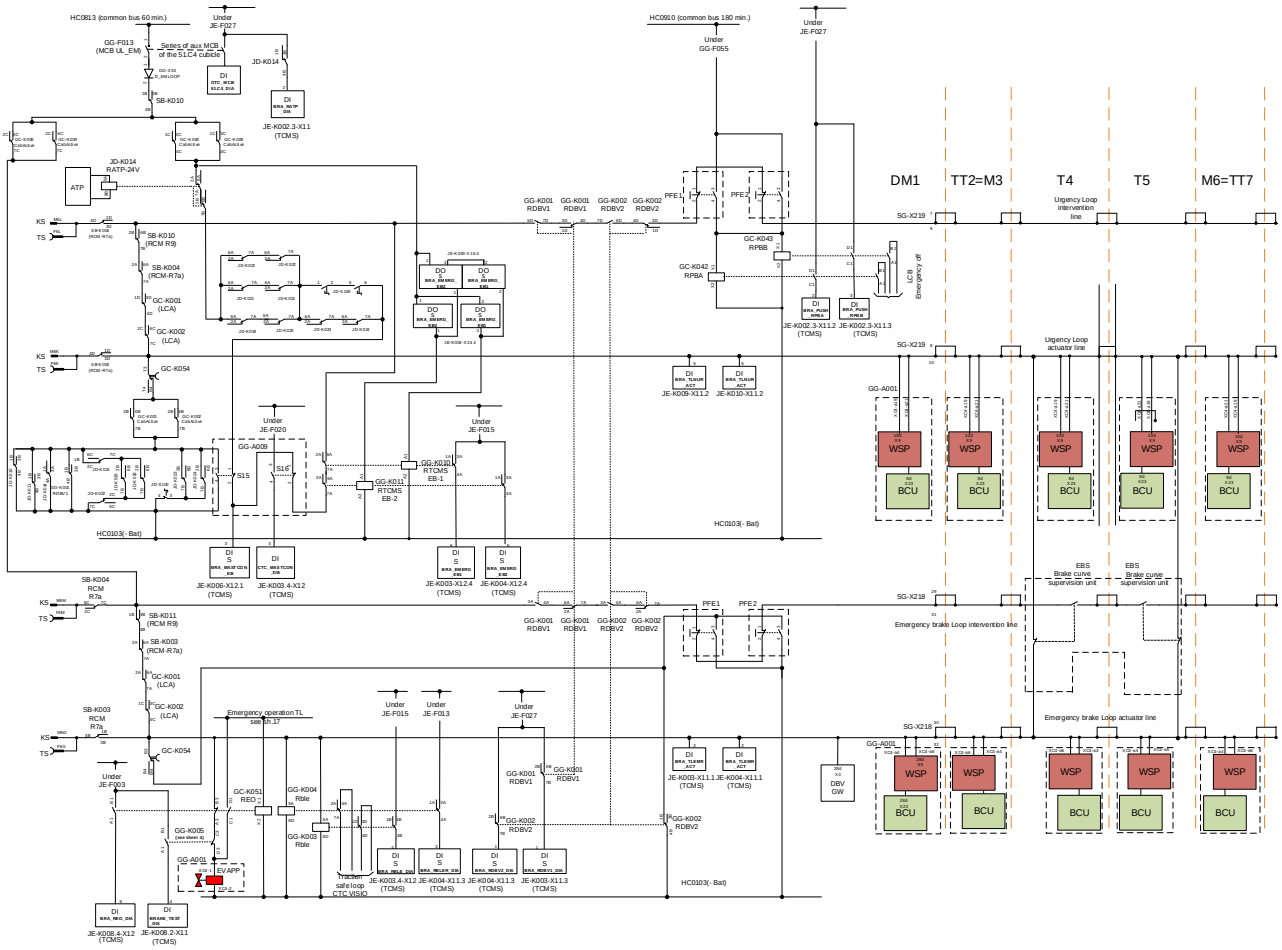
The Emergency brake is controlled and activated in a fail-safe way in case of:

- Fatal failure in the systems related to emergency brake
- Power supply to the Emergency brake control is cut off
- Train part is lost (unintentionally de-coupled)
- ATP Emergency brake
- Master Controller (Drive/Brake Handle) via urgency brake loop
- TCMS via urgency brake loop
- Emergency push button
- Driver's brake valve (DBV)

The electrical emergency brake control loop is fed from the active cab. The EB loop actuator line is connected to all the emergency brake application valves located in the DM1/8-cars and all BCU, WSP-CU and DBV-GW units.

If no EBS unit is present the urgency brake loop is directly connected to the emergency brake loop. In case the urgency brake loop is de-energising directly the actuator line of the EB loop will de-energise (the EBS controlled contacts shown in Figure 3.5.1 – D.1 are represented in the position without EBS device installed on the train, i.e. with the predisposition of the urgency brake but urgency brake function not active).

<3.5.1 - D2: Urgency/Emergency brake loop design>



The refilling of the brake pipe is controlled by energising the neutral valve EVNEU on the cabin panel 01.01 located in driver's cab. The command of the neutral valve is directly connected to emergency brake relay Rble.
In case the emergency brake loop is de-energised the neutral valve is energised by a contact of Rble to inhibit the refill of the brake pipe.

For a redundant path the neutral valve EVNEU is commanded by TCMS output in case of problem by the hardware chain.

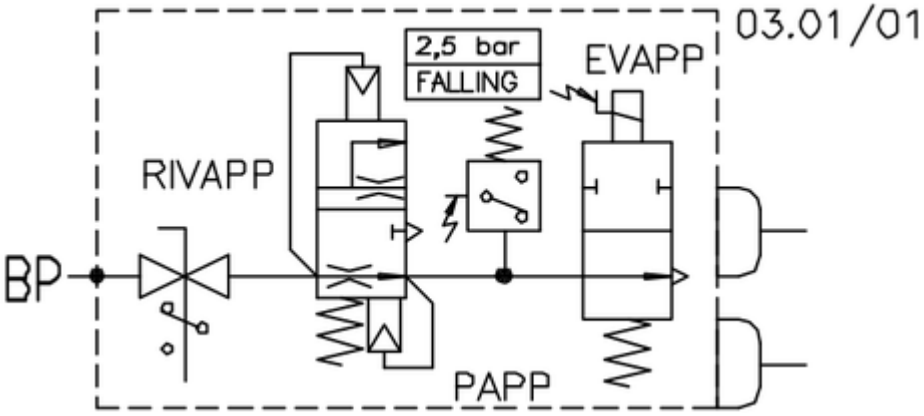
3.5.1.1 *Emergency Brake Application and Control*

Emergency brake application venting the brake pipe:

The emergency brake loop is connected the emergency brake application valve EVAPP group 03.01/01 located in each DM car. In case the emergency brake loop is de-energised the application magnet valve EVAPP will vent the pre-control pressure of the pneumatic application valve which opens the venting port for the brake pipe.

The emergency brake application valve group 03.01/01 has following components:
- EVAPP magnet valve for venting pre-control pressure
- Pneumatic application valve for venting the brake pipe
- PAPP pressure switch 250 kPa for supervision of pneumatic valve application
- RIVAPP emergency brake application valve isolation cock

<3.5.1.1 - D1: Emergency application valve pneumatic circuit>



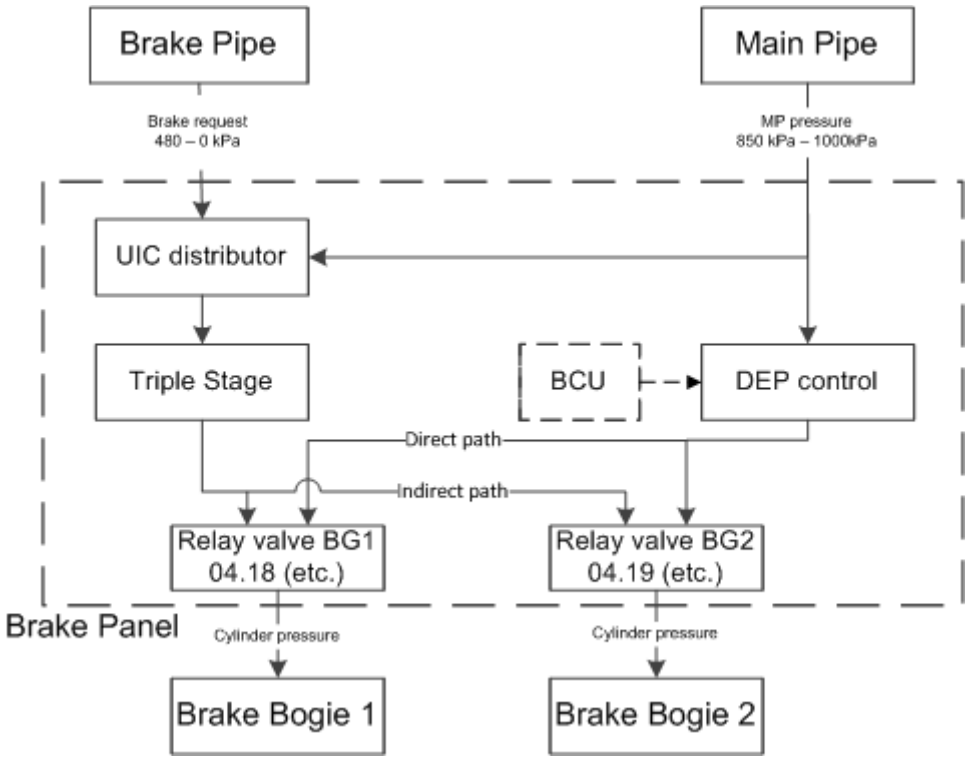
The control pressure to move the pneumatic application valve portion is supervised by the PAPP pressure switch to be sure the valve is moved to the brake pipe venting position in case of emergency brake. This pressure switch signal is used during the emergency brake test as part of the daily brake test and for diagnostic purpose.

The emergency brake application valve can be isolated by isolation cock RIVAPP in case of maintenance or failure. The isolation cock has an electrical switch, the position is supervised by the TCMS.

Emergency brake control on car level:

The emergency brake function has two different pneumatic application channels to generate the cylinder pressure on car level:
- Cylinder pressure by indirect brake control
- Cylinder pressure by direct brake control

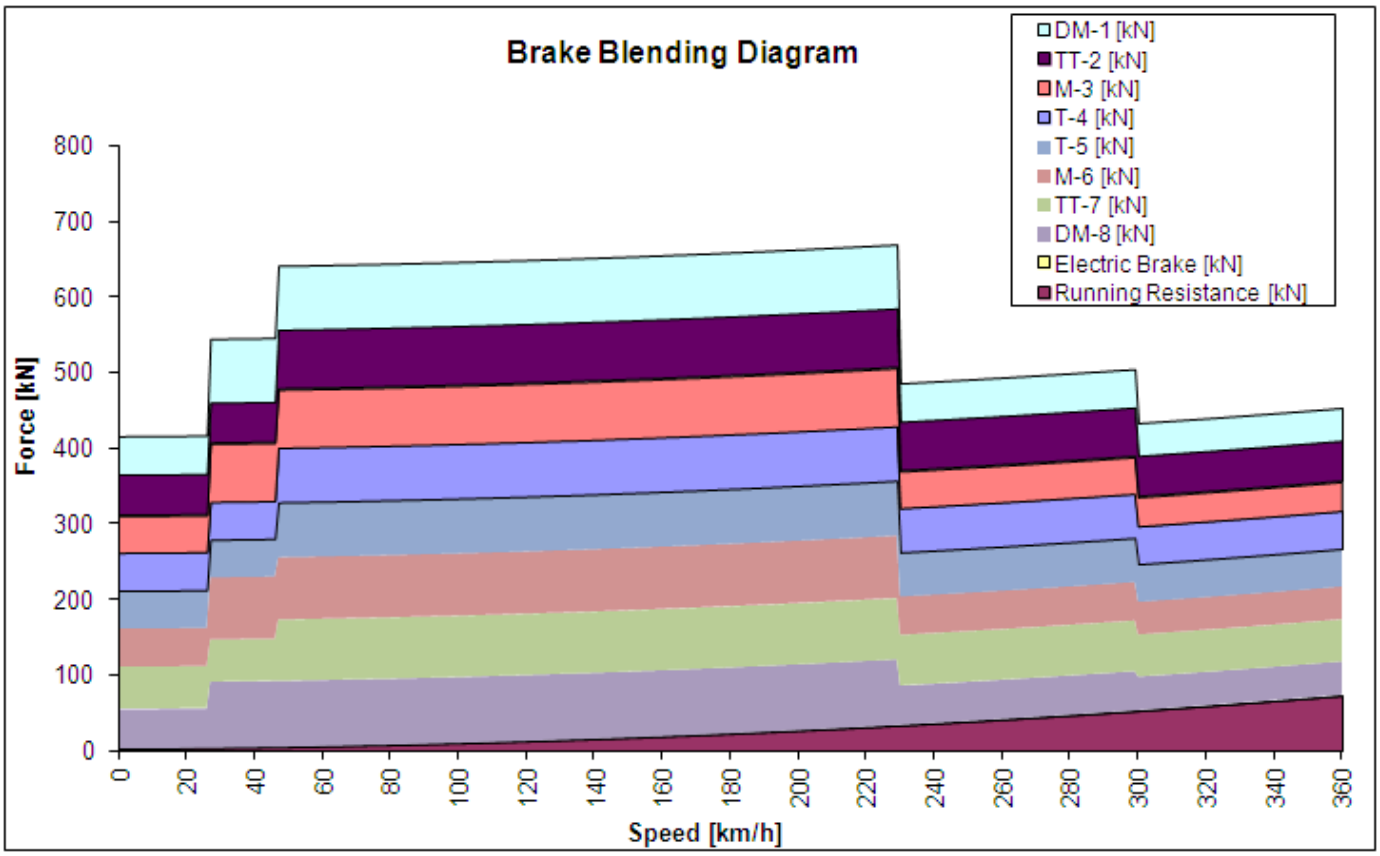
<3.5.1.1 - D2: Emergency brake pneumatic flow chart - two channels>



Indirect emergency brake application:

The indirect emergency brake application is guaranteed based on venting the brake pipe pressure to 0 kPa. The distributor valve 04.24/01 will translate the drop of brake pipe pressure to the emergency brake pilot control pressure. The pilot control pressure will be adjusted based on the actual speed to achieve three different pressure steps as described in chapter 3.2.1.4 by the triple stage relay valve 04.44 and related command valves EVTS1/EVTS2. The triple stage relay valve 04.44 is setting the cylinder pressure. The indirect cylinder pressure goes to the first port of the relay valve 04.18 for bogie 1 and 04.19 bogie 2.

<3.5.1.1 - D3: Emergency brake force diagram indirect brake application>

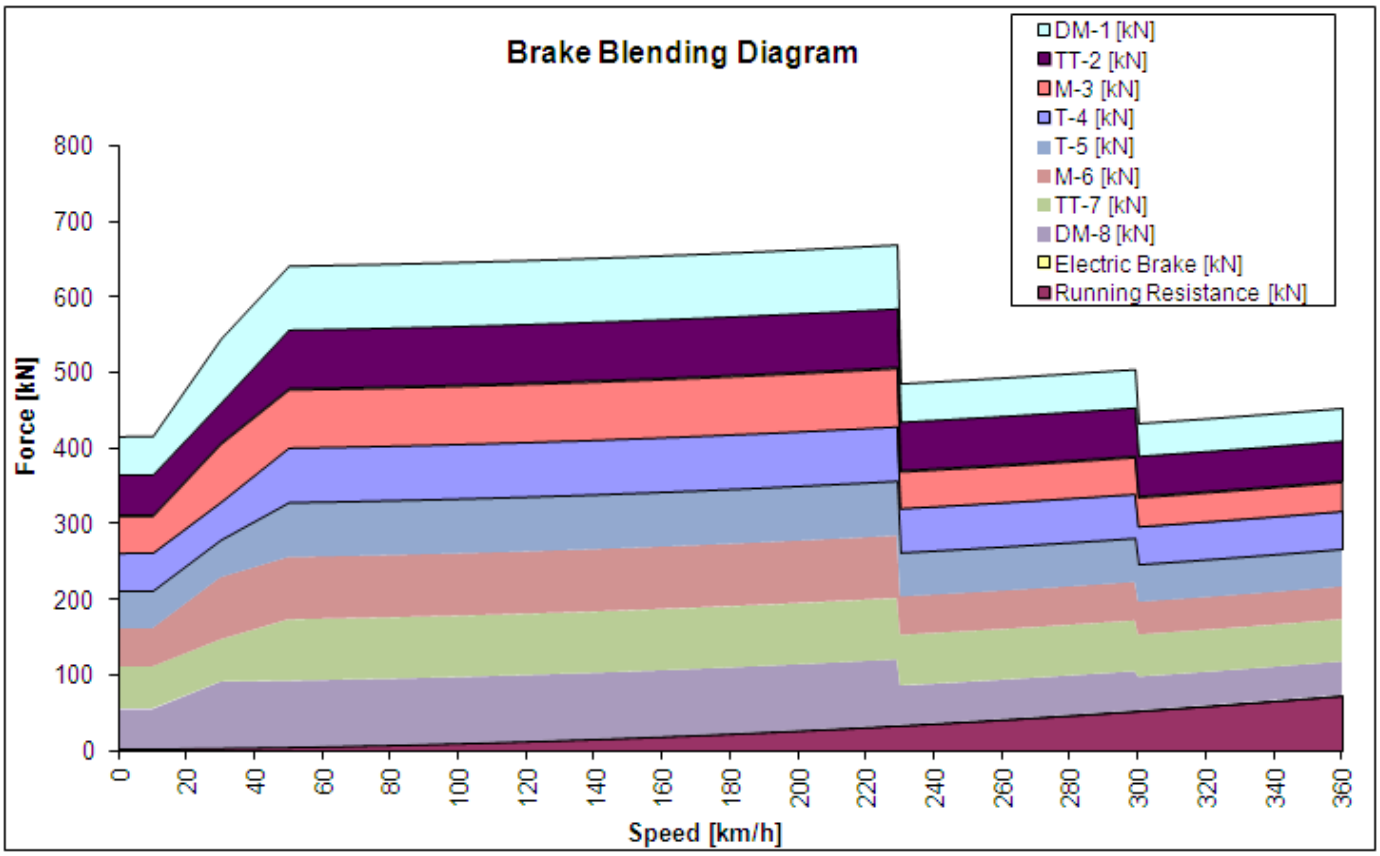


Direct emergency brake application:

The direct emergency brake application is controlled by the BCU via the direct braking group described in chapter 3.1.1. Based on the emergency brake loop and IP status the BCU commands the same triple stage cylinder pressure above the 4 stage speed range, in the 4 stage speed range a pressure ramp is introduced to reflect the pad friction behaviour to achieve a constant deceleration.

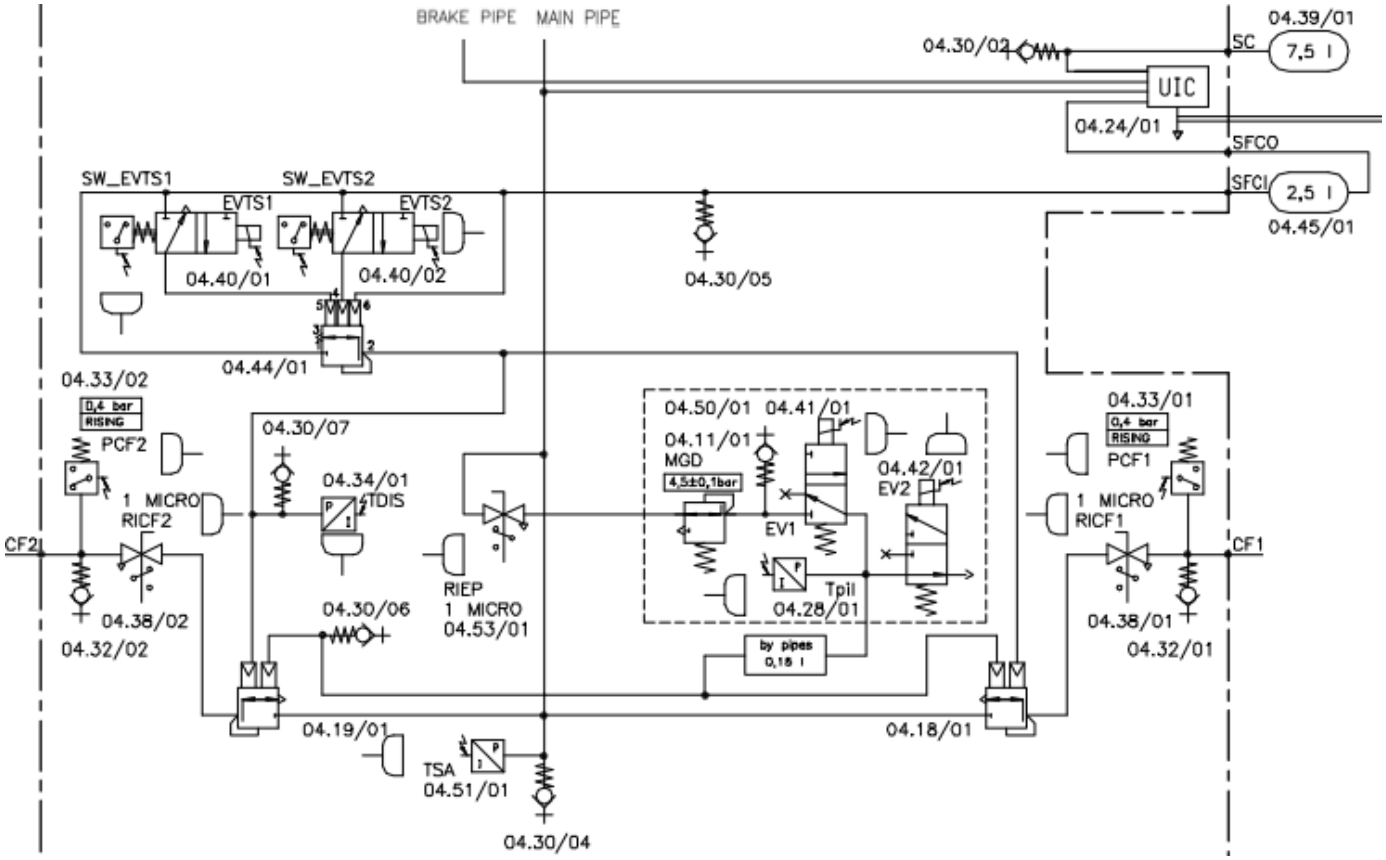
Above the DEP 4 stage speed threshold 30 km/h on motor and 50 km/h on trailer car the BCU sets the same pressures as for indirect brake by the direct braking group setting the valves EV1 and EV2. Below the speed threshold the BCU is ramping in a linear way the pressures from high pressure stage to the medium pressure stage on motor until 10 km/h and low pressure stage on trailer cars until 30 km/h.
The direct cylinder pressure goes to the second port of the relay valve 04.18 for bogie 1 and 04.19 bogie 2.

<3.5.1.1 - D4: Emergency brake force diagram direct brake application>



The higher pressure on the first or second port of the bogie relay valve 04.18/04.19 is used to set the cylinder pressure to the bogie CF1/CF2 and build-up the needed brake force. This design results in a condition that in case one cylinder pressure channel is missing the correct brake force is build-up to stop the train in a safe way on the redundant path.
At low speed for 4 stage brake design the two channels have different pressures as explained above.

<3.5.1.1 - D5: Emergency brake pneumatic control>



3.5.1.2 *Emergency Brake Traction Block/Safe Command*

The traction block/safe is generated in two different ways, the traction block is based on a software signal generated by TCMS and traction safe is commanded by hardware logic.

Traction block:

Traction block is set at each brake condition by TCMS logic and only inhibits the positive traction demand. The negative traction demand used for regenerative and rheostatic braking is still possible if traction block is set.
Traction block is set in the following brake conditions:

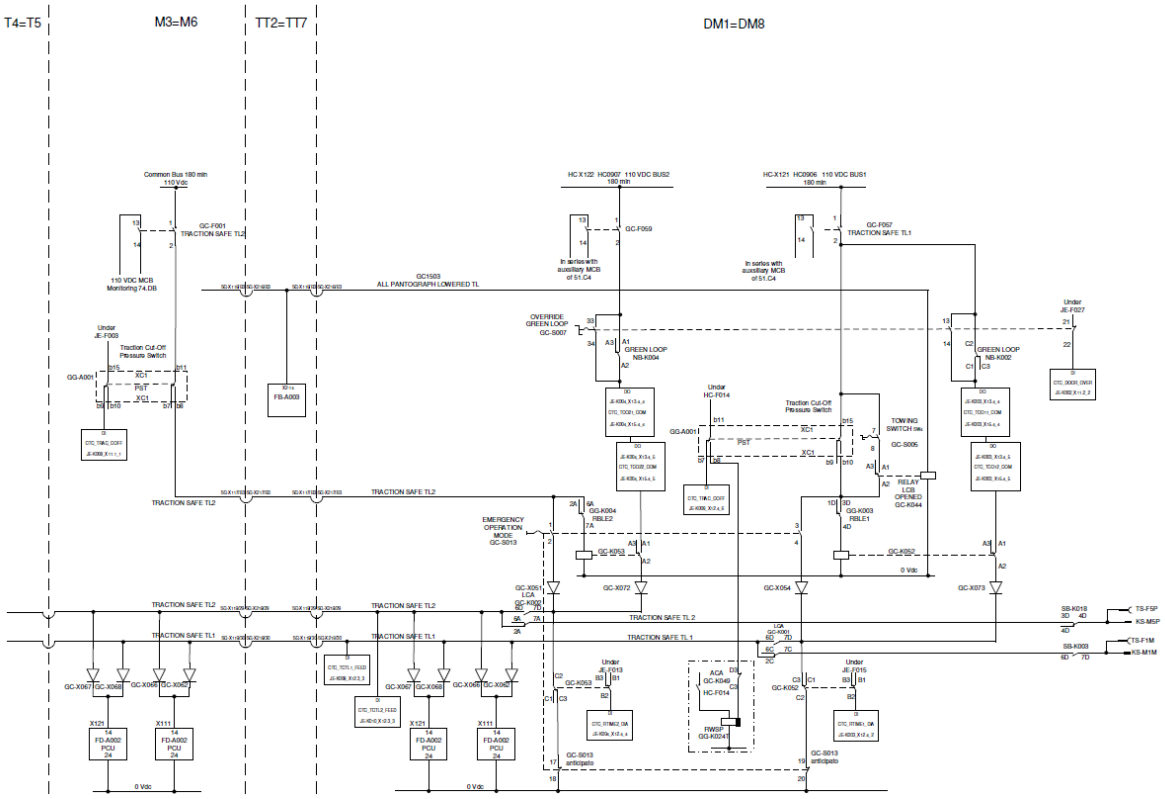
- Service brake demand active
- Emergency/Urgency brake active

Traction safe:

To inhibit the traction demand in a safe way during the emergency brake application the traction safe input set by hardware logic. Therefore, the pressure switch PST 03.04 monitoring the brake pipe in the DM cars and a contact of the emergency brake loop relay Rble is connected to the traction safe train line. In addition, two safe DOs commanded by the TCMS open the loop by the brake pipe information acquires by the pressure transducers.
In addition to the brake functions a green loop (all doors are closed) relay contact is connected to the traction safe train line. Therefore, following elements can open the traction safe train line:

- PST 03.04 pressure switch brake pipe
- Rble emergency brake loop relay
- Green loop relay
- TCMS DOs

<3.5.1.2 - D1: Traction safe train line / traction safe override train line>



The traction safe train line is connected to a safe digital input of all motor converters to inhibit the traction, in this case the positive and negative traction used for braking is switched-off if the digital input is low.

During an emergency brake application and pressure drop of the brake pipe the pressure switch PST 03.04 has following switching levels:

- Pressure drop < 350 kPa traction safe train line de-energised and traction safe active
- Pressure increase > 450 kPa traction safe train line energised and traction safe not active

The PST 03.04 pressure switch output is monitored by TCMS input for diagnostic purpose. In case the PST input is not align to the brake pipe pressure level an event is set:

- TCMS event PST not align to brake pipe pressure

Degraded mode:

In case of failure of the traction safe train line, for example the train line is permanent low and traction safe is set, an override function is present. The override is permitted based on the status of the PST 03.04 pressure switch in the M3/M6 car and emergency operation mode activated.
In case the brake pipe is > 450 kPa the traction safe override train line is energised and the inputs to the motor converter are set high. If the brake pipe pressure drops < 350 kPa the override is reset by the pressure switch PST 03.04 as long the brake pipe pressure is low.

3.5.1.3 *Emergency Brake Loop*

The emergency brake loop design is in conjunction with urgency brake loop design and following the same design principles. The emergency brake loop is feed in the active cab. The emergency loop has an intervention line and actuator line as the return of the train end.

The emergency brake loop has no intervention in the active cab feeding chain, as the loop is connected to the urgency brake loop. In case no emergency brake supervision (EBS) unit is foreseen all active chain contacts are opening the urgency loop and therefore directly the emergency brake loop. In this case the urgency brake loop is directly connected to the emergency brake actuator line.

All interventions of the emergency brake loop intervention line have a second contact used for the loop shorting for safety reason to guarantee that the loop will be de-energised.

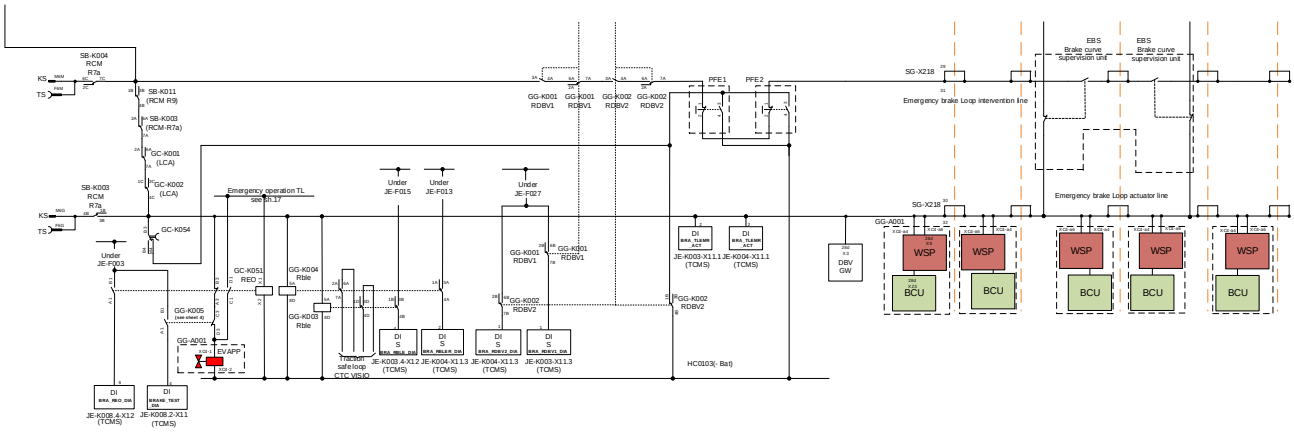
In the emergency brake loop intervention line contacts of following devices are implemented, these commands are always active as the devices are venting the brake pipe in parallel:

- PCF 1/2 Emergency push button 01.13
- Driver's brake valve 01.02

If the emergency brake loop is de-energised the emergency brake application valve group 03.01 located in each DM car (EVAPP in brake frame) is disconnected from the power supply. The emergency brake application valve EVAPP is venting the brake pipe by the pre-controlled pneumatic application valve to allow a fast venting time via a large venting port.

The EVAPP supply from the emergency brake loop can be disconnected by brake pipe continuity test relay contact GC-K005. This allows the venting of brake pipe by single EVAPP valve regardless of the status of emergency brake loop for automatic and manual brake test function described in 3.11.

<3.5.1.3 - D1: Emergency brake loop electrical design>



The emergency brake actuator line is connected to the DBW-GW, BCU and WSP-CU for direct brake back-up triple stage function and diagnostic. In case IEP brake pipe control is active Minitrol will de-energise brake pipe safety valve EVSICU based on the emergency brake loop input in redundancy to the neutral valve EVNEU. In IP brake this function is not active.

The emergency brake relay Rble connected to the emergency brake loop opens by a contact the traction safe train line in case the EB loop is de-energised. This guarantees the safe power-off of the traction converters to avoid traction during the emergency brake application.

The emergency brake loop actuator line is supervised by redundant TCMS inputs in each DM car. In case of failure or implausibility of the emergency brake loop an alarm will be presented to the driver.

In addition, each BCU and WSP-CU compares the status of the emergency brake loop input to the IP signal send by TCMS for diagnostic purpose.

3.5.1.4 *Emergency Operation Mode - Degraded Mode*

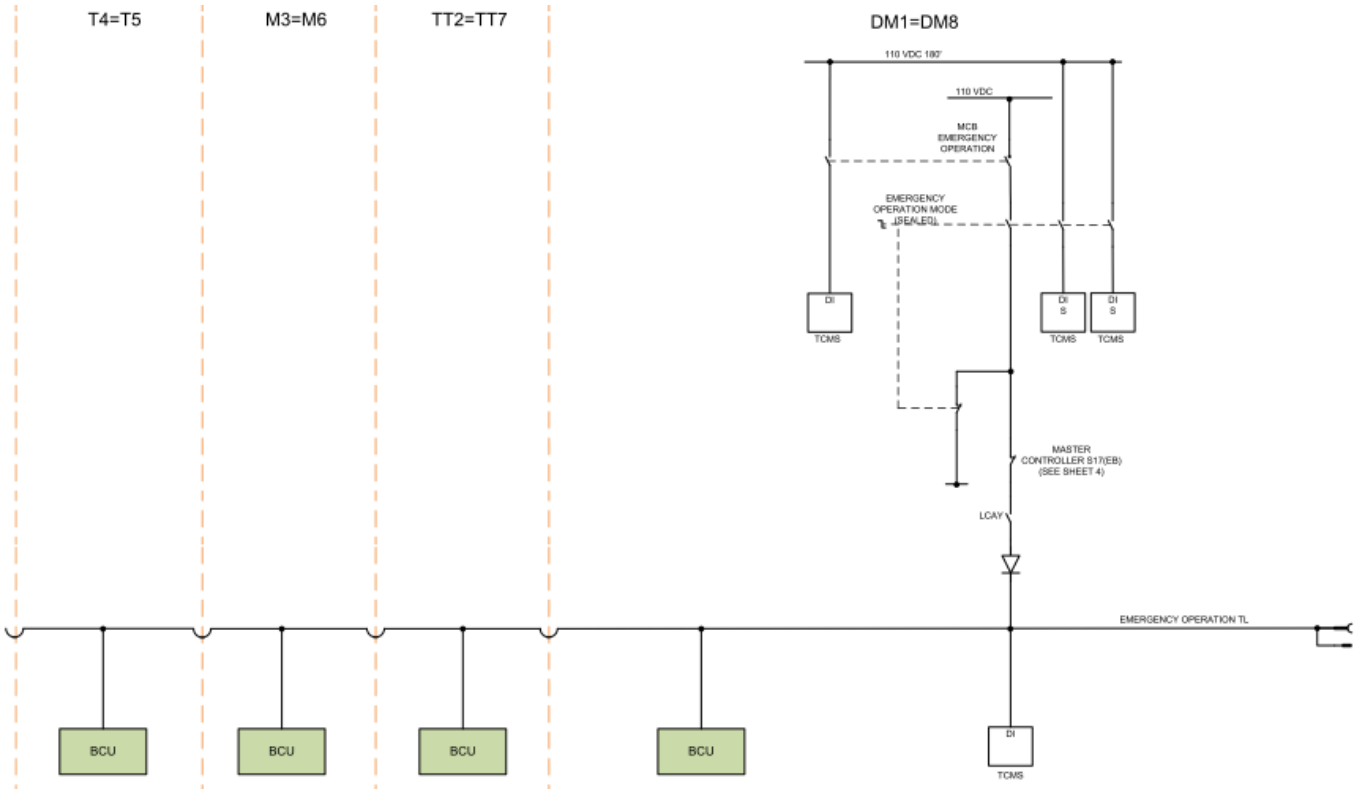
In case of failure of the emergency brake loop the emergency operation mode is used to bypass the electrical emergency control chain to allow a fast clearance of the track. In this mode a speed limit is set of 80 km/h by TCMS.
The emergency operation mode is activated by turning the sealed rotary selector switch "Emergency operation mode" in emergency operation mode position.

The emergency operation mode is to be used in following conditions:
- Emergency brake loop failure (permanent low)
- Contact failure of emergency push button or DBV to emergency brake loop
- Urgency brake loop failure
- Contact failure of ATP, master controller and TCMS to urgency brake loop
- TCMS failure of urgency/emergency brake demand

If the emergency operation mode is active and the emergency operation train line is energised, the emergency operation relay REO is energised.
In this case one REO relay contact disconnects the emergency brake loop to the emergency brake application valve EVAPP. The second REO relay contact activates the connection between the emergency operation train line to the application valve EVAPP. The EVAPP is energised directly by the emergency operation train line and the venting of the brake pipe is stopped.

The REO relay is supervised by TCMS for diagnostic purpose.

<3.5.1.4 - D1: Emergency operation train line>



The emergency operation train line is connected to the BCU, if the train line is energised the BCU is not considering the urgency and emergency brake loop inputs connected to the brake control unit.

In this condition the redundant direct brake path of the emergency brake control is switched-off by the BCU to guarantee the brake release in the affected car with faulty emergency or urgency brake loop.

The BCU monitors only the brake not train line described in chapter 3.1.1.1, in case the brake not train line is de-energised a pre-defined service brake force is applied by the BCU. The brake demand from TCMS via the IP bus will be not considered as long the emergency operation mode is active.

The safe emergency brake application during emergency operation mode is guaranteed by the indirect brake pneumatic path, activated by the following components:
- PCF1/2 EB push button 01.13
- Driver's brake valve 01.02
- ATP pneumatic interface 03.10

As the master controller has no pneumatic interface and this control element is the preferred one to initiate a driver's emergency brake, a contact of the master controller is provided in the feeding chain of the emergency operation train line.
This assures a emergency brake application by de-energise the emergency operation train line and therefore to de-energise the EVAPP application valve, as this is restored to the emergency brake loop feed by the REO relay contact.
In addition, the second direct brake bath is immediately reset and working as in normal operation to apply the emergency brake by the BCU.

Impact on normal operation:
To avoid a negative effect on the normal operation in case the emergency operation mode is set without request, the emergency operation train line has a shorting as long the rotary switch is in normal operation position.
In addition, a cab active relay contact is introduced in the feeding chain of the emergency operation train line to avoid an unwanted voltage in a non active cab.

<3**.5.1.5 – T2: ASFA Inhibition and diagnostic>**

3.5.2 Failure Information

3.5.2.1 *Normal condition*

In normal conditions the brake pipe is at 500 kPa level and in case of emergency brake application the brake pipe pressure shall be dropped to < 250 kPa.

In addition, all application elements for emergency brake will open the emergency brake loop which commands the emergency brake application valves EVAPP.

The brake pipe is vented by the emergency brake application valves EVAPP in DM cars and other devices like Minitrol on cabin panel, EB push button, DBV and ATP based on the train brake mode and EB application mode.

3.5.2.2 *Failure cases*

1. Failure on emergency brake application valve (EVAPP)
2. Failure on emergency brake loop
3. Failure of neutral valve 01.27 (VNEU) (no stop of refill of brake pipe)
4. Failure of local emergency brake demand by distributor valve
5. Failure of emergency operation train line (unwanted voltage)

3.5.2.3 *Failure detection*

1. Application of EVAPP valve is supervised by pressure switch PAPP read by TCMS and BCU, in case of mismatch driver event on IDU. Emergency brake activation chain test is part of automatic basic brake test.
2. Emergency brake loop actuator line is supervised by two TCMS control points, in case of mismatch an event is set.
   Second direct emergency brake command (DEP) via IP interface to BCU.
3. Application of EVNEU valve is supervised by DBV-GW and TCMS, in case of implausibility driver event on IDU.
4. In case of local failure at indirect brake chain or distributor valve, emergency brake is commanded in parallel by the BCU direct brake.
   In case of inconsistency between emergency brake loop and distributor pre-control pressure an event is set.

5. In case of unwanted voltage on emergency operation train line inconsistency between emergency operation train line status and emergency operation mode selector switch TCMS input. Driver event on IDU for inconsistency.

3.5.2.4 *Failure handling*

1. Isolation of defective EVAPP application valve. Train has to be stopped to isolate affected valve.
   Operation permitted until next unplanned workshop visit to repair during maintenance (at least two days) as one EVAPP valve is sufficient to vent the brake pipe.
2. Degraded mode stopping fault (major fault). Driver to switch in emergency operation mode to clear the track and stop the train at next convenient position.
   Repair during maintenance.
3. No failure handling possible, no affect on train level performance due to redundancy by direct brake.
   Repair during maintenance.
4. No failure handling possible, no affect on train level performance due to redundancy by direct brake.
5. No failure handling possible, affect on train level as complete loss of electronic emergency brake application (emergency brake loop and IP interface for direct brake). Only pneumatic application possible.
   Repair during maintenance.

3.5.3 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.172
Description	If the brake pipe is vented below 2,5 bar the emergency brake shall be applied.
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.237	Emergency brake shall be applied if the brake pipe pressure <= 250 kPa.	Derived Requirement	4S_07.-.Braking System		SIL 4
2F_05.02.Zefiro-Europe.CONCEPT.640	If brake pipe pressure drop < 250 kPa at any TCG sensor based on BCU IP analogue signal "Brake pipe pressure TCG" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.637	In case of emergency brake friction brake without any ED-brake shall be applied based on brake pipe pressure. The brake force shall be controlled to realize three brake steps.	Derived Requirement	4S_07.-.Braking System		SIL 4

ID	2F_05.02.Zefiro-Europe.TRS.151
Description	For emergency brake, the brake pipe is vented (<=2,5bar). The ED brake shall be disabled (like IP brake) and not considered.
Safety level	SIL 4

ID	2F_05.02.Zefiro-Europe.TRS.168
Description	The emergency brake shall also inhibit the traction.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1048	Traction block shall be set at emergency brake active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.432	Traction and ED brake shall be disabled by opening the traction safe loop if the EB loop is de-energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.1635	The pressure switch PST shall open if the brake pipe pressure is less than 350 +/-10 kPa. It shall close the contacts if the brake pipe pressure is above 450 +/-10 kPa.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3856	The brake pipe pressure switch PST in DM and M car shall be connected to TCMS input	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.3857	The brake pipe pressure switch PST shall be verfied by TCMS comparing the brake pipe pressure derived by TCG brake pipe pressure sensor and the pressure switching level of the PST as described in concept requirement #1635. In case of implausibility an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.175
Description	At emergency brake, the re-filling of the brake pipe shall be interrupted.
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.258	If emergency brake is activated, refilling of brake pipe shall be blocked as long emergency brake is activated by energise neutral magnet valve EVNEU 01.27.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2053	If the emergency brake loop is de-energised the neutral valve EVNEU (01.27) shall be energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3

ID	2F_05.02.Zefiro-Europe.TRS.184
Description	The brake pipe shall be vented directly from the stop push buttons (mushroom) in the cab.
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.231	There shall be two emergency push buttons (mushroom) PFE1 and PFE2 on driver's desk for activation of emergency brake.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.639	The driver's emergency stop push button shall remain engaged until released by pulling the push button in vertical direction.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.433	If the emergency stop push button is pushed the emergency brake shall be activated by directly venting the brake pipe.	Derived Requirement	4S_07.-.Braking System		SIL 4
2F_05.02.Zefiro-Europe.CONCEPT.1959	If one of the emergency stop push-button is pressed the emergency brake loop shall be opened. One contact of each emergency push-button shall be connected to a relay, one normally closed contact of the relay shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1960	The contact to the emergency brake loop shall be supervised by TCMS input, if an emergency push button is pressed and EB push button relay contact input is set and the emergency brake loop is remain energized (control point 1 and 2 on actuator line) an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.29
Description	The emergency brake shall provide braking with highest level of integrity. It shall be controlled and activated in a fail-safe way by the following safety related systems venting the brake pipe: – ATP initiated brake – Driver's Brake Valve ("DBV") – Train disconnection
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.234	Emergency brake shall be applied if ATP pneumatic interface is venting the brake pipe.	Derived Requirement	4S_07.-.Braking System		SIL 4

2F_05.02.Zefiro-Europe.CONCEPT.435	When the driver's brake valve is in emergency brake position both EB contacts of switch 6 and 7 (normaly closed) shall de-energise the two DBV EB1 and EB2 relays. The EB1 and EB2 relay contact shall be connected to emergency brake loop in series (normally open). Normaly closed contact of DBV EB 2 relay driven by DBV switch 6 shall be connected to the shorting and normaly open contact of DBV EB 2 relay driven shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.1275	Brake pipe shall be vented if the driver's brake valve (DBV) is pulled into the emergency brake position.	Derived Requirement	4S_07.-.Braking System		SIL 4
2F_05.02.Zefiro-Europe.CONCEPT.436	Emergency brake loop shall be de-energised if the electrical connections between two trains are incidentially disconnected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.1547	Emergency brake shall be activated if the pneumatic connection between two trains are incidentially disconnected and the brake pipe is vented.	Derived Requirement	4S_07.-.Braking System		SIL 4

ID	2F_05.02.Zefiro-Europe.TRS.315
Description	The emergency brake shall provide braking with highest level of integrity. It shall be controlled and activated in a fail-safe way by the following safety related systems opening the electrical emergency brake loop or venting the brake pipe: – Driver's safety (vigilance) device ("DSD") – Power supply to the emergency brake control is cut off
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.434	Driver's vigilance device shall open the emergency brake loop, for Italian base configuration refer to ATP brake interface.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.438	Emergency brake shall be activated when power supply to emergency brake loop control is cut off.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3

ID	2F_05.02.Zefiro-Europe.TRS.185
Description	In order to reduce the thermal load on the brake disc, the pressure on the emergency braking shall be managed with a double stage valve, which reduces the pressure at high speed. This management also allows the system to respect the adherence requested by the par. 4.2.4.1 of the HS Rolling Stock TSI.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.240	Emergency brake shall be managed through trible stage valve to reduce thermal load on brake disc. The triple stage valve shall reduce brake cylinder pressures at high speed. Refer to chapter 3.2.2.4.	Derived Requirement	4S_07.-.Braking System		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.441	Minimum deceleration in TSI case A with speed 350-300 km/h shall be 0,82 m/s² Minimum deceleration in TSI case A with speed 300-230 km/h shall be 0,91 m/s² Minimum deceleration in TSI case A with speed 230- 0 km/h shall be 1,23 m/s²	Derived Requirement	4S_07.-.Braking System		N/A

2F_05.02.Zefiro-Europe.CONCEPT.442	Brake deceleration in degraded mode with one car isolated: Minimum deceleration in TSI case B with speed 350-300 km/h shall be 0,66 m/s² Minimum deceleration in TSI case B with speed 300-230 km/h shall be 0,71 m/s² Minimum deceleration in TSI case B with speed 230- 0 km/h shall be 0,93 m/s²	Derived Requirement	4S_07.-.Braking System		N/A

3.5.3.1 *Emergency Brake Loop*

ID	2F_05.02.Zefiro-Europe.TRS.160
Description	For emergency brake, the emergency loop shall open to de-energise the application valves at least in the DM cars.
Safety level	SIL 3

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3328	The emergency brake loop shall be considered as de-energised if the emergency brake loop actuator line control points to TCMS input are set all to low = 0.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3318	If the emergency brake loop is de-energised (TCMS control points 1 and 2 on EB loop actuator = 0) the IP signal "Emergency brake active" shall be send by TCMS to BCU and WSP-CU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1649	If the emergency brake loop is de-energised (control point 1 and 2) an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1853	The 110V battery feed of the emergency brake loop shall be in the active cab. The intervention line is going from the active front cab to the non-active rear cab and the return is the actuator line going from the non-active rear cab to the active front cab. The EB loop actuator line in the DM cars shall be connected to TCMS input (energised = 1).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1846	Whenever the emergency brake is ordered from following devices the emergency brake loop shall be de-energized. – Emergency stop push-button – Driver's brake valve The contacts to open the emergency brake loop shall be in the intervention line.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.243	The electric application valves for pneumatic emergency brake control EVAPP (located in brake frame in DM-cars) shall be energised via the emergency brake loop actuator line if emergency brake is not activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.447	If the emergency brake loop is de-energised, the EVAPP emergency electric application valves in DM cars for emergency brake function shall be de-energised and shall activate emergency brake by venting the brake pipe.	Derived Requirement	4S_07.-.Braking System		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.629	Brake modules in DM1 and DM8 shall be equipped with an emergency brake valve (EVAPP).	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.630	All cabin panels shall be equipped with the emergency brake valve (EVSICU). This is used with IEP brake in emergency brake.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.628	A digital input on each BCU, WSP-CU and DBV-GW shall be connected to the emergency brake loop. The brake control units shall read the status of the EB loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1562	At cab deactivation without parking mode, the EB loop shall be de-energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

	At cab deactivation with parking mode, the EB loop shall be kept energised.				
2F_05.02.Zefiro-Europe.CONCEPT.1858	The actual reason for applying Emergency brake shall be displayed on the IDU. The status of the emergency brake loop shall be displayed on the IDU train status menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1843	If the control points 1 and 2 for the actuator line of the emergency brake loop are not all set (1) a maintainer event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2096	The BCU and WSP-CU shall read the emergency brake loop status per car, if the emergency brake loop and the IP signal by TCMS "Emergency brake activated" do not have the same status, the IP signal "DIA EB loop implausible" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2097	If the signal "DIA EB loop implausible" is sent by BCU or WSP-CU an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2098	If the emergency brake loop is de-energised without any order an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.199
Description	At emergency braking also maximal DEP shall be applied via the BCU:s on the train.
Safety level	SIL 2

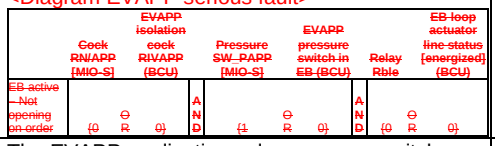
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.518	At emergency braking DEP brake shall be applied via the BCUs on the train with the same brake performance as the triple stage UIC brake. The direct brake emergency brake shall be used as a redundancy in case the emergency brake loop input is low (=0). In emergency operation mode is active (emergency brake operation train line energised) AND the emergency brake loop is de-energised the DEP emergency brake shall be not applied.	Derived Requirement	4S_07.-.Braking System		SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.200
Description	The emergency brake application shall be recorded in the data recorder (Belgian standard).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.521	The emergency brake application shall be sent to the data recorder (DIS).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.5.3.2 *Emergency Brake Diagnostic*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.642	The status of the electrical Emergency brake loop shall be presented in the cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3191	The EVAPP emergency brake application valve pressure switch PAPP shall be connected to TCMS input. Country Option: TCMS input shall be designed as safe input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.1059	If the EVAPP application valve pressure switch PAPP is not indicating that the valve has vented the brake pipe if the emergency brake loop is de-energised by TCMS input PAPP OR sent by BCU via IP signal "EB application valve pressure switch in emergency position (PAPP)", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2676	If one of the not isolated EVAPP valve is not opening on order detected by the emergency loop Rble relay TCMS input or BCU EB loop actuator line feedback and the PAPP TCMS or BCU input status in any DM car, a serious fault shall be set. The driver shall isolate the affected EVAPP valve. In case serious fault is triggered (and corresponding valve is not isolated) , fault shall be kept till when a new Emergency brake is commanded AND trigged condition are not true anymore. Status not opening: (Rble TCMS input = 0 OR signal "EB loop actuator line status (energized)" from one or more valid BCU = 0) AND (PAPP TCMS input = 1 OR signal "EB application pressure switch in EB position" from one or more valid BCU = 0) If RIVAPP is closed (EVAPP isolated) the corresponding EVAPP shall be disregarded. Refer to diagram in concept 3937.	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 2
~~2F_05.02.Zefiro-Europe.CONCEPT.3937~~	~~<Diagram EVAPP serious fault>~~ 	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~		~~N/A~~
2F_05.02.Zefiro-Europe.CONCEPT.2674	The EVAPP application valve pressure switch PAPP shall be read by the BCU, the BCU shall sent the IP signal "EB application pressure switch in emergency position (PAPP)" if the pressure switch is indicating that the EVAPP valve has vented the brake pipe correctly.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2675	If the status of the TCMS input of EVAPP pressure switch PAPP and the BCU IP signal "EB application pressure switch in emergency position (PAPP)" is not align, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2292	The isolation cock on brake frame RIVAPP (03.01) shall be connected to TCMS input, the input shall be set if the isolation cock is in isolated position. Country Option: TCMS input shall be designed as safe input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3192	The isolation cock on brake frame RIVAPP (03.01) shall be read by the BCU. The BCU shall sent the IP signal "EB application valve isolation cock isolated (RIVAPP)"	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1789	If the isolation cock RIVAPP for emergency brake application valve on brake frame is in isolated position sent by the BCU IP signal "EB application valve isolation cock isolated (RIVAPP)" OR by TCMS input of RIVAPP, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.3320	The BCU shall verify the inconsistency between the emergency brake loop input and the pre-control pressure of the distributor valve measured by TDIS sensor. In case of inconsistency the BCU shall set IP signal "DIA Emergency brake integrity failure" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3321	If the BCU detects an inconsistency between the emergency brake loop input and the pre-control pressure of the distributor valve and sets IP signal "DIA Emergency brake integrity failure" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1788	If the isolation cock RIPCG (01.31) for brake pipe test valve on cabin panel is in isolated position, the DBV-GW shall sent IP signal "Isolation cock RIPCG BP exhaust valve in isolated position".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2293	If the DBV-GW sent IP signal "Isolation cock RIPCG BP exhaust valve in isolated position" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3193	A contact of the emergency brake loop relay Rble shall be connected to safe TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3194	If the emergency brake loop relay contact of Rble in TCMS is not according the status of the emergency brake loop status an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3826	The TCMS digital output for brake test shall be connected to EVAPP relay. The power supply for the TCMS output to command the EVAPP relay shall be only active at standstill.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3827	The EVAPP valve shall be connected to an EVAPP relay contact. The EVAPP valve shall be possible to command (de-energise) by the EVAPP relay regardless the status of the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3828	The PAPP pressure switch of EVAPP valve shall be connected to TCMS digital input for supervision purpose.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3829	TCMS shall supervise the EVAPP relay status by the TCMS supervision relay contact input. An event shall be set in case the TCMS command is not align to the TCMS input: – TCMS EVAPP relay output 1 AND TCMS supervision input 0 – TCMS EVAPP relay output 0 AND TCMS supervision input 1	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.5.3.3 *Emergency Operation Mode / Emergency Brake Loop Bypass*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2942	If the emergency operation trainline is energised the relay contact of REO shall close the contact to directly feed the emergency brake application valve EVAPP by the emergency operation trainline. Another contact of REO shall disconnect the feed of the emergency brake loop to the emergency application valve.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.3207	During emergency operation mode TCMS shall set a speed limitation, refer to SFRS CTC.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3205	For supervision of the emergency operation relay REO contact shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3206	If the emergency operation train line is status energised/denergised and the TCMS input of	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	REO relay contact is not plausible an event shall be set.				
2F_05.02.Zefiro-Europe.CONCEPT.3609	If the emergency operation train line is status is set by BCU via IP signal "Emergency operation trainline energised" and the TCMS input of emergency operation train line is not set (=0) an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.6 Parking Brake

3.6.1 Function Design

The parking brake is able to prevent the ETR1000 VZI-R1 train in design mass in working order and with maximum wheel diameter from starting to roll on a slope of 4% according [16].

The parking brake caliper is a spring-applied type, the parking brake is released by a certain air pressure level.
Parking brake calipers are provided on all trailer bogies to achieve the needed parking brake force on train level.

The parking brake is automatically applied if the driver's cab is deactivated. In case the auxiliary brake reservoir pressure drops below 650 kPa the parking brake starts to apply automatically as the spring force starts to become effective. At a parking brake control pressure level of about 430 kPa the parking brake spring force is fully applied.

Parking brake is released via the illuminated "Parking brake" push button located on driver's desk. The parking brake push button is illuminated based on the parking brake status:
- Parking brake applied: Illuminated
- Parking brake released: Not Illuminated
- Parking brake failure (not all released/minimum not applied): Flashing

The train parking brake status information is shown on the IDU brake menu. The parking brake can have following status defined by TCMS diagnostic on bogie level:
- Parking brake released
- Parking brake applied
- Parking brake isolated and manual released
- Parking brake status inconsistent/failure

The control of the parking is done by TCMS, the apply or release command is set by a safe redundant TCMS output to the bi-stable control magnet valve.

The release command is given to the EVFS1 release magnet valve, this can only be done if one cab is active AND the protection versus undue release of the parking brake by the TCMS is de-activated (GG-K028). If one cab is active the train line "any cab active train line" is energised which close the contact of RER relay to command the release pulse.

If the cab is active, when the parking brake application is commanded by the push-button GG-A024 on the driver's desk, the train line "safe parking brake application" is de-energized opening the contact of GG-K018 relay. As consequence, the contact of GG-K028 relay is opened to protect the release pulse by TCMS.

Each time a parking brake demand (application OR release) has been generated (manually by the illuminated push-button OR automatically by the TCMS itself), TCMS shall compare the status of the GG-K028 relay, acquired by two redundant safe DIs (BRA_EVFS1_PRO), with the parking brake status in each trailer car. In case of inconsistency, between parking brake demand/request and GG-K028 relay status a maintenance diagnostic event shall be generated by TCMS for each trailer car.

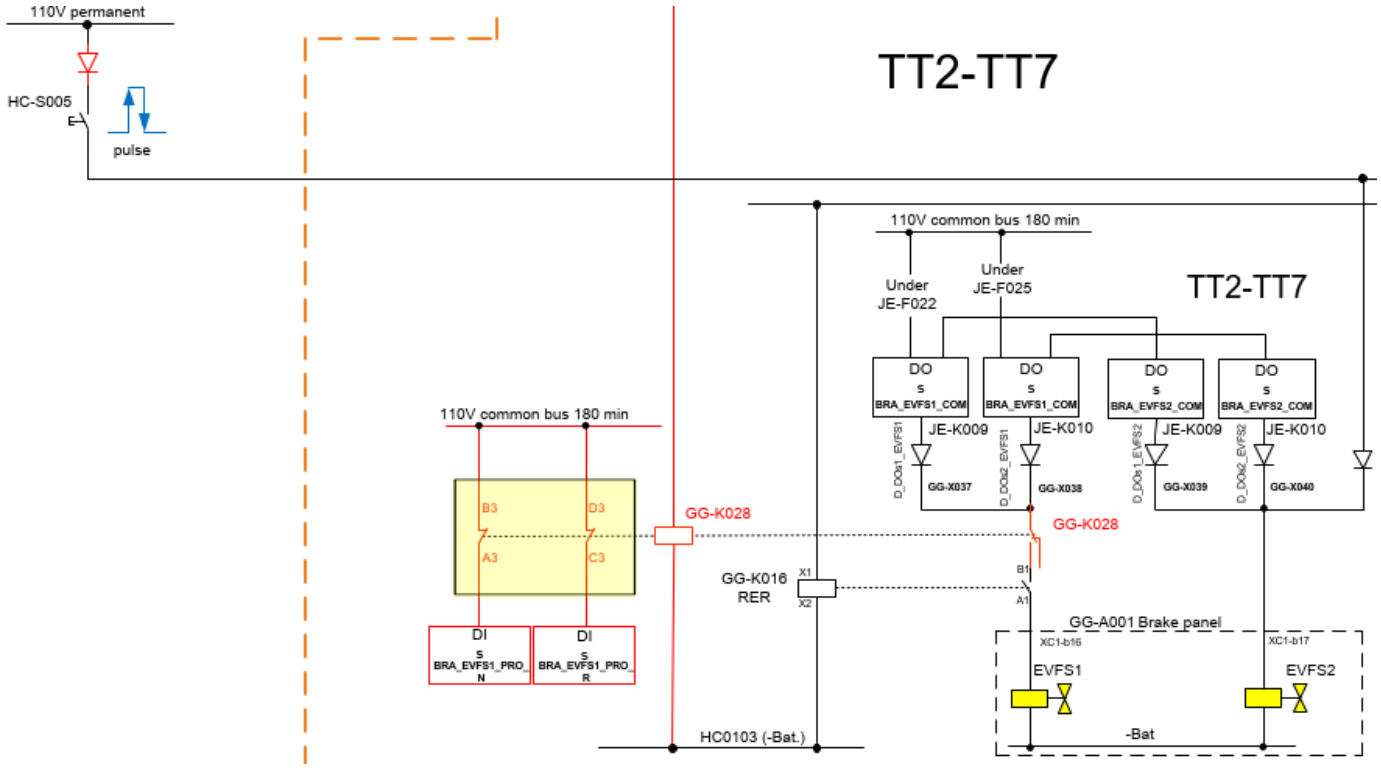
If the cab is active, when the parking brake release is requested by the push-button GG-A024 on the driver's desk, the train line "safe parking brake application" is energized closing the contact of GG-K018. As consequence, the contact of GG-K028 relay is closed to command the release pulse by TCMS.

GG-K018 is a bi-stable relay actioned by GG-A024 on each DM cars.

The apply command is given to the EVFS2 apply magnet valve by the TCMS safe output.

In case no pressurized air in the main pipe or auxiliary brake reservoir is available the parking brake can be released manually by manual release handle on each side of the bogie frame.

<3.6.1 - D1: Parking brake apply/release design>



The parking brake pneumatic control consist of following hardware:

- EVFS1/EVFS2 06.01: Bi-stable application valve of the parking brake apply/release
- TFS 06.03: Pressure transducer for parking brake read by the BCU
- 06.07: Pressure reducing valve 750 kPa
- RIFS1/2 06.02: Parking brake isolation cock per bogie

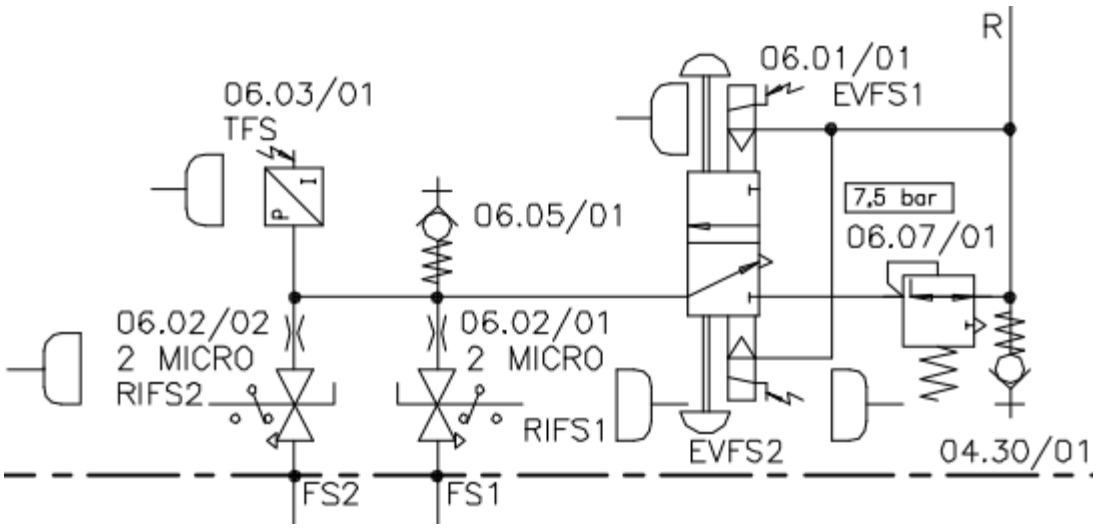
The parking brake control pressure is supplied from the auxiliary brake reservoir protected by a check valve against the main pipe. In case of pressure drop in the main pipe the parking brake is released as long the auxiliary brake reservoir is > 700 kPa. The pressure reducing valve 06.07 is installed to protect the parking brake caliper against excessive stress on the cylinder.

After the bi-stable magnet valve 06.01 for apply or release the parking brake pressure one choke per bogie is installed before the two isolations cocks RIFS1/2. In case of broken parking brake hose to the parking brake cylinder the pressure air flow is limited by this choke to avoid venting of auxiliary brake reservoir and big pressure loss.
As the auxiliary brake reservoir pressure is protected by the choke and the refill by main reservoir is able to compensate the venting air flow. Due to this the build-up of emergency brake cylinder pressure is possible with a broken parking brake hose and the emergency brake force is normally applied on the cylinder.

The TFS pressure transducer is monitoring the control pressure of the bi-stable magnet valve and read by the BCU. Based on this pressure the BCU will define the status of the parking brake and send this information to TCMS.

The parking brake can be isolated on bogie level by parking brake isolation cock RIFS1/2. The isolation cock has a double electric contact and is supervised by TCMS and BCU.

<3.6.1 - D2: Parking brake pneumatic control and monitoring circuit>



3.6.1.1 *Parking Brake Monitoring*

On each parking brake caliper unit micro switches are installed to indicate the applied or released parking brake status derived from the caliper movement. The double contact switch-over micro switches PB_SW are read by TCMS and used for external parking brake status indication.
The PBR_SW is used to set parking brake manual release status on bogie level read by TCMS and to command external parking brake status indication independent of PB_SW.
In case the parking brake is manually released the PB_SW is still indicating parking brake applied, refer to table 3.6.1.1 - T1.

The status of the parking brake in TCMS is defined based on the BCU pressure sensor TFS (06.03) on car level and the parking brake caliper switches PB_SW on bogie level.
The status of the parking brake applied or released is set per bogie based on the parking brake caliper switches PB_SW status and BCU signal applied/released.

The following logic is used to define the parking brake status in TCMS based on the caliper switches PB_SW connected to TCMS input and BCU IP signal, refer to table 3.6.1.1. - T1:

- Applied: All four caliper switches PB_SW contacts are opened to set apply status and BCU signal apply is set
- Released: All four caliper switches PB_SW contacts are closed to set release status and BCU signal release is set
- Isolated and manual released: Isolation cock signal RIFS is set and all calipers per bogie are manually released indicated by PBR_SW
- Failure/Inconsistent: The parking brake failure status is defined in case the different logical signals are not consistent

<3.6.1.1 - T1: Parking brake status definition>

Parking brake train level status / Parking brake status	*Parking brake status (per bogie)* TCMS Applied	TCMS Released	BCU Applied	BCU Released	Isolated (BCU OR TCMS)
Parking brake released	0	1	0	1	0
Parking brake applied at standstill (zero speed)	1	0	1	0	0
Parking brake applied at speed v>3 km/h	1	X	X	X	0
	X	X	1	X	0
	X	0	X	X	0
Parking brake isolated and manually released	X	1	X	X	1
Parking brake failure/unkown	All other logic combinations				

(X) = Not applicable/to be considered

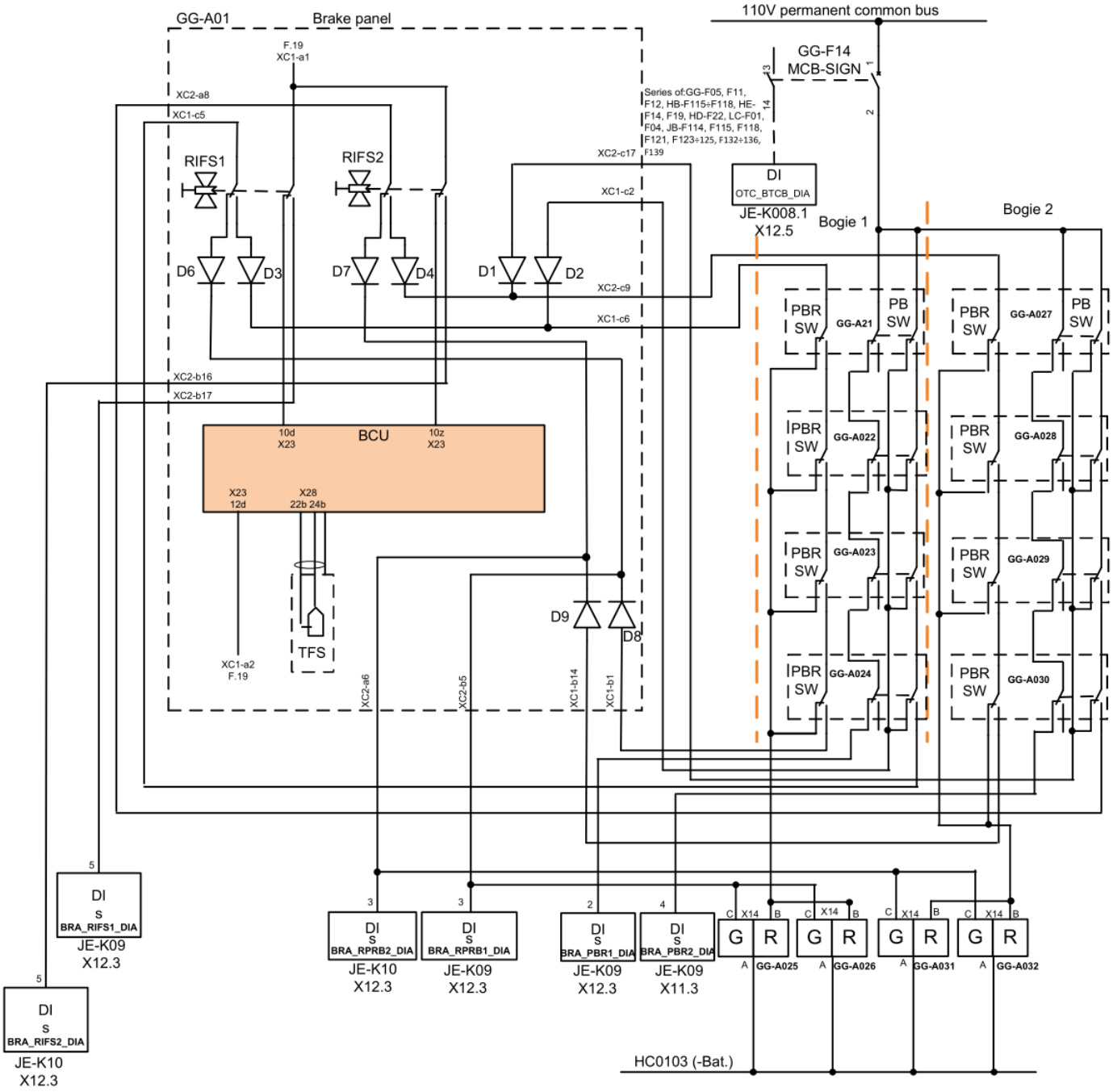
IDU indication

The four different parking brake status described above are shown on the IDU brake menu on bogie level:
- Applied (red)
- Released (green)
- Isolated and manual released (blue)
- Failure condition (cross indication)

TCMS compares the valid BCU IP signal based on pressure transducer TFS and the valid TCMS input signal by PB_SW to define the actual parking brake status. In case of implausibility between the signals a dedicated maintenance event is set by TCMS. In case one of the signals is not valid the parking brake supervision is based on the available valid unit BCU TFS sensor or TCMS input PB_SW.

The isolation condition is based on signal of BCU isolation cock RIFS input send via IP or direct TCMS isolation cock RIFS input.

<3.6.1.1 - D1: Parking brake monitoring logic>



The minimum required amount of parking brake is supervised by TCMS. In case the not enough parking brakes are available to keep the train at an inclination of 3.5 % the parking brake push button starts to flash and an event is presented to the driver. The alarm is reset during driving at speed > 5 km/h.

The tolerable loss (isolated or not available) of parking brake is more than two bogies in normal load conditions, which results in a loss of 25% parking brake performance. If this minimum parking brake performance is available normal operation of the train is possible. Following condition are considered to set the respective alarm or event.

– One bogie isolated or not available -> tolerable
– Two bogies isolated or not available -> tolerable, TCMS maintenance event

– More than two bogies -> not tolerable, parking brake lamp flashing and TCMS event

In addition to the minimum required parking brakes, TCMS supervise the release status of all parking brake in the train. In case the status is not consistent and not all parking brake are released, the parking brake push button starts to flash to inform the driver. Isolated parking brakes will be not considered for the release supervision.

External parking brake indication:

The electrical controlled external parking brake indicators are located for each bogie on both train sides. The electrical indicators are set by a combination of the caliper switches PB_SW, the manual release switch PBR_SW and the isolation cock status RIFS1/2. The indicator has to two defined status, a green and a red flag.

In normal conditions the external indicator are set by the following logic:
– Apply red flag: At least one of the four caliper switches PB_SW contact is opened
– Release green flag: All four caliper switches PB_SW contacts are closed to set release status

In case the parking brake is applied no traction block or traction safe is active. The driver can apply traction while the parking brake is applied to provide the possibility for a manual parking brake test against a certain traction force in case the automatic test is not available.

3.6.1.2 *Parking Brake Applied at Speed*

Based on the diagnostic described in chapter above, in case an undue parking brake apply signal is set during driving (train speed > 3 km/h) TCMS applies immediately a direct brake application if the speed is above 90 km/h. The brakes are applied until the target speed of 80 km/h is reached. Beyond the target speed of 80 km/h the direct brake application is automatically released by TCMS to allow the driver to clear the track in a not safe situation at low speed (tunnel, etc.).

In addition to the automatic brake application an alarm is presented to the driver on the IDU if the parking brake is applied at speed (complete speed range).

The automatic TCMS brake application and the alarm to the driver are performed if either the BCU reports parking brake applied at speed or the TCMS input of the parking brake caliper switches PB_SW is set applied.

The TCMS automatic brake application is supervised by an internal safe supervision logic. In case the target speed of 80 km/h is not reached in a specified time, TCMS applies the emergency brake to ensure the train is braked in a safe condition.
The aim of the automatic brake function is to protect the disc against overheating and prevention of fire.

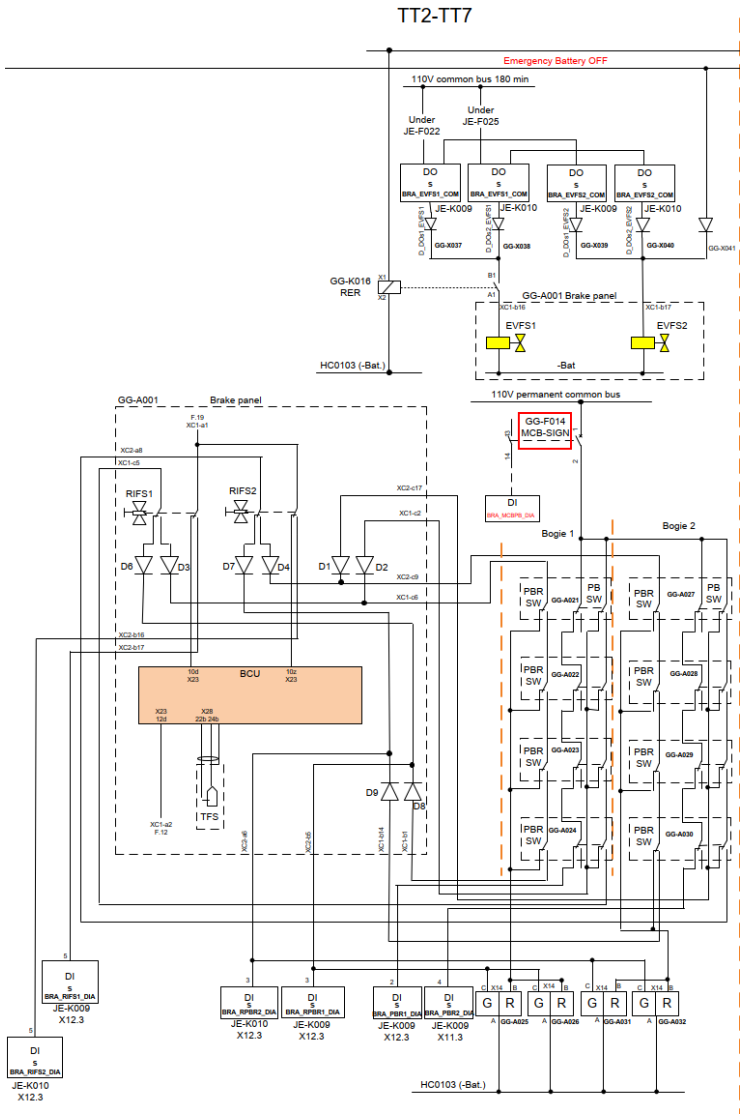
3.6.1.2.1 *Undue "Parking Brake apply at speed": remove speed limit*

Speed limitation management – Parking brake undue activation alarm

In the event the TCMS detects a parking brake application while running, it activates a speed limitation at 80 km/h. In particular, if the vehicle speed, when the alarm occurs, is greater than 80 km/h the TCMS applies a Maximum Service Brake so to reach this target speed value. The Maximum Service Brake can be replaced by the Emergency Brake if the target speed of 80 km/h is not achieved in a predefined time interval.

The intervention of the magnetic circuit breaker GG-F014, related to the electric circuit of the parking brake external indicators, in one of the trailer cars of the vehicle (T or TT), triggers the undue activation of the TCMS supervision (diagnostic event “parking brake applied while running”) and consequently the speed limitation at 80 km/h.

<3.6.1.2.1 - D1: Parking brake electric circuit (cars: T and TT)>



Once the vehicle is stopped, re-armed (if possible) the magnetic circuit breaker GG-F014 (located in the 59.C1 zone of the car-body) and with both bogies of the identified car isolated and manually released, the Driving Personnel can start excluding the TCMS supervision of the parking brake applied while running on the identified car, so that following activations of the same magnetic circuit breaker GG-F014 will not cause undue speed limitations and then further problems to the commercial service.

All the other cars of the vehicle, not being excluded, continue normally to be monitored by the TCMS supervision of the parking brake applied while running, until they are possibly subject to a following Exclusion.

The car excluded from the TCMS supervision of the parking brake applied while running will not be monitored by the supervision, until the TCMS is not powered off, after that it will be reinserted in the list of the cars of the vehicle supervised by the Function itself.

3.6.1.3 *Parking Brake Manual Release*

On each parking brake cylinder there is a mechanical release mechanism (14.14) with handle. Hence there are four parking brake release handles located on each trailer bogie frame.

The mechanical release handle is supervised by the mechanical switch PBR_SW. This switch works in the same principle as the PB_SW and is monitoring the cylinder movement. In case the mechanical release handle is pulled the cylinder reaches the manual release position and the contact of PBR_SW is closed.

The parking brake has to be manual released by the driver if the train is at standstill. In any case the manual release is performed the affected bogie has to be isolated as well pneumatically by isolation cock RIFS1/2 to avoid to re-arm the parking brake during driving.

The parking brake has to be manual released in case of:
- Train gets towed and no sufficient air supply is available to release the parking brake
- Failure of pneumatic control or the parking brake is isolated by isolation cock RIFS1/2

The parking brake is re-armed by apply the normal release pressure > 650 kPa.

3.6.1.4 *Failure Information*

3.6.1.4.1 *Normal condition*

In normal conditions parking brake applies when main pipe pressure is low or parking brake is activated by driver (toggling push button) or automatically after deactivation of the cab.

3.6.1.4.2 *Failure cases*

1. Parking brake does not apply
2. Parking brake does not release
3. Parking brake applied at train speed
4. Isolation cock RIFS (06.02) status is isolated in non-isolated position
5. Isolation cock RIFS (06.02) status is non-isolated in isolated position
6. Faulty pressure sensor TFS (06.03)

3.6.1.4.3 *Failure detection*

1. Pressure sensor TFS (06.03) detects if parking brake pipe is pressurized. Additionally, brake caliper contact status and parking brake indicator status provide information if parking brake is not applied when demanded.
2. Pressure sensor TFS (06.03) detects if parking brake pipe is pressurized. Additionally, brake caliper contact status and parking brake indicator status provide information if parking brake has undue application.
3. Pressure sensor TFS (06.03) detects if parking brake pipe pressure is vented. Additionally, brake caliper contact status and parking brake indicator status provide information if parking brake has undue application at running speed > 3 km/h.
4. Status of Isolation cock RIFS (06.02) is available on IDU and diagnostic by TCMS and BCU.
5. Indication is done by status of parking brake applied (see case 1./2.). Parking brake will not release and diagnostic by TCMS and BCU.
6. Pressure sensor TFS internal diagnostic by BCU, if pressure value does not correspond to the parking brake status given by caliper contacts an event will be set by TCMS.

3.6.1.4.4 *Failure handling*

1. Not applied parking brake (not more than two bogies) will create an maintenance event on IDU, no effect on train level.
   In case more than two bogies without parking brake applied an alarm is shown to the driver by TCMS. The holding must be applied until the train is secured against rolling by other means.
2. Not released parking brake will create an event on IDU. Manual release of parking brake necessary.
3. Undue application of parking brake at speed will create an alarm and event on IDU, at speed > 90km/h automatic brake application by TCMS.
4. In case parking brake status given by pressure sensor TFS and caliper contactor corresponds to parking brake demand, failure can be repaired at next workshop and no further failure handling is necessary. No effect on train level.

5. Manual release of parking brake and visual check of isolation cock is necessary.
6. Parking Brake caliper contactors are verified against the pressure sensor status.

3.6.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.37
Description	The parking brake is a pure mechanical function which shall be installed on certain axles where a spring applies the brake pads if air pressure is low. This shall prevent the train from rolling when parked.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.95	Two spring applied parking brake units shall be provided per Trailer car (TT2,T4,T5 and TT7). Each unit shall consist of 4 parking brake cylinders (8 parking brake cylinders per Trailer car).	Derived Requirement	4S_07.-.Braking System		SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.34
Description	Parking brake shall be applied: – automatically when main reservoir pressure drops below threshold.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.264	Parking brake cylinder will start to mechanical apply automaticly when main reservoir pressure drops below 650kPa. Parking brake will be fully applied at a pressure of 430kPa. Note: The parking brake can be released by certain service brake pressure.	Derived Requirement	4S_07.-.Braking System		SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.135
Description	If the main reservoir air pressure is below threshold, and the parking brake is applied, a mechanical release shall be possible.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.270	The manual release of the parking brake can be done by the manual release handle on bogie frame (14.14).	Derived Requirement	4S_07.-.Braking System		N/A

ID	2F_05.02.Zefiro-Europe.TRS.196
Description	Parking brake shall be applied: – manually by a push-button on the driver's desk – automatically when deactivating driver's cab.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.561	It shall not be possible to apply the parking brake if the train speed is greater than 10 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3963	It shall not be possible to apply the parking brake if the train speed is greater than 1,5 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1975	If parking brake is applied at speed > 3 km/h the BCU shall sent the IP signal "Parking brake applied at speed v > 3 km/h"	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.569	If parking brake is reported applied at speed > 3 km/h by BCU OR TCMS input PB status is set applied at speed v > 3 km/h an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.2107	If the parking brake is reported applied at speed by BCU IP signal "Parking brake applied at speed v > 3 km/h" OR TCMS status PB applied at speed is set (concept 1492 + 1760) TCMS shall activate full direct service brake independent of service brake mode if the speed ≥ 90 km/h. If the Parking brake alarm is active a traction limit shall be set at a speed of 80 km/h. If the train speed is < 80 km/h OR the parking brake applied by BCU and TCMS input is reset the full service brake request shall be reset and the brake application shall be released.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3442	TCMS shall supervise the deceleration time of the full direct service brake application described in concept 2107 for different function: – Parking brake applied at speed – DNRA/Hot box request active Based on the initial speed and the linear deceleration value of 0.5 m/s² to reach the target speed of 80 km/h the supervision shall be performed. If the deceleration is out of the range TCMS shall apply urgency brake until the target speed < 80 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.557	Parking brakes shall be applied when the battery contactor open sequence is started and the train is at standstill.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.556	Parking brakes shall be applied automatically if the driver's desk key is turned to position "0", i.e. the cab is de-activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3950	Parking brake shall be applied by hardware impulse to bi-stable magnet valve EVFS2 in case the battery emergency off train line is activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3980	Parking brake command (manual, automatic) shall electrically inhibit an undue TCMS command to the release bi-stable magnet valve EVFS1.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.134
Description	Parking brake shall be possible to be released only if the main reservoir pressure is above threshold.
Safety level	SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.197
Description	Parking brake shall be released manually by a push-button on the drivers desk but only – if the cab from where the release is commanded is activated AND – if the main reservoir pressure is above threshold.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.133
Description	A push button function shall be provided in the cab for apply/release of the parking brake.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.481	Parking brake shall only be allowed to be released if push button "Parking brake" is pressed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3395	Parking brake shall be possible to be released manually by push button "Parking brake" when following conditions are true: – Cab active AND – Main pipe pressure is above 650kPa	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.1974	The TCMS output to command parking brake release magnet valve shall be in series with one cab active train line relay contact (hardware relay normally open).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.482	There shall be an illuminated push button "Parking brake" in the cab for apply/release of parking brake. The button shall be spring loaded for function apply/release.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1500	Parking brake bistable valve (EVFS) shall be controlled and wired to TCMS using redundant digital outputs: – Parking brake release magnet valve EVFS1 – Parking brake apply magnet valve EVFS2	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1069	If the push button on driver's desk for parking brake is pressed, the parking brake shall be – applied if it has been in released status before or – released if it has been in applied status before.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1779	The lamp of the parking brake button shall have following functionality: – Illuminated if parking brake minimum amount is applied in the train – Not illuminated if parking brake is released on all cars – Lamp shall flash if parking brake is not released in all cars in which the parking brake is not isolated and manual released OR minimum applied parking brake amount not reached	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3716	The minimum required amount or maximum allowed loss of parking brake to trigger the parking brake lamp flashing alarm shall be the follow: – More than two bogies or more than four axles without parking brake applied status OR – More than two bogies OR more than four axles are isolated or manual released	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.6.2.1 *Monitoring/Diagnostics*

ID	2F_05.02.Zefiro-Europe.TRS.320
Description	Parking Brake monitoring: Diagnostics from the brake computer shall be stored in the train diagnostic system.
Safety level	N/A

ID	2F_05.02.Zefiro-Europe.TRS.321
Description	The status (applied, released, cut-out, faulty) shall be indicated for the parking brake on an axle basis.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1720	The BCU shall read pressure sensor TFS and sent the status on car level via IP signal "Parking brake applied" and "Parking brake released" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2 & SIL0 (as redundancy)
2F_05.02.Zefiro-Europe.CONCEPT.2021	The normally closed contact of switch 1 of mechanical parking brake switches PBR_SW shall be in parallel connected to the red flag of parking brake indicator.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

<table>
<tr><td></td><td>The normally closed contact of switch 2 of mechanical parking brake switches PBR_SW shall be in serious connected to TCMS input applied.</td><td></td><td></td><td></td><td></td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.2067</td><td>The normally open contact of switch 1 of mechanical parking brake switches PBR_SW shall be in series connected to the green flag of parking brake indicator and to TCMS input release.</td><td>Derived Requirement</td><td>4S_09.02.09.-.Conventional Train Control - Circuits</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.3852</td><td><PB status matrix>
<table><tr><td>Parking brake train level status</td><td colspan="5">Parking brake status (per bogie)</td></tr><tr><td>Parking brake status</td><td>TCMS Applied</td><td>TCMS Released</td><td>BCU Applied</td><td>BCU Released</td><td>Isolated (BCU OR TCMS)</td></tr><tr><td>Parking brake released</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Parking brake applied at standstill (zero speed)</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td rowspan="3">Parking brake applied at speed v>3 km/h</td><td>1</td><td>X</td><td>X</td><td>X</td><td>0</td></tr><tr><td>X</td><td>X</td><td>1</td><td>X</td><td>0</td></tr><tr><td>X</td><td>0</td><td>X</td><td>X</td><td>0</td></tr><tr><td>Parking brake isolated and manually released</td><td>X</td><td>1</td><td>X</td><td>X</td><td>1</td></tr><tr><td>Parking brake failure/unkown</td><td colspan="5">All other logic combinations</td></tr></table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.3846</td><td>The TCMS shall set the status parking brake applied bogie 1 at zero speed if:
– TCMS input applied bogie 1 is set (1) AND
– BCU IP signal "Parking brake applied" is set AND
– PB bogie 1 is considered as not isolated (concept #2081) AND not manual released

The status applied shall only be set based on valid signals.
In case one bogie is isolated signal coming from BCU in this car shall not be set as valid.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.3847</td><td>The TCMS shall set the status parking brake applied at zero speed bogie 2 if:
– TCMS input applied bogie 2 is set (1) AND
– BCU IP signal "Parking brake applied" is set AND
– PB bogie 2 is considered as not isolated (concept #2081) AND not manual released

The status applied shall only be set based on valid signals.
In case one bogie is isolated signal coming from BCU in this car shall not be set as valid.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.4095</td><td>The TCMS shall set the status parking brake applied bogie 1 at speed v > 3 km/h if:
– TCMS input applied bogie 1 is set (1) OR
– TCMS input released bogie 1 is reset (0) AND
– PB bogie 1 is considered as not isolated (concept #2081) AND not manual released

The status applied shall only be set based on valid signals and on concept #4088.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.4096</td><td>The TCMS shall set the status parking brake applied bogie 2 at speed v > 3 km/h if:
– TCMS input applied bogie 2 is set (1) OR
– TCMS input released bogie 2 is reset (0) AND
– PB bogie 2 is considered as not isolated (concept #3018) AND not manual released

The status applied shall only be set based on valid signals and on concept #4088.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_05.02.Zefiro-Europe.CONCEPT.2068</td><td>The TCMS shall set the status parking brake released bogie 1 if:
– TCMS input released bogie 1 is set (1) AND</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
</table>

	– BCU IP signal "Parking brake released" is set AND – PB bogie 1 is considered as NOT isolated (concept #2081) The status released shall only be set based on valid signals. In case one bogie is isolated signal coming from BCU in this car shall not be set as valid.				
2F_05.02.Zefiro-Europe.CONCEPT.2069	The TCMS shall set the status parking brake released bogie 2 if: – TCMS input released bogie 2 is set (1) AND – BCU IP signal "Parking brake released" is set AND – PB bogie 2 is considered as NOT isolated (concept #2081) The status released shall only be set based on valid signals. In case one bogie is isolated signal coming from BCU in this car shall not be set as valid.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3848	The TCMS shall set the status parking brake isolated AND manullay released bogie 1 if: – TCMS input released bogie 1 is set (1) AND – PB bogie 1 is considered as isolated (concept #2081) The status released shall be set based on valid signals.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3849	The TCMS shall set the status parking brake isolated AND manullay released bogie 2 if: – TCMS input released bogie 2 is set (1) AND – PB bogie 2 is considered as isolated (concept #2081) The status released shall be set based on valid signals.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3850	The TCMS shall set the status parking brake failure for bogie 1/2 in case the logic is not following the status applied, released, isolated and manual released according matrix in concept 3851.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1723	The TCMS shall check the plausibilty of parking brake caliper status TCMS input applied bogie 1 AND BCU IP signal "Parking brake applied". If the status is not align AND main reservoir pressure > 650 kPa AND parking brake is not considered isolated and manually released an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2070	The TCMS shall check the plausibilty of parking brake caliper status TCMS input released bogie 1 AND BCU IP signal "Parking brake released". If the status is not align AND main reservoir pressure > 650 kPa AND parking brake is not considered isolated and manually released an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2071	The TCMS shall check the plausibilty of parking brake caliper status TCMS input applied bogie 2 AND BCU IP signal "Parking brake applied". If the status is not align AND main reservoir pressure > 650 kPa AND parking brake is not considered isolated and manually released an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2072	The TCMS shall check the plausibilty of parking brake caliper status TCMS input released bogie 2 AND BCU IP signal "Parking brake release".	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	If the status is not align AND main reservoir pressure > 650 kPa AND parking brake is not considered isolated and manually released an event shall be set.				
2F_05.02.Zefiro-Europe.CONCEPT.1726	If Parking Brake release status is not set after the release order is set an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2078	If Parking Brake applied status is not set after the apply order is set an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3655	If Parking Brake is released without order an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3656	If Parking Brake is applied without order an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.771	The event not apply/release on order shall be blocked in case the respective bogie is isolated by isolation cock RIFS (06.02).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1660	The pressure sensor TFS (06.03) shall be read by the BCU. The BCU shall sent TFS pressure via IP signal "Parking brake pressure (TFS)" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2888	If the parking brake sensor TFS is faulty the BCU shall sent IP signal "TFS PB transducer faulty" to TCMS and set the related safety signal to invalid.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2889	If the TFS senor is faulty and IP signal "TFS PB transducer faulty" is sent by the BCU an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1068	The isolation cock signal "Isolation cock RIFS1/2 isolated" shall be provided to TCMS by the BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2491	RIFS1 and RIFS2 parking brake isolation cock shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2081	If parking brake on bogie 1 is reported isolated by BCU (RIFS1) or TCMS input is set to isolated (1) the isolation cock shall be considered as isolated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.762	If parking brake on bogie 1 is reported isolated by BCU (RIFS1) or TCMS input is set to isolated (1) an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3018	If parking brake on bogie 2 is reported isolated by BCU (RIFS2) or TCMS input is set to isolated (1) the isolation cock shall be considered as isolated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1552	If parking brake on bogie 2 is reported isolated by BCU (RIFS2) or TCMS input is set to isolated (1) an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2493	If the parking brake isolation cock status RIFS1 and RIFS2 read by TCMS and BCU IP signal is not consistent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.763	When parking brake is considered as isolated in more than one bogie an event shall be set that parking brake is not more able to prevent the train from rolling at 35 promilles slope.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.765	When parking brake is considered as isolated in more than two bogies an event shall be set parking brake is able to prevent the train from rolling at 30 promilles slope.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.766	The status of the parking brake cut out cocks shall be indicated on the IDU brake menu: – parking brake cut out cock bogie 1 (RIFS1) open/closed – parking brake cut out cock bogie 2 (RIFS2) open/closed	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1695	The parking brake status shall be indicated on the IDU brake menu for each bogie individually: – PB applied – PB released – PB isolated and manual released – PB status inconsistent / in failure	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.3844	The parking brake pressure derived from TFS pressure sensor shall be indicated on TOD brake menu pressure overview on car level in T-cars.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3654	An event shall be set if the parking brake push button has been pressed continously > 30s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.343
Description	A protection versus and Undue Parking Brake Release Command by TCMS at vehicle level which could generated a total loss of Parking Brake Effort once Parking Brake has been applied shall be available.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3979	**GIVEN** (Driver Desk Disabled) **WHEN** (BRA_EVFS1_PRO == 0) **THEN** (Alarm "Enable/disable parking brake release not available" shall be set by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.4075	**GIVEN** (Driver Desk Disabled) **WHEN** (BRA_EVFS1_PRO == 1) **THEN** (Alarm "Enable/disable parking brake release not available" shall be reset by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.4076	**GIVEN** ((Driver Desk Enabled) **AND** (BRA_PARK_COM == 1)) **WHEN** (After *T_down* s BRA_EVFS1_PRO has not changed status from "1" to "0") **OR** (After *T_up* s BRA_EVFS1_PRO has not changed status from "0" to "1") **THEN** (Alarm "Enable/disable parking brake release not available" shall be set by TCMS) *T_down* = 1 s *T_up* = 5 s	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.4077	**GIVEN** ((Driver Desk Enabled) **AND** (BRA_PARK_COM == 1)) **WHEN** (After *T_down* s BRA_EVFS1_PRO has changed status from "1" to "0") **OR** (After *T_up* s BRA_EVFS1_PRO has changed status from "0" to "1") **THEN** (Alarm "Enable/disable parking brake release not available" shall be reset by TCMS) *T_down* = 1 s *T_up* = 5 s	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.341
Description	In case of undue parking brake applied at speed, driver shall be able to by-pass speed limitation.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_05.02.Zefiro-Europe.CONCEPT.4087	**GIVEN** (At least one "*Parking Brake applied at speed*" valid event is active in the consist) **WHEN** ((Vehicle is at "*Standstill*") **AND** (All the bogies in the cars in which "*Parking Brake applied at speed*" events have been activated, are isolated and manually released)) **THEN** ("*Bypass PB*" "*Soft Key*" shall be represented in the "*Brake Page*" of IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4093	**GIVEN** ("*Bypass PB*" "*Soft Key*" on IDU pushed) **WHEN** ("*Return*" button on IDU pushed) **THEN** (IDU shall go back in the previous page and TCMS shall not perform any action)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4088	**GIVEN** ("*Bypass PB*" "*Soft Key*" on IDU pushed) **WHEN** ("*Hard Key*" of IDU acknowledged and accepted by TCMS) **THEN** ((Monitoring Function of "*Parking Brake applied at speed*" shall be maintained active only in the cars of the consist in which the "*Parking Brake applied at speed*" events are not active) **AND** (Car is in "*Parking Brake Bypassed*" status))	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.4089	**GIVEN** (Car is in "*Parking Brake Bypassed*" status) **THEN** ((All the "*Parking Brake applied at speed*" active events shall be reset) **AND** (All the "*Undue Parking Brake applied*" active events in the car shall be reset and disabled) **AND** (All the "*Unknown Parking Brake*" active events in the car shall be reset and disabled))	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4090	**GIVEN** (Car is in "*Parking Brake Bypassed*" status) **THEN** (Event "*Parking Brake Speed Limit Reset*" shall be stored)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4091	**GIVEN** (Car is in "*Parking Brake Bypassed*" status) **THEN** (Cars of the consist in which the Monitoring Function of "*Parking Brake applied at speed*" are not active anymore shall be stored as Event "*Parking Brake cars not monitored anymore*")	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4099	**GIVEN** (	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

	(Car is in "*Parking Brake Bypassed*" status) **AND** (Vehicle is not in "*Emergency Operation*") **AND** (NO "*Parking Brake applied at speed*" events active) **AND** ("*DNRA OK*" icon is not represented to the driver on IDU) **AND** (NO "*Unknown Parking Brake*" events active) **AND** (All the "*not isolated*" Parking Brake are "*released*")) **THEN** ("*PB OK*" icon shall be represented by TCMS to the driver on IDU)				
2F_05.02.Zefiro-Europe.CONCEPT.4154	**GIVEN** ((Car is in "*Parking Brake Bypassed*" status) **AND** (Vehicle is not in "*Emergency Operation*") **AND** (NO "*Parking Brake applied at speed*" events active) **AND** ("*DNRA OK*" icon is represented to the driver on IDU)) **THEN** ("*DNRA & PB OK*" icon shall be represented by TCMS to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4155	**GIVEN** ("*PB OK*" icon is represented on IDU) **WHEN** (At least one "*Parking Brake applied at speed*" event is active) **THEN** ("*PB OK*" icon shall not be represented anymore by TCMS to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4157	**GIVEN** ("*DNRA & PB OK*" icon is represented on IDU) **WHEN** (At least one "*Parking Brake applied at speed*" event is active) **THEN** ("*DNRA OK*" icon shall be represented by TCMS to the driver on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4163	**GIVEN** (("*PB OK*" icon is represented on IDU) **OR** ("*DNRA & PB OK*" icon is represented on IDU)) **WHEN** (Vehicle is in "*Emergency Operation*") **THEN** (Icon "*Emergency Operation*" is represented on IDU)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4085	**GIVEN** (TCMS ON) **WHEN** ("*Parking Brake Main Circuit Braker*" tripped) **THEN** ("*Alarm MCB tripped*" event only in trailer car shall be set by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.4086	**GIVEN** (TCMS ON) **WHEN** ("*Parking Brake Main Circuit Braker*" not tripped anymore) **THEN** ("*Alarm MCB tripped*" event only in trailer car shall be reset by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.7 Wheel Slide Protection

3.7.1 Function Design

The WSP system is designed to protect the vehicle against wheel flats and to achieve the best possible stopping distance based on the available adhesion. The system is fully compliant to EN 15595:2009+A1 (apart § 4.2.4.3.3 feature which is recommended, but not mandatory).

The WSP ensures the maximum use of available adhesion, avoid locking of the wheel sets and prevent wheel flats during braking regardless of brake mode, speed and rail conditions. The WSP function is controlled by the WSP-CU and is separated from the brake control unit (BCU).

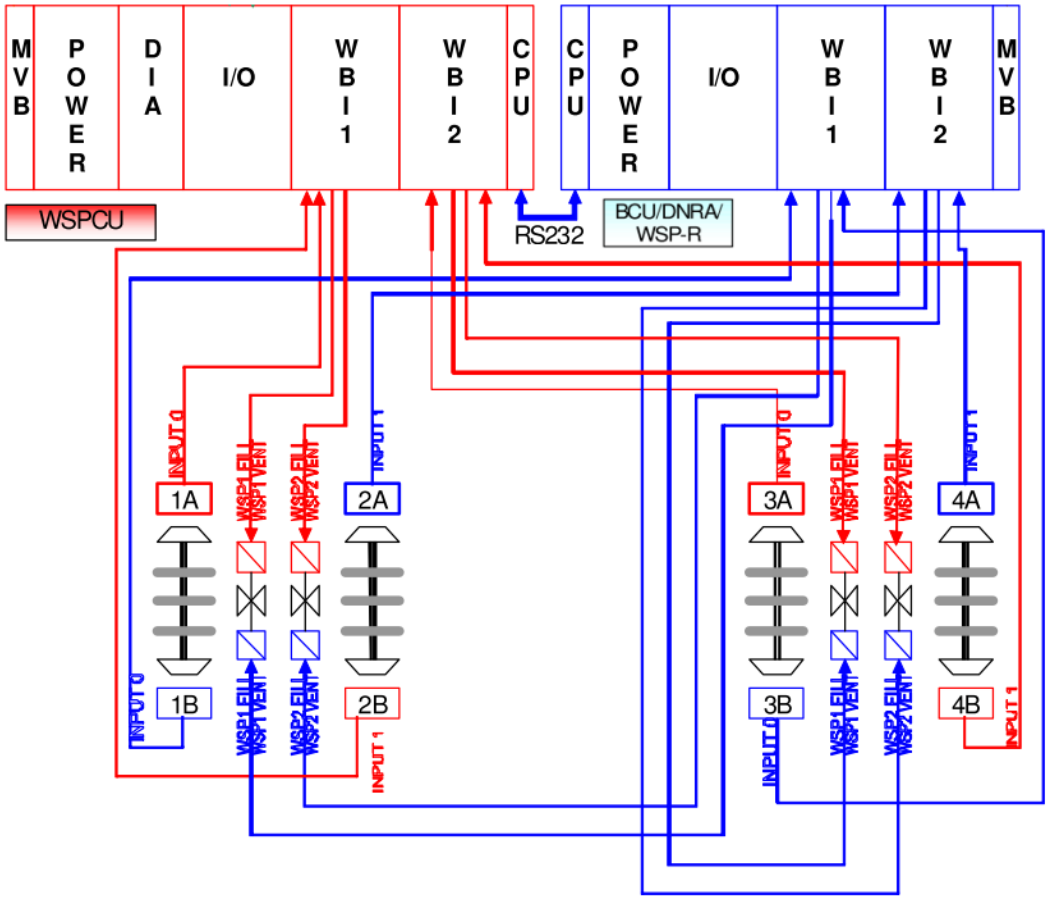
The WSP function is realised in a redundant way, the BCU is taking over the responsibility in case the WSP-CU has a major failure. The speed sensor channels are redundant, the WSP-CU and BCU acquire separate speed sensor channel from one physical speed senor mounted on each axle of the train.

In case the WSP detects sliding the driver is informed on the IDU as long the sliding control is active in any section of the train.

WSP ED controlled by brake system and WSP ED optimized

During braking the complete WSP control of the electro-dynamic brake and friction brake is managed by the WSP-CU as first WSP ED control mode. To control the ED brake during sliding two hardwired control lines are installed to the propulsion control unit to reduce, hold, increase or switching-off the ED brake demand (the ED brake demand for the WSP control, see figure 3.7.1.1 - D1). The control logic of the ED brake is realised on axle level. If the hardwired signals are not working (diagnostic during the brake test) or if the WSP ED time in control is higher than 10s, the WSP ED control is swapped by BCU/WSP to traction control (**WSP ED optimized**). With WSP ED optimized the WSP ED control and WSP pneumatic control are independent, working separately.

<3.7.1 - D1: WSP control layout>



The WSP system is composed of:

- WSP-CU: Electronic control unit (WSP-CU) primary responsible to optimize the braking performance in function of wheel adhesion condition.
- BCU: Brake control unit, able to take in charge the WSP control in case of WSP-CU failure
- SS1-4 07.06: One double speed sensor per axle (sensors with double channels for the WSP-CU and BCU)
- EWS 07.01: One dump valve group per axle (double coil on each valve for redundancy)

Each dump valve group EWS is composed of:

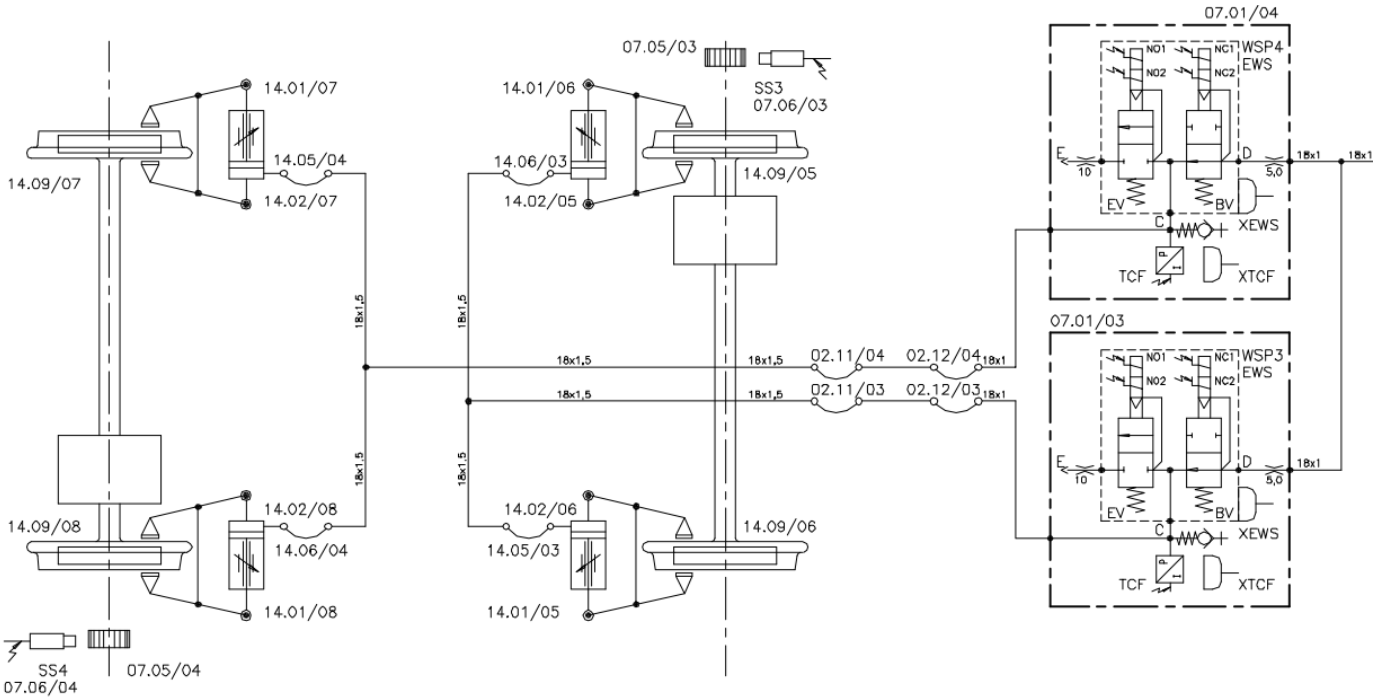
- EWS-BV: Anti-slide apply magnet valve
- EWS-EV: Anti-slide release magnet valve
- TCF: Pressure transducer for cylinder pressure per axle
- Test fitting of cylinder pressure
- Chokes in venting 10mm and supply 5mm of cylinder pressure

The WSP system measures each axle speed in order to define the instantaneous deceleration based on the average reference speed build up by the four speed signals.

When the speed signal received from one axle is over a certain threshold, the WSP logic operates on the WSP dump valves or commands the WSP ED brake reduction signals in order to obtain an acceptable sliding value which regenerates the available adhesion. The dump valves are controlled by the two signals fill and vent per axle to command the EWS-BV (fill) and the EWS-EV (apply) magnet valve.

Pneumatic regulation by the WSP shall be inhibited:
if the commanded braking effort (target < 0,2 bar) AND the brake pipe pressure value is higher than 4.65 bar AND the brake cylinder pressure values are below 0.25 bar at least for 500 ms.

<3.7.1 - D2: Pneumatic WSP valve control>



WSP ED brake interface

For a fast reaction time by controlling the ED-brake by WSP-CU or BCU during sliding a hardware interface on axle basis is installed with two binary coded signal lines representing 4 states:

- WSP build up ED brake force
- WSP hold ED brake force (keep)
- WSP vent ED brake force (reduce)
- WSP switch-off ED brake force (also defined as ED Cut Off)

<3.7.1 - T1: ED brake reduction signals for WSP control>

Command	WSP_Hold	WSP_Vent
Build-up ED brake force	0	0
Keep ED brake force	1	0
Reduce ED brake force	1	1
Switch-off ED brake force	0	1

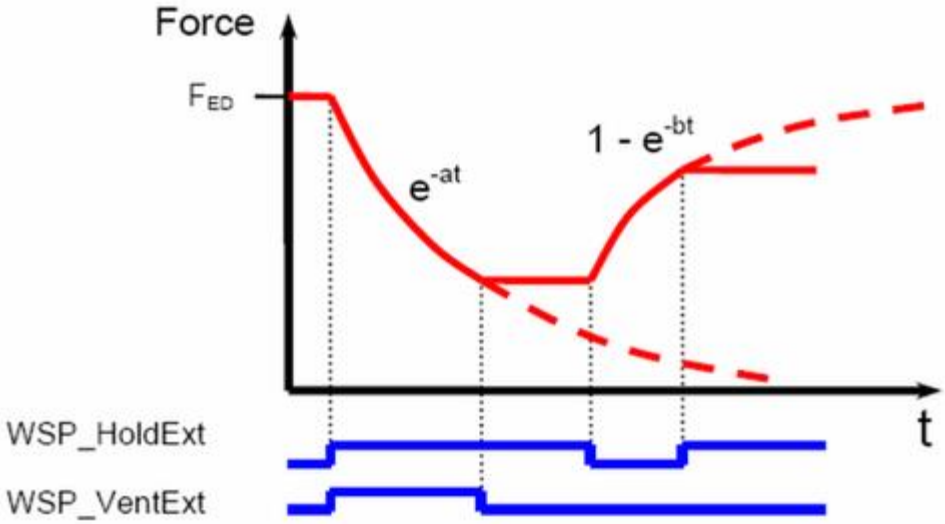
The ED-brake WSP control of each axle will be managed by the master WSP control unit (WSP-CU or BCU).

In case the two axles of one bogie are controlled by different control units (one axle by BCU and one axle by WSP-CU) both control units BCU and WSP-CU can control the ED-brake force during sliding for the respective axle.

On propulsion system the ED brake is reduced and build-up with e-function gradient in case respective hardware control signals are received by propulsion control unit. The WSP ED brake control is realised on axle level.

WSP ED control is performed per axle. In case of ED Cut Off on one axle, the ED brake is set to 0 on that axle.
The ED brake is still active on other axles.

<3.7.1 - D3: ED brake force reduction based on WSP ED hold and vent signals>



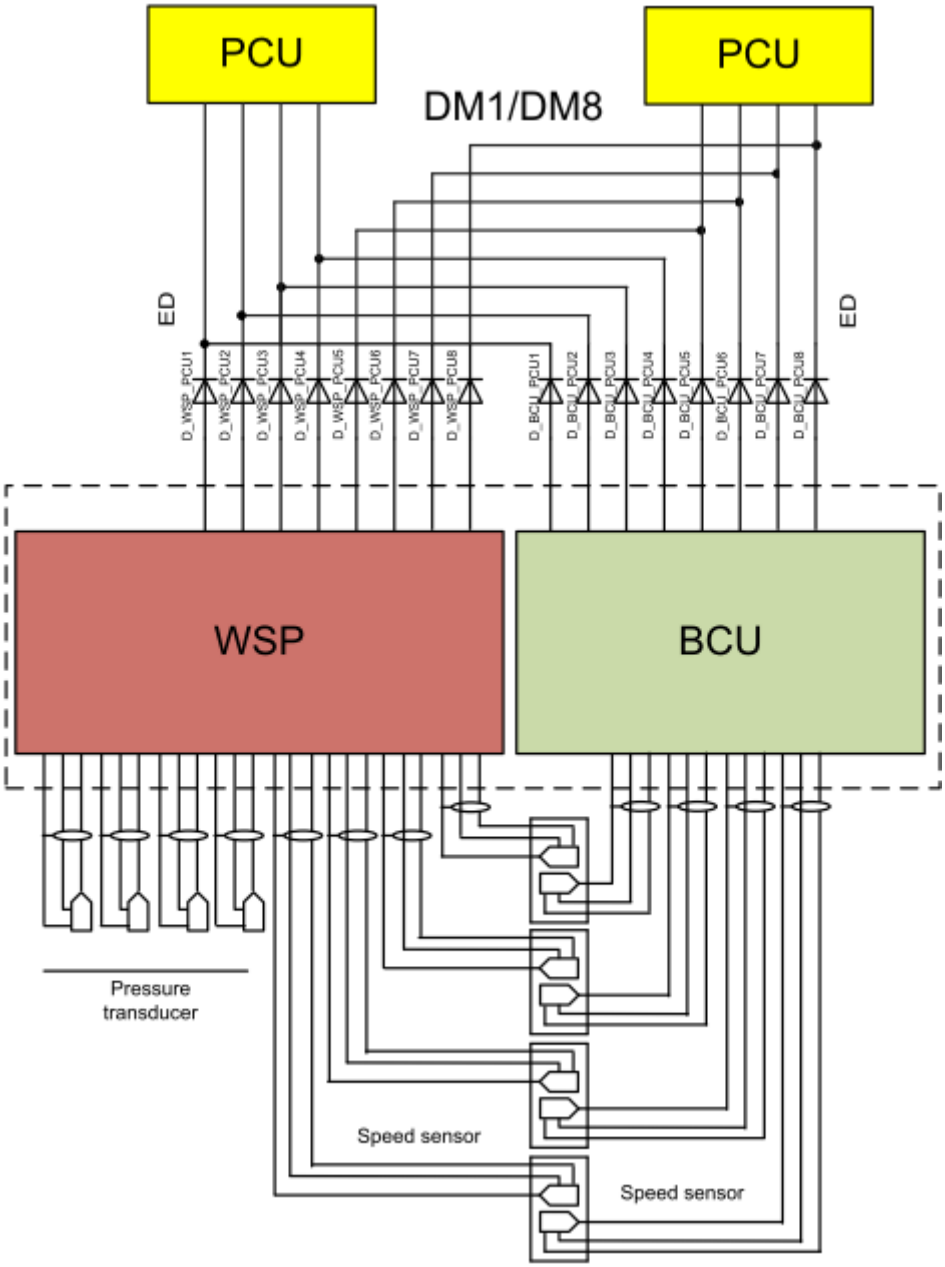
3.7.1.1 *WSP Monitoring*

When one axle can not be controlled by the primary WSP-CU for the slide control, the control per axle in failure condition is taken over by the BCU as redundant WSP. The BCU is acquiring speed information directly on the controlled axles by the second speed channel and is able to act on the redundant dump valve coils to command the fill and vent signals. The WSP system architecture is complete electrical redundant.

In case the BCU takes over the WSP function on a certain axles a maintenance event is set to the TCMS.

In addition to the redundant control of the dump valves the two hardware binary signals to the propulsion control unit in the motor cars are designed in a redundant way. In case the BCU takes over of the slide control of one bogie or one axle, the ED-brake WSP control signals are sent on the redundant path by the BCU instead of the WSP-CU.

<3.7.1.1 - D1: WSP ED control and monitoring>



If both redundant WSP control units WSP-CU and BCU are in failure in the same car not able to control the WSP function a serious fault is set and an event is presented to the driver to isolate the brake in the affected car by RIEP-DIST.

The summary failure of WSP function disturbed is generated based on following monitoring functions:

- DV electrical failure (output to dump valve)
- Speed sensor faulty
- Acquisition channel failure
- Dump Valve pressure incoherence failure
- Dump Valve hw watchdog activated
- Dump Valve hw security circuit failure
- ED control watchdog intervention
- ED control DO in open or short circuit

3.7.1.1.1 *WSP Watchdog Function*

In WSP standards UIC 541-05 and EN15595 watchdog functions are required to monitor the correct control of brake force reduction during slide. It is only allowed to reduce the brake force, pneumatic friction brake and ED brake force, for certain amount of time to guarantee the safe stopping of the train.

To be compliant to above standards independent watchdog functions are implemented by the brake systems and propulsion system.

Brake system watchdog functions:
- 10s of permanent WSP dump valve hold command active -> Brake force build-up
- 10s of permanent WSP ED brake hold command active -> Brake force build-up.

Brake system watchdog functions shall be reset:
- vehicle in traction AND the brake pipe pressure value is higher than 4.65bar AND all the brake cylinder pressure values are below 0.25 bar at least for 4 s;

OR
- vehicle in coasting AND the brake pipe pressure value is higher than 4.65 bar AND the brake cylinder pressure values are below 0.25 bar at least for 6 s.

Propulsion system watchdog functions:
- 12s of permanent WSP ED brake hold command received -> Activation internal ED slide control for entire brake request

The reset condition for propulsion system to switch back to nominal ED brake slide control by WSP-CU or BCU is to activate traction.

In case the watchdog functions are in failure dedicated maintenance events are set for the affected axle and summary event WSP function faulty is set on car level to indicate failure condition to the driver.

3.7.1.2 *WSP Function in Degraded Mode*

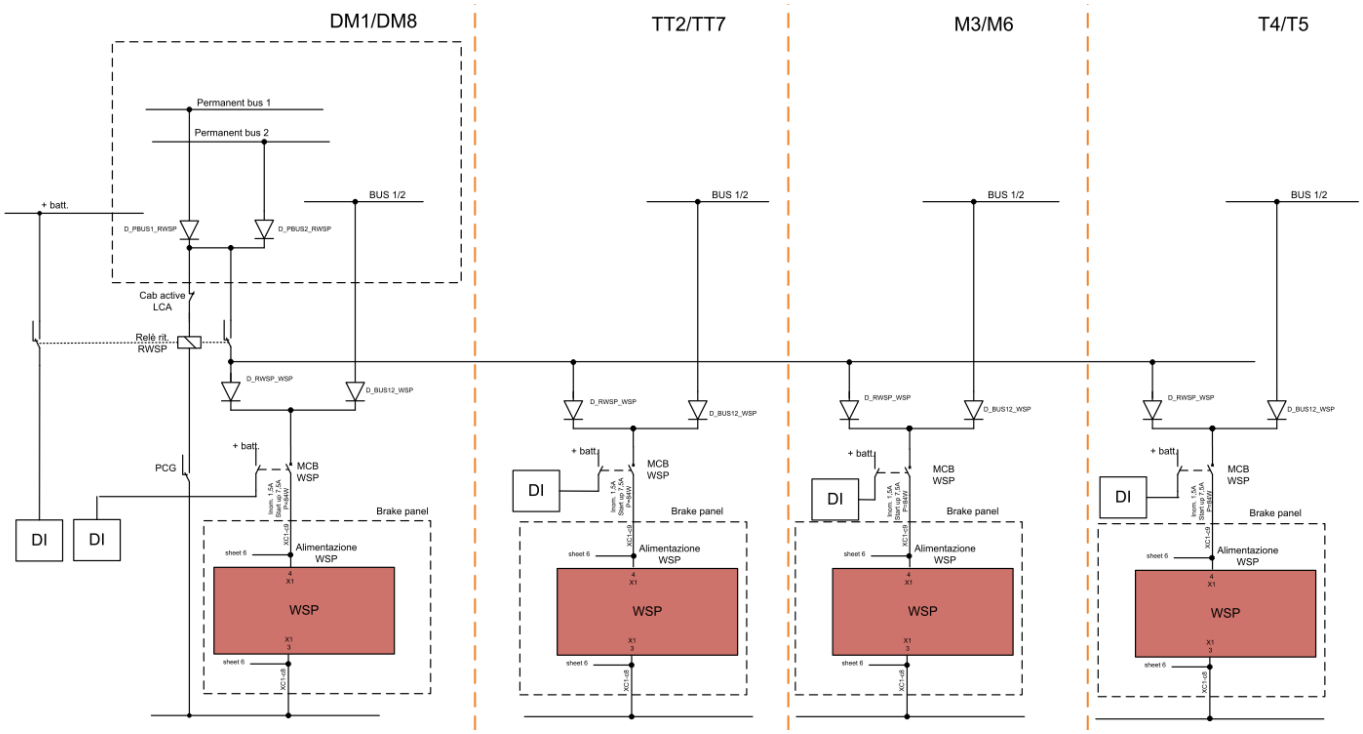
The WSP-CU is connected to the permanent battery bus and the operational battery bus to allow the WSP working as long as sufficient battery voltage is present. In case of loss of complete battery power the WSP function is not available.

In case of degraded mode without normal battery bus available the WSP is feed by the two redundant permanent bus from the DM car in a four car train segment. This occurs mainly if the ETR1000 VZI-R1 train set is shut down and is towed without TCMS available in pure indirect pneumatic brake mode.

The permanent feed is given as long the RWSP relay is energised and the contact in the permanent feed train line is closed. The feed of the RWSP relay is based on the contact of the brake pipe pressure switch PST. As long the brake pipe is maintained at a pressure > 350 kPa the pressure switch contact is closed.

If the pressure in the brake pipe is less than < 250 kPa and an emergency brake is triggered a time relay is set by the PST contact to switch-off the permanent WSP-CU power supply after a defined time to avoid a complete discharge of the battery at standstill.

<3.7.1.2 - D1: WSP power supply in degraded and normal condition>



3.7.1.3 *Failure Information*

3.7.1.3.1 *Normal condition*

In normal conditions WSP system is preventing that wheels are locked during braking and that maximum brake effort is used related to adhesion conditions.

3.7.1.3.2 *Failure cases*

1. Failure on WSP control valve DV-17 (one per axle)
2. Failure on WSP control valve command chain
3. Failure on speed sensor
4. Failure on WSP control unit (WSP-CU)

3.7.1.3.3 *Failure detection*

1. Event indication on IDU that WSP-CU control is lost on affected axle (pressure incoherence)
2. Event indication on IDU that WSP-CU control is lost on affected axle
3. Event indication on IDU of faulty speed sensor and loss of WSP-CU control of affected axle
4. Event indication on IDU that WSP-CU is faulty or communication is lost

3.7.1.3.4 *Failure handling*

1. a) Failure of one dump valve coil DV-17: Redundancy by BCU. Failed WSP valve has to be repaired during maintenance. No effect on train level.
   b) Complete dump valve faulty DV-17 (both coils): Driver has to isolate affected bogie which has failed WSP valve. Train has to be stopped.
2. a) WSP-CU control valve chain redundant by BCU. Failed WSP valve or command chain has to be repaired during maintenance. No effect on train level.
   b) If both control chains are faulty (WSP-CU and BCU) driver has to isolate bogie which has failed WSP valve. Train has to be stopped.
   c) If dump valve command is undue set to vent cylinder pressure, safety circuit is activated to stop vent command by WSP-CU or BCU. No effect on train level.
   d) If dump valve command is permanent set to vent cylinder pressure and no safety circuit is activated to stop vent command by WSP-CU. Circuit breaker of WSP-CU control to be opened. Train has to be stopped.
3. a) Speed sensor channel failure: Redundancy concept between WSP-CU and BCU on axle level. No effect on train level.
   b) In case both WSP-CU and BCU sensor channels on the same axle are faulty the driver has to isolate the bogie. Train has to be stopped.
   Failed speed sensor has to be changed during maintenance.
4. a) Failure of complete WSP-CU: Redundancy concept between WSP-CU and BCU on axle level.
   b) In case both WSP-CU and BCU are faulty the driver has to isolate the affected car.
   WSP-CU and BCU to be repaired during maintenance.

1-4) Unplanned workshop visit is necessary after first failure for minimizing of service time with less braked bogies and reduction of risk for additional failure. In case of additional failure(s) requiring isolation of > 2 bogies the remaining brake performance is not sufficient for normal service => cancellation at next opportunity.

3.7.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.137
Description	The WSP shall effect each axle separately
Safety level	SIL 3

ID	2F_05.02.Zefiro-Europe.TRS.44
Description	Through anti-slide valves, controlled by WSP control unit (UIC 514-05 compliance), the system shall be able to make an optimum use of available adhesion during braking, optimize stopping distances and prevent locked (sliding) wheels and the resulting damages.
Safety level	SIL 3

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.105	WSP function shall control each axle separately. WSP function shall contain double coil dump and apply valve for each axle.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1191	WSP function shall be controlled by WSP control unit as primary controller, in case of WSP control unit failure the WSP function shall be controlled by the BCU as redundant device.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1863	In case of fault on WSP control by WSP-CU (at axle/bogie/vehicle level) the BCU shall be able to immediately take in charge the control of the WSP on affected axle/bogie/vehicle.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1192	The WSP control valves (EWS) shall be designed with double coil.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.646	The correct function of the wheel slide protection according UIC541-05 shall be available in all braking modes as long as power supply of the brake control unit is available.	Derived Requirement	4S_07.-.Braking System		SIL 3

2F_05.02.Zefiro-Europe.CONCEPT.2495	If the power supply is loss during a brake application and WSP venting action, the braking operation and cylinder pressure shall be restored and applied again.	Derived Requirement	4S_07.-.Braking System		SIL 3
2F_05.02.Zefiro-Europe.CONCEPT.1864	Each axle shall have a pressure transducer to be used for diagnostic purpose during WSP test. The pressure transducer TCF shall be read by WSP-CU. The pressure signal shall be transferred via the RS232 interface to the BCU to calculate the achieved braking force.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1865	The pressure transducer TCF shall be connected to the WSP-CU in brake frame connector XC3.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1866	The cylinder pressure of each axle shall be sent by the WSP-CU to TCMS via IP signal "Brake cylinder pressure 1...4"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.484	WSP activity shall be sent per axle to TCMS via IP signal "WSP activity axle 1...4 (sliding detected)" by WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3222	WSP activity shall be sent per axle to TCMS via IP signal "Redundant WSP activity axle 1...4 (sliding detected)" by BCU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.661	WSP activity shall be displayed on the IDU. The indication "WSP active" shall be shown if at least one axle per train consits has active WSP.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.486	Speed signal channel from WSP-CU shall be used for DNRA redundant function as backup.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1868	The WSP-CU shall build a reference speed signal elaborating the measured 4 speed signals of axle 1-4.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.645	The axle speed shall be compared to the calculated reference signal for train speed to detect a slide.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.279	The WSP system shall measure each axle speed in order to define the instantenous deceleration and when the speed received from one axle is below a critical speed, the logic shall operate the dump valve to keep acceptable sliding value which regenerates the available adhesion.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.754	Difference between train speed and instantenous axle speeds shall be according to UIC 541-05 during braking and WSP intervention.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1870	ED brake WSP control and pneumatic brake WSP control shall be managed always by the same control unit.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1869	The control unit BCU, which is in charge of the WSP control on a axle level in case of failure of WSP-CU, shall send the IP signal "BCU is controlling WSP function axle 1...4" to TCMS in case the WSP functionality has been taken over from WSP-CU for the affected axle. 2F_05.02.Zefiro-Europe.CONCEPT.3539	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1871	TCMS shall set an event if the WSP function is changed on axle level from WSP-CU to BCU if the IP signal "BCU is controlling WSP function axle 1...4" is set per axle.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1878	A watchdog counter shall be sent by WSP-CU to TCMS via IP signal "WSP-CU life sign".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR44	**Automatic drive mode** TCMS shall provide the Automatic Drive Mode to BCU TCMS shall send a boolean to BCU to define if the train is command is automatic drive 1 = automatic drive 0 = not automatic drive	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	All status where the traction and/or braking are driven without changing train line status, it shall considered automatic drive (also called speed mode; power mode; reference mode; etc.)				
2F_05.02.Zefiro-Europe.CONCEPT.HR45	**[Rev. 3.0] WSP status in automatic drive mode** In automatic drive mode WSP shall use the Ethernet signals if all the wired signals are in traction status. If wired signals are in a non-traction status, WSP shall use the wired signals as priority. TC_BI_#WSP_S_StatusAuto	Requirement	4S_07.-.Braking System		Event SIL 0 Diagnosticated in SIL 2

3.7.2.1 *Speed Signal*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1662	The axle speed shall be sent by the WSP-CU to TCMS via the IP signal "WSP speed axle 1...4" based on medium wheel diameter 850mm.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1806	At train standstill (v = 0 km/h) the digital output "zero speed" shall be set by BCU and WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1867	For redundancy the zero speed shall be sent by BCU and WSP-CU to TCMS via IP signal "Speed zero (v=0km/h)"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1872	At speed above 10 km/h (v > 10 km/h) the digital output "Speed v > 10 km/h" shall be set by BCU and WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1873	For redundancy the speed v > 10 km/h shall be sent by BCU and WSP-CU to TCMS via IP signal "Speed v > 10 km/h"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1874	The IP signals "Zero speed (v=0 km/h)" from BCU and WSP-CU shall be compared to the BCU and WSP-CU speed zero relay contact to the supervision TCMS inputs. If IP and TCMS input is not consistent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2336	The IP signals "speed v > 10 km/h" from BCU and WSP-CU shall be compared to the BCU and WSP-CU speed > 10 km/h relay contact to the supervision TCMS inputs. If IP and TCMS input is not consistent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3934	The WSP-CU and BCU shall transfer the WSP reference speed to TCMS via IP signal "WSP reference speed".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR46	**Blocked Axle Diagnostic** **F_SS_INCOH_X (minor)** Fault description: This failure is detected at standstill and during train coasting conditions due to an incoherence between BCU and WSP speed signals. AT STANDSTILL: If VWSPX < 3 km/h (WSP_SPD_THR_INCOH_STOP parameter) AND - VBCU j, AND VBCU k < 3 km/h (BCU_SPD_THR_INCOH_STOP parameter) (with j/k axles in the same coach but different	Derived Requirement	4S_07.-.Braking System	EVENT	SIL 2

	from axle x) for 5 s (SENS_VAL_TIME_STOP PARAMETER) consecutive, the following check shall be performed: |ΔV1| = | VWSPX - VBCUX | ≥ 5 km/h (DELTA_THR_INCOH_STOP parameter) If the above condition is true for more than 300 ms (VAL_TIME_INCOH_STOP parameter) the BCU axle x speed sensor shall be declared fault without any changes in the precondition). This check is inhibited on the corresponding axle if : - A SS failure OR Speed Acquisition Channel OR Incoherence OR Disturbed is detected on the axle X by BCU; OR - A SS failure OR Speed Acquisition Channel OR Incoherence OR Disturbed is detected on the axle 1 by WSP. **~~In case of RS232 failure, it shall be used Ethernet signals.~~** IN COASTING: If |VWSPx – VBCU j| < Kventry km/h and - If |VWSPx – VBCU k| < Kventry For 5 s (VAL_TIME_COAST_COND PARAMETER) consecutive, where: j/k different from axle x Kventry = 8 km/h (ENTRY_COND_INCOH_RUN parameter), the following check shall be performed: Kvcheck = 12 km/h (DELTA_THR_INCOH_RUN parameter) |ΔV1| = | VBCUX - VWSPX | ≥ Kvcheck km/h If the above condition is true for more than 500 ms (VAL_TIME_INCOH_RUN PARAMETER) the fault shall be declared on the BCU axle x speed sensor (without any changes in the precondition). This check is inhibited on the corresponding axle if : - The train exit from coasting OR - A SS failure OR Speed Acquisition Channel OR Incoherence OR Disturbed is detected on the axle 1 by BCU. OR - A SS failure OR Speed Acquisition Channel OR Incoherence OR Disturbed is detected on the axle 1 by WSP. In case of RS232 failure, the WSP speed sensor shall be considered in failure and the check shall be inhibited. System consequences Fault type 8984 (minor) The following functionalities are inhibited or modified by the BCU on the specific axle: - Axle X shall be excluded by the Vref calculation; - DNRA check on axle X is inhibited A rescue request is sent to WSP to control the axle X. Fault reset This fault is TEMPORARY and the fault shall be reset at standstill; after 5 consecutive declaration the fault shall be PERMANENT. It is manually erased when the fault cause is removed. BCU shall send to TCMS: The specific counter refreshed				

2F_05.02.Zefiro-Europe.CONCEPT.HR47	**F_SS_DISTURBED_X (minor)** Fault description: This failure is detected for each train conditions (coasting, traction or braking status). This failure shall check the train acceleration parameter against the axle X variation. The following conditions shall be considered: 1. The diagnostic shall be enabled when: - Vref ≥ 50 km/h for the activation of the control (SPD_DISTURBED_LWR_THR parameter) - An hysteresis of - 5 km/h from Vref shall be considered to reset the condition (DISTURBED_HYS_SPD parameter). 2. When the diagnostic is enabled the BCU shall monitoring the Axle X acceleration: - Acceleration accaxle X is monitored if is greater than the following threshold |accaxle X| > 25 m/s2 (DISTURBED_ACC_THR parameter) -After any occurrence of the above conditions in order to enable again the check, the acceleration shall become | accaxle X| < 6,5 m/s2 (ACC_STABLE_DISTURBED parameter) - The error shall be set after 3 occurrences (TIMES_VAL_DISTURBED parameter) in 8 sec. (TM_WIND_VAL_DISTURBED) This check is inhibited on the axle X if: • A SS failure OR Speed Acquisition Channel OR Incoherence OR Disturbed is detected on the axle X by WSP. OR • A SS failure OR Speed Acquisition Channel OR Incoherence OR Disturbed is detected on the axle X by BCU. ~~**In case of RS232 failure, it shall be used Ethernet signals.**~~ System consequences Fault type 8984 (minor) The following functionalities are inhibited or modified by the BCU on the specific axle: - Axle X shall be excluded by the Vref calculation; - DNRA check on axle X is inhibited A rescue request is sent to WSP to control the axle X. Fault reset This fault is TEMPORARY and the fault shall be reset at standstill; after 5 consecutive declaration the fault shall be PERMANENT. It is manually erased when the fault cause is removed BCU shall send to TCMS: The specific counter refreshed The refreshed acceleration value The maximum acceleration achieved	Derived Requirement	4S_07.-.Braking System	EVENT	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR48	**BCU Speed 2vs1 diagnostics** Active in coasting at least by 5 seconds, reference speed > 50 km/h For motorized cars BCU shall compare the TCU speed shared in the Ethernet bus with actual speed as input to the BCU and input to the WSP:	Derived Requirement	4S_07.-.Braking System	EVENT	SIL 0

	If **F_SS_INCOH_X = 0 & F_SS_DISTURBED = 0 (for WSP and BCU)** If |TCU axle speed - BCU axle speed| > |15 km/h| for at least 5 seconds TACU_BSS_X = 1				
2F_05.02.Zefiro-Europe.CONCEPT.HR49	**RS232 redundancy** Signals exchanged by BCU and WSP via RS232that are used to guarantee WSP, DNRA and brake functionalities (not diagnostics) shall be provided redundant to Ethernet. If RS232 fails, it shall be necessary to use Ethernet redundancy.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR50	**TCMS management of the failure** TCMS shall send failures HR-BRK-VZI6-00117 HR-BRK-VZI6-00119 HR-BRK-VZI6-00120 to maintenance and record as side variables: Train status, speed of the train, speed of the axle, WSP activity, driver brake interfaces status (manipulator, master controller, etc.), brake pipe status, main pipe status, ED achieved, the specific counter refreshed by BCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR51	**BCU and WSP management of the speed channel in failure** In the case where there is a failure F_SS_INCOH_X OR F_SS_DISTURBED_X, BCU OR WSP detecting this failure shall: - Not considering the speed valid and not considering the signal for functionalities or diagnostics or reference speed calculation	Derived Requirement	4S_07.-.Braking System		SIL 2

3.7.2.2 *WSP ED-Brake Control*

ID	2F_05.02.Zefiro-Europe.TRS.138
Description	On the motor axles, the slide control for ED brake shall be adjusted so that the slide control for ED is intervening first. Communication of slide status shall be exchanged between WSP and MOCU/M. It shall be possible to reduce, keep or eliminate the ED depending on the WSP slide control logic in the brake computer.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.276	The two slide control signals to propulsion per axle shall be hardwired directly from WSP-CU and BCU to propulsion controller (MOCU/M).	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1884	For WSP test the two slide control signals per axle shall be sent to TCMS via IP signal "ED bit 1 active axle 1...4" and "ED bit 2 active axle 1...4" by WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3545	For WSP test the two slide control signals per axle shall be sent to TCMS via IP signal "ED bit 1 active axle 1...4 redundancy" and "ED bit 2 active axle 1...4 redundancy" by BCU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.351	The two slide control signals to propulsion per axle shall be hardwired directly from WSP-CU and BCU to propulsion controller (MOCU/M).	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1663	The two hardwired slide regulation signals per axle shall be connected between the WSP-CU and BCU in the brake panel and the MOCU/M in the motor converter box. The signals axle1, bogie 1 are: WSP_ED11 ED WSP_ED12 ED Axle 2, bogie 1: WSP_ED21 ED WSP_ED22 ED Axle 3, bogie 2: WSP_ED31 ED WSP_ED32 ED Axle 4, bogie 2: WSP_ED41 ED WSP_ED42 ED	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1290	When MOCU/M receives "reduce (Bit1 = 1, Bit2 = 1)" from WSP-CU and BCU the ED brake shall be reduced.	Derived Requirement	4S_05.-.Propulsion		SIL 1
2F_05.02.Zefiro-Europe.CONCEPT.1293	When MOCU/M receives "keep (Bit1 = 1, Bit2 = 0)" from WSP-CU and BCU the ED brake force shall remain the same.	Derived Requirement	4S_05.-.Propulsion		SIL 1
2F_05.02.Zefiro-Europe.CONCEPT.1294	When MOCU/M receives "switch off ED brake (Bit1 = 0, Bit2 = 1)" from WSP-CU and BCU the ED brake force shall reduced to 0 and the MCM shall stop pulsing of converter.	Derived Requirement	4S_05.-.Propulsion		SIL 1

2F_05.02.Zefiro-Europe.CONCEPT.1381	When MOCU/M receives "build up (Bit1 = 0, Bit2 = 0)" from WSP-CU and BCU the ED brake force shall be increased as demanded by TCMS.	Derived Requirement	4S_05.-.Propulsion		SIL 1
2F_05.02.Zefiro-Europe.CONCEPT.1989	The status of the two slide regulation hardware signals from WSP-CU and BCU shall be provided to TCMS via IP signal to be evaluated during the WSP test.	Derived Requirement	4S_05.-.Propulsion		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3935	If IP communication is lost from both WSP-CU and BCU "BCU/WSP-CU Life sign" OR both control units BCU and WSP-CU set IP signal "DIA - WSP function disturbed" in the same car TCMS shall activate slide control of propulsion control units via IP signal "Optimised slide control".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3936	If TCMS sets IP signal "Optimised slide control" the propulsion control units the PCU shall activate slide control for ED brake.	Derived Requirement	4S_05.-.Propulsion		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR122	If the hardwired signals are not working (diagnostic during the brake test; 2F_05.02.Zefiro-Europe.CONCEPT.1891) or if the WSP ED time in control is higher than 12s, the WSP ED control shall be swapped by BCU/WSP to traction control (**WSP ED optimized**). With WSP ED optimized the WSP ED control and WSP pneumatic control are independent, working separately.	Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR126	If the hardwired signals are not working (diagnostic during the brake test; 2F_05.02.Zefiro-Europe.CONCEPT.1891) or if the WSP ED time in control is higher than 12s, the WSP ED control is swapped by BCU/WSP to traction control (**WSP ED optimized**). With WSP ED optimized the WSP ED control and WSP pneumatic control are independent, working separately. Traction shall take the control of the WSP ED accordingly.	Requirement	4S_05.-.Propulsion		SIL 0

3.7.2.3 *Wheel Slide Protection Diagnostic*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1877	The WSP-CU shall be able to perfom internal WSP control self test at start-up. If no failures are detected the WSP-CU shall send IP signal "WSP start-up ok" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1037	If WSP-CU or BCU is resetting signal "WSP-CU start-up ok" the WSP function shall be considered as faulty and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1765	The following errors for ED-brake control shall be indicated by WSP-CU or BCU via IP signals to TCMS: "DIA ED_11_WSP/BCU_B1 short circuit" "DIA ED_12_WSP/BCU_B1 short circuit" "DIA ED_21_WSP/BCU_B1 short circuit" "DIA ED_22_WSP/BCU_B1 short circuit" "DIA ED_31_WSP/BCU_B2 short circuit" "DIA ED_32_WSP/BCU_B2 short circuit" "DIA ED_41_WSP/BCU_B2 short circuit" "DIA ED_42_WSP/BCU_B2 short circuit"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1766	The following errors for ED-brake control shall be indicated by WSP-CU or BCU via IP signals to TCMS: "DIA ED_11_WSP/BCU_B1 open circuit" "DIA ED_12_WSP/BCU_B1 open circuit" "DIA ED_21_WSP/BCU_B1 open circuit" "DIA ED_22_WSP/BCU_B1 open circuit"	Derived Requirement	4S_07.-.Braking System		SIL 0

	"DIA ED_31_WSP/BCU_B2 open circuit" "DIA ED_32_WSP/BCU_B2 open circuit" "DIA ED_41_WSP/BCU_B2 open circuit" "DIA ED_42_WSP/BCU_B2 open circuit"				
2F_05.02.Zefiro-Europe.CONCEPT.1776	If a short circuit of the WSP Vent or Hold 1...4 is indicated by WSP-CU or BCU IP signal an event shall be set for each axle individually. "DIA ED_11_WSP/BCU_B1 short circuit" "DIA ED_12_WSP/BCU_B1 short circuit" "DIA ED_21_WSP/BCU_B1 short circuit" "DIA ED_22_WSP/BCU_B1 short circuit" "DIA ED_31_WSP/BCU_B2 short circuit" "DIA ED_32_WSP/BCU_B2 short circuit" "DIA ED_41_WSP/BCU_B2 short circuit" "DIA ED_42_WSP/BCU_B2 short circuit"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1777	If an open circuit of the WSP Vent or Hold 1...4 is indicated by WSP-CU or BCU IP signal an event shall be set for each axle individually. "DIA ED_11_WSP/BCU_B1 open circuit" "DIA ED_12_WSP/BCU_B1 open circuit" "DIA ED_21_WSP/BCU_B1 open circuit" "DIA ED_22_WSP/BCU_B1 open circuit" "DIA ED_31_WSP/BCU_B2 open circuit" "DIA ED_32_WSP/BCU_B2 open circuit" "DIA ED_41_WSP/BCU_B2 open circuit" "DIA ED_42_WSP/BCU_B2 open circuit"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.658	If the WSP dump valve is opened for more than 10s the WSP-CU or BCU shall send IP signal on axle basis "DIA dump valve axle 1...4 active for more than 10 sec security circuit activated" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1038	If WSP fault is set by WSP-CU or BCU via IP signal "DIA dump valve axle 1...4 active for more than 10 sec security circuit activated" an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.659	The dump valve short circuit errors shall be indicated by WSP-CU and BCU via IP signal "DIA Dump valve axle 1...4 short circuit" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1039	In case of a short circuit in an dump valve reported by WSP-CU or BCU via IP signal "DIA Dump valve axle 1...4 short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1904	The dump valve open circuit errors shall be indicated by WSP-CU and BCU via IP signal "DIA Dump valve axle 1...4 open circuit" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1852	In case of a open circuit in an dump valve reported by WSP-CU or BCU via IP signal "DIA Dump valve axle 1...4 open circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3223	The WSP security circuit failure shall be indicated by WSP-CU and BCU via IP signal "DIA Dump valve axle 1...4 security circuit failure (WSP test)" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3224	In WSP security circuit failure reported by WSP-CU or BCU via IP signal "DIA Dump valve axle 1...4 security circuit failure (WSP test)" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.794	If the WSP-CU detects a faulty speed sensor (short or open circuit) the IP signal "DIA Faulty supply speed sensor WSP-CU axle 1...4" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1041	In case of a faulty speed sensor (short or open circuit) is reported by WSP-CU via IP signal "DIA Faulty supply speed sensor WSP-CU axle 1...4" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.795	If the WSP-CU detects a speed sensor for an axle has too much diviation from reference speed the IP signal "DIA WSP-CU faulty speed sensor acquisition channel axle 1...4" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1671	If a speed sensor is indicated by WSP-CU via IP signal "DIA WSP-CU faulty speed sensor acquisition channel axle 1...4" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1908	If pneumatic problem has been detected on dump valves (pressure rise and or fall) detected on pressure sensor TCF the WSP control for that axle shall be disabled and the IP signal "DIA pressure incoherence on DV1...4" shall be sent to TCMS by WSP-CU or BCU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1909	If one dump valve is pneumaticaly detected as faulty by WSP-CU or BCU by IP signal "DIA pressure incoherence on DV1...4" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1193	If WSP-CU control unit detects a faulty speed sensor for WSP control, the BCU shall take over the WSP functionality with DNRA speed sensors on bogie level.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1905	If WSP-CU or BCU has lost the life signal from other control unit the IP signal "BCU/WSP internal bus failure (RS232)" shall be set by WSP-CU or BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1906	If one of the control units report via IP signal "BCU/WSP internal bus failure (RS232)" life sign lost an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.345	If the WSP function is disturbed due to any reason the WSP-CU shall sent the IP signal "DIA - WSP function disturbed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1284	If the WSP function is disturbed indicated by WSP-CU via IP signal "DIA - WSP function disturbed" (summary event) an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3546	If the WSP function is disturbed due to any reason the BCU shall sent the IP signal "DIA - WSP function disturbed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3547	If the WSP function is disturbed indicated by BCU via IP signal "DIA - WSP function disturbed" (summary event) an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3562	If the WSP-CU/BCU detects open circuit on speed zero signal output the IP signal "DIA zero speed open circuit" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3563	If the WSP-CU or BCU indicates open circuit on speed zero signal output via IP signal "DIA zero speed open circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3564	If the WSP-CU/BCU detects short circuit on speed zero signal output the IP signal "DIA zero speed short circuit" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3565	If the WSP-CU or BCU indicates short circuit on speed zero signal output via IP signal "DIA zero speed short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3566	If the WSP-CU/BCU detects open circuit on speed v > 10 km/h signal output the IP signal "DIA v > 10 km/h speed output open circuit" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3567	If the WSP-CU or BCU indicates open circuit on speed v > 10 km/h signal output	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	via IP signal "DIA v > 10 km/h speed output open circuit" an event shall be set.				
2F_05.02.Zefiro-Europe.CONCEPT.3568	If the WSP-CU/BCU detects short circuit on speed v > 10 km/h signal output the IP signal "DIA v > 10 km/h speed output short circuit" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3569	If the WSP-CU or BCU indicates short circuit on speed v > 10 km/h signal output via IP signal "DIA v > 10 km/h speed output short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1907	If IP communication is lost from both WSP-CU and BCU "BCU/WSP-CU Life sign" OR both control units BCU and WSP-CU set IP signal "DIA - WSP function disturbed" in the same car a serious fault alarm shall be set to isolate the affected car by isolation cock RIEP-DIST.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3794	If WSP-CU detects a minor fault (summary event) the IP signal "WSP-CU minor fault" shall be set to TCMS.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3916	If WSP-CU sends the IP signal "WSP-CU minor fault" TCMS shall set an event.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_05.02.Zefiro-Europe.CONCEPT.HR39	WSP Control Unit shall provide to TCMS the information related to the axle wheel slide protection if it is in charge to WSP CU TC_BI_#WSP_S_axleprotwsp_x	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR114	An event shall be provided for 2F_05.02.Zefiro-Europe.CONCEPT.HR39	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR113	WSP Control Unit shall provide to TCMS the information related to the WSP watchdog (tripping) triggered, per axle BCU_S_ExhaustVlvAx1 BCU_S_ExhaustVlvAx2 BCU_S_ExhaustVlvAx3 BCU_S_ExhaustVlvAx4 WSP_S_ExhaustVlvAx1 WSP _S_ExhaustVlvAx2 WSP _S_ExhaustVlvAx3 WSP _S_ExhaustVlvAx4	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR115	An event shall be provided for 2F_05.02.Zefiro-Europe.CONCEPT.HR113	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR114	WSP Control Unit shall provide to TCMS the information related to the WSP watchdog (tripping) reset, per axle	Derived Requirement	4S_07.-.Braking System		SIL2
2F_05.02.Zefiro-Europe.CONCEPT.HR116	An event shall be provided for 2F_05.02.Zefiro-Europe.CONCEPT.HR113	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.HR112	TCMS shall record (store) the signal of 2F_05.02.Zefiro-Europe.CONCEPT.HR39 in case of: Heavy slide event (as per 2F_05.02.Zefiro-Europe.CONCEPT.HR91) OR TC_BI_#WSP_S_DnraWhSetx OR TC_BI_#WSP_S_WspActAxx OR BCU_S_ExhaustVlvAx1 BCU_S_ExhaustVlvAx2 BCU_S_ExhaustVlvAx3 BCU_S_ExhaustVlvAx4 WSP_S_ExhaustVlvAx1 WSP _S_ExhaustVlvAx2 WSP _S_ExhaustVlvAx3 WSP _S_ExhaustVlvAx4	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR91	WSP shall provide an event to TCMS in case of heavy slide for each axle controlled. If the axle control for WSP functionality is in charge to BCU the signal shall be set as not valid	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HR125	TCMS shall set an event in case of WSP heavy slide signal received (2F_05.02.Zefiro-Europe.CONCEPT.HR91) directed for maintenance	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.8 Detection of Non-Rotating Axle (DNRA)

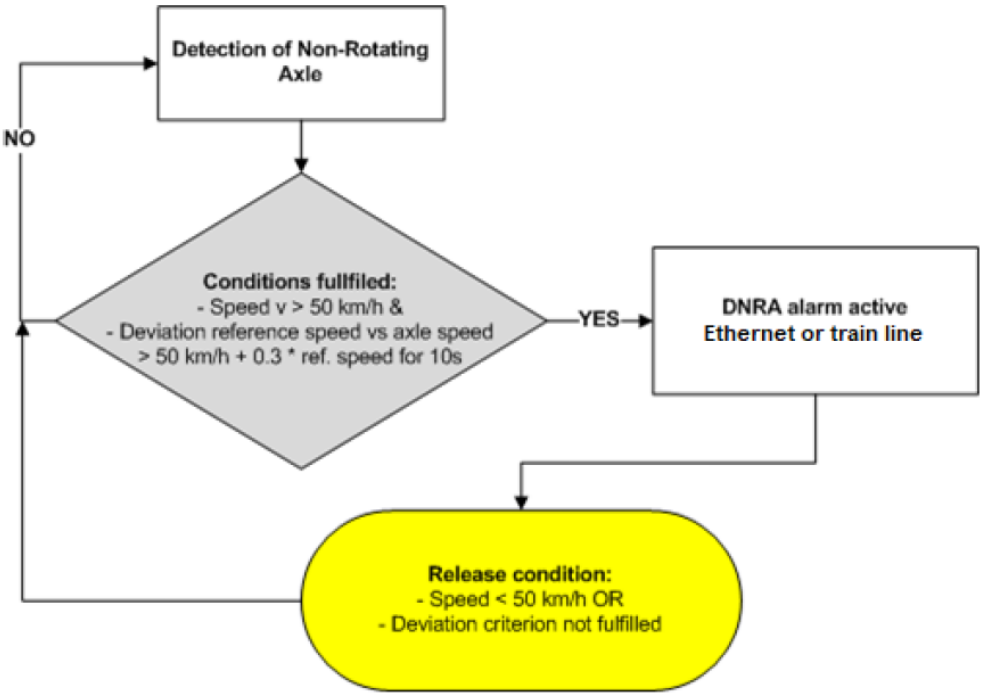
3.8.1 Function Design

The DNRA function is used to detect a non-rotating axle on the V300ZEFIRO. The DNRA system is a separated system independent from WSP control. The DNRA function is in charge of the BCU and WSP-CU (as hot redundancy).
To guarantee the independence of WSP function the DNRA function performed in the BCU uses the speed sensor channels independent from WSP-CU. The DNRA function is completely redundant as the WSP function described in chapter 3.7 with the only difference that a hot redundancy is used.
In case of failure of the BCU or the related speed acquisition channel the DNRA function is active in the WSP-CU on car level. DNRA supervision is performed by BCU and as redundancy by WSP-CU on the same time on car level.
If a non-rotating axle is detected, a warning is sent to driver through a dedicated train line and via the IP by the BCU or WSP-CU.
The DNRA alarm is shown by an indication element "DNRA alarm" on driver's desk, if the alarm is active the indication element is steady illuminated. In addition, the DNRA alarm is presented on the IDU.
The automatic brake application based on DNRA alarm is described in chapter 3.8.1.2.

The DNRA system is composed of:
- BCU: Brake control unit DNRA function per car
- WSP-CU: WSP control unit for redundant DNRA function
- SS1-4: Double speed sensor per axle (sensors with double channels one for BCU and the other for WSP-CU)
- DNRA train line
- DNRA alarm indication lamp

Any axle is considered locked if the difference between reference speed (calculated) based on the speed signals of all four axles per car and the relevant axle speed (measured) is greater than 50 km/h + 0.3 * reference speed [km/h] for more than 10 s (UIC standards parameters).

<3.8.1 - D1: DNRA function flow chart>



3.8.1.1 *DNRA Train Line*

The DNRA train line is feed from the non-active cab and can be opened by following contacts:
- DNRA function by BCU or WSP-CU

In case of locked axle condition and detection the BCU or the WSP-CU in case of redundancy of the relevant car energises the DNRA output. The output energises the relay Rss1 or Rss2, the relay contacts open and de-energise the DNRA train line.

3.8.1.2 *DNRA Monitoring*

The DNRA output channel is supervised by the BCU or in redundant cases the WSP-CU. In case of failure the BCU or WSP-CU sends a dedicated IP signal to TCMS.
TCMS supervises the two DNRA relays Rss1 and Rss2 by digital input for correct switching of the relay on car level.
In case of any failure in the DNRA chain a maintenance event is set. During the complete brake test the output of the BCU and WSP-CU is set to de-energise the DNRA train line to test the complete signal transfer to the driver.

3.8.1.3 *Failure Information*

3.8.1.3.1 *Normal condition*

In normal conditions the DNRA function is supervising that the axles are not locked for more than 10s during running speed > 50 km/h.

3.8.1.3.2 *Failure cases*

1. Failure of speed sensor
2. Failure of BCU or BCU speed acquisition channel

3.8.1.3.3 *Failure detection*

1. Event indication on IDU of faulty speed sensor and alarm in case both units are affected BCU and WSP-CU
2. Event indication on IDU that BCU is faulty

3.8.1.3.4 *Failure handling*

1. WSP-CU redundancy of DNRA function in case only one channel is affected. No effect on train level. If one channel of the speed sensor is defective the channel is excluded by DNRA diagnostic and the functionality switch to the other channel.
   In case complete sensor is faulty (both channels for BCU and WSP-CU) an alarm is set and DNRA supervision not more available on this axle, speed limitation to 200 km/h.
   Faulty speed sensor to be replaced during maintenance.
2. In case of BCU failure the DNRA function is redundant by WSP-CU, no driver action is required. No effect on train level.
   In case the BCU failure set output to DNRA train line permanent high and opens DNRA train line, the affected BCU has to be switched-off (powerless).

3.8.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.51
Description	The DNRA control shall be separated from the Wheel Slide Protection System (WSP)
Safety level	N/A

ID	2F_05.02.Zefiro-Europe.TRS.147
Description	The DNRA shall be integrated in the brake control unit and shall be active in all conditions
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.110	DNRA control in BCU shall be physically separated from WSP control unit.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2499	BCU shall be powered and DNRA function shall be active when the battery bus has voltage available.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.02.Zefiro-Europe.CONCEPT.282	During start-up of BCU or WSP-CU the DNRA function shall be checked and IP signal "DNRA function ok" shall be sent to TCMS if no error is detected. The signal shall be reset during service if the DNRA function is disturbed due to following reasons: – Speed sensor failure – Speed sensor channel aquisition failure	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1910	BCU and WSP-CU shall use one separate speed sensor channel on each axle.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1958	The BCU shall send the speed for each axle via IP signal "BCU speed axle 1...4" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1194	If BCU detects a failure in any of the speed channels signals the WSP-CU shall take over the DNRA function and use the WSP-CU speed sensor channel.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1918	If the signal "DNRA function ok" is reset to false during running at v>0 km/h an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.1917	If the BCU selftest of DNRA control is not successful and the IP signal "DNRA function ok" is reset an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2000	If the WSP-CU selftest of DNRA control is not successful and the IP signal "DNRA function ok" is reset an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.490	In case of a locked axle detection, BCU and/or WSP-CU shall energise the output to relay to open contact in the DNRA train line. Both units shall be indepedent to control the DNRA function.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2001	In case of a locked axle detection, BCU or WSP-CU shall send the IP signal "DNRA request active" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1911	The BCU/WSP-CU shall build a reference speed signal elaborating the measured 4 speed signals for DNRA function.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.752	Any axle shall be considered as locked if the difference between reference speed and axle speed is greater than 50 km/h + 0,3*reference speed (km/h) for more than 10 s.	Derived Requirement	4S_07.-.Braking System		SIL 2

3.8.2.1.1 *DNRA Diagnostic*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.662	If the BCU detects a faulty speed sensor the IP signal "DIA faulty supply speed sensor axle 1...4" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1046	If the BCU reports IP signal "DIA faulty supply speed sensor axle 1...4", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2892	If the BCU detects a faulty speed sensor acquisition channel the IP signal "BCU faulty speed sensor acquisition channel 1...4" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2893	If the BCU reports the IP signal "BCU faulty speed sensor acquisition channel 1...4", an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3549	BCU or WSP-CU shall set IP signal "DIA DNRA function disturbed" in case the DNRA function is not available (summary event).	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.660	If the BCU/WSP detects a locked axle the IP signal "DIA wheel set 1...4 is detected as non-rotating" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3555	If the BCU/WSP-CU detects an open circuit on DNRA hardware output the IP signal "DIA DNRA output from BCU/WSP-CU open circuit" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3556	If the BCU/WSP-CU indicates via the IP signal "DIA DNRA output from BCU/WSP-CU open circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3557	If the BCU/WSP-CU detects an short circuit on DNRA hardware output the IP signal "DIA DNRA output from BCU/WSP-CU short circuit" shall be sent to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3558	If the BCU/WSP-CU indicates via the IP signal "DIA DNRA output from BCU/WSP-CU short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.9 Passenger Emergency Brake

3.9.1 Function Design

The passenger emergency brake request is ordered by pulling the passenger emergency brake (PEB) handle located in each car of the train. The brake request and override possibility depend on the train PEB Control Mode.

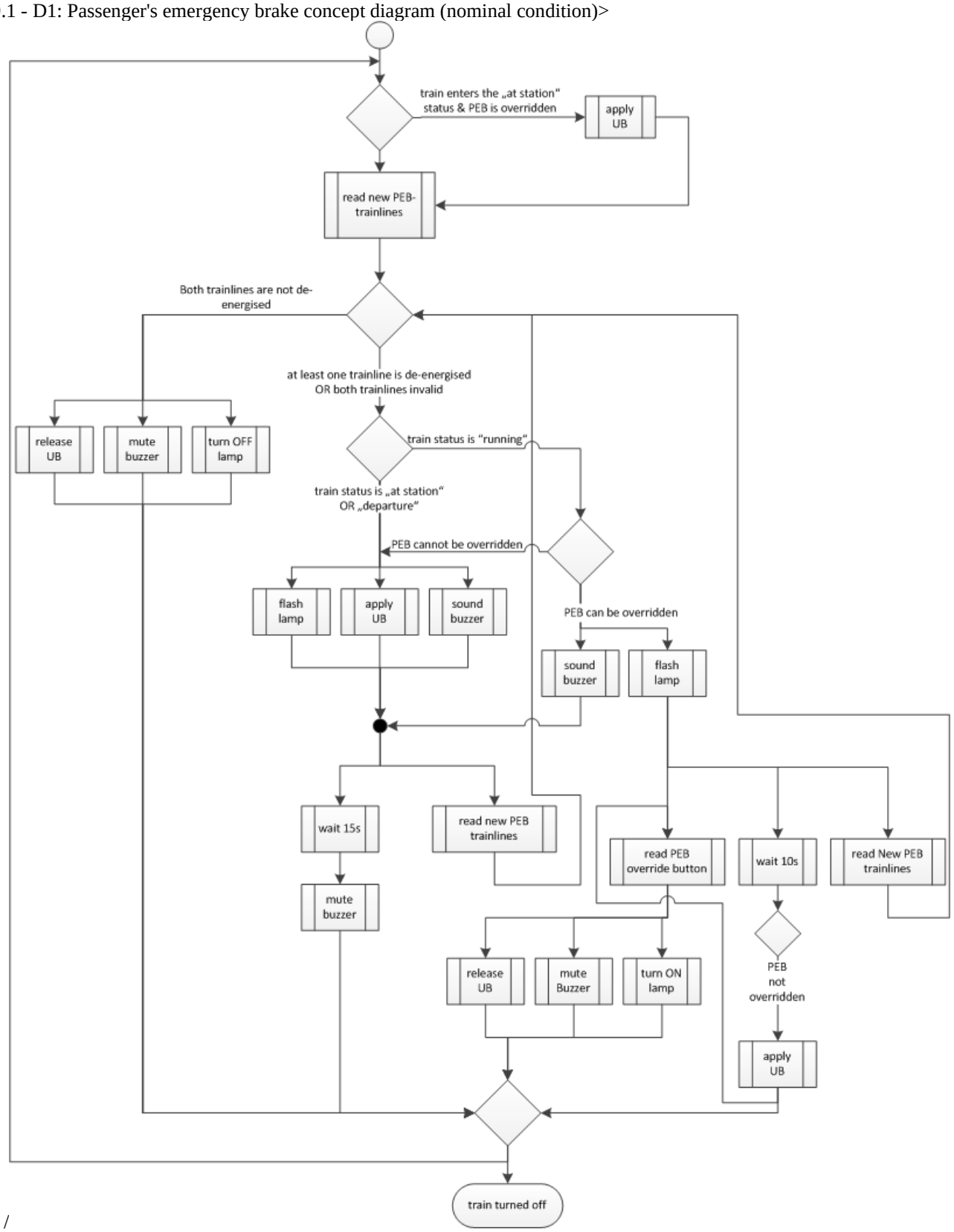
The PEB request is transferred via two PEB train lines to the DM car with the active cab, the PEB function is controlled by TCMS.

The driver is notified by the "Passenger emergency brake request active" lamp located on driver's desk inside the "Passenger's emergency brake override" push button (POVR) and an acoustical buzzer alarm in case of active PEB demand.
In addition, the PEB request is shown on the IDU with the information in which car the PEB handle has been pulled derived from the PIS system input.

The PEB demand can be overridden by the "Passenger's emergency brake override" push button (POVR, refer to diagram 3.9.1.2 - D2).

The train status to control the correct PEB Control Mode is distinguished between:
- PEB override active or not-active
- Train condition: Stop with doors released / Train departures / Train is running

<3.9.1 - D1: Passenger's emergency brake concept diagram (nominal condition)>



/

Driver's Cab Elements

Passenger emergency brake indication lamp

The "Passenger emergency brake request active" indication lamp has the following conditions:

- Not illuminated = No PEB request active
- Flashing = PEB request active
- Steady illuminated = PEB override pressed and active and brake released

The acoustical alarm buzzer has the following conditions:

- Alarm buzzer active for 15s if PEB request is active
- Alarm buzzer will mute immediately if PEB override is active

PEB override enable rotary switch

PEB override function can be deactivated with a sealed two position rotary switch (PEOR, refer to diagram 3.9.1.2 - D2) placed on the electrical cabinet in driver's cab for emergency reasons. In nominal operation the emergency override function is always active. By a redundant digital input, the PEOR selector is read by TCMS which is in charge to manage the complete passenger emergency brake algorithm according 3.9.1 - D1 diagram.

<3.9.1 – D2: sealed selector rotary switch "PEB override enable">



3.9.1.1 *PEB Control Modes*

The PEB control mode has to be divided in PEB override function active or not active and depends on the train status.

PEB override not-active (degraded condition)

The PEB override function can be disabled by turning the sealed selector rotary switch "PEB override enable" (PEOR) to the position PEB override not-active. It normally happens in case of failure of the system itself. In this case the driver has no possibility to override the TCMS brake request until the passenger emergency handle is re-armed:

1. Emergency brake is applied by TCMS immediately when passenger alarm handle is pulled
2. Reset of emergency brake and re-arming of the handles are allowed only at standstill
3. The PEB alarm is indicated to the driver acoustically and visually

PEB override active (nominal condition)

Sealed selector rotary switch "PEB override enable" (PEOR) in the position PEB override active.

Case A) "Train stop with doors released"
If the doors are released and the train is standstill the train status is set to "Train stop with doors released".

In this status, the PEB automatic brake application cannot be overridden by the driver with PEOR illuminated push-button and the PEB request is defined in the following way:

1. Emergency brake is applied by TCMS immediately after the activation of the brake handle and cannot be overridden by the driver

2. The passenger emergency brake alarm is indicated to the driver acoustically and visually.

The brake request remains active until the PEB handle has been reset.

Case B) "Train departure"
If the platform has not been left by the train (speed ≤ 10 km/h or travel distance is ≤ than 100 m) the train status is set to "Train departure". It means that if the train is at standstill with doors closed and locked it's already in this status.

In this status, the PEB automatic brake application cannot be overridden by the driver with PEOR illuminated push-button and the PEB request is defined in the following way:

1. Emergency brake is applied by TCMS immediately after the activation of the brake handle and cannot be overridden by the driver
2. The passenger emergency brake alarm is indicated to the driver acoustically and visually.

The brake request remains active until the PEB handle has been reset.

Case C) "Train running"
If the platform has been already left by the train (speed > 10 km/h or travel distance is > than 100 m) the train status is set to "Train running".

The PEB automatic brake application can be overridden by the driver with PEOR illuminated push-button and the PEB request is defined in the following way:

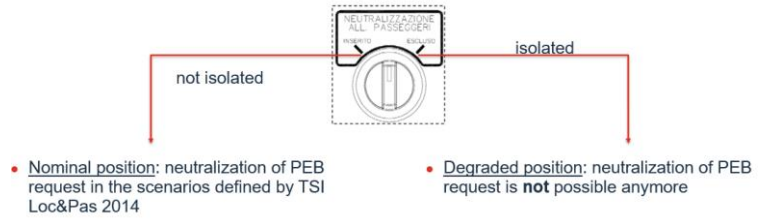
1. The passenger emergency brake alarm is indicated to the driver acoustically and visually
2. If the driver has not overridden the PEB request 10 s after initiation, an emergency brake shall be applied
3. The driver has the possibility to override the PEB brake application during the complete brake request

The override of the brake request is active until the PEB handle has been reset or if train comes to standstill (zero speed).

Permanent indication to the driver of the degraded mode

As described above, the activation of the degraded mode can be obtained only by the driver action, that can use the PEOR rotatory selector in order to overcome the fault in the PEB override button (see below).

<3.9.1.1 – D1: sealed selector rotary switch "PEB override enable" – functional view>



The selector described in the above figure is in a panel visible to the driver located in the driver cab above the driver assistant seat. No cover is present, all the selectors are visible and accessible to the driver, each selector having a dedicated function. The PEOR selector is circled in red in the figure below and the degraded position is associated to the "ESCLUSO" position (isolated as shown in the figure above).

Having this kind of arrangement, the driver has permanently the indication of the degraded mode activation looking at the position of the selector (it must be highlighted that the selector is sealed and the status of the selector is checked by the driver in order to see the foreseen status of the selectors).

<3.9.1.1 – D2: sealed selector rotary switch "PEB override enable" – location in the driving cab.>



It is further noted that the selector is used to manage the fault of the button in the specific cabin (DM1 or DM8). Therefore, its position is read by the TCMS only for the active cab. This means that:

- In case the other cabin has the push button healthy, it is not necessary to pass in the degraded mode using the PEOR of the related cab
- In case the other cabin is not enabled, the position of the selector is irrelevant and, then, the nominal mode is established when the driver is changing the cab (change of direction at the terminus station).

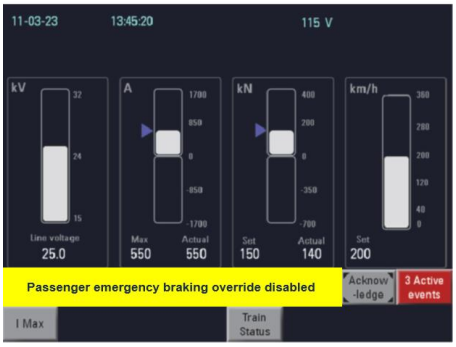
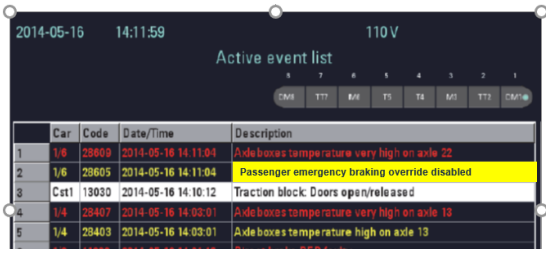
Considering also the scenario of the vehicle activation, the driver is instructed to check the status of the selectors in the panel in the cab. Thus, putting the selectors in the nominal mode at the start of mission is asked to the driver. Then, in case of fault the driver will command again the selector being aware of the degraded status.

Considering the scenario of the change of the driver, having said that the driver can check the position of the selector without specific access or other manoeuvre, also the taking over driver can see the position of the selector and, then, re-establishing the nominal position (it will be, then, managed the exclusion in case of fault of the button requesting the activation of the degraded mode).

It is worth noted that in case the selector is put in isolated position (i.e. degraded mode active) a dedicated diagnostic event is firstly shown in the first driving page of the IDU and the event is always retained as active and listed in the “active event list”, that is indicated in the “Active events” soft key used by the driver to see all the active events present during the mission.

This can be also seen as complementary mean to have the degraded mode permanently indicated to the driver.

<3.9.1.1 – D3: sealed selector rotary switch "PEB override enable" – message on the IDU>

	
First driving page with message label (activated by the selector command in degraded mode). This message will be then put in the "active events" secondary page, that is always accessible to the driver through the "Active events" softkey (this softkey is reporting the message if active events are present) – The "Active events" softkey is shown, too.	*In case of active events (in this case we highlight the specific message linked to the PEB degraded mode), the driver is instructed to access to the secondary driving page. In this page there is the detailed event list where the message is permanently indicated as soon as the selector remains in isolated position.*

3.9.1.2 *Passenger's emergency brake train line*

The passenger's emergency brake train line is composed of two redundant hardware lines. The two PEB train lines are feed from the last non active and not coupled rear cab and transfer the PEB request signal to the active cab.

All PEB handles have three contacts, one contact is respectively used to interrupt the two train lines. The third contact is used for the passenger information system to activate the communication system in case of emergency.

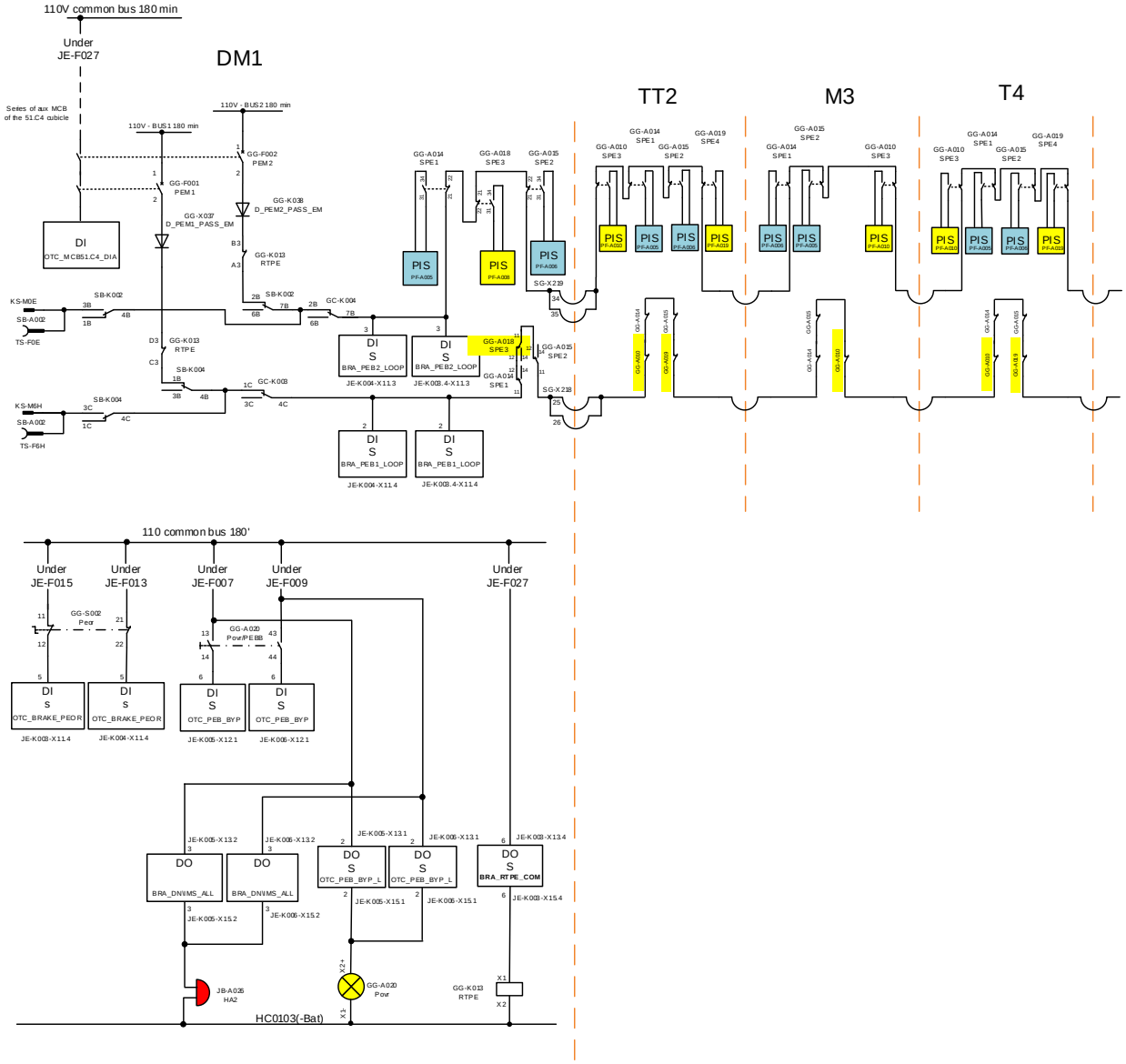
Once the PEB handle has been activated, the only way to move back it in the original position and reset the PEB request, it is with a special tool (square key) available only to the staff.

3.9.1.2 – D1: PEB handle design>



Each train line is supervised by safe TCMS control points located in each DM1/8 car. For testing purpose, a TCMS output can open the PEB train line during brake test (3.11.1.4.7) by energising the RTPE relay. The relay contact will open the train line in the non active and not coupled last rear cab.

<3.9.1.2 – D2: PEB train line design>



3.9.1.3 *PEB Monitoring and Diagnostic*

The two redundant passenger emergency brake train lines are permanently supervised by TCMS diagnostic inputs located in each DM car.

In case of failure of one train line or one contact of PEB handle the affected train line is unwanted de-energised and a permanent high PEB request is active. To avoid a degradation of the train TCMS will verify during automatic PEB test at standstill the defect PEB chain.

After the PEB test at standstill TCMS de-selects the affected faulty transfer PEB chain and the function is guaranteed and active by the redundant second train line. The TCMS logic to define the PEB degraded mode is shown in table 3.9.1.3 - T1.

<3.9.1.3 - T1: PEB train line design>

Status	PEB train line 1	PEB train line 2
Valid, no PEB demand	1	1
Valid, PEB demand active	0	0
Not valid, PEB demand active ; Automatic test required	0	1
Not valid, PEB demand active ; Automatic test required	1	0

The identified faulty train line is not used any more to transfer the PEB alarm. In this degraded mode a maintenance event is set by TCMS. The driver can start by the dedicated softkey on IDU the TCMS recovery procedure able to identify and exclude the faulty train line.

This PEB redundancy concept allows continuing the train operation with active passenger emergency brake function until the next planned maintenance visit.

3.9.1.3.1 *Failure Information*

3.9.1.3.1.1 Normal condition

In normal conditions when passenger emergency brake is activated the driver gets a warning through illuminated push button on driver's desk (override push button) and the audible signal sounds in the cab.

Override of automatic brake request possible in train running mode.

3.9.1.3.1.2 Failure cases

1. Passenger emergency brake alarm is not activated when passenger emergency brake handle is pulled.
2. Passenger emergency brake alarm is set without passenger handle request (train line de-energised).
3. TCMS is not applying the emergency brake based on PEB alarm.
4. Override function is not possible at train running mode (button failure)
5. Override function request permanent active (button failure)

3.9.1.3.1.3 Failure detection

1. Visual check while maintenance work. No failure detection while operation possible in case of defect handle.
2. a) Two train lines de-energised: Inconsistency event between train lines and PEB handle input by PIS system.
   b) One train line de-energised: Inconsistency event between the two PEB train lines.
3. No failure detection possible.
4. No failure detection possible.
5. Event on IDU in case the PEB override button is continuously pressed > 15s.

3.9.1.3.1.4 Failure handling

1. Regular maintenance or functional test intervals to be defined according safety analysis for PEB handle. No impact on train operation.
2. a) Override of PEB alarm possible during train running mode by driver. No impact on operation. At train station mode emergency operation mode to be activated.
   b) In case only one PEB train line affected self test de-selects faulty train line if all handles are re-armed. No effect on train level.
3. Driver to react and apply brake application at convenient time on visual and acoustical PEB alarm. No effect on train level.
4. In case failure during operation the emergency operation mode to be activated.
   PEB override push button part of daily brake test.
5. In case failure during operation the emergency operation mode to be activated.
   PEB override push button part of daily brake test.

3.9.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.53
Description	It shall be possible for passengers or on board personnel to initiate an emergency braking. As requested by the HS Rolling Stock TSI para. 4.2.5.3, when an emergency handle is pulled a service brake shall be applied with a defined deceleration rate.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.971	There shall be two redundant PEB train lines feed of from the non-active rear cab. If any passenger emergency brake handle is activated the two passenger emergency brake train lines shall be de-energized. The train lines shall be connected to a redundant safe TCMS MIO input in each DM car.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2542	The sealed PEB selector switch "PEB override enable PEOR" shall be located on driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3208	The PEB selector switch "PEB override enable" shall be connected to TCMS input, the TCMS input shall be set (1) if the selector switch is in PEB override disabled position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2007	The driver shall have the possibility to change between "PEB override enabled"=0 and "PEB override disabled"=1 mode. This shall be done by a mechanical selector switch "PEB override enable".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1061	If the PEB override is disabled "PEB override disabled" mode and PEB train lines are de-energised the TCMS shall de-energise the urgency brake loop.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3949	In case the sealed selector switch "PEB override disabled PEOR" is in position override disabled (=1) an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.174
Description	The driver shall be notified visually and audibly about an active Passenger Emergency brake. The audible part shall mute if the brake application is inhibited. The visual part will disappear when active brake handles are restored (rearmed).
Safety level	SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.183
Description	Activation of one or more additional passenger alarms shall have no additional effect until all alarm handles have been restored (rearmed).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.294	If PEB request is activated by de-energised PEB train line 1 OR de- energised PEB train line 2, the "Passenger emergency brake request active" indication lamp inside the "Passenger's emergency brake override" push button shall start to flash.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3216	If PEB request is activated by de-energised PEB train line 1 OR de-energised PEB train line 2, the acoustic alarm shall start to sound in the driver's cab for 15s.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.497	If passenger emergency brake is overriden by the PEB emergency brake override push button if PEB request is active, the "Passenger's	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

	emergency brake request active" indication element shall be steady illuminated.				
2F_05.02.Zefiro-Europe.CONCEPT.3217	If passenger emergency brake is overriden by the PEB emergency brake override push button if PEB request is active, the acoustic alarm shall mute.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2023	If the driver triggers an emergency brake OR the PEB emergency override has been pressed the audible alarm signal shall mute.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1063	The passenger emergency train line shall be energised if all PEB handles are re-armed (passenger emergency brake not activated).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.751	In case of any passenger emergency brake handle is pulled and the two passenger's emergency train line are de-energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2543	In case of any passenger emergency brake handle is activated an event shall be set based on the PIS IP (Ethernet) signal. The status of each handle will be provided by PIS to the train crew compartment monitor.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.291	The passenger emergency brake request shall be reset if the PEB train lines are energised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2882	For testing purpose of the two PEB train lines a TCMS output shall be connected to PEB test relay RTPE.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2884	The test relay shall be connected to the feed of the two PEB train lines 1 and 2 in the cab cars to de-energise the train line during PEB test.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2883	The PEB TCMS test output shall be set if the PEB function is going to be tested, refer to brake test chapter 3.12.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.9.2.1 *Passenger Emergency Brake Override Function*

ID	2F_05.02.Zefiro-Europe.TRS.139
Description	The passenger emergency brake application shall be able to be inhibited manually by the driver depending on the position of the train.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.494	There shall be an illuminated push button on the driver's desk for "Passenger's emergency brake override".	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3209	The "Passenger's emergency brake override" push button shall be connected to safe redundant TCMS input. "1"=override "0"=no override	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3210	There shall be an indication element on the driver's desk for the passenger's emergency brake request inside the "Passenger's emergency brake override" push button.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3211	The indication element "Passenger's emergency brake request active" inside the "Passenger's emergency brake override" push button shall be connected to safe redundant TCMS output.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.495	If the PEB override is enabled and the train status is "train stop with doors released" or "train departure" then the urgency brake shall be applied by TCMS immediately and cannot be overridden by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1699	If the PEB override function is enabled and the train status is "train is running" then:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

	– The passenger emergency brake alarm shall be indicated to the driver visually – If the driver has not overridden the PEB request 10 seconds (TSI SRT) after initiation, TCMS shall open the urgency brake loop. – The driver shall have the possibilty to override the PEB urgency brake request during the complete brake application. MIO DI "1=override", "0=not override"				
2F_05.02.Zefiro-Europe.CONCEPT.2012	If the train is standstill (v = 0km/h) and the doors are released, the train shall have the status "Train stop with doors released".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2013	If the train has been in status "Train stop with doors released" and the train starts to accelerate the train status shall be set from "Train stop with doors released" to "Train departure" until the train reached > 10 km/h OR the distance travelled by the train from standstill > 100m (TSI SRT).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2014	If the train speed is > 10 km/h OR the distance travelled by the train from standstill > 100m (TSI SRT) the train status shall be set to "Train is running".	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.498	The passenger emergency override command shall be reset if the PEB train lines are energised OR at standstill with doors released.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

3.9.2.2 *Passenger Emergency Brake Diagnostic*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3213	If both PEB train lines have not the same status, TCMS input of PEB train line 1 and TCMS input of PEB train line 2 are incoherent, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3214	If both PEB train lines have not the same status, TCMS input of PEB train line 1 and TCMS input of PEB train line 2 are incoherent, the single PEB test will be necessary to discover the train line is still correctly working, if any PEB handle is NOT pulled.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3215	If both train lines PEB1/2 have not the same status at the end of a successfully single PEB test, TCMS shall de-select the train line which remains de-energized. Only the selected train line will be used in case of PEB request until the next single PEB test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3657	An event shall be set if the PEB override push button "Passenger's emergency brake override" is pressed continously > 30s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.10 Sanding

3.10.1 Function Design

The sanding system is installed to increase the adhesion coefficient between wheel and rail in case of bad adhesion condition. Sanding is only activated during braking, no sanding is foreseen for propulsion purpose.

The sanding system is provided on leading axle in DM1/8 and on one axle in T4/T5 cars, in total four axles in the 8 car train are equipped with sanding. During driving only two axles are active depending on the driving direction of the train, the axle 1 on DM car and the axle 15 on the T-car.

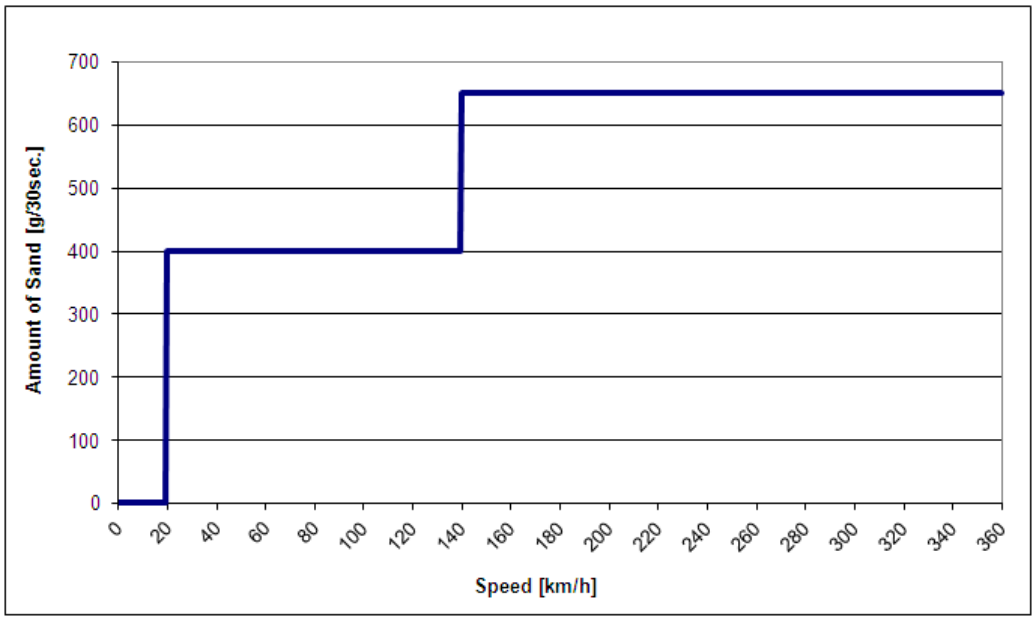
Two levels of sand flow are implemented depending on the train speed [15]:

– speed < 140 km/h sand quantity = 400g/30s

– speed ≥ 140 km/h sand quantity = 650g/30s

In the VZI-R1 train 2 sanding modes are available:
- Automatic sanding during emergency brake and WSP slide intervention controlled by WSP-CU
- Manual sanding control by driver pressing the "Sanding active" push button on driver's desk

<3.10.1 - D1: Sanding amount curve>



The sanding is controlled by the WSP-CU on the above conditions for 30 s.

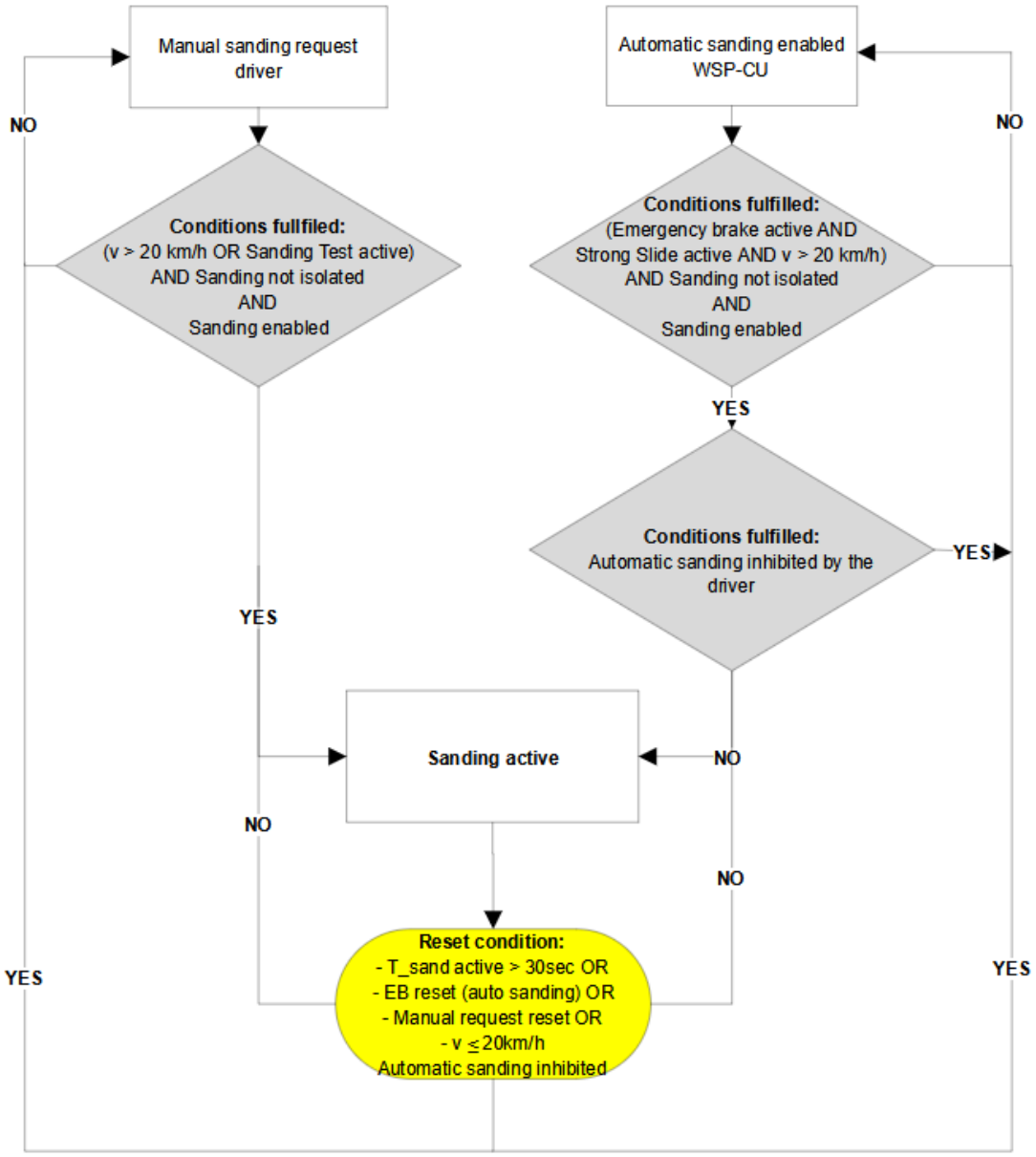
If automatic sanding is active the lamp “Sanding active” on the selector MB-A009 SS selector is illuminated (flashing), it can be interrupted anytime by the driver moving the selector MB-A009 SS on the driver desk in "EXC" unstable position.

The automatic sanding function is controlled based on following conditions for 30 sec:
- Emergency brake active and
- WSP slide interaction active and
- speed > 20km/h

In addition to the automatic sanding a manual sanding can be performed by the driver moving the selector MB-A009 SS on driver's desk in "MAN" unstable position for speed above to 20 km/h. As long the sanding function is active the lamp "Sanding active" on the selector MB-A009 SS is illuminated on driver's desk as part of the illuminated bi-unstable selector.
During sanding test at standstill only sanding can be activated regardless of the defined driving direction of the train. The sanding test can be activated via the brake test menu on the IDU, refer to chapter 3.11.
The driving direction is selected manually by the driver on driver's desk (forward, neutral, reverse). Only the sanding devices in the selected driving direction are enabled. TCMS sends a driving direction dependent IP signal to the WSP-CU sanding control units which allows the WSP-CU to distinguish which sanding unit must be activated.

Only two sanding units (for one direction) on an 8 car train are active during operation on the same time.

<3.10.1 - D2: Sanding function flow chart>

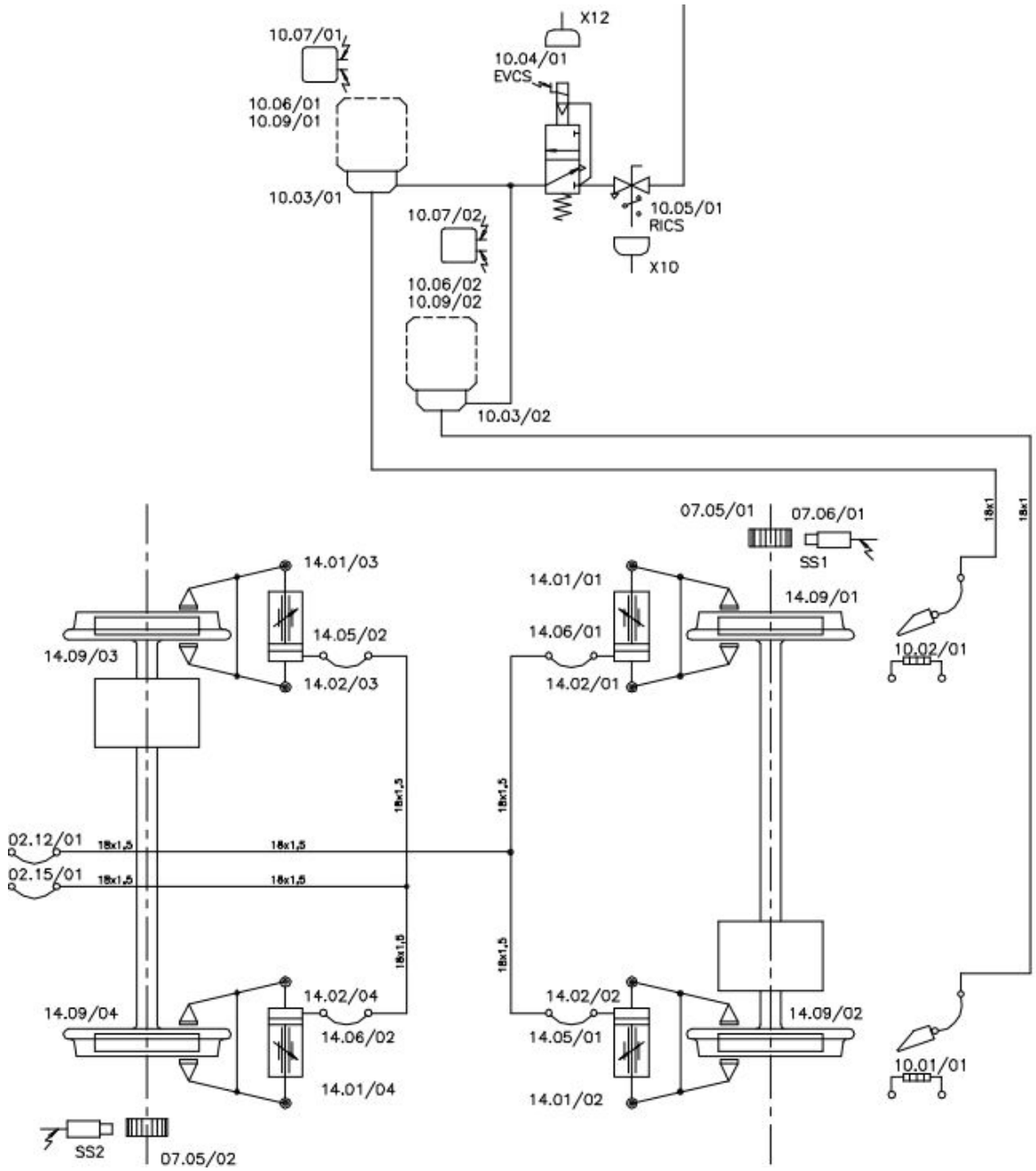


3.10.1.1 *Sanding Control Hardware*

The sanding system is composed of following components installed in DM1/8 and T4/5 cars:

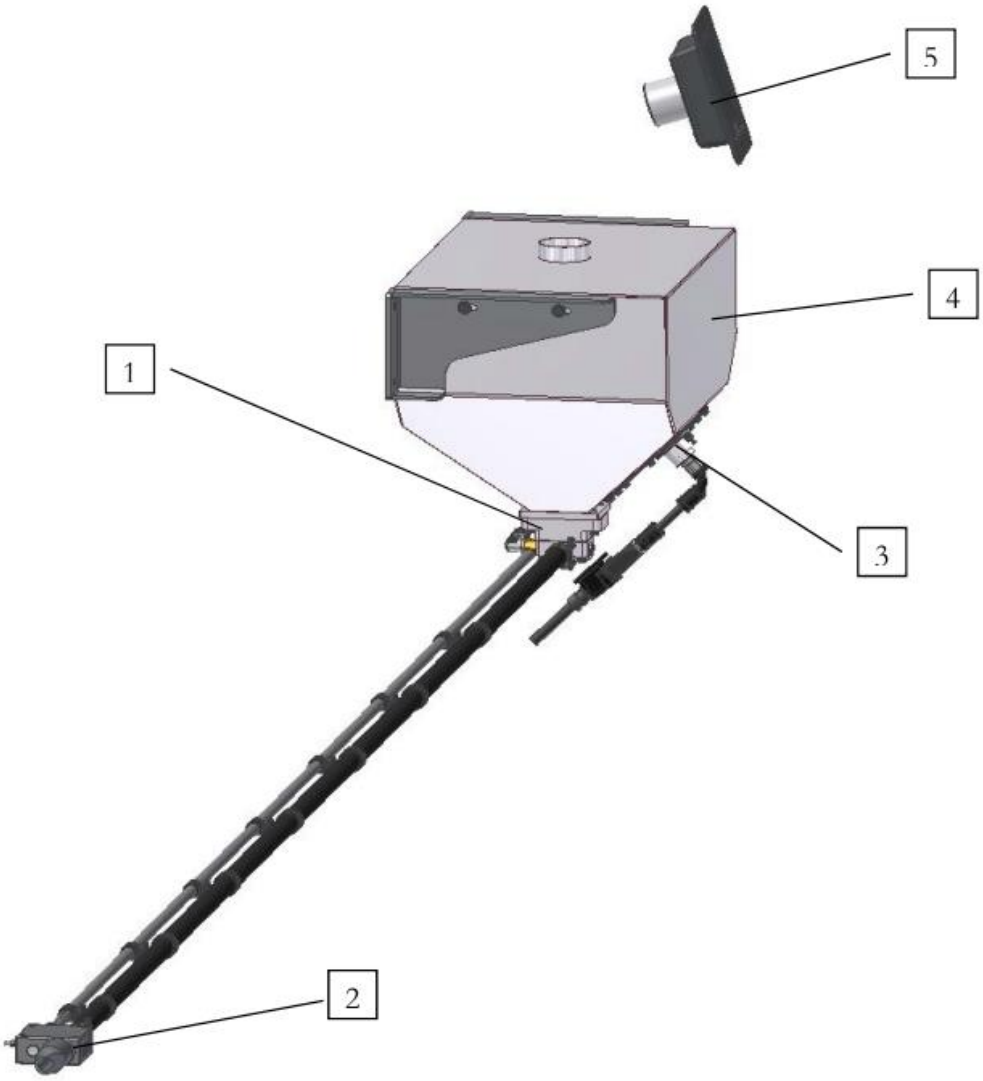
- EVCS 10.04: Electro-pneumatic control valve per axle
- 10.06: Sand box including sand level indicator 10.07
- 10.03: Dosing system including double dosage sanding injector
- Sanding diffuser per wheel 10.01/10.02 for DM-cars and 10.10/10.11 for T-cars
- RICS 10.05: Sanding system isolation cock with double electrical contact

<3.10.1.1 - D1: Sanding system pneumatic control DM1/8>



The sanding is activated by the control valve EVCS to pressurize the sanding dosage system to initiate the sand flow. Based on the speed the sand is distributed by sanding dosage system and transported to the sanding diffusor by the air flow which provides the sand in the gap between wheel and rail.

<3.10.1.1 - D2: Sanding components>



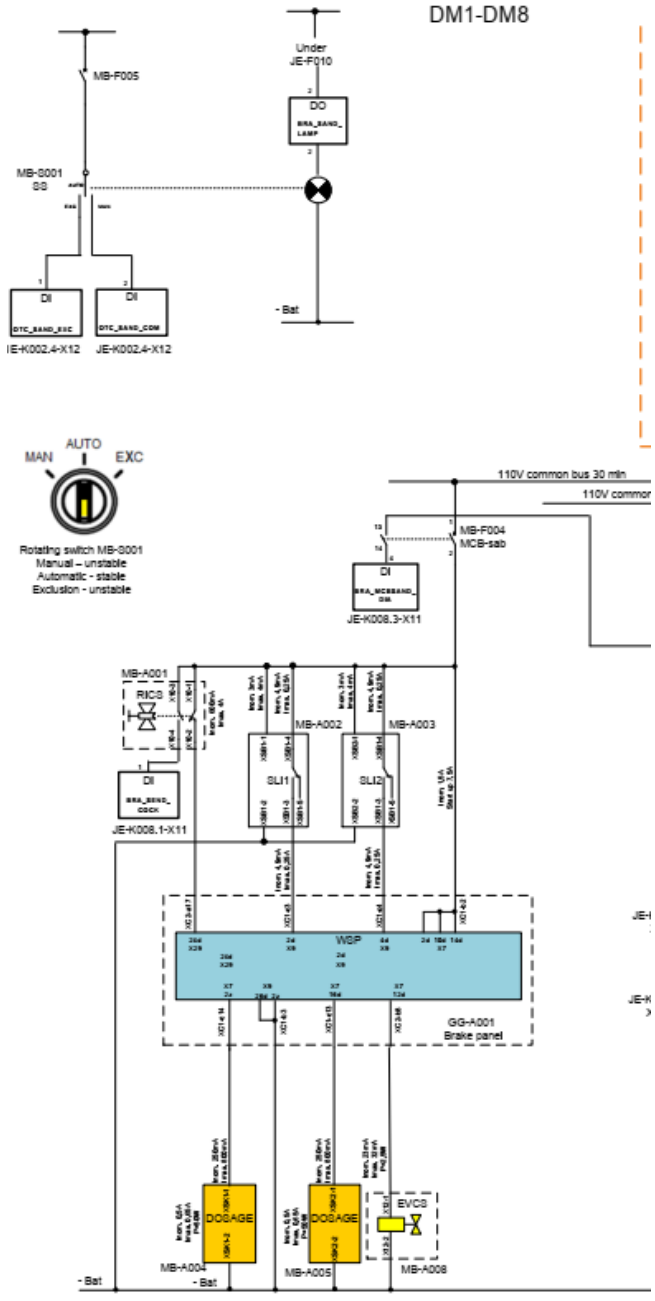
The sanding systems components in addition to the pneumatic controls are listed below and shown in diagram 3.11.1.1 - D2:
[1]. Sanding dosage
[2]. Distributor pipe
[3]. Sand level switch
[4]. Sand box
[5]. Filling flap

The sand outlet is heated by the sanding diffuser in case of low temperatures less 5°C to avoid any disturbance due to frozen nozzle.

The sanding system can be isolated per axle by sanding isolation cock RICS 10.05 for maintenance purpose or in case of failure of the system.

The sanding control and sand flow regulation by WSP-CU is done by a digital output duty cycle signal to the sanding dosage system. Depending on the duration of a high output (ms) the sanding dosage regulates the sand flow. In the same time the WSP-CU sets the magnet valve EVCS 10.04 to start the sand flow by air pressure.
After the sanding distribution is finished, a cleaning function of 5s is activated to remove the remaining sand/dirt in the sand pipe.

<3.10.1.1 - D3: Sanding control logic>



3.10.1.2 *Sanding System Monitoring*

The sand level in the sand box is continuously monitored by the sand level indicator switch. In case a certain limit is reached a maintenance, signal will be generated to refill the sand box at next regular interval.
The level indicator is read by the WSP-CU, in case of low sand level the WSP-CU sets a maintenance event to the TCMS.

The EVCS control valve is supervised by the WSP-CU, in case of failure a maintenance message is generated by TCMS.

The heating of the sanding diffuser pipes is controlled by TCMS output based on the outside temperature given by the HVAC system, if the temperature drops under 5°C the TCMS output is set and the relay RH closes the contacts for the heating elements.

The correct switching of the relay contact is supervised by TCMS input, in case of implausibility a maintenance event is set.

The isolation cock RICS is read by TCMS and WSP-CU, in case the isolation cock is in isolated position the sanding function is disabled in the related car by the WSP-CU. A warning message is generated by TCMS in case of isolation.

3.10.1.3 *Failure Information*

3.10.1.3.1 *Normal condition*

Sanding can be activated manually by the “Sanding Selector” MB-A009 SS or automatically by the WSP-CU in case of sliding depending on the country the train is operating in.
The sanding is controlled by WSP control unit.

3.10.1.3.2 *Failure cases*

1. Sand is not distributed when demanded.
2. Sand is distributed when not demanded.
3. Isolation cock 10.05 status is indicated as "isolated" in non-isolated position
4. Isolation cock 10.05 status is indicated as "non-isolated" in isolated position
5. Failure of WSP-CU for sanding control

3.10.1.3.3 *Failure detection*

1. Visible check at sanding brake test. No detection possible during driving.
2. Visible check at sanding brake test. No detection possible during driving.
3. Status of isolation cock is indicated on IDU, the real status only detected while manually opening of the isolation cock RICS during maintenance. No detection possible during driving.
4. No sand is distributed, only be checked visually during sanding brake test. No detection possible during driving.
5. Event on IDU in case of WSP-CU failure or communication.

3.10.1.3.4 *Failure handling*

1. No failure handling possible, second sanding system in train is active in parallel. No effect on train level.
2. No failure handling possible during driving, isolation cock RICS 10.05 must be isolated after visual check. No impact on train operation.
3. Affected sanding unit is deactivated until reset of status in next scheduled workshop. No impact on operation, second sanding unit in train is active in parallel.
4. No failure handling possible. Visual checks at sanding brake test must prove functionality of sanding unit and position of isolation cock.
5. No failure handling possible, loss of sanding performance in this car. No effect on train level.

3.10.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.190
Description	Sanding shall be activated manually by the driver pressing a button on the driver's desk.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.327
Description	The driver shall be able to switch off sanding in case automatic sanding function is provided.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.4006	There shall be an not latching illuminated selector for sanding in the driver's desk	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4007	Sanding shall be activated as long as the sanding selector (MB-A009 SS) on the driver's desk is in "MAN" position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4008	The sanding active status shall be reported by WSP-CU to TCMS via IP signal "Sanding is active".	Derived Requirement	4S_07.-.Braking System		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.4009	The lamp in the sanding selector (MB-A009 SS) shall be illuminated if sanding is active (reported from WSP-CU IP signal "Sanding is active").	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2545	Heating cartridge of distributor pipe shall be connected to TCMS output.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1504	Heating of distributor pipe shall be enabled if ambient temperature is below 5°C.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2033	Heating of distributor pipe shall be disabled if ambient temperature is above 8°C.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1768	If filling level in sandbox is low, the status shall be reported by WSP-CU to TCMS: – Sanding level low wheel 1 IP signal "Sanding level low wheel 1" – Sanding level low wheel 2 IP signal "Sanding level low wheel 2"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1769	If sand filling level is low reported by WSP-CU via IP signal "Sanding level low wheel 1" or "Sanding level low wheel 2" an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1770	The isolation cock RICS shall be monitored by WSP-CU. The isolated position shall be reported to TCMS via IP signal "Sanding isolation cock (RICS) isolated".	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1771	If isolation cock RICS is in position "isolated" reported by WSP-CU via IP signal "Sanding isolation cock (RICS) isolated" an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.188
Description	Sanding devices shall only be activated in driving direction.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.309	Only the sanding devices matching the selected driving direction shall be enabled. The sanding enable is given by TCMS depending on driving direction via IP signal "Sanding device enable" to WSP-CU.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1733	TCMS shall provide IP signal "Sanding device enable" per car to WSP-CU in driving direction. If DM1 is forward, the sanding enable shall be sent to DM1 and T4. If DM8 is forward, the sanding enable shall be sent to DM8 and T5.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.189
Description	Sanding shall be disabled – within switching zones (optional, if possible) – during braking at speeds below 20 km/h.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.325
Description	Sanding shall be disabled at standstill
Safety level	SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.327
Description	The driver shall be able to switch off sanding in case automatic sanding function is provided.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_05.02.Zefiro-Europe.CONCEPT.499	Automatic and manual sanding shall be disabled at train speed below 20 km/h. If brake test active command is set the sanding shall be allowed only at v = 0 km/h.	Derived Requirement	4S_07.-.Braking System		SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.191
Description	Sanding shall be activated automatically if any sliding is detected on the train.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.327
Description	The driver shall be able to switch off sanding in case automatic sanding function is provided.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.743	During urgency and emergency braking, sanding shall be activated by WSP-CU on local sanding units if any WSP activity is indicated on the car.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4054	Automatic sanding shall be set by TCMS to WSP-CU via IP signal "Automatic sanding enable" if no failure on sanding unit is present AND the Manual Sanding Selector is in "AUTO" Position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.4012	An automatic applied sanding shall be indicated to the driver by a flashing light of the manual sanding selector reported from WSP-CU IP signal "Sanding is active" in case no manual sanding is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.194
Description	The amount of sand distributed along the rail is specified in clause 4.1.1 and the maximum number of active sanding devices is defined in clause 4.1.2 of Annex A Appendix 1 of the Control-Command and Signalling TSI 2006
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.327	The amount of sand distributed along the rails shall be as following: Train speed < 140 km/h:maximum amount of sand/axle 400g+100g during 30s Train speed >=140 km/h : maximum amount of sand/axle 650g+150g during 30s	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.742	The total number of active sanding units must not exceed two for the 8 car train. The first and 15th axle (depending on driving direction) shall have an active sanding unit.	Derived Requirement	4S_07.-.Braking System		N/A

3.10.2.1 *Sanding Diagnostic*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1773	The fault status of the sanding contol valve EVCS shall be reported by the WSP-CU to TCMS: – EVCS short circuit IP signal "DIA Sanding valve short circuit (EVCS)" – EVCS open circuit IP signal "DIA Sanding valve open circuit (EVCS)"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1774	If WSP-CU reports "DIA Sanding valve short circuit (EVCS)" an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1775	If WSP-CU reports "DIA Sanding valve open circuit (EVCS)" an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3589	The fault status short/open circuit of the sanding control command to sanding unit wheel 1 shall be set by the WSP-CU to TCMS:	Derived Requirement	4S_07.-.Braking System		SIL 0

	– Sanding command open circuit IP signal "DIA Sanding command wheel 1 open circuit" – Sanding command short circuit IP signal "DIA Sanding command wheel 1 short circuit"				
2F_05.02.Zefiro-Europe.CONCEPT.3590	If WSP-CU reports "DIA Sanding command wheel 1 open circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3591	If WSP-CU reports "DIA Sanding command wheel 1 short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3592	The fault status short/open circuit of the sanding control command to sanding unit wheel 2 shall be set by the WSP-CU to TCMS: – Sanding command open circuit IP signal "DIA Sanding command wheel 2 open circuit" – Sanding command short circuit IP signal "DIA Sanding command wheel 2 short circuit"	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3593	If WSP-CU reports "DIA Sanding command wheel 2 open circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3594	If WSP-CU reports "DIA Sanding command wheel 2 short circuit" an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_05.02.Zefiro-Europe.TRS.327
Description	The driver shall be able to switch off sanding in case automatic sanding function is provided.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3985	An event shall be set if the manual sanding selector (MB-A009 SS) has been moved in "EXC" position for more than 30s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3986	An event shall be set if the manual sanding selector (MB-A009 SS) has been moved in "MAN" position for more than 30s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11 Brake Test

3.11.1 Function Design

The brake test layout of VZI-R1 is divided in three main test categories. The full semi-automatic brake test, the basic semi-automatic brake test and single brake function tests which can be selected from the IDU brake test menu.

Therefore, following different semi-automatic brake tests can be performed:

- Full semi-automatic brake test
- Basic semi-automatic brake test
- Single brake functions tests

The brake tests are performed according [11] and in line with [13].

Following relation between [11] and [13] is considered:

- Full brake test of [11] correspond to Type A test of [13]
- Basic brake test of [11] correspond to Type D test of [13]

In case of failures to the semi-automatic brake test function, manual brake tests can be performed according procedure in [13]:

- Brake Pipe continuity test
- Brake Pipe tightness test (using BP manometer in the cab)

Single brake functions test can be performed in case of maintenance or by operative instruction requests for the driver.

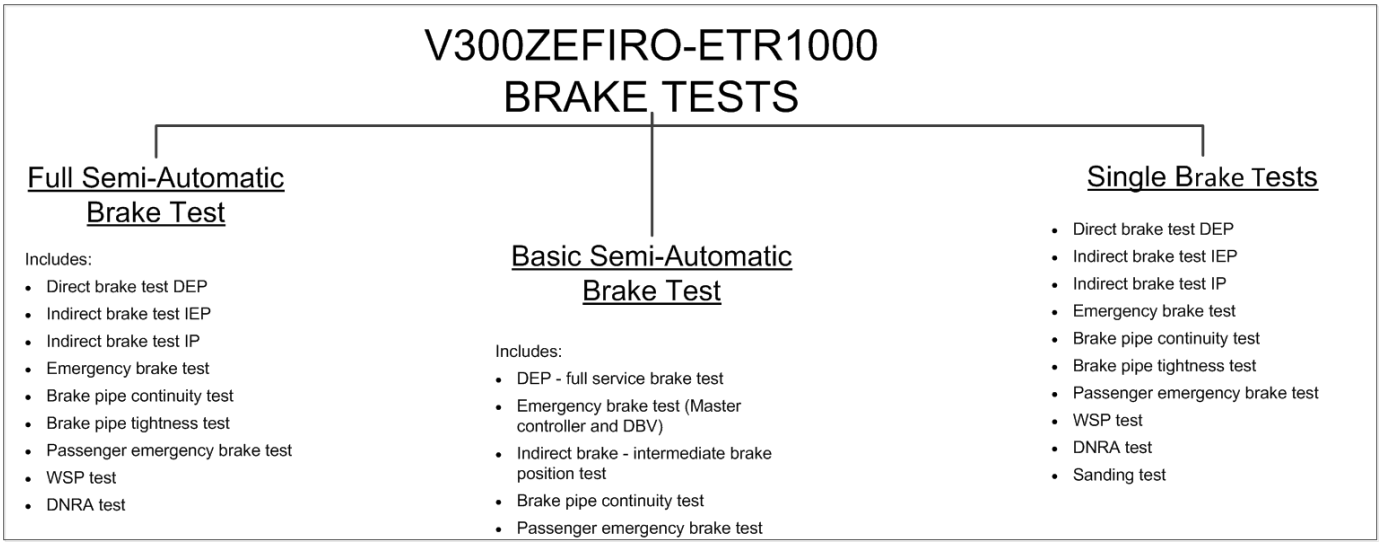
For the brake test, following operation elements are foreseen in the driver's cab:

- IDU interface brake test selection menu and driver guidance during semi-automatic brake test
- Selector switch for manual BP continuity test
- Manometer for manual BP tightness test
- Safe acknowledge push button (integrated in IDU)

The different semi-automatic brake tests are designed for different purpose.
The basic brake test should be performed by the driver before to enter in commercial service, at each cab activation and after a coupling/de-coupling sequence, this test is activated via the menu on the IDU. The test objective is to verify the basic functions as DEP or Indirect service brake, emergency brake and passenger emergency brake.
The full brake test is recommended to start daily during the preparation service in front of normal operation of the train, at minimum every 24h. The full brake is started via the IDU brake test menu. The test objective is to verify the functionality of the full brake system and each control element on driver's desk. It's a test for the maintainer.

Anyway, it's up to the customer to define the tests to put in place in the different operating conditions.

<3.11.1 - D1: Brake test overview>



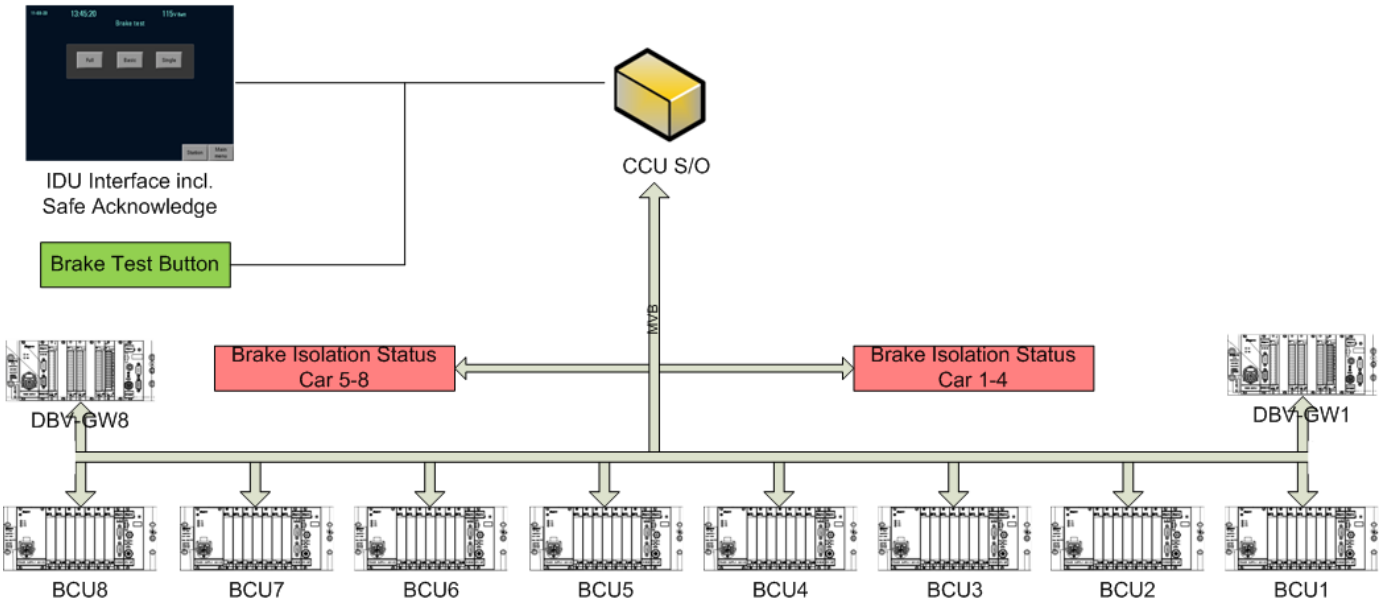
The brake test is managed by TCMS. TCMS defines the task to BCU, WSP-CU and DBV-GW of each vehicle and receives the local result by each brake control unit via IP .
Some brake test tasks are directly performed by TCMS by setting I/O signals and/or IP signals to the BCU, WSP-CU and DBV-GW.

The BCUs and WSP-CUs are responsible for the pass/fail criteria of the single vehicle brake tests, local car test results are reported by BCU to TCMS.
The DBV-GW is involved for Indirect Electro-Pneumatic (IEP) test by Minitrol.

TCMS determinate the global test result collecting the entire local car test results by the brake control units. Each single brake test is split in several steps, for each of them TCMS will wait for the result by the local brake control units.
Only if all local test results of the current step are received and successful the test step is considered passed.

The global result is acceptable and successful only if all the single steps are successful. In case on single brake test step is failed the overall global test result is failed with information which step was not passed.

<3.11.1 - D2: Automatic Brake Test Hardware>



3.11.1.1 *Brake Test Execution*

The brake tests full semi-automatic brake test, basic semi-automatic brake test and single brake tests will be executed according the following process:

1) Selection of the brake test menu page on IDU
   - The IDU shows the whole overview of available brake tests:
     Full semi-automatic brake test
     Basic semi-automatic brake test
     Single brake tests (sub-menu on IDU with all single brake tests)

If the preconditions are not valid for a specific brake test the IDU softkey is presented in a different colour.

2a) Selection of the test to be executed on IDU
   - The driver has to selected the test to be executed on IDU

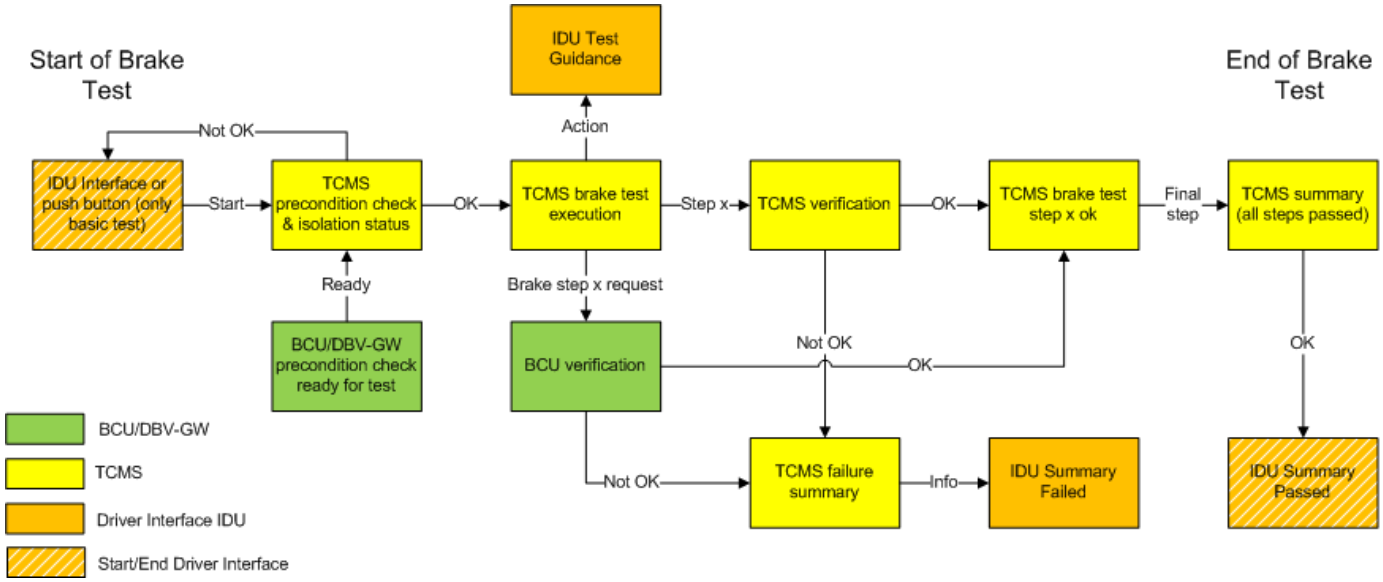
2b) Pressing push button "Brake test"
   - The basic semi-automatic brake test (ABT) is started immediately as long all preconditions are fulfilled

3) Precondition check to start brake test
   - Precondition check is done by TCMS based on feedback by BCU, WSP-CU and DBV-GW
   - Preconditions are defined in chapter 3.11.1.1.1

If the preconditions are not all valid the driver is informed about the missing preconditions to start the test on IDU.

4) The driver confirm the start of brake test selected on IDU by acknowledge button (except for brake test push button where the test is started immediately)

<3.11.1.1 - D1: Brake Test Execution Flow Chart>



3.11.1.1.1 *Brake Test Start Preconditions*

The precondition to allow the start of the automatic brake test depends on the kind of brake test to be performed. The preconditions are defined at train level by TCMS and locally per car by BCU.

In addition to the precondition the isolation and available status of certain equipment is verified. TCMS will not test the respective isolated or not available cars (preconditions locally not fulfilled by BCU, see below BCU preconditions).

The preconditions for the brake control units (BCU and DBV-GW) are listed below, if all preconditions are locally on car level fulfilled the brake control units setting the respective brake test ready IP signal to TCMS.

BCU preconditions:
- Speed v = 0 km/h
- Main reservoir pipe pressure > 700 kPa
- Emergency brake not active
- Urgency brake not active
- DNRA output not active
- Brake pipe pressure at 500 kPa
- RIEP-DIST, RICP and RICF1 AND RICF 2 in not isolated position

DBV-GW preconditions:
- RIM cock in not isolated position
- IEP mode available and active
- DBV in released position (RP)
- Brake pipe pressure at 500 kPa

The needed train level brake test preconditions verified by TCMS are shown in the overview table.

<3.11.1.1.1 - T1: TCMS Brake Test Preconditions>

TCMS Train Level Condition	Main Brake Tests		Single Brake Tests											
Type of test (= X)	Full Test	Basic Test	DEP	IEP	IP	EB	BP continuity	BP tightness	PEB	WSP	DNRA	Sanding	UB (O)	MTB (O)
Test ID	F	B	1	2	3	5	6	7	8	9	10	12	4	13
Driver's desk activated	X	X	X	X	X	X	X	X	X	X	X	X	X	X
MVB communication with all available BCU ok	X	X	X	X	X	X				X	X		X	X
MVB communication with all available WSP-CU ok	X	X	X	X	X	X				X	X		X	X
BCU MVB signal "Ready for brake test" is set per available car	X	X	X	X	X	X				X	X		X	X
Train is at standstill (v=0 km/h)	X	X	X	X	X	X	X	X	X	X	X	X	X	X
MR-pressure is above 700 kPa	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Parking brake is applied (minimum required amount)	X	X	X	X	X	X	X	X	X	X	X	X	X	X
No emergency brake is active (emergency loop energized)	X	X	X	X	X	X	X	X		X			X	X
No urgency brake is active (urgency loop energized)	X	X	X	X	X	X	X	X		X			X	X
DNRA train line is energized	X										X			
Master controller in released position	X	X	X	X	X	X	X						X	
Driver's brake valve in released position	X	X	X	X	X	X	X	X					X	
Brake pipe at 500 kPa +/- 5 kPa	X	X	X	X	X	X	X	X					X	
DEP service brake mode active (for ABT)	X (DEP)	X (DEP)	X											
IEP brake pipe control active				X										
IP brake pipe control active (RIBK in IP position)					X									
Isolation condition														
Brake pipe control panel is not isolated on active cabin (RIM)	X	X	X	X	X	X	X	X					X	
Brake pipe control panel is not isolated on not-active cabin panels (RIM)	X													
Brake is not isolated on brake frames (RIEP-DIST, RICP)														X
Direct brake is not isolated on brake frames (RIEP)														
Bogie brakes are not isolated (RICF1/2)														
Sanding systems are not isolated (RICS)												X		

(O) = OPTIONAL

The status of the TCMS precondition has to be set true before the selected brake test can be activated.
If any precondition is not fulfilled, the TCMS informs the driver to perform the action in order to fulfil the precondition to start the brake test. If the preconditions are not all valid, detailed information are shown in which cars the precondition are not fulfilled and which equipment has to be isolated.

Each brake test can be performed with certain brake equipment isolated. If the friction brake is isolated in a complete car by isolation cock RICP or RIEP-DIST and on bogie level by RICF1 AND RICF2 cock, this car is not considered during the brake test. The brake test can be executed with one bogie isolated per car.

In this case the overall train level test result is successful if all other not complete isolated cars are passed the brake test.
For the remaining available cars or bogies the test is performed and the results are displayed on the brake test summary screen during and at the end of the test. The driver has to acknowledge the overall final test result including the amount of isolated cars not considered during the test.

If major TCMS preconditions which are not changing during the execution are reset during a running test the brake test will abort:
- The lamp "Brake test active" switches off
- IDU message of abortion is shown to the driver (missing precondition is part of the message)
- Train status is defined as standstill and holding brake is applied
- The system is ready again to start another brake test

The above listed preconditions are checked continuously by TCMS on train level and locally by BCUs and DBV-GW.

The degraded modes are managed in case of following conditions are true:
- Isolation of certain brake equipment
- BCU or WSP-CU or DBV-GW missing IP communication

In case of certain equipment is isolated the driver is responsible after the global test result to calculate the brake performance based on the amount of available cars. The acknowledgement by the driver confirms that the degraded brake performance is noticed.

3.11.1.2 *Full Semi-Automatic Brake Test*

The full semi-automatic brake test of main brake system functions should be done daily in front of the operation of the train and during maintenance.

The full semi-automatic brake test contains several single brake system function tests which can be also started individually via the IDU brake test menu (refer to 3.11.1.4).

The full semi-automatic brake test is lead by TCMS, performed by TCMS and BCUs in conjunction with the driver and train staff guided by the IDU instructions.

The full brake test contains (FABT) following single brake tests:
- DEP brake test
- Indirect brake IEP test
- Indirect brake IP test
- Brake pipe continuity test
- Brake pipe tightness test
- Emergency brake test and safety loop functionality test
- Passenger emergency brake test
- WSP test
- DNRA test

3.11.1.3 *Basic Semi-Automatic Brake Test*

The basic semi-automatic brake test (ABT) is designed according the requirements of the EN15734-1 [11], the design is covering as well the functional test requested in brake test type D described in [13]. One basic brake tests is considered for the two different type of brake demand and application present on the train:
- Direct brake (DEP)
- Indirect brake mode

The basic brake test is performed at train cab activation, cab change and after coupling or de-coupling procedure. The basic brake test has a test duration of less than 3 min (see [11] par. 5.15.2 and [14] par. 1.2.8 requirement).

The basic semi-automatic brake test is lead by TCMS, performed by TCMS and BCUs in conjunction with the driver guidance by the IDU instructions.

In order to guarantee the execution of the basic brake test in case of identified degraded modes (brake isolations) and in case of failures which are not affecting significantly the brake performances, the regular basic brake test can be performed by the driver to test the remaining available not isolated brakes.
In this condition the overall global test result will pass with the summary which cars are not available/isolated to be acknowledged by the driver.

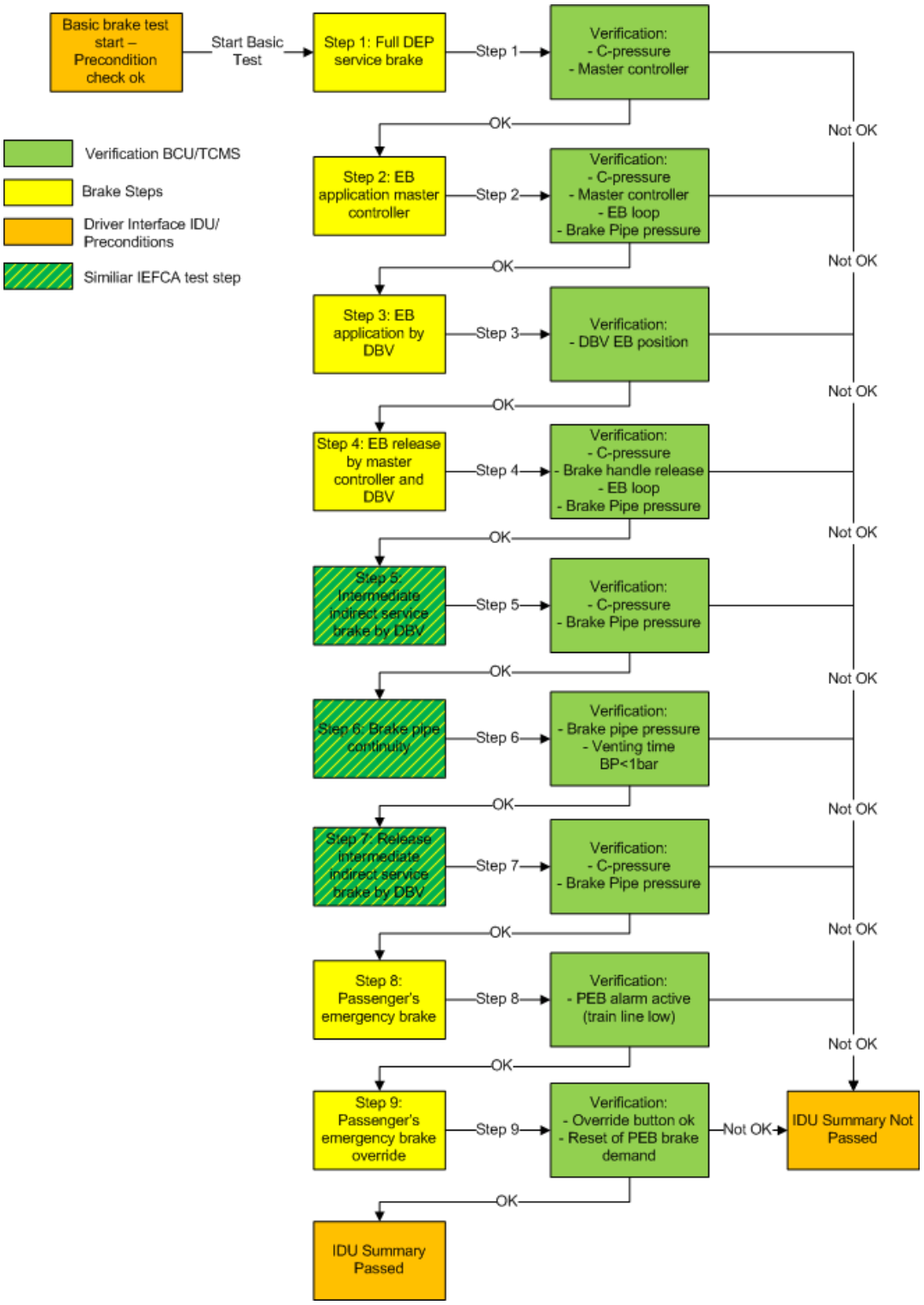
Redundant monitoring inputs connected to the brake control units (BCU and WSP-CU) and to TCMS inputs are used during the several steps of the basic brake test.
The content of the basic test is described in the following chapter.

3.11.1.3.1 *Basic Semi-Automatic Brake Test (ABT) Content Overview*

The Basic Semi-automatic Brake Test (ABT) is used in nominal condition when the service brake is in DEP mode. The test has following content:

- DEP achieved full service brake force check by master controller (full service brake position)
- Emergency brake and safety loop functionality test by master controller
- Emergency position check of driver's brake valve
- Indirect service brake IEP (Intermediate position, venting BP 60 - 80 kPa)
- Brake pipe continuity test (including check of brake pipe isolation in non-active cab)
- Passenger emergency brake test

<3.11.1.3.1 - D1: Basic Brake Test Flow Chart>



3.11.1.4 *Single Brake Test Functions*

Each single brake test function can be started separately via the IDU brake test menu.
Following single brake test cases are present for the brake system.

Single brake function tests:

– DEP brake test

- IEP brake test
- IP brake test
- Emergency brake test
- Brake pipe continuity test
- Brake pipe tightness test
- Passenger emergency brake test
- WSP test
- DNRA test
- Sanding test

Test case DEP, IEP, IP and Emergency brake test are single tests with precondition to start the test as BCUs, WSP-CU and DBV-GW are involved.
All other tests can be performed also if the system is in degraded conditions, in this case the driver is informed about the degradation.

In the following chapters each single brake test is described more in detail with information about general preconditions and the sequence with pass/fail criteria in the different test steps.

3.11.1.4.1 *DEP brake test*

The objectives of the direct electro-pneumatic brake test are:
- Check functionality of the master controller brake request to TCMS
- Check the correct transmission of brake demand by TCMS via IP to BCUs
- Brake and release application by direct brake on each vehicle and cylinder pressure verification
- Brake pipe maintained at 500 kPa

Signals to be monitored during the test:
- Cylinder pressure
- Brake pipe pressure (TCG sensor)
- Master controller position

General
TCMS verifies that the brake pipe pressure is at 500 +/- 10 kPa in all cars.

Step 1 – DEP maximum service brake application
- The driver is requested by IDU to move the master controller to maximum service brake position.
- Verification: Brake is applied according brake demand by TCMS, the BCU verifies the correct amount of cylinder pressure on car level. TCMS checks the master controller full service brake position.
- BCU sets a passed signal on IP to TCMS if step is successful.

Step 2 - DEP maximum service brake release
- The driver is requested by IDU to move the master controller to zero position (coasting) position.
- Verification: Brake demand by TCMS is reset and brakes are released, the BCU verifies the cylinder pressure is released. TCMS checks the master controller release position.
- BCU sets a passed signal on IP to TCMS if step is successful.

If the step 1 and step 2 is passed the complete DEP brake test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event is set during the DEP brake test.

3.11.1.4.2 *IEP brake test*

The objectives of the indirect electro-pneumatic brake test are:
- Check the functionality of the Driver's Brake Valve PWM position signal to Minitrol
- Check the functionality of the BP control by DBV Minitrol
- Brake and release application by indirect brake on each vehicle

Signals to be monitored during the test:
- DBV position
- Cylinder pressure
- Brake pipe pressure (TCG sensor)
- Brake refill inhibition neutral valve position (VNEU)

General
TCMS verifies that the service brake control mode is indirect brake mode and IEP brake pipe control active (RIBK cock position).

Step 1 – IEP intermediate service brake application
- The driver is requested by IDU to move the driver's brake valve to service brake position 03.
- Brake demand is transferred by lowering the brake pipe pressure, the BCU verifies the correct brake pipe pressure and cylinder pressure (4stage level, medium/low pressure level of triple stage group) on car level. If the values are in range and applied in a certain time the BCU sets the step as passed via IP to TCMS.
- TCMS verifies the brake pipe pressure.

Step 2 – IEP maximum service brake application
- The driver is requested by IDU to move the driver's brake valve to maximum service brake position MB.
- Brake demand is transferred by lowering the brake pipe pressure, the BCU verifies the correct brake pipe pressure and cylinder pressure (4stage level, medium/low pressure level of triple stage group) on car level. If the values are in range and applied in a certain time the BCU sets the step as passed via IP to TCMS.
- TCMS verifies the brake pipe pressure.

Step 3 – IEP maximum service brake release
- The driver is requested by IDU to move the driver's brake valve to release position RS.
- Brake pipe is filled and brakes are released, the BCU verifies the correct release brake pipe pressure and cylinder pressure. If the values are in range and brakes are released in a certain time the BCU sets the step as passed via IP to TCMS.
- TCMS verifies the brake pipe pressure.

If the step 1, 2 and step 3 is passed the complete IEP brake test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event is set during the IEP brake test.

3.11.1.4.3 *IP brake test*

The objectives of the indirect pure pneumatic brake test are:
- Check the functionality of the Driver's Brake Valve position
- Check the functionality of the BP control by Driver's Brake Valve pneumatic control
- Brake and release application by indirect brake on each vehicle

Signals to be monitored during the test:
- Isolation cock RIBK to activate IP brake pipe control
- DBV position
- Cylinder pressure
- Brake pipe pressure (TCG sensor)

General
TCMS verifies that the service brake control mode is indirect brake mode and the RIBK isolation cock is in IP brake pipe control position.

Step 1 – IP intermediate service brake application
- The driver is requested by IDU to move the driver's brake valve to service brake position 03.
- Brake demand is transferred by lowering the brake pipe pressure, the BCU verifies the correct brake pipe pressure and cylinder pressure (4stage level, medium/low pressure level of triple stage group) on car level. If the values are in range and applied in a certain time the BCU sets the step as passed via IP to TCMS.
- TCMS verifies the brake pipe pressure.

Step 2 – IP maximum service brake application
- The driver is requested by IDU to move the driver's brake valve to maximum service brake position MB.
- Brake demand is transferred by lowering the brake pipe pressure, the BCU verifies the correct brake pipe pressure and cylinder pressure (4stage level, medium/low pressure level of triple stage group) on car level. If the values are in range and applied in a certain time the BCU sets the step as passed via IP to TCMS.
- TCMS verifies the brake pipe pressure.

Step 3 – IP maximum service brake release
- The driver is requested by IDU to move the driver's brake valve to release position RS.
- Brake pipe is filled to 500 kPa and brakes are released, the BCU verifies the correct release brake pipe pressure and cylinder pressure. If the values are in range and brakes are released in a certain time the BCU sets the step as passed via IP to TCMS.
- TCMS verifies the brake pipe pressure.

If the step 1 to step 3 is passed the complete IEP brake test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event is set during the IP brake test.

3.11.1.4.4 *Emergency brake test*

The objectives of the emergency brake test are:
- Intervention of emergency brake loop by emergency operation elements
- Brake pipe pressure venting
- Emergency loop diagnostic
- EVAPP valve application
- TCMS input from devices (DBV, relays, etc) and emergency/urgency loop
- Brake and release application by indirect brake
- Brake and release application by direct brake

Signals to be monitored during the test:
- Signals from different control elements to open emergency circuit
- Brake pipe pressure signals
- Emergency brake loop inputs to BCU/WSP-CU
- Cylinder pressure and pressure switches
- Status of neutral valve (SW-NEU)
- TDIS indirect pressure signal
- TPIL direct pressure signal (not considered in case RIEP isolation cock in isolated position)
- Pressure switches PAPP (application valve)

Optional: In case of option Emergency Brake (EB) activated from Urgency Brake (UB) by Emergency Brake Supervision Unit: The test orders the Emergency Brake Supervision Unit to go into fail safe state, i.e. EB and UB activated simultaneously (same as without option).

Step 1 – Emergency brake application master controller
- The driver is requested by IDU to move the master controller to emergency brake position.
- Brake demand is transferred by lowering the brake pipe pressure and EB loop, the BCU verifies the correct brake pipe pressure.
- TCMS verifies the correct switching of emergency application valve EVAPP by pilot pressure switch PAPP feedback
- The BCU verifies the pre-control pressure (weak comparison) the two independent emergency brake channels.
- The BCU check the emergency/urgency brake loop input, the correct pre-control DEP brake pressure at TPIL sensor and the correct indirect brake pre-control pressure at TDIS sensor.
- TCMS verifies the WSP-CU emergency brake loop input feedback.
- The cylinder pressure (4stage level, medium/low pressure level of triple stage group) on car level is verified by the BCU.

If the values are in range and applied in a certain time the BCU sets the step as passed via IP to TCMS.

3.11.1.4.4.1 BCU pressure switch check (per bogie)

BCU shall check PCF in the brake test in emergency, with emergency brake applied, when pressure are stabilized, after the trasducer check:

PCF (shared by TCMS to BCU)	TCF1	TCF2	Effect
Braking (1)	< 0,2 bar	< 0,2 bar	Event PCF failure (check not ok) – type 1
Not braking (0)	> 0,8 bar	> 0,8 bar	Event HR-BRK-VZI6-00124 PCF failure (check not ok) – type 2
Not braking (0)	< 0,2 bar	< 0,2 bar	Reset type 1
Braking (1)	> 0,8 bar	> 0,8 bar	Reset type 2

Filter if BCU transducers check at the brake test (per bogie) = Check ok, 15 seconds
This failure shall NOT prevent a brake test start or block/fail a brake test

3.11.1.4.4.2 BCU transducers check at the brake test (per bogie)

BCU will check in the brake test in emergency, with emergency brake applied, when pressure are stabilized:

|TCF1 - TCF2| = |0,15| = Check ok
Else
BCU: maintenance fault “Transducer bogie 1 faulty” OR BCU: maintenance fault “Transducer bogie 2 faulty” (TCF bogie X failure)

This failure shall NOT prevent a brake test start or block/fail a brake test

Step 2 – Emergency brake release master controller
- The driver is requested by IDU to move the master controller to zero position (coasting),
- Emergency brake is released.
- Brake demand is reset by refill of brake pipe pressure and EB loop closure, the BCU verifies the correct brake pipe pressure.
- TCMS verifies the correct reset of emergency application valve EVAPP by pilot pressure switch PAPP feedback.
- The BCU verifies the release of the pre-control pressure of the two independent emergency brake channels.
- TCMS verifies the emergency loop input is reset.
- The cylinder pressure release on car level is verified by the BCU.

If the values are in range and applied in a certain time the BCU sets the step as passed via IP to TCMS.

Note: The direct brake chain verification at TPIL pre-control pressure sensor will not be performed in case the direct brake isolation cock RIEP is in isolated position

Step 3 – Emergency application DBV
- The driver is requested by IDU to move the driver's brake valve to emergency brake position EB.
- Brake demand is transferred by lowering the brake pipe pressure and EB loop, TCMS verifies the brake pipe pressure and DBV emergency position input.
- TCMS verifies the brake loop status reset by BCUs and WSP-CUs emergency brake loop input feedback.

If the values are in range and applied in a certain time TCMS sets the step as passed.

Step 4 - Emergency release DBV
- The driver is requested by IDU to move the driver's brake valve to release position RS.
- Emergency brake is released
- Brake demand is reset by refill of brake pipe pressure and EB loop closure, TCMS verifies the brake pipe pressure and DBV release position input.
- TCMS verifies the brake loop status set by BCUs and WSP-CUs emergency brake loop input feedback.

If the values are in range and applied in a certain time TCMS sets the step as passed.

If the step 1 to 4 is passed the complete emergency brake test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event for EB loop and application is set during the emergency brake test.

The complete emergency brake test step 1 to 4 is done during the full semi-automatic brake test.

3.11.1.4.5 *Brake pipe continuity test*

The objectives of the brake pipe continuity test are:
- Check the continuity of the brake pipe
- Check the isolation of brake pipe refill in the not activated cabs

Signals to be monitored during the test:
- Brake pipe pressure signals TCG
- Status of isolation valve and neutral valve (SW-ISO and SW-NEU)

General
TCMS verifies that in all not active cabs the isolation valve is in isolated position by SW_ISO**.**

Step 1 – Brake pipe isolation in active cab and venting
- TCMS commands the EVISO and EVNEU brake pipe isolation valves and checks the SW_ISO and SW_NEU position feedback.
- TCMS commands the emergency brake application relay in the last rear cab to open the brake pipe by emergency brake application valve EVAPP.
- Verification: TCMS verifies the brake pipe pressure drop in the active cab car until the brake pipe has reached 100 kPa.

If the values are in range and the pressure drop is inside the specified time TCMS sets step 1 as passed.

Step 2 – Brake pipe release and reset of brake pipe isolation
- TCMS resets command to the EVISO and EVNEU brake pipe isolation valves and checks the SW_ISO and SW_NEU position feedback in refill position.
- TCMS resets the command to EVAPP relay in the last rear cab to close the brake pipe application valve EVAPP.
- Verification: TCMS verifies the brake pipe pressure refill in the active cab car until the brake pipe has reached 500 kPa.

If the values are in range and the pressure increase is inside the specified time TCMS sets step 2 as passed.

If the step 1 and step 2 is passed the complete brake pipe continuity test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event for brake pipe isolation valves is set during the continuity test.

3.11.1.4.6 *Brake pipe tightness test*

The objectives of the brake pipe tightness test are:
- Check the tightness of the brake pipe
- Check the isolation of brake pipe refill in all cabs

Signals to be monitored during the test:
- Brake pipe pressure signals TCG

- Status of isolation and neutral valve (SW-ISO and SW-NEU)

General
TCMS verifies that in all not active cabs the isolation valve is in isolated position by SW_ISO.

Step 1 – Brake pipe isolation and tightness test
- TCMS commands the EVISO and EVNEU brake pipe isolation valves in active cab and checks the SW_ISO and SW_NEU is in isolated position.
- TCMS verifies the brake pipe leakage in all DM-cars within 60 seconds.

If the values are in range and the pressure drop is inside the specified time TCMS sets step 1 as passed.

Step 2 – Brake pipe isolation reset and refill
- TCMS resets command to the EVISO and EVNEU brake pipe isolation valves and checks the SW_ISO and SW_NEU position feedback in refill position.
- TCMS verifies the brake pipe pressure refill in the active cab car until the brake pipe has reached 500 kPa.

If the values are in range and the pressure increase is inside the specified limit TCMS sets step 2 as passed.

If the step 1 and step 2 is passed the complete brake pipe tightness test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event for brake pipe isolation valves is set during the tightness test.

3.11.1.4.7 *Passenger's emergency brake test*

The objectives of the passenger's emergency brake test are:
- Check the integrity of the train lines used to activate the passenger emergency brake
- Check the lamp and audible signal are working correctly
- Check the emergency brake application due to PEB TCMS command
- Check the PEB override function

Signals to be monitored during the test:
- Passenger emergency brake train line status 1 and 2
- Alarm indication and buzzer (by driver)
- Emergency brake loop opening by TCMS

Step 1 – Passenger emergency alarm intervention
- TCMS opens the RTPE test relay contacts on the last rear cab to de-energise the PEB train line 1 and 2.
- TCMS verifies that at least one PEB train line input is set (de-energised).
- TCMS shall apply EB and verifies the TCMS emergency brake relays supervision inputs.

If the TCMS considers at least one train line as de-energised and TCMS emergency brake relays are opened step 1 is considered as passed.

Step 2 – Override by the driver and reset of PEB train line
- The driver is requested by IDU to override the passenger emergency brake by the PEB override push button.
- The driver is requested to verify and acknowledge that the lamp is steady illuminated.
- TCMS closes the RTPE test relay contacts in the last rear cab after the acknowledgement of the driver to energise both PEB train lines.
- TCMS verifies that at least one PEB train line inputs is reset (energised) and no emergency brake is applied.

If the driver confirms that lamp is active and TCMS considers at least one train lines as energised after the acknowledgement step 2 is considered as passed.

If the step 1 to step 2 is passed the complete PEB test shall be considered as successful.

3.11.1.4.8 *WSP test*

The objectives of the WSP test are:
- Check the correct function of WSP system
- Check WSP control interface between WSP-CU/BCU and propulsion control

Signals to be monitored during the test:
- Cylinder pressures
- Dump valve command
- WSP electro-dynamic control signals

General
TCMS verifies that the parking brake has been applied correctly before starting the test.

Step 1 – WSP ED brake interface test
- TCMS sets the two ED control test bits 1 and 2 in a sequence via IP to WSP-CU.
- WSP-CU and BCU sets the output ED1 and ED2 to the propulsion control unit based on the sequence.
- Propulsion control units set the status via IP to TCMS. TCMS verifies the complete chain and set the step as passed.

Step 2 – WSP valve test WSP-CU
- TCMS commands the WSP test start via IP to all WSP-CUs in the complete train.
- WSP-CU executes the internal test and verifies the dump valve application times, if the test is passed WSP-CU sets signal via IP to TCMS.

Step 3 – WSP valve test BCU redundant
- TCMS commands the WSP test start via IP to all BCUs in the complete train.
- After WSP-CU the BCU executes the internal test and verifies the dump valve application times, if the test is passed BCU sets signal via IP to TCMS.

If the step 1 to step 3 is passed the complete WSP test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event for WSP function is set during the WSP test.

During WSP test the WSP control unit manage the WSP ED signals. Traction control unit check if both signals are acquired and change 0-1-0 during brake test. If these signals are not changed, a failure is provided to TCMS for maintenance and the WSP ED control is changed to WSP ED optimized control. Failure reset at the following WSP test with WSP ED signals Hold and Vent changed 0-1-0.

Command	WSP_Hold	WSP_Vent
Build-up ED brake force	0	0
Keep ED brake force	1	0
Reduce ED brake force	1	1
Switch-off ED brake force	0	1

3.11.1.4.9 *DNRA test*

The objectives of the DNRA test are:
- Check the correct functionality of the DNRA train line
- Check indication lamp on the driver's desk

Signals to be monitored during the test:
- Single DNRA Relays RSS on each vehicle
- Lamp command relays RDNRA

Step 1 – DNRA alarm test BCU
- TCMS sets command via IP in sequence to every BCU to activate a DNRA alarm at standstill.
- Based on the DNRA test command BCU activates the output to RSS2 relay.
- TCMS verifies for each train line opening on car level if the relays RDNRA1 and RDNRA2 in activated cab are de-energized.

The step is considered as passed if all interventions have been tested successful and the driver confirmed the lamp activity.

Step 2 – DNRA alarm test WSP-CU
- TCMS sets command via IP in sequence to every WSP-CU to activate a DNRA alarm at standstill.
- Based on the test command WSP-CU activates the output to RSS1 relay.
- TCMS verifies for each train line opening on car level if the relays RDNRA1 and RDNRA2 in activated cab are de-energized.

The step is considered as passed if all interventions have been tested successful.

Step 3 – DNRA alarm test
- TCMS verifies for each train line opening on car level if the relays RDNRA1 and RDNRA2 in activated cab are de-energized.

The step is considered as passed if all interventions have been tested successful.

If the step 1, 2 and step 3 is passed the complete DNRA test shall be considered as successful. The test is failed in case any criteria have not passed and a diagnostic event for DNRA function is set during the DNRA test.

3.11.1.4.10 *Sanding test*

The objectives of the sanding test are:
- Check the correct function of sanding system
- Check the delivery of sand on nozzle at train

Signals to be monitored during the test:
- None

Step 1 – Activate sanding test
- TCMS sets the sanding test command via IP to WSP-CU.
- WSP-CU activates the sanding test at standstill with lowest sand amount according TSI values on all sanding units independent of the driving direction as in normal operation.

The driver/train personal has to visual check the sand delivery between wheel and rail on train.

3.11.1.5 *Manual brake test*

If the automatic brake test function is not available, manual brake test can be performed without TCMS functionality. Following manual tests are possible to be performed:
- Indirect brake test
- Brake pipe continuity test
- Brake pipe tightness test
- WSP test on car level (FT service terminal tool or direct access to brake control unit required)

For the manual brake tests following control elements are present in the driver's cab:
- Driver's brake valve 01.02
- Manual brake test selector switch GG-S03
- Manometer BP pressure 01.15
- Service brake indicators 14.11 per car

The manual brake test selector switch has two active positions, one position is to inhibit the brake pipe refill to perform the brake pipe tightness test or to perform the brake pipe continuity test. In this position the EVISO brake pipe isolation valve is directly energised without any software function involved.

The second position is to vent the brake pipe for the brake pipe continuity test in the local cab where the manual brake test selector switch is present. In this position the emergency brake application valve relay GG-K005 is energised which opens a contact in the feed to the EVAPP valve to vent the brake pipe. The command supply to the EVAPP test relay is only possible at standstill, a zero speed relay contact is used to inhibit the command supply during driving to avoid unwanted stopping failures.

Indirect brake test

The driver can perform the indirect brake test manually by using the driver's brake valve in IP mode in the driver's cab.
To verify the active brake demand in driver's cab the brake pipe manometer 01.15 and cylinder pressure manometer 01.14 of the first bogie can be checked. The local brake demand on car level can be verified by the outside service brake indicators.

Step 1: Intermediate brake application
The driver has to move the driver's brake valve in intermediate service brake position 03.
Brake pipe pressure at 420 kPa and cylinder pressure > 150 kPa on manometer has to be verified.
The outside service brake indicators on car level to be monitored switching to red.

Step 2: Full brake application
The driver has to move the driver's brake valve in full service brake position MB.
Brake pipe pressure at 330 kPa and cylinder pressure of > 300 kPa on manometer has to be verified.
The outside service brake indicators on car level to be monitored remaining red.

Step 3: Full brake release
The driver has to move the driver's brake valve in release position RP.
Brake pipe pressure at 500 kPa and cylinder pressure of < 20 kPa on manometer has to be verified.
The outside service brake indicators on car level to be monitored switching to green.

Brake pipe continuity test

The driver can perform the brake pipe continuity test manually by putting the manual brake test rotary selector switch in the driver's cab to brake pipe continuity test position. For that the driver and train crew have to perform the following steps.

Step 1: Brake pipe refill inhibition
The driver has to move the manual brake test rotary selector switch in active front cab into brake pipe isolated position to ensure the brake pipe refill is stopped, this position energises the brake pipe isolation valve EVISO to avoid the brake pipe refill.

Step 2: Venting of brake pipe
In the not-active last rear cab of the train or coupled composition the assistant train personal has to move the manual brake test rotary selector switch in position "BP continuity test" to start the venting procedure of the brake pipe by the emergency brake application valve.
The driver in the active front cab has to supervise the brake pipe pressure drop on the manometer 01.15 located on driver's cab to verify the correct venting behaviour.

Step 3: Verification
If the brake pipe pressure drops from 500 kPa to 300 kPa in a specified time the continuity test is successful.
The train crew and driver have to reset the control elements.

Brake pipe tightness test

The driver can perform the brake pipe tightness test manually by using the manual brake test selector switch in the driver's cab. Therefore the driver has to perform following steps.

Step 1: Brake pipe refill inhibition

The driver has to move the manual brake test selector switch in active cab to start the BP filling inhibition, this will energise the neutral and isolation valve to avoid the brake pipe refill.

Step 2: Tightness verification
If the pressure in the brake pipe shown on the BP manometer drops not more than 10 kPa in 60 seconds the test has passed.

WSP manual test

The WSP manual test on car level can be started via the FT service terminal software tool or directly on brake control unit mounted on the brake frame located in the underframe assembly.

In case the FT service terminal software tool is used it has to be connected to the train IP network to access the local brake control units separately and build up a connection, a joined test start command for all brake control units is not possible.

The description to start directly the manual WSP test on brake control unit is described in brake control unit service description and maintenance manual.

3.11.1.6 *Failure Information*

3.11.1.6.1 *Normal condition*

The full, basic and single semi-automatic brake tests can be activated by the driver via IDU. The basic brake test is activated via IPB "Brake test".

The basic brake test is used for the daily test. The several single system brake tests can be activated via IDU by the driver or maintenance staff.

3.11.1.6.2 *Failure cases*

1. The basic brake test is not starting when activated via "Brake test" push button.
2. The full and single brake tests are not starting when activated via IDU.
3. Part of brake test are not passed successfully but overall result is indicated passed
4. All parts of brake test are passed successfully but overall result is indicated as not passed

3.11.1.6.3 *Failure detection*

1. No failure detection possible.
2. No failure detection possible.
3. No failure detection possible, at the end of brake test summary which part of test has not passed on IDU.
4. No failure detection possible.

3.11.1.6.4 *Failure handling*

1. Driver to start basic brake test via IDU. No effect on train operation.
2. a) Only IDU failure: Driver to start basic brake test via "Brake Test" push button.
   b) TCMS failure automatic brake test: Driver to perform manual brake test as described in operational manual. No impact on train operation.
3. No failure handling possible, wrong indication by TCMS.
4. Driver to acknowledge not passed brake test and to perform manual brake test. No effect on train operation.

3.11.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.142
Description	Brake tests shall be preformed at least every 24 hours or prior entering service.
Safety level	N/A

ID	2F_05.02.Zefiro-Europe.TRS.206
Description	The following test shall be provided:

	– service brake by master controller – parking brake
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.141
Description	The Driver shall be reminded automatically by the control system to perform the applicable manual part of the brake test.
Safety level	SIL 0

ID	2F_05.02.Zefiro-Europe.TRS.11
Description	The following test shall be provided: – brake pipe venting – emergency brake by electrical brake loop – Emergency brake position of DBV handle and master controller – magnet track brake.
Safety level	SIL 2

ID	2F_05.02.Zefiro-Europe.TRS.144
Description	The status of the cut out switches shall be monitored by the diagnostic system and the driver shall be alerted in case of any cut out switch in "cut out" position.
Safety level	SIL 0

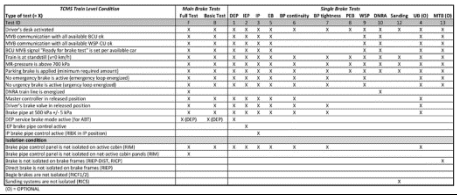
3.11.2.1 *General brake test requirements*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2677	The general process of any brake test shall be the following: – Test selection on IDU or pressing brake test push button – Precondition and device isolation check – Acknowledgement by the driver to start the test (not valid for push button "brake test") – Different test phase execution with partial results – Overall test result acknowledgement by the driver – Overall test result storage	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2462	The driver shall be able to choose on IDU at first level brake test page following main categories: – Basic brake test – Full brake test – Single brake tests The single brake test sub-menu on IDU shall contain following softkeys to start each test individually: – DEP brake test – IEP brake test – IP brake test – Urgency brake test (optional) – Emergency brake test – Brake pipe continuity test – Brake pipe tightness test – WSP test – DNRA test – Passenger's emergency brake test – Sanding test – Magnetic track brake test (option)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2687	When test X is selected (X is the type of test), on IDU a dedicated page shall be shown, with different fields:	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

	– Type of test – Message to driver – Test status – Acknowledgement by driver				
2F_05.02.Zefiro-Europe.CONCEPT.2500	If a test has been selected on the IDU and the preconditions are fulfilled TCMS shall start the selected test sequence after the driver has acknowledged to start the test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.710	There shall be a not-latching illuminated push button "Brake test" at driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2612	The "Brake Test" push button on the driver's desk shall be connected to safe TCMS input. The "Brake Test" push button lamp shall be connected to TCMS output.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2678	The basic brake test shall be started pressing the "Brake Test" push button present on the driver's desk if no other brake test is active and the preconditions are fulfilled.	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1103	The brake test lamp shall have following status: • Flashing slowly = Test status is on going • Flashing fast (for 10s) = Test is aborted or missing preconditions / Test not performed within 24 hours • Steadly illuminated = Test result is passed (without failure) • Not illuminated = Test status is not active or after acknowledgement of test results	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2508	If TCMS during full and basic brake test will not receive the IP feedback signal "Brake test step X passed" from BCU or WSP-CU related to the single brake test function within the specified time TCMS shall abort the actual test step and shall set the message Brake test step x aborted due to missing BCU feedback on IDU. The driver shall have the possibilty to continue with the next brake test step after he has acknowledged via softkey the missing BCU feedback.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2509	If the brake test step x time out is achieved due to missing BCU feedback an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2688	If the driver doesn't acknowledge any TCMS IDU request or doesn't perform the action required (IDU driver action) in 30s a message shall be set on IDU "Test aborted due to time out".	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1640	An event shall be set in case the brake test is aborted due to driver time out.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2469	For any test the **test status** shall change its value as follow: – Ongoing: if it was not active AND if any test is active AND precondition are successfully checked – Passed: if the test status from all the involved local BCU is successfull (result depending from specific test procedure of each test) AND all internal criteria check made by TCMS are true – Failed: if the test status from any involved local BCU/WSP-CU is not successfull AND any internal check performed by TCMS is not true – Aborted: if it was "ongoing" AND (if a timeout abortion occurr OR if a missing precondition abortion occurr OR any abort condition occurred)) – Test end: if the brake test status was passed OR failed AND the test result has been acknowledged by the driver the test result shall be stored as events	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.2477	The overall **test result** shall be shown and acknowledged by the driver for full, basic and single brake tests.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2468	If the test status is *ongoing* a message shall be displayed on the IDU "Brake test X ongoing".	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2478	When local test status signal of any BCU is set to failed OR any internal TCMS check is failed, the test step status shall be set failed. Refer to single brake test step requirements.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2607	When local test status signal of all BCU are set to passed and the internal check performed by TCMS are passed, the test step status shall be set to passed. Refer to single brake test step requirements. The intermediate steps of basic or single brake test steps are not needed to be acknowledged by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2601	The brake test shall be aborted if the IDU screen is changed by the driver (leave the brake test menu).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3887	In case the TCMS IP signal "Train brake test active" is reset during test execution the brake test shall be aborted by BCU.	Derived Requirement	4S_07.-.Braking System		SIL 2

3.11.2.1.1 *Brake Test Precondition/Isolation Condition*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2467	TCMS shall check the related preconditions for each single brake test before the dedicated test can been selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3669	The below precondition shall be valid for each selected brake test to start the test and set the IP signal "Train brake test active" IP signal to BCU, WSP-CU and DBV-GW. 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

<T1 - TCMS Precondition Brake Test>

TCMS Train Level Condition	*Main Brake Tests*		*Single Brake Tests*											
Type of test (= X)	**Full Test**	**Basic Test**	**DEP**	**IEP**	**IP**	**EB**	**BP continuity**	**BP tightness**	**PEB**	**WSP**	**DNRA**	**Sanding**	**UB (O)**	**MTB (O)**
Test ID	F	B	1	2	3	5	6	7	8	9	10	12	4	13
Driver's desk activated	X	X	X	X	X	X	X	X	X	X	X	X	X	X
MVB communication with all available BCU ok	X	X	X	X	X	X				X	X		X	X
MVB communication with all available WSP-CU ok	X	X	X	X	X	X				X	X		X	X
BCU MVB signal "Ready for brake test" is set per available car	X	X	X	X	X	X				X	X		X	X
Train is at standstill (v=0 km/h)	X	X	X	X	X	X	X	X	X	X	X	X	X	X
MR-pressure is above 700 kPa	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Parking brake is applied (minimum required amount)	X	X	X	X	X	X	X	X	X	X	X	X	X	X
No emergency brake is active (emergency loop energized)	X	X	X	X	X	X	X	X		X			X	X
No urgency brake is active (urgency loop energized)	X	X	X	X	X	X	X	X		X			X	X
DNRA train line is energized	X										X			
Master controller is released position	X	X	X	X	X	X	X						X	
Driver's brake valve in released position	X	X	X	X	X	X	X	X					X	
Brake pipe at 500 kPa +/- 5 kPa	X	X	X	X	X	X	X	X					X	
DEP service brake mode active (for ABT)	X (DEP)	X (DEP)	X											
IEP brake pipe control active				X										
IP brake pipe control active (RIBK in IP position)					X									
Isolation condition														
Brake pipe control panel is not isolated on active cabin (RIM)	X	X	X	X	X	X	X	X					X	
Brake pipe control panel is not isolated on not-active cabin panels (RIM)	X													
Brake is not isolated on brake frames (RIEP-DIST, RICP)														X
Direct brake is not isolated on brake frames (RIEP)														
Bogie brakes are not isolated (RICF1/2)														
Sanding systems are not isolated (RICS)												X		

(O) = OPTIONAL

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2470	If the related preconditions are not fullfilled to start "Brake test step X" TCMS shall not start the test sequence.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2686	If any precondition is missing a message to driver "Test not started due to missing precondition" shall be set showing on the IDU the missing preconditions.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2496	During each single brake test step x the above precondition shall be checked permanently if not differently specified in single brake test step x. The brake test shall be completly aborted if any of the preconditions are reset during test execution.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1021	When a brake test is selected AND the precondition check is successfull TCMS shall start the test with defined amount of cars available after the driver's acknowledgement to start the test. For isolated cars, see definition concept 3671, TCMS shall ignore the missing preconditions.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3671	TCMS shall consider the isolation status of each vehicle (TCMS input), if at least one of the following isolation cocks is in isolated position TCMS shall not consider this car for the brake test result: – RICP – RIEP-DIST – RICF 1 AND RICF 2	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3951	In case one of the RICF isoloation cock is in isolated position the car shall be considered during brake test, TCMS shall indicate the isolation condition on bogie level on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3672	TCMS shall consider the isolation status of each DEP brake functionality per vehicle (IP signal), if the isolation cock RIEP is in isolated position TCMS shall not consider this car for the DEP brake test result.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3673	The overall brake test result shall be considered as failed if DEP brake is isolated in more than two cars by RIEP isolation cock and direct service brake mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2695	When all internal brake system preconditions are true the BCU shall set the "Ready for brake test" IP signal to TCMS. – Speed v < 0.5 km/h – Main reservoir pressure available > 700 kPa – Emergency brake not active – Urgency brake not active – Brake pipe pressure 500 kPa +/- 10 kPa – RIEP-DIST, RIEP, RICP and RICF1 and 2 isolation cocks in not isolated position – DNRA output not active In case the preconditions are not set true and brake test start trigger is set by TCMS signal "Train brake test active" the BCU shall not start the brake test.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.3845	When all internal brake system preconditions are true the DBV-GW shall set the "Ready for brake test" IP signal to TCMS. – RIM cock not isolated – IEP brake pipe control active – DBV in release position – Brake pipe pressure 500 kPa +/- 10 kPa	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.666	The BCU shall reset the IP signal "Ready for brake test" (=0) if any preconditions has been reset (not true) during test execution.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2792	On isolated vehicles the IP signal "Ready for brake test" shall not be set. Isolation cock monitored by BCU: – RICP in isolated position – RIEP-DIST in isolated position – RICF1 and 2 in isolated position Isolation cock monitored by DBV-GW: – RIM in isolated position	Derived Requirement	4S_07.-.Braking System		SIL 2

3.11.2.1.1.1 Brake Test Result Acknowledgement

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2605	If the single test status is passed or failed, the single test status shall be displayed on the IDU for each test as test result summary.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2536	If the overall test status is passed or failed it shall be displayed on the IDU and the amount of isolated car/bogie, the driver shall be requested to acknowledge the overall test result and isolated cars by the safe acknowledge button. In case the test is successful with more than one car isolated a warning shall be shown on the IDU that the train has low brake performance.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2484	The overall brake test result summary shall include the information which car/bogie is isolated (not considered in brake test) and which car was passed/not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2486	If the driver acknowledge the summary test result, the brake test is finished and the status shall be set to not active and TCMS IP signal "Train brake test active" shall be reset (=0) to BCU and WSP-CU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2480	If the driver has acknowledged the overall test result and the test result is passed, an event "Test result passed" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2483	If the test result is failed, an event "Test result failed" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2488	If the driver has not acknowledged the test result within the specified time an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.1.1.2 Brake Test Status (24 hours reminder)

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.330	The driver shall be reminded automatically to perform the basic brake tests by the brake test button indication lamp flashing:	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0

	– 24 hours have passed since last test in actual active cab AND – Train at standstill				
2F_05.02.Zefiro-Europe.CONCEPT.703	An event shall be set if the basic brake test has not been performed in last 24h and 10 minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.669	Traction block shall be set as long as any brake test is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.11.2.2 *Full Brake Test*

Following brake test shall be performed during full brake test as specified in single brake test chapter:

- DEP brake test
- IEP brake test
- IP brake test
- Emergency brake test
- Brake pipe continuity test
- Brake pipe tightness test
- Passenger emergency brake test
- WSP test
- DNRA test

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.679	The driver can activate the full brake test if the preconditions are according concept 3669.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.675	The full brake test shall be started if the full brake test button is pressed on the IDU brake test menu. If all pre-conditions are set true and the test is started the IP signal "Train brake test active" shall be set to all BCU, WSP-CU and DBV-GW units.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3877	If the full brake test is selected by IDU, the preconditions are set true and the test is started the holding brake shall be released during the complete test except during WSP test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2301	Following single brake test with all steps shall be performed in a sequence during full brake test: – DEP brake test – IEP brake test – IP brake test – Emergency brake test – Brake pipe continuity test – Brake pipe tightness test – Passenger emergency brake test – WSP test – DNRA test	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2640	At the end of the full brake test a complete test summary of all single brake tests shall be presented on IDU with an overview which single brake test has been passed or failed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.11.2.3 *Basic Brake Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.665	The basic semi-automatic brake test of the systems shall be able to be performed by the driver if the preconditions in concept 3669 are true.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.1088	The basic brake test shall be started if the basic brake test button is pressed on the IDU brake test menu or by brake test push button. If all pre-conditions are set true and the test is started the IP signal "Train brake test active" shall be set to all BCU and DBV-GW units.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3879	If the basic brake test is selected by IDU or by brake test push button, the preconditions are set true and the test is started the holding brake shall be released during the complete test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.678	Following single brake test shall be performed in a sequence during basic brake test, refer to detailed requirements below: – DEP full service brake test – Emergency brake test master controller and safety loop – Emergency brake test driver's brake valve – Indirect brake intermediated position (IEP or IP) – Brake pipe continuity test – - Passenger emergency brake test	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 1: Full DEP service brake application master controller

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3718	TCMS shall present driver text on the IDU for basic brake test step 1: Move master controller in full service brake position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3719	TCMS shall set IP signal "Brake test step 1" to all BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3720	If the BCU receives the IP signal "Brake test step 1" and "Train brake test active", the BCU shall verify following criterias: – Direct brake pre-control pressure (Tpil) according full service brake pressure AND – Cylinder pressure (TCF) according full service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3721	The BCU shall abort basic brake test step 1 if the criterias are not true within 15s in both composition.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3722	TCMS shall verify following criterias for basic brake test step 1: – Master controller grey code in full service brake position AND – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3723	TCMS shall consider the basic brake test step 1 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 1 passed" – TCMS criterias of basic brake test step 1 are fulfiled If one of the condition is not set in 30s the basic brake test step 1 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3724	TCMS shall reset IP signal "Brake test step 1" to all BCUs if the basic brake test step 1 has been passed or not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	TCMS shall continue with basic brake test step 2 if the step has been passed.				

Basic brake test step 2: Emergency brake application master controller

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3726	TCMS shall present driver text on the IDU: Move master controller in emergency position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3727	TCMS shall set IP signal "Brake test step 4" to all BCUs in basic brake test step 2.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3728	If the BCU receives the IP signal "Brake test step 4" and "Train brake test active", the BCU shall verify following criterias: – BCU AND WSP-CU EB loop input reset (= 0) AND – BCU AND WSP-CU UB loop input reset (= 0) AND – Indirect brake pre-control pressure (TDIS) according emergency brake pressure AND – Direct brake pre-control pressure (Tpil) according emergency brake pressure AND – Cylinder pressure (TCF) according emergency brake pressure via RS232 from WSP-CU AND – Brake pipe pressure (TCG) < 100 kPa If all criterias are true the BCU shall set the IP signal "Brake test step 4 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3729	The BCU shall abort basic brake test step 2 if the criterias are not true within 15s valid for single and double composition.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3730	TCMS shall verify following criterias in basic brake test step 2: – Master controller grey code in emergency position AND – {BCU IP signal "EB application pressure switch in emergency position (PAPP)" is set by BCU AND TCMS input EB application valve PAPP in emergency position at least in one DM-car per consist (if isolated by RIVAPP cock local PAPP shall be not considered)} AND – BCU IP signal "Brake pipe pressure (TCG)" < 100 kPa in any DM car AND – BCU and WSP-CU emergency brake loop input feedback reset IP signal (=0) AND – Rble relay TCMS input is reset (= 0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3731	TCMS shall consider the basic brake test step 2 as passed if the following conditions are set within 35s: – BCU set the IP signal "Brake test step 4 passed" – TCMS criterias of basic brake step 2 are fulfiled If one of the condition is not set in 35s the basic brake test step 2 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3732	TCMS shall reset IP signal "Brake test step 4" to all BCUs if the basic brake test step 2 has been passed or not passed. TCMS shall continue with basic brake test step 3 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 3: Emergency brake application driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3734	TCMS shall present driver text on the IDU: Move driver's brake valve in emergency brake position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3735	TCMS shall verify following criterias basic brake test step 3: – BCU and WSP-CU emergency brake loop input feedback reset IP AND – Driver's brake valve supervision relay inputs RDBV1 and RDBV2 are both reset (=0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3736	TCMS shall consider the basic brake test step 3 as passed if the following conditions are set within 30s: – TCMS criterias of basic brake step 3 are fulfiled If one of the condition is not set in 30s the basic brake test step 3 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3737	TCMS shall continue with basic brake test step 4 if the basic brake step 3 has been passed sucessful.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 4: Emergency brake release master controller and DBV

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3739	TCMS shall present driver text on the IDU: First move master controller in release position Second move driver's brake valve in quick release position (after RP released position)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3740	TCMS shall set IP signal "Brake test step 4.1" to all BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3741	If the BCU receives the IP signal "Brake test step 4.1" and "Train brake test active", the BCU shall verify following criterias: – BCU AND WSP-CU EB loop input reset (= 1) AND – BCU AND WSP-CU UB loop input reset (= 1) AND – Cylinder pressure (TCF) according release pressure ≤ 20 kPa via RS232 from WSP-CU AND – Brake pipe pressure (TCG) > 480 kPa If all criterias are true the BCU shall set the IP signal "Brake test step 4.1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3742	The BCU shall abort basic brake test step 4 if the criterias are not true within 30s in single composition. In double composition the time-out value is 40s.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.3743	TCMS shall verify following criterias basic brake test step 4: – Master controller grey code in release position AND – {BCU IP signal "EB application pressure switch in emergency position (PAPP)" is reset by BCU (=0) AND TCMS input EB application valve PAPP in not emergency position at least in one DMcar (if isolated by RIVAPP cock local PAPP shall be not considered)} AND – BCU IP signal "Brake pipe pressure (TCG)" 495 kPa (+/- 15 kPa) in any DM car AND – BCU and WSP-CU emergency brake loop input feedback set IP signal (=1) AND – Rble relay TCMS input is set (= 1) AND – Driver's brake valve supervision relay inputs RDBV1 and RDBV2 are both set (=1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3744	TCMS shall consider the basic brake test step 4 as passed if the following conditions are set within 75s: – BCU set the IP signal "Brake test step 4.1 passed" – TCMS criterias of basic brake test step 4 are fulfiled If one of the condition is not set in 75s the basic brake test step 4 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3745	TCMS shall reset IP signal "Brake test step 4.1" to all BCUs if the basic brake test step 4 has been passed or not passed. TCMS shall continue with basic brake test step 5 if the step has been passed sucessful.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 5: Intermediate indirect service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3747	TCMS shall present driver text on the IDU for basic brake test step 5: Move driver's brake valve in intermediate service brake position 03 to achieve brake pipe pressure of 420 kPa	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3748	TCMS shall set IP signal "Brake test step 5" to all BCUs in basic brake test step 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3749	If the BCU receives the IP signal "Brake test step 5" and "Train brake test active", the BCU shall verify following criterias: – Indirect brake pre-control pressure (TDIS) according intermediate service brake pressure AND – Cylinder pressure (TCF) according intermediate service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3750	The BCU shall abort basic brake test step 5 if the criterias are not true within 15s in both composition.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.3751	TCMS shall verify following criterias for basic brake test step 5: – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 420 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3752	TCMS shall consider the basic brake test step 5 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 5 passed" – TCMS criterias of basic brake step 5 are fulfiled If one of the condition is not set in 30s the basic brake test step 5 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3753	TCMS shall reset IP signal "Brake test step 5" to all BCUs if the IEP service brake test step 1 has been passed or not passed. TCMS shall continue with basic brake test step 6 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 6: Brake pipe continuity venting

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3755	If the preconditions (except brake pipe at 500 kPa) are fullfiled TCMS shall energise the EVAPP relay in the last rear cab of the train consist to vent the brake pipe via EVAPP valve.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3756	On the same time during basic brake test step 6 as the brake pipe continuity test command to EVAPP relay is set TCMS shall energise the neutral valve EVNEU to isolate the brake pipe.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3758	TCMS shall compare the time to decrease the brake pipe pressure starting at EVNEU command. As long the test is active for a maximum of 40s: if the pressure in the brake pipe + 22kPa detected by the BCU in the enabled cab is greater than the pressure in the brake pipe detected by the BCU in the rear cab. AND the pressure switch PAPP on the enabled cab. switches 1s after the other in the rear cab. (in the case of RIVAPP not closed) AND the emergency loop is not energized, test is successful. As long the test is active for a maximum of 40s: if the pressure in the brake pipe + 22kPa detected by the BCU in the enabled cab. is less than the pressure in the brake pipe detected by the BCU in the rear cab. OR the pressure switch PAPP on the enabled cab. switches 1s before the other on the rear cab. (in case of RIVAPP not closed) OR the emergency loop is energized, test fails.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.3770	If the brake pipe pressure is < 100 kPa and time criteria in concept 3758 is passed TCMS shall reset command to EVAPP relay in last DM car (0) to maintain the BP pressure at this level. The command to neutral valve to isolate the brake pipe shall be reset and brake pipe pressure shall be able to be released.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3771	On the same time as the command to EVAPP relay is reset (0) TCMS shall de-energise the neutral valve EVNEU to isolate the brake pipe.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3769	TCMS shall continue with basic brake test step 7 if the basic brake step 6 has been passed sucessful. If the condition is not set in 20s TCMS shall consider brake test step 6 as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 7: Release indirect service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3762	TCMS shall present driver text on the IDU for basic brake test step 7: Move driver's brake valve in brake release position RP	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3763	TCMS shall set IP signal "Brake test step 5.2" to all BCUs in basic brake test step 7.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3764	If the BCU receives the IP signal "Brake test step 5.2" and "Train brake test active", the BCU shall verify following criterias: – Cylinder pressure (TCF) according release pressure 0 ≤ 20 kPa via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5.2 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3765	The BCU shall abort basic brake test step 7 if the criterias are not true within 30s in single and 40s in double compositon.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3766	TCMS shall verify following criterias in basic brake test step 7: – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3767	TCMS shall consider the basic brake test step 7 as passed if the following conditions are set within 45s: – BCU set the IP signal "Brake test step 5.2 passed" – TCMS criterias of basic brake step 7 are fulfiled If one of the condition is not set in 45s the basic brake test step 7 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3768	TCMS shall reset IP signal "Brake test step 5.2" to all BCUs if the basic brake test step 7 has been passed or not passed. If the basic brake step 7 is passed TCMS shall continue if test step 8.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Basic brake test step 8: PEB alarm intervention

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3781	TCMS shall present driver text on the IDU: Passenger emergency brake is going to be tested	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.3782	To start basic brake test step 8, TCMS shall open the RTPE test relay contacts on the last rear cab to de-energise the PEB train line 1 and 2 AND TCMS shall not set PEB train running mode to test UB application by TCMS during brake test active	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3783	TCMS shall verify following criterias to pass basic brake test step 8: – TCMS verifies that at least one PEB train line inputs is not set (=0) AND – TCMS shall verifiy the TCMS emergency brake relays RTCMS 1/2 supervision inputs are not set (=0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3784	TCMS shall consider the basic brake test step 8 as passed if the following conditions are set: – TCMS criterias of basic brake test step 8 are fulfiled If one of the condition is not set in 5s the basic brake test step 8 shall be considered as not passed. If the basic brake step 8 is passed TCMS shall continue if test step 9.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

Basic brake test step 9: Override function

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3785	TCMS shall present driver text on the IDU during basic brake test step 9: Passengers emergency brake override button to be pressed if flashing	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3786	If the PEB override button has been pressed in basic brake test step 9 TCMS shall reset RTPE test relay output (=0) AND close contacts in the last rear cab to energise both PEB train lines.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3787	TCMS shall verify following criterias during basic brake test step 9: – PEB override push button has been pressed TCMS input is set (=1) AND – TCMS verifies that at least one PEB train line inputs is set (=1) AND – TCMS shall verifiy the TCMS emergency brake relays RTCMS 1/2 supervision inputs are set (=1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3788	TCMS shall consider the basic brake test step 9 as passed if the following conditions are set within 15s: – TCMS criterias of basic brake test step 9 are fulfiled If one of the condition is not set in 15s the basic brake test step 9 shall be considered as not passed. TCMS shall finish basic brake test if the step 9 has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3761	An event shall be set if the basic brake test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.680	At the end of the basic brake test a complete test summary of all single brake tests shall be presented on IDU with an overview which brake test has been passed or failed.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.11.2.4 *Single Brake Tests*

3.11.2.4.1 *DEP Service Brake Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2714	If the IDU direct service brake test button has been pressed on the IDU brake test menu AND the IP signal "Ready for brake test" is set by the BCU AND the preconditions are fulfiled, TCMS shall send the IP signal "Train brake test active" to BCU and WSP-CU AND TCMS shall start the test sequence. (The precodition to start the direct brake test as defined concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3880	If the DEP single brake test is selected by IDU, the preconditions are set true and the test is started the holding brake shall be released during the complete DEP test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Direct service brake test step 1: Full service brake application master controller

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3611	TCMS shall present driver text on the IDU for DEP brake test step 1: Move master controller in full service brake position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2513	TCMS shall set IP signal "Brake test step 1" to all BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2514	If the BCU receives the IP signal "Brake test step 1" and "Train brake test active", the BCU shall verify following criterias: – Direct brake pre-control pressure (Tpil) according full service brake pressure AND – Cylinder pressure (TCF) according full service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2735	The BCU shall abort DEP service brake test step 1 if the criterias are not true within 30s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2715	TCMS shall verify following criterias for DEP Step 1: – Master controller grey code in full service brake position AND – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2525	TCMS shall consider the direct service brake test step 1 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 1 passed" – TCMS criterias of DEP step 1 are fulfiled If one of the condition is not set in 30s the direct brake test step 1 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2521	TCMS shall reset IP signal "Brake test step 1" to all BCUs if the test step 1 has been passed or not passed. TCMS shall continue with direct brake test step 2 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Direct service brake test step 2: Full service brake release master controller

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2524	TCMS shall present driver text on the IDU for DEP brake test step 2: Move master controller to released position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2528	TCMS shall set IP signal "Brake test step 1.1" to all BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2531	If the BCU receives the IP signal "Brake test step 1.1" and "Train brake test active", the BCU shall verify following criterias: – Cylinder pressure (TCF) according release pressure ≤ 20 kPa via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 1.1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2515	The BCU shall abort DEP service brake test step 2 if the criterias are not true within 30s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2716	TCMS shall verify following criterias for direct service brake test step 2: – Master controller grey code in release position AND – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2529	TCMS shall consider the direct service brake test step 2 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 1.1 passed" – TCMS criterias of step 2 are fulfiled If one of the condition is not set in 30s the direct brake test step 2 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2532	TCMS shall reset IP signal "Brake test step 1.1" to all BCUs if the test step 2 has been passed or not passed. TCMS shall finish direct brake test if the step 2 has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2534	An event shall be set if the DEP brake test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.2 *Indirect IEP Brake Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2575	If the IDU indirect - IEP service brake test button has been pressed on the IDU brake test menu AND the IP signal "Ready for brake test" is set by the BCU AND the preconditions are fulfiled, TCMS shall send the IP signal "Train brake test active" to BCU, WSP-CU and DBV-GW AND TCMS shall start the test sequence. (The precodition to start the IEP indirect brake test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3881	If the IEP single brake test is selected by IDU, the preconditions are set true and the test is started the holding brake shall be released during the complete IEP test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Indirect - IEP service brake test step 1: Intermediate service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2577	TCMS shall present driver text on the IDU for IEP brake test step 1: Move driver's brake valve in order to achieve a brake pipe pressure value of 390 kPa	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2578	TCMS shall set IP signal "Brake test step 5" to all BCUs in IEP brake test step 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2576	If the BCU receives the IP signal "Brake test step 5" and "Train brake test active", the BCU shall verify following criterias: – Indirect brake pre-control pressure (TDIS) according full service brake pressure AND – Cylinder pressure (TCF) according intermediate service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2579	The BCU shall abort IEP service brake test step 1 if the criterias are not true within 30s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2580	TCMS shall verify following criterias for IEP service brake test step 1: – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 420 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2621	TCMS shall consider the IEP service brake test step 1 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 5 passed" – TCMS criterias of IEP service brake step 1 are fulfiled If one of the condition is not set in 30s the IEP brake test step 1 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2622	TCMS shall reset IP signal "Brake test step 5" to all BCUs if the IEP service brake test step 1 has been passed or not passed. TCMS shall continue with IEP brake test step 2 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Indirect - IEP service brake test step 2: Full service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3676	TCMS shall present driver text on the IDU for IEP brake test step 2: Move driver's brake valve in full service brake position MB	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3677	TCMS shall set IP signal "Brake test step 5.1" to all BCUs in IEP brake test step 2.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3678	If the BCU receives the IP signal "Brake test step 5.1" and "Train brake test active", the BCU shall verify following criterias: – Indirect brake pre-control pressure (TDIS) according full service brake pressure AND – Cylinder pressure (TCF) according full service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5.1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3679	The BCU shall abort IEP service brake test step 2 if the criterias are not true within 30s.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.3680	TCMS shall verify following criterias IEP service brake step 2: – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 330 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3681	TCMS shall consider the IEP service brake test step 2 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 5.1 passed" – TCMS criterias of IEP service brake step 2 are fulfiled If one of the condition is not set in 30s the IEP brake test step 1 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3682	TCMS shall reset IP signal "Brake test step 5.1" to all BCUs if the IEP service brake test step 2 has been passed or not passed. TCMS shall continue with IEP brake test step 3 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Indirect - IEP service brake test step 3: Release service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2595	TCMS shall present driver text on the IDU for IEP brake test step 3: Move driver's brake valve in brake release position RP	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2581	TCMS shall set IP signal "Brake test step 5.2" to all BCUs in IEP brake test step 3.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2583	If the BCU receives the IP signal "Brake test step 5.2" and "Train brake test active", the BCU shall verify following criterias: – Cylinder pressure (TCF) according release pressure 0 ≤ 20 kPa via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5.2 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2719	The BCU shall abort IEP service brake test step 3 if the criterias are not true within 45s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2587	TCMS shall verify following criterias in IEP brake test step 3: – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2628	TCMS shall consider the IEP service brake test step 3 as passed if the following conditions are set within 45s: – BCU set the IP signal "Brake test step 5.2 passed" – TCMS criterias of IEP service brake step 3 are fulfiled If one of the condition is not set in 45s the IEP brake test step 3 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2630	TCMS shall reset IP signal "Brake test step 5.2" to all BCUs if the IEP service brake test step 3 has been passed or not passed. TCMS shall finish IEP brake test if the step 3 has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2585	An event shall be set if the IEP brake test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.3 *Indirect IP Brake Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2721	If the IDU indirect - IP service brake test button has been pressed on the IDU brake test menu AND the IP signal "Ready for brake test" is set by the BCU AND the preconditions are fulfiled, TCMS shall send the IP signal "Train brake test active" to BCU and WSP-CU AND TCMS shall start the test sequence. (The precodition to start the IP indirect brake test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3882	If the IP single brake test is selected by IDU, the preconditions are set true and the test is started the holding brake shall be released during the complete IP test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Indirect - IP service brake test step 1: Intermediate service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.697	TCMS shall present driver text on the IDU: Move driver's brake valve in intermediate service brake position 03 to achieve brake pipe pressure of 420 kPa	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2535	TCMS shall set IP signal "Brake test step 5" to all BCUs in IP brake test step 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3683	If the BCU receives the IP signal "Brake test step 5" and "Train brake test active", the BCU shall verify following criterias: – Indirect brake pre-control pressure (TDIS) according full service brake pressure AND – Cylinder pressure (TCF) according intermediate service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3684	The BCU shall abort IEP service brake test step 1 if the criterias are not true within 30s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3685	TCMS shall verify following criterias IP brake test step 1: – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 420 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2447	TCMS shall consider the IP service brake test step 1 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 5 passed" – TCMS criterias of IP service brake step 1 are fulfiled If one of the condition is not set in 30s the IEP brake test step 1 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2594	TCMS shall reset IP signal "Brake test step 5" to all BCUs if the IP service brake test step 1 has been passed or not passed. TCMS shall continue with direct brake test step 2 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Indirect - IP service brake test step 2: Full service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2586	TCMS shall present driver text on the IDU for IP brake test step 2: Move driver's brake valve in full service brake position MB	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2591	TCMS shall set IP signal "Brake test step 5.1" to all BCUs in IP brake test step 2.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2633	If the BCU receives the IP signal "Brake test step 5.1" and "Train brake test active", the BCU shall verify following criterias: – Indirect brake pre-control pressure (TDIS) according full service brake pressure AND – Cylinder pressure (TCF) according full service brake pressure via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5.1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2446	The BCU shall abort IEP service brake test step 2 if the criterias are not true within 30s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2589	TCMS shall verify following criterias IP brake step 2: – BCU IP signal "Brake reservoir pressure available" is set AND – BCU IP signal "Brake pipe pressure (TCG)" 330 kPa (+/- 10 kPa) in any DM car	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2592	TCMS shall consider the IP service brake test step 2 as passed if the following conditions are set within 30s: – BCU set the IP signal "Brake test step 5.1 passed" – TCMS criterias of IP service brake step 2 are fulfiled If one of the condition is not set in 30s the IP brake test step 2 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2593	TCMS shall reset IP signal "Brake test step 5.1" to all BCUs if the IP service brake test step 2 has been passed or not passed. TCMS shall continue with IP brake test step 3 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Indirect - IP service brake test step 3: Release service brake driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.698	TCMS shall present driver text on the IDU for IP brake test step 3: Move driver's brake valve in brake release position RP	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2448	TCMS shall set IP signal "Brake test step 5.2" to all BCUs in IP brake test step 3.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2733	If the BCU receives the IP signal "Brake test step 5.2" and "Train brake test active", the BCU shall verify following criterias: – Cylinder pressure (TCF) according release pressure 0 ≤ 20 kPa via RS232 from WSP-CU If all criterias are true the BCU shall set the IP signal "Brake test step 5.2 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2734	The BCU shall abort IP service brake test step 3 if the criterias are not true within 45s.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.700	TCMS shall verify following criterias in IP brake test step 3:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

	– BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car				
2F_05.02.Zefiro-Europe.CONCEPT.701	TCMS shall consider the IP service brake test step 3 as passed if the following conditions are set within 45s: – BCU set the IP signal "Brake test step 5.2 passed" – TCMS criterias of IP service brake step 3 are fulfiled If one of the condition is not set in 45s the IP brake test step 3 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2449	TCMS shall reset IP signal "Brake test step 5.2" to all BCUs if the IP service brake test step 3 has been passed or not passed. TCMS shall finish IP brake test if the step 3 has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.2461	An event shall be set if the IP brake test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.4 *Emergency Brake Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3525	If the IDU emergency brake test button has been pressed on the IDU brake test menu AND the IP signal "Ready for brake test" is set by the BCU AND the preconditions are fulfiled, TCMS shall send the IP signal "Train brake test active" to BCU and WSP-CU AND TCMS shall start the test sequence. (The precodition to start the emergency brake test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3884	If the emergency brake single brake test is selected by IDU, the preconditions are set true and the test is started the holding brake shall be released during the complete emergency brake test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Emergency brake test step 1: Emergency brake application master controller

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2788	TCMS shall present driver text on the IDU: Move master controller in emergency position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2789	TCMS shall set IP signal "Brake test step 4" to all BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2790	If the BCU receives the IP signal "Brake test step 4" and "Train brake test active", the BCU shall verify following criterias: – BCU AND WSP-CU EB loop input reset (= 0) AND – BCU AND WSP-CU UB loop input reset (= 0) AND – Indirect brake pre-control pressure (TDIS) according emergency brake pressure AND – Direct brake pre-control pressure (Tpil) according emergency brake pressure AND – Cylinder pressure (TCF) according emergency brake pressure via RS232 from WSP-CU AND – Brake pipe pressure (TCG) < 100 kPa If all criterias are true the BCU shall set the IP signal "Brake test step 4 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2791	The BCU shall abort emergency brake test step 1 if the criterias are not true within 40s.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.2793	TCMS shall verify following criterias in EB test step 1: – Master controller grey code in emergency position AND – {BCU IP signal "EB application pressure switch in emergency position (PAPP)" is set by BCU AND TCMS input EB application valve PAPP in emergency position at least in one DM-car per consist (if isolated by RIVAPP cock local PAPP shall be not considered)} AND – BCU IP signal "Brake pipe pressure (TCG)" < 100 kPa in any DM car AND – BCU and WSP-CU emergency brake loop input feedback reset IP signal (=0) AND – Rble relay TCMS input is reset (= 0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2801	TCMS shall consider the emergency brake test step 1 as passed if the following conditions are set within 35s: – BCU set the IP signal "Brake test step 4 passed" – TCMS criterias of step 1 are fulfiled If one of the condition is not set in 35s the emergency brake test step 1 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2802	TCMS shall reset IP signal "Brake test step 4" to all BCUs if the test step 1 has been passed or not passed. TCMS shall continue with emergency brake test step 2 if the step has been passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Emergency brake test step 2: Emergency brake release master controller

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2795	TCMS shall present driver text on the IDU: Move master controller in released position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2796	TCMS shall set IP signal "Brake test step 4.1" to all BCUs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2844	If the BCU receives the IP signal "Brake test step 4.1" and "Train brake test active", the BCU shall verify following criterias: – BCU AND WSP-CU EB loop input reset (= 1) AND – BCU AND WSP-CU UB loop input reset (= 1) AND – Cylinder pressure (TCF) according release pressure ≤ 20 kPa via RS232 from WSP-CU AND – Brake pipe pressure (TCG) > 480 kPa If all criterias are true the BCU shall set the IP signal "Brake test step 4.1 passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2797	The BCU shall abort emergency brake test step 2 if the criterias are not true within 45s.	Derived Requirement	4S_07.-.Braking System		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.2805	TCMS shall verify following criterias EB test step 2: – Master controller grey code in release position AND – {BCU IP signal "EB application pressure switch in emergency position (PAPP)" is reset by BCU (=0) AND TCMS input EB application valve PAPP in not emergency position at least in one DMcar (if isolated by RIVAPP cock local PAPP shall be not considered)} AND – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car AND – BCU and WSP-CU emergency brake loop input feedback set IP signal (=1) AND – Rble relay TCMS input is set (= 1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2807	TCMS shall consider the emergency brake test step 2 as passed if the following conditions are set within 75s: – BCU set the IP signal "Brake test step 4.1 passed" – TCMS criterias of step 2 are fulfiled If one of the condition is not set in 75s the emergency brake test step 2 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2808	TCMS shall reset IP signal "Brake test step 4.1" to all BCUs if the test step 2 has been passed or not passed. TCMS shall continue with emergency brake test step 3 if the step has been passed sucessful.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Emergency brake test step 3: Emergency brake application driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2809	TCMS shall present driver text on the IDU: Move driver's brake valve in emergency brake position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2810	TCMS shall verify following criterias EB test step 3: – {BCU IP signal "EB application pressure switch in emergency position (PAPP)" is set by BCU AND TCMS input EB application valve PAPP in emergency position at least in one DMcar (if isolated by RIVAPP cock local PAPP shall be not considered)} AND – BCU IP signal "Brake pipe pressure (TCG)" < 100 kPa in any DM car AND – BCU and WSP-CU emergency brake loop input feedback reset IP signal (=0) AND – Rble relay TCMS input is reset (= 0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2815	TCMS shall consider the emergency brake test step 3 as passed if the following conditions are set within 35s: – TCMS criterias of step 3 are fulfiled If one of the condition is not set in 35s the emergency brake test step 3 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2816	TCMS shall continue with emergency brake test step 4 if the EB step 3 has been passed sucessful.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Emergency brake test step 4: Emergency brake release driver's brake valve

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2818	TCMS shall present driver text on the IDU: Move driver's brake valve in filling position to fill up brake pipe, afterwards to release position	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2819	TCMS shall verify following criterias EB step 4: – {BCU IP signal "EB application pressure switch in emergency position (PAPP)" is reset by BCU (=0) AND TCMS input EB application valve PAPP in not emergency position at least in one DMcar (if isolated by RIVAPP cock local PAPP shall be not considered)} AND – BCU IP signal "Brake pipe pressure (TCG)" 500 kPa (+/- 10 kPa) in any DM car AND – BCU and WSP-CU emergency brake loop input feedback set IP signal (=1) AND – Rble relay TCMS input is set (= 1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2822	TCMS shall consider the emergency brake test step 4 as passed if the following conditions are set within 75s: – TCMS criterias of step 4 are fulfiled If one of the condition is not set in 75s the emergency brake test step 4 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2823	TCMS shall finish single emergency brake test if the step 4 has been passed /not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3703	An event shall be set if the emergency brake test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.5 *BCU pressure Transducer test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR52	When the emergency brake is applied and the pressure is stabilized, the BCU checks the TCFs: IF |TCF1 - TCF2| > |0,15| THEN TCF1 OR TCF2 of bogie X is setting out of range		4S_07.-.Braking System		2
2F_05.02.Zefiro-Europe.CONCEPT.HR53	If BCU detects TCF fault an event is created:" "Transducer bogie x setting out of range". Class: minor fault, directed to maintenance. Ethernet safe SIL 2 list of signals. TC_BI_#BCU_S_Trans_outranb_X This failure shall NOT prevent a brake test start or block/fail a brake test Event is N/A, signal is SIL 2 because it is used for dragging brake SIL 2 diagnostic		4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR109	IDU shall record the Transducer bogie x setting out of range failure and define it as: Class: minor, Visible to: maintenance only Record: date and time, MRP pressure, BP pressure, Speed, brake demand.			IDU	N/A

3.11.2.4.6 *BCU pressure switch test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_05.02.Zefiro-Europe.CONCEPT.HR54	If TCF1x AND TCF2x are not faulty AND are NOT out of range IF on bogie x TCF1x < 0,2 bar AND TCF2x < 0,2 bar AND PCF1x = 1 (braking) for 15 sec THEN PCF1 bogie x is Faulty – type1		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR55	If TCF1x AND TCF2x are not faulty AND are NOT out of range IF TCF1x < 0,2 bar AND TCF2x < 0,2 bar AND PCF1x = 0 (NOT braking) for 6 sec THEN PCF1 bogie x is Faulty – type 1 shall be reset		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR56	IF PCF1 bogie x is faulty an event shall be created Pressure switch bogie 1 faulty – type 1 Class: minor Visible to: maintenance only Ethernet safe SIL 2 list of signals. Event is N/A, signal is SIL 2 because it is used for dragging brake SIL 2 diagnostic		4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR57	If TCF1x AND TCF2x are not faulty AND are NOT out of range IF TCF1x < 0,2 bar AND TCF2x < 0,2 bar PCF2x = 1 (braking) for 15 sec THEN PCF2 bogie x is Faulty – type1		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR58	If TCF1x AND TCF2x are not faulty AND are NOT out of range IF TCF1x < 0,2 bar AND TCF2x < 0,2 bar AND PCF2x = 0 (NOT braking) for 6 sec THEN PCF2 bogie x is Faulty – type1 shall be reset		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR59	IF PCF2 bogie x is faulty type 1 an event shall be created Pressure switch bogie 1 faulty – type 1 Class: minor Visible to: maintenance only Ethernet safe SIL 2 list of signals. TC_BI_#BCU_S_PCFbtcheckb_X (port safe) Event is N/A, signal is SIL 2 because it is used for dragging brake SIL 2 diagnostic		4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR60	If TCF1x AND TCF2x are not faulty AND are NOT out of range AND IF TCF1x > 0,8 bar AND TCF2x > 0,8 bar AND PCF1x = 0 (NOT braking) for 6 sec THEN PCF1 bogie x is Faulty – type 2		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR61	If TCF1x AND TCF2x are not faulty AND are NOT out of range AND IF TCF1x > 0,8 bar AND TCF2x > 0,8 bar AND PCF1x = 1 (braking) for 6 sec THEN PCF1 bogie x is Faulty – type 2 shall be reset		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR62	IF PCF1 bogie x is faulty type 2 an event shall be created Pressure switch bogie 1 faulty – type 2 Class: minor Visible to: maintenance only Ethernet safe SIL 2 list of signals. TC_BI_#BCU_S_PCFbtcheckb_X (port safe) TC_BI_#BCU_S_PCFbtcheckt2b_X (non safe) Event is N/A, signal is SIL 2 because it is used for dragging brake SIL 2 diagnostic		4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_05.02.Zefiro-Europe.CONCEPT.HR63	If TCF1x AND TCF2x are not faulty AND are NOT out of range AND IF TCF1x > 0,8 bar AND TCF2x > 0,8 bar PCF2x = 0 (NOT braking) for 6 sec THEN PCF2 bogie x is Faulty – type 2		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR64	If TCF1x AND TCF2x are not faulty AND are NOT out of range AND IF TCF1x > 0,8 bar AND TCF2x > 0,8 bar AND PCF2x = 1 (braking) for 6 sec THEN PCF2 bogie x is Faulty – type 2 shall be reset		4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.HR65	IF PCF2 bogie x is faulty type 2 an event shall be created Pressure switch bogie 1 faulty – type 2 Class: minor Visible to: maintenance only Ethernet safe SIL 2 list of signals. Event is N/A, signal is SIL 2 because it is used for dragging brake SIL 2 diagnostic		4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.7 *Brake Pipe Continuity Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3390	If the IDU brake pipe continuity test button has been pressed on the IDU brake test menu TCMS shall start the brake pipe continuity test if the preconditions are fulfilled. (The precodition to start the brake pipe continuity test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Brake pipe continuity test step 1: Venting

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3391	See concept 3755 and 3756	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3688	see concept 3758	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

2F_05.02.Zefiro-Europe.CONCEPT.2872	TCMS shall compare the time to decrease the brake pipe pressure starting at EVNEU command to BP pressure = 100 kPa, if the time x < 15s the test step 1 "Venting" shall be succesful.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

Brake pipe continuity test step 2: Releasing

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2873	If the brake pipe pressure is < 100 kPa and time criteria in concept 2872 is passed TCMS shall reset command to EVAPP relay in last DM car (0) to maintain the BP pressure at current level.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3895	When the brake pipe continuity test step 2 command to EVAPP relay is reset TCMS shall reset the command to neutral valve EVNEU to isolate the brake pipe.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2876	TCMS shall verify the brake pipe pressure in the front DM car with active cab derived from the TCG brake pipe sensor from BCU. TCMS shall consider the BP continuity test as passed if the brake pipe pressure has reached 500 kPa +/- 15 kPa starting to count at the reset of EVNEU valve command in < 20s.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3709	An event shall be set if the brake pipe continuity test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.8 *Brake Pipe Tightness Test*

ID	2F_05.02.Zefiro-Europe.TRS.173
Description	A leak indicator shall detect and report leaks on the brake control pipe. The leakage shall be reported to the driver via the diagnostics system. A luminous indication and an acoustic signal shall indicate the leakage if the cab is active and a driving direction is selected.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3439	If the IDU brake pipe tightness test button has been pressed on the IDU brake test menu AND the precondition to start the test are fulfiled TCMS shall start the BP tightness test sequence. (The precodition to start the BP tightness test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Brake pipe tightness test step 1: Isolation

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.213	If the preconditions are fulfiled TCMS shall command the isolation/neutral valve TCMS output (1) to stop the refill of the brake pipe in the active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.477	TCMS shall measure the leakage of the brake pipe after the command to the neutral valve is active, if the brake pipe pressure TCG received by BCU in DM cars is going down less < 10 kPa within 60 seconds the brake tightness test step 1 shall be considered as passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.797	If the brake pipe leakage test step 1 is not passed as defined in concept 477 TCMS shall abort the test and reset the command to isolation/neutral valve (0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

Brake pipe tightness test step 2: Refill active

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2880	If the brake pipe tightness test is considered as passed TCMS shall reset the command to isolation/neutral valve (0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2881	TCMS shall verify the isolation and neutral valve is reset to refill position. TCMS shall consider the BP tightness test step 2 as passed if both valves are reset, feedback by mechanical switch: – EVISO command reset SW_ISO = 1 – EVNEU command reset SW_NEU = 1	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3440	TCMS shall consider the brake pipe tightness test as passed if step 1 and step 2 are passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3710	An event shall be set if the brake pipe tightness test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.9 *Passenger Emergency Brake Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3636	If the IDU passenger's emergency brake test button has been pressed on the IDU brake test menu AND the IP signal "Ready for brake test" is set by the BCU AND the preconditions are fulfiled, TCMS shall start the PEB test sequence. (The precodition to start the PEB test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Passenger emergency brake step 1: PEB alarm intervention

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.695	TCMS shall present driver text on the IDU: "Passenger emergency brake is going to be tested".	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3638	To start PEB test step 1, TCMS shall open the RTPE test relay contacts on the last rear cab to de-energise the PEB train line 1 and 2 AND TCMS shall not set PEB train running mode to test UB application by TCMS during brake test active	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3637	TCMS shall verify following criterias to pass PEB test step 1: – TCMS verifies that at least one PEB train line inputs is not set (=0) AND – TCMS shall verifiy the TCMS emergency brake relays RTCMS 1/2 supervision inputs are not set (=0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2424	TCMS shall consider the passenger emergency brake test step 1 as passed if the following conditions are set: – TCMS criterias of step 1 are fulfiled If one of the condition is not set in 5s the PEB test step 1 shall be considered as not passed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

Passenger emergency brake step 2: Override function

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.696	TCMS shall present driver text on the IDU during PEB test step 2: Passengers emergency brake override button to be pressed if flashing	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3642	If the PEB override button has been pressed TCMS shall reset RTPE test relay output (=0) and close contacts in the last rear cab to energise both PEB train lines.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3641	TCMS shall verify following criterias during PEB test step 2: – PEB override push button has been pressed TCMS input is set (=1) AND – TCMS verifies that at least one PEB train line inputs is set (=1) AND – TCMS shall verifiy the TCMS emergency brake relays RTCMS 1/2 supervision inputs are set (=1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2425	TCMS shall consider the PEB brake test step 2 as passed if the following conditions are set within 15s: – TCMS criterias of step 2 are fulfiled If one of the condition is not set in 15s the PEB brake test step 2 shall be considered as not passed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3711	An event shall be set if the passenger emergency brake test has been passed successful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.10 *WSP Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2300	If WSP test precondition are true TCMS shall start the WSP test sequence: – ED brake interface bit 1 WSP-CU – ED brake interface bit 2 WSP-CU – ED brake interface bit 1 BCU – ED brake interface bit 2 BCU – WSP internal test (The precodition to start the WSP test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Internal WSP test of WSP-CU (and BCU):

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1882	On the IDU there shall be a possibilty to start the WSP test in each car separatley for maintenance purpose.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1881	If WSP test button is pressed on IDU brake test menu TCMS shall set the IP signal "Train brake test active" and "WSP test active" to the WSP-CU and BCU. The signal shall be reset when the IP signal "WSP test passed" OR "WSP test not passed" is received from both units WSP-CU and BCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3509	The complete WSP test shall be failed if following conditions are not true: – no response from either the propulsion control units for communication test OR – no response from the WSP-CU and BCU for internal test is present for t > 180s OR	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	– the WSP-CU AND BCU set IP signal "WSP test not passed" OR – the precondition to start the test are reset				
2F_05.02.Zefiro-Europe.CONCEPT.1879	When the WSP-CU receives the "Train brake test active" and "WSP test active" IP signal and the precondition is true a complete internal WSP test including the control of dump valves and related cylinder pressure venting times shall be perfromed.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3809	When the BCU receives the "Train brake test active" and "WSP test active" IP signal and the precondition is true a complete internal WSP test including the control of dump valves and related cylinder pressure venting times shall be perfromed in the redundant unit.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3500	The WSP-CU and BCU shall sent IP signal "WSP test passed" to TCMS after the test is completed succesfully. If the test has failed the WSP-CU or BCU shall set IP signal "WSP test not passed" to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2

Test of WSP ED brake interface ED bit 1:

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1885	On the same time of the internal WSP test, TCMS shall set IP signal “WSP test bit1 ED” to all WSP-CU (BCU) to activate the HW output "WSP_Bit1 ED axle 1...4" to the motor converter on axle basis.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3502	As long the WSP-CU receives the IP signal "WSP test bit1 ED" from TCMS during the WSP test active status, the WSP-CU shall set the hardware output "WSP test bit1 ED axle 1...4" for all axles.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3503	TCMS shall verify the IP signal "WSP reduction signal Hold 1...4" from the motor converters 1...4 in all cars = 1, the test shall proceed automatically to test the ED bit 2. If after t = 3 seconds from when the “WSP test bit1 ED” command is set (= 1) not all "WSP reduction signal Hold 1...4" status from the motor converters are set (= 1) the test shall be aborted and failed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Test of WSP ED brake interface ED bit 2:

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1888	TCMS shall set IP signal “WSP test bit2 ED” to all WSP-CU (BCU) to activate the HW output "WSP_Bit2 ED" to the motor converter on axle basis.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3506	As long the WSP-CU receives the IP signal "WSP test bit2 ED" from TCMS during the WSP test active status, the WSP-CU shall set the hardware output "WSP test bit2 ED axle 1...4" for all axles.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1889	When the status "WSP reduction signal Vent 1...4" of motor converters 1...4 in all car = 1, there shall be indication on the IDU "WSP communication test has passed". If after t = 3 seconds from when the "WSP test bit2 ED" command is set (= 1) not all IP signal "WSP reduction signal Vent 1...4"" status from the motor converters are set (= 1) the test shall be aborted and failed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_05.02.Zefiro-Europe.CONCEPT.3697	The same WSP interface ED brake test sequence shall be done for the redundant BCU after the WSP-CU interface test is finished.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1886	The WSP communication test shall be considered as passed if both ED bit1 and ED bit2 tests has been successful for both WSP-CU and BCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1892	The complete WSP test shall be considered as passed when the internal WSP test in the brake control units are successfull by IP singal "WSP test passed" from WSP-CU OR BCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.1890	If the WSP test is passed an event shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1891	If the WSP test has failed an event shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.11 *DNRA Test*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3510	If the IDU DNRA test button has been pressed on the IDU brake test menu AND the IP signal "Ready for brake test" is set by the BCU AND the preconditions are fulfiled, TCMS shall send the IP signal "Train brake test active" to BCU and WSP-CU AND TCMS shall start the DNRA test sequence. (The precodition to start the DNRA test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Step 1: BCU/WSP-CU DNRA test

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3514	At DNRA test TCMS shall set the IP signal "Brake test DNRA" to all BCUs in sequence and afterwards to all WSP-CU in sequence.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3515	If the BCU receives the IP signal "Brake test DNRA" and "Train brake test active", the BCU shall energise the DNRA output to RSS1 relay AND IP signal "DNRA request active" as long the TCMS IP signal "Brake test DNRA" is active. If the WSP-CU receives the IP signal "Brake test DNRA" and "Train brake test active", the WSP-CU shall energise the RSS2 relay AND IP signal "DNRA request active" as long the TCMS IP signal "Brake test DNRA" is active.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3516	TCMS shall verify if the DNRA train line relays RDNRA 1 and RDNRA 2 supervision inputs are both reset (= 0) for each single sequence step from BCU OR WSP-CU on car level.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.3518	The DNRA test is considered as passed if the relay supervision TCMS inputs RDNRA 1 and RDNRA 2 are all reset for each step of the test sequence. The test step 1 is not passed if not all BCU and WSP-CU units correctly opened the DNRA train line and TCMS supervision input was not reset accordingly.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3519	If the test not passed the DNRA test shall be aborted and TCMS shall indicate in which car the test has failed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Step 2: BMS/DNRA test

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3524	If the DNRA test has been passed succesful an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.2.4.12 *Sanding Test*

ID	2F_05.02.Zefiro-Europe.TRS.193
Description	The sanding devices shall be tested at standstill
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.2119	If the IDU sanding test button has been pressed on the IDU brake test menu AND the preconditions are fulfiled, TCMS shall send the IP signal "Train brake test active" to WSP-CU and TCMS shall start the test sequence. (The precodition to start the sanding test as defined in concept 3669.)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2547	TCMS shall present driver text on the IDU: Press manual sanding push button and acknowledge sanding flow.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2549	Sanding command shall be activated by WSP-CU in case the TCMS set "Train brake test active" = 1 with following conditions: – v = 0 km/h AND – Sanding active for 30s at lowest sand amount (command only acceptance once during test) AND – Sanding shall be active on all sanding units (independent of direction signal) AND – No sand cleaning function during test	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.2548	There shall be an acknowledge softkey on IDU that sand flow was present.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.961	TCMS shall consider the sanding test as passed if the following conditions are set: – Driver has confirmed on IDU sanding flow present The sanding test shall be considered as not passed if the driver has not acknowleged the sanding flow.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.962	An event shall be set if the sanding test has not been acknowledged succesful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.3700	An event shall be set if the sanding test has been acknowledged succesful.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.11.3 Manual Brake Test

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.3821	A rotary selector switch shall be present in driver's cab with two active position: – Brake pipe continuity (BP venting) – Brake pipe isolation (BP no refill)	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3822	The rotary selector switch manual brake test shall be connected to EVISO valve. In position brake pipe isolation (BP no refill) the EVISO shall be energised to inhibit the refill of the brake pipe.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3823	The rotary selector switch manual brake test shall be connected to EVAPP command relay. In position brake pipe continuity (BP venting) the EVAPP test relay shall be energised to de-	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

	energise the EVAPP valve in case the train is at standstill (zero speed contact).				
2F_05.02.Zefiro-Europe.CONCEPT.3824	The rotary selector switch manual brake test shall be connected to TCMS digital input. In position neutral the input shall be de-energised. In position brake pipe isolation or brake pipe discharge the input shall be energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3825	An event shall be set in case the TCMS digital input of manual brake test rotary switch is set (1).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.12 Towing

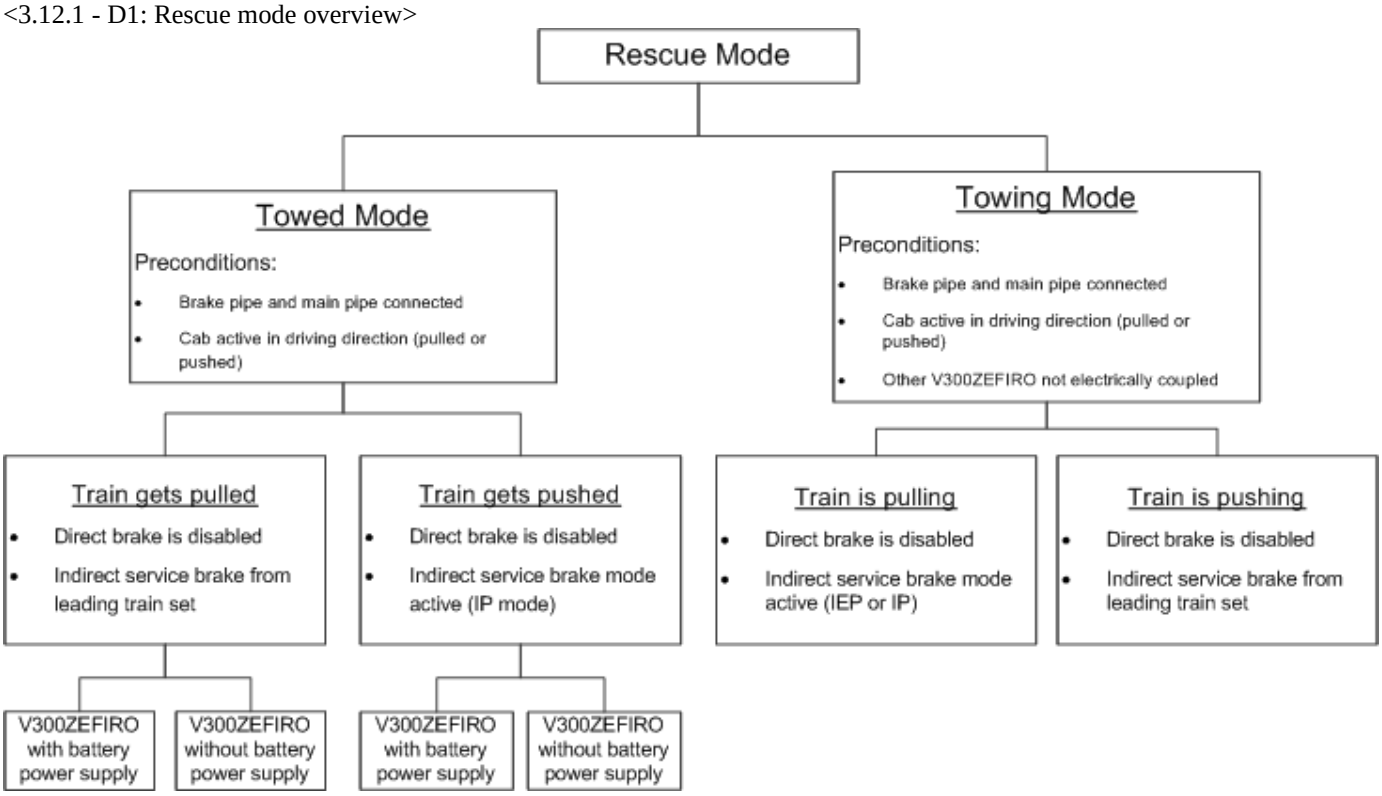
3.12.1 Function Design

The VZI-R1 is able to rescue another train or to get rescued via the emergency coupler and the brake and main pipe connection. Therefore, different functional towing cases have to be considered.

For towing, the train needs to operate in IP brake mode, to ensure that the brake pipe is used to control the service brake. As precondition the respective cab in driving direction has to be always active. Operational it is not allowed to tow a train with a active cab not in driving direction.

In case two VZI-R1 are electrically coupled, this scenario is not a towing/towed situation. For towing purpose the electrical heads has to be disconnected.

<3.12.1 - D1: Rescue mode overview>



Following rescue scenarios are valid:

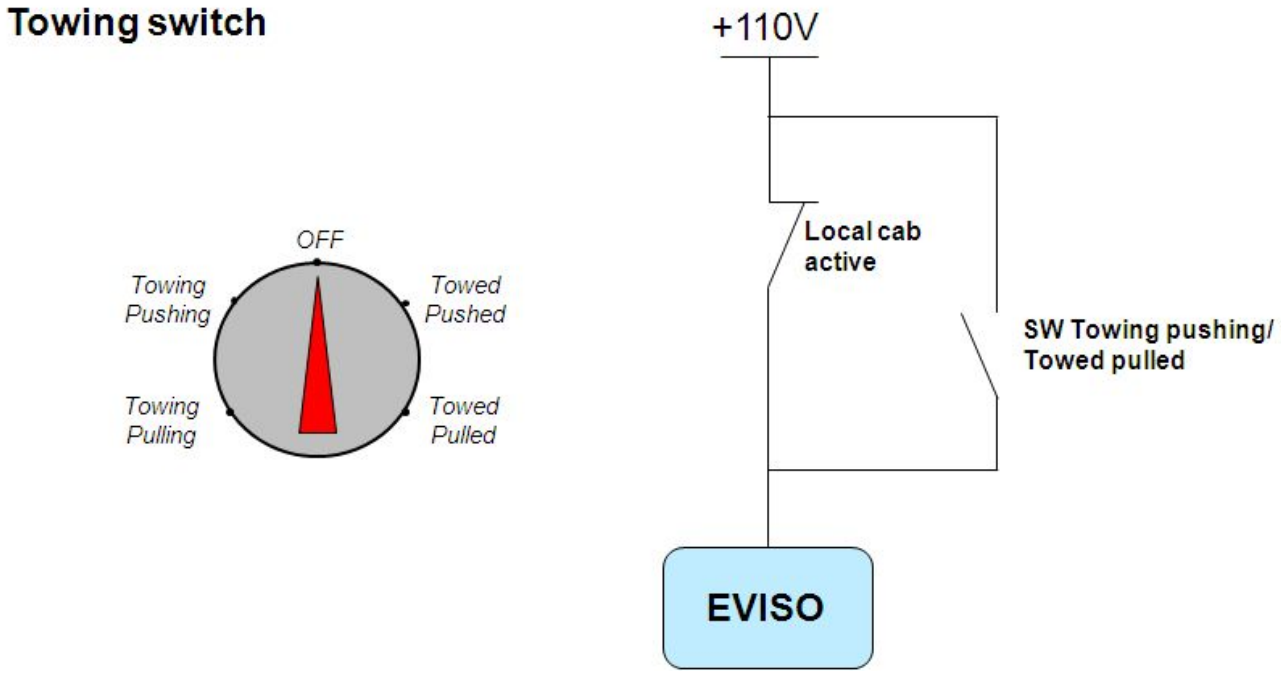
1. The VZI-R1 train is towed by another vehicle
   a) The demobilised VZI-R1 gets pulled
   b) The demobilised VZI-R1 gets pushed and is controlling the brake pipe

2. The VZI-R1 train is towing another vehicle or another V300ZEFIRO

a) The VZI-R1 is pulling another train
b) The VZI-R1 is pushing another train

In the driver's cab a towing/towed switch is present with 5 positions. Zero position towing/towed function is not active. The remaining four positions are as described above 1a, 1b, 2a and 2b.

<3.12.1 - D2: Towing/towed switch layout>



3.12.1.1 *VZI-R1 Towed Mode*

VZI-R1 train is towed by another vehicle:

General preconditions:
- Brake pipe and main pipe connected via towing coupler
- Cab active in driving direction (pulled or pushed) to close emergency brake loop
- Battery voltage available on V300ZEFIRO

a) VZI-R1 gets pulled - towing/towed scenario 1
   - Towing switch is in position towed/pulled to energise brake pipe isolation valve EVISO to inhibit control of brake pipe in active cab (driving direction)
   - Holding brake is automatically released in case of towing switch in position towed/pulled, parking brake to be released by push button when the VZI-R1 train is coupled to rescue train
   - Direct brake disabled in V300ZEFIRO
   - Brake pipe pressure control in towing vehicle to release/apply indirect brake

<3.12.1.1 - T1: Towing/towed scenario 1 - towed pulled >

Nr.	Driver/crew actions	Reactions on V300ZEFIRO
1	• Lower all pantographs • Open LCB • Activate cab in driving direction • Apply parking brake • Master controller in neutral position • Driver's brake valve in release position • Select driving direction in towing direction • Select "Towed pulled"- mode with tow-ing/towed selector switch in active cab	• Holding brake automatically re-leased • Brake pipe automatically isolated (no refill)
2	Rescue loco in front of V300ZEFIRO (active cab) • Mount towing coupler to rescue loco • Check MRP and BP connection of towing coupler to rescue locomotive. • Check that E-Coupler on top of automatic coupler is in uncoupled position • Isolate E-Coupler (Close isolation cock on the side of coupler head) In case of low main pipe pressure or TCMS failure: • Isolate RIVAPP isolation cock to bypass safe electrical emergency brake application valve (EVAPP) in all DM cars • Isolate RIEP isolation cock to diable the direct brake in all cars	
3	• Couple Loco with V300ZEFIRO by moving the loco	"Towed mode active" event is set • Failure reaction: "Towed mode not active" is set
4	• Apply parking/holding brake in rescue loco	
5	• Fill BP from rescue locomotive • Verify BP pressure on manometer in active cab (BP pressure approx. 5bar)	
6	• Perform manual brake test from rescue loco	• Verify indirect brake application
7	• Define brake weight for consist • Release parking brake in V300ZEFIRO	

b) VZI-R1 gets pushed - towing/towed scenario 2

- Towing switch is in position towed/pushed in active cab (driving direction)
- Indirect service brake mode is activated, brake pipe control mode is IP by turning RIBK isolation cock in IP position
- Holding brake is automatically released in case of towing switch in position towed/pushed, parking brake to be released by push button when the VZI-R1 train is coupled to rescue train
- Direct brake disabled in V300ZEFIRO
- Brake pipe pressure control in VZI-R1 to release/apply indirect brake via the driver's brake valve
- In case only the brake pipe is connected, the parking brakes has to be released manually before rescue the V300ZEFIRO.

<3.12.1.1 - T2: Towing/towed scenario 2 - towed pushed>

Nr.	Driver/crew actions	Reactions on V300ZEFIRO
1	• Ensure communication from Rescue vehicle to leading DM car • Lower all pantographs • Open LCB • Activate cab in driving direction • Apply parking brake • Master controller in neutral position • Driver's brake valve in release position • Select driving direction in pushing direction • Select "Towed pushed"-mode with tow-ing/towed selector switch in active cab	• Holding brake automatically re-leased
2	• Activate indirect brake mode with service brake selector switch • Activate IP brake pipe control mode by turn-ing RIBK in IP position In case of low main pipe pressure or TCMS failure: • Isolate RIVAPP isolation cock to bypass safe electrical emergency brake application valve (EVAPP) in all DM cars • Isolate RIEP isolation cock to diable the direct brake in all cars	• Indirect brake control active by Driver's Brake Valve
3	Rescue loco on rear end of V300ZEFIRO (not active cab) • Mount towing coupler to rescue loco • Check MRP and BP connection of towing coupler to rescue loco. • Check that E-Coupler on top of automatic coupler is in uncoupled position • Isolate E-Coupler (Close isolation cock on the side of coupler head)	
4	• Couple locomotive with V300ZEFIRO by moving the loco	"Towed mode active" event is set • Failure reaction: "Towed mode not active" is set
5	• Apply parking/holding brake in rescue loco	
6	• Fill BP from active cab DVB in V300ZEFIRO • Verify BP pressure on manometer in pushing loco (BP pressure approx. 5bar)	
7	• Perform manual brake test from V300ZEFIRO	• Verify indirect brake application
8	• Define brake weight for consist • Release parking brake in V300ZEFIRO	• Verify indirect brake application

3.12.1.1.1 *VZI-R1 Towed Mode Out of Battery*

In case the towed VZI-R1 has insufficient battery voltage, the train is towed as described in chapter 3.14.1.1 with indirect brake pipe control.

All electronic functions are not available during towing, especially:

- Wheel slide protection not available
- Urgency/Emergency brake loop control inactive
- ATP system disabled
- TCMS not available
- Brake pipe isolation not available
- IEP brake not available

The driver has to take following actions to ensure the train can be towed without battery:
- If parking brake are not released a manual release has to be performed on bogie frame
- ATP ERTMS pneumatic valves to be isolated (to inhibit permanent venting of brake pipe)
- Emergency application valves EVAPP to be isolated (to inhibit permanent venting of brake pipe)

a) VZI-R1 gets pulled - towing/towed scenario 3
   - Both cabin panels 01.01 with brake pipe control located in each DM car to be isolated by RIM 01.03 isolation cock

b) VZI-R1 gets pushed
   - Cabin panel 01.01 with brake pipe control located in coupled DM car to be isolated by RIM 01.03 isolation cock
   - IP brake pipe control mode by turning RIBK 01.26 isolation cock in IP position

<3.12.1.1 - T3: Towing/towed scenario 3 - towed pulled (insufficient battery voltage)>

Nr.	Driver/crew actions	Reactions on V300ZEFIRO
1	• Lower all pantographs • Open LCB • Master controller in neutral position • Driver's brake valve in release position • Open battery contactor • Ensure safe immobilisation position by either manual application of parking brake (parking brake isolation cock RIFS per bogie) or with brake shoes	V300ZEFIRO status: • Insufficient battery power • Indirect brake applied
2	• Isolate RIVAPP isolation cock to bypass safe electrical emergency brake application valve (EVAPP) in all DM cars • Isolate RIEP isolation cock to diable the direct brake in all cars • Activate IP brake pipe control mode by turning RIBK in IP position • Isolate RIM isolation cock for brake pipe control cabin panel in both DM cars • Isolation of ATP emergency brake valve	• ATP disabled • TCMS/Emergency brake loop venting of BP disabled • Direct brake disabled • DBV disabled
3	Rescue locomotive in front of V300ZEFIRO • Mount towing coupler to rescue locomotive • Check MRP and BP connection of towing coupler to rescue locomotive • Check that E-Coupler on top of automatic coupler is in uncoupled position • Isolate E-Coupler (Close isolation cock on the side of coupler head)	
4	• Couple locomotive with V300ZEFIRO by moving the locomotive	
5	• Apply parking/holding brake in rescue locomotive	
6	• Fill BP from rescue locomotive • Verify BP pressure on manometer in rescue locomotive and ETR1000-V300ZEFIRO (BP pressure approx. 5bar)	• Indirect and direct brake released
7	• Perform manual brake test from rescue locomotive	• Verify indirect brake application
8	• Define brake weight for consist • In case of sufficient air supply (feeding from rescue locomotive or min MR pressure of 7 bar) release parking brake by isolation cock RIFS - Otherwise manual release parking brake via bogie mounted parking brake release handles and isolate all RIFS isoaltion cock	• Verify parking brake released

3.12.1.2 *VZI-R1 Towing Mode*

VZI-R1 train is towing another vehicle or another V300ZEFIRO:

General preconditions:
- Brake pipe and main pipe connected
- Cab active in driving direction (pulling or pushing) to allow traction control
- Other VZI-R1 not electrically coupled

a) VZI-R1 is pulling another train
   - Towing switch is in position towing/pulling in active cab (driving direction)
   - Parking brake to be released by push button when the VZI-R1 train is coupled to towed train
   - Indirect service brake mode is activated, brake pipe control mode is IEP or IP mode
   - Direct brake disabled in V300ZEFIRO
   - Brake pipe pressure control in VZI-R1 to release/apply indirect brake via the driver's brake valve

b) VZI-R1 is pushing another train
   - Towing switch is in position towing/pushing to energise brake pipe isolation valve EVISO to inhibit control of brake pipe in active cab (driving direction)
   - Parking brake to be released by push button when the VZI-R1 train is coupled to towed train
   - Direct brake disabled in V300ZEFIRO
   - Brake pipe pressure control in towed vehicle to release/apply indirect brake

The maximum towing speed in towing mode is to be controlled and in responsibility by the driver. TCMS is not limiting the speed in towing/towed mode.

In case of incoherent towing/towed switch position to brake pipe control mode an event is set by TCMS.

When towing another train in towing mode equipped with a indirect brake pipe control, the overcharge button located on driver's desk (described in chapter 3.2) shall be used to assimilate the brake pipe pressure in the towed train.

3.12.1.3 *Failure Information*

3.12.1.3.1 *Normal condition*

The two vehicles for the towing/towed operation are connected mechanically and pneumatically.
The mechanical coupler is connected together with the main reservoir pipe and the brake pipe.

The cab of the towed train has to be activated to energise the safety urgency and emergency brake loops. The towing/towed switch has to be placed in the respective position to inhibit the brake pipe refill command in not leading cab.

3.12.1.3.2 *Failure cases*

1. Main reservoir pipe blocked between the towing and the towed vehicle.
2. Brake pipe blocked between the towing and the towed vehicle.
3. Towing/towed switch not put into "towed mode", but mechanically and pneumatically connected.
4. Towing/towed switch put into "towed mode" but brake pipe control mode is not set into IP brake pipe control.
5. Towing/towed switch put into towing/towed mode in normal operation.
6. Failure of towing/towed switch.

3.12.1.3.3 *Failure detection*

1. The low main reservoir pressure will be detected in the towed vehicle by failure events, below 600kPa emergency brake will be applied (see description in SFRS Air supply) at V300ZEFIRO/ETR1000.
2. Indirect brake in the towed vehicle cannot be controlled by the brake pipe, brakes is applied in towed train. Brake pipe pressure supervision in V300ZEFIRO/ETR1000 to detect possible failures.
3. An event is set to the driver that the train is mechanically coupled with another unknown vehicle (no electrical connection).
4. An event is set to the driver to activate the IP mode in towed vehicle.
5. An event is set if the towing/towed switch is in towing or towed position but no vehicle is mechanically coupled.
6. No further failure detection possible, refer to item 3.

3.12.1.3.4 *Failure handling*

1. Failure handling as described in SFRS Air Supply, train cannot move if pressure below 600kPa
2. Train cannot be towed by one driver using the brake pipe. Two alternatives:
   – A second driver can operate the brake of the towed train OR.
   – The brakes of the towed train need to be released mechanically and the train may be towed unbraked.
3. The driver has to activate the respective towing/towed mode and IP brake mode.
4. The driver has to activate the IP mode in towed vehicle.
5. The driver has to put the towing/towed switch in normal operation position.
6. The train can be towed in degraded condition with speed limitation of 40 km/h.

3.12.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.88
Description	In towing the brake command is given by the UIC brake pipe pressure manipulated from a towing vehicle. The braking function is like in IP braking.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.720	In towed mode the DEP brake mode shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.3036	If the rotary selector switch Towing/Towed is in position towing pushing or towing pulling the towing mode shall be activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.3037	If the rotary selector switch Towing/Towed is in position towed pushed or towed pulled the towed mode shall be activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.1646	If the train is in towing mode and the service brake DEP mode is still active, TCMS shall set an event to inform the driver to go in indirect brake mode (IEP or IP).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.1505	In case the towed mode is selected the pressure switch PST 03.09 in DM car connected to the traction safe loop shall be bypassed. The traction safe loop shall not be opened during towed.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_05.02.Zefiro-Europe.CONCEPT.540	Holding brake shall be released automatically if train is put in towed mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2610	The five position rotary selector switch Towing/Towed shall be located in cabinet in driver's cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_05.02.Zefiro-Europe.CONCEPT.2611	The towing switch shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.2613	The towing/towed switch shall be supervised by TCMS, if the towing/towed switch is in position towed or towing an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.13 Fire Safety & Running Capability

3.13.1 Function Design

Requirements for fire safety and running capability to fulfil the running capability of the train for 15 minutes. The electrical and pneumatical control lines must be designed accordingly.
Vital functions are either realised by fire safe cabling/piping or possible to be overridden in case of false indication by e.g. emergency operation mode.

3.13.2 Requirements

ID	2F_05.02.Zefiro-Europe.TRS.161
Description	The train shall be able to run under attack of any fire for 15min, this corresponds to a distance 20km at speed 80 km/h.
Safety level	N/A

ID	2F_05.02.Zefiro-Europe.TRS.322
Description	Fire shall never cause an immediate automatic brake application.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.1802	The control elements and communication links which have direct effect on the UB and EB loop or any local brake application shall be fire resistant to endure 15min of fire. This applies to following: – urgency brake loop – emergency brake loop – DNRA train line – PEB train line – local brake control lines which would result in a brake application if interrupted (WSP) – DBV and UIC brake pipe	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.02.Zefiro-Europe.CONCEPT.2544	The control elements and communication links which have direct effect on the local internal brake functions shall be fire resistant to endure 15min of fire.	Derived Requirement	4S_07.-.Braking System		N/A
2F_05.02.Zefiro-Europe.CONCEPT.1803	The control elements and the communication links which have effect on running capability shall be fire resistant to endure 15 min of fire. This applies to following: – traction and braking demand trainlines for emergency operation – emergency operation trainline	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.14 FIRE system leakage test enabling

If the Main Pipe pressure (main pipe pressure, byte 38, SEL_CCU_BRK_CMD_x) is higher than 8 bar & no brake test is running & no signalling test is running, TCMS shall send to the antifire system the antifire tight test permission signal = 1.

2 = antifire tight test not permitted
1 = antifire tight test permitted
0 = not valid

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR38	IF the Main Pipe pressure > 8 bar AND brake test is NOT running AND signalling test is NOT running THEN TCMS shall set the fire system tight test tight test enabling		4S_09.03.05.-.TCMS Software		SIL 0

3.15 TCMS diagnostic events

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR66	If TCMS detects at least 50% of brake isolation valves in 1 unit (for single or multiple composition) .TCMS shall trigger an event of **Too many bogies isolated in one unit** e.g. trigger if at least 8/16 bogies for a single unit, for multiple unit at least 8 of one unit.	Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR67	If TCMS detects 100% of parking brake isolation valves in an unit. In multiple unit if the parking brakes are isolated at least 50% of all bogies the alarm will , TCMS shall trigger an event of **Too many parking brake isolated** e.g. trigger if at least 4 of 8 bogies isolated in single unit trigger if at least 8 of 16 bogies isolated in single unit	Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HR68	If the Cab active car have all bogies isolated OR the End of Train car have all bogies isolated, set the event End car with braks isolated	Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.16 BCU status in automatic drive mode

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In automatic drive mode BCU shall use the Ethernet signals to acquire the traction/braking status

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR69	In automatic drive mode the BCU acquires the running mode (braking/traction) by TCMS via IP bus only if the not braking TL is 0 or Emergency loop is 0, the wired lines have greater priority		4S_07.-.Braking System		SIL 0

3.17 Enviromental data in the braking system diagnostic event

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_05.02.Zefiro-Europe.CONCEPT.HR70	In case of brake events or in case of an event specified in this document the following environmental data shall be appended to the event signal: • Heavy slide and slide from BCU and WSP. • Brake Pipe pressure • Main Pipe pressure • Manipulators and brake driver interfaces status (emergency pushbuttons, master controller, etc.) • Brake cylinder pressures (by BCU) • PCF • WSP function in charge to • DNRA function in charge to • Compressor command • Sanding command • BCU axle speed • WSP axle speed • BCU reference speed • WSP reference speed • Train lines status	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

4.1.1.1 *Inputs*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1118	Motoring (Traction) demand trainline	L	2F_05.02.Zefiro-Europe.CONCEPT.165
2F_05.02.Zefiro-Europe.CONCEPT.1119	Braking demand trainline	L	2F_05.02.Zefiro-Europe.CONCEPT.165
2F_05.02.Zefiro-Europe.CONCEPT.1862	Emergency operation active trainline	L	2F_05.02.Zefiro-Europe.CONCEPT.165
2F_05.02.Zefiro-Europe.CONCEPT.2662	Urgency brake loop	L	
2F_05.02.Zefiro-Europe.CONCEPT.2663	Emergency brake loop	L	
2F_05.02.Zefiro-Europe.CONCEPT.2664	Neutral valve energised if EB loop de-energised	L	
2F_05.02.Zefiro-Europe.CONCEPT.2665	Isolation valve energised if cab is de-activated	L	

4.1.1.2 *Outputs*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1146	Slip slide regulation axle 1 bit 1	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1148	Slip slide regulation axle 1 bit 2	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1486	Slip slide regulation axle 2 bit 1	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1147	Slip slide regulation axle 2 bit 2	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1739	Slip slide regulation axle 3 bit 1	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1740	Slip slide regulation axle 3 bit 2	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1741	Slip slide regulation axle 4 bit 1	L	2F_05.02.Zefiro-Europe.CONCEPT.276
2F_05.02.Zefiro-Europe.CONCEPT.1742	Slip slide regulation axle 4 bit 2	L	2F_05.02.Zefiro-Europe.CONCEPT.276

2F_05.02.Zefiro-Europe.CONCEPT.1780	DNRA detected (trainline)	L	2F_05.02.Zefiro-Europe.CONCEPT.490
2F_05.02.Zefiro-Europe.CONCEPT.1805	Speed zero	L	2F_05.02.Zefiro-Europe.CONCEPT.1806
2F_05.02.Zefiro-Europe.CONCEPT.2666	Speed > 10 km/h	L	
2F_05.02.Zefiro-Europe.CONCEPT.2648	Pressure switch PST brake pipe vented	L	

4.1.2 TCMS SW

4.1.2.1 *Inputs:*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1632	RIEP-DIST isolation cock isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1588
2F_05.02.Zefiro-Europe.CONCEPT.1628	RICF1 isolation cock bogie 1 isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1583
2F_05.02.Zefiro-Europe.CONCEPT.1629	RICF2 isolation cock bogie 2 isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1584
2F_05.02.Zefiro-Europe.CONCEPT.2082	RIFS1 isolation cock parking brake isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.771
2F_05.02.Zefiro-Europe.CONCEPT.2083	RIFS2 isolation cock parking brake isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.771
2F_05.02.Zefiro-Europe.CONCEPT.1630	RICP isolation cock isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1586
2F_05.02.Zefiro-Europe.CONCEPT.1970	RISOSP 05.02/01 isolation cock isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1518 2F_05.02.Zefiro-Europe.CONCEPT.2376
2F_05.02.Zefiro-Europe.CONCEPT.1792	RIVAPP isolation cock not isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1789
2F_05.02.Zefiro-Europe.CONCEPT.2646	RIBK isolation cock not isolated	L	
2F_05.02.Zefiro-Europe.CONCEPT.2089	RICS sanding isolation cock isolated	L	
2F_05.02.Zefiro-Europe.CONCEPT.2085	Parking brake applied bogie 1	L	
2F_05.02.Zefiro-Europe.CONCEPT.2086	Parking brake applied bogie 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.2087	Parking brake release bogie 1	L	
2F_05.02.Zefiro-Europe.CONCEPT.2088	Parking brake release bogie 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.1174	Passenger emergency brake override button	L	2F_05.02.Zefiro-Europe.CONCEPT.494

2F_05.02.Zefiro-Europe.CONCEPT.1810	Passenger emergency brake train line energized	L	2F_05.02.Zefiro-Europe.CONCEPT.971
2F_05.02.Zefiro-Europe.CONCEPT.1178	Parking brake apply/release button	L	2F_05.02.Zefiro-Europe.CONCEPT.482
2F_05.02.Zefiro-Europe.CONCEPT.1176	Holding brake release push button	L	2F_05.02.Zefiro-Europe.CONCEPT.222 2F_05.02.Zefiro-Europe.CONCEPT.429
2F_05.02.Zefiro-Europe.CONCEPT.1180	Sanding button	L	2F_05.02.Zefiro-Europe.CONCEPT.315
2F_05.02.Zefiro-Europe.CONCEPT.1183	Brake test button	L	2F_05.02.Zefiro-Europe.CONCEPT.710
2F_05.02.Zefiro-Europe.CONCEPT.1187	Emergency stop buttons PFE1 and PFE2 (one or both are pressed)	L	2F_05.02.Zefiro-Europe.CONCEPT.1960
2F_05.02.Zefiro-Europe.CONCEPT.1811	Emergency brake loop energised	L	2F_05.02.Zefiro-Europe.CONCEPT.521 2F_05.02.Zefiro-Europe.CONCEPT.1649
2F_05.02.Zefiro-Europe.CONCEPT.1610	Brake applied bogie 1 (pressure switch PCF1)	L	2F_05.02.Zefiro-Europe.CONCEPT.1612
2F_05.02.Zefiro-Europe.CONCEPT.1611	Brake applied bogie 2 (pressure switch PCF2)	L	2F_05.02.Zefiro-Europe.CONCEPT.1613
2F_05.02.Zefiro-Europe.CONCEPT.1715	DNRA trainline status "energised".	L	2F_05.02.Zefiro-Europe.CONCEPT.963 2F_05.02.Zefiro-Europe.CONCEPT.967
2F_05.02.Zefiro-Europe.CONCEPT.1812	MTB supervision circuit ok	L	2F_05.02.Zefiro-Europe.CONCEPT.1656
2F_05.02.Zefiro-Europe.CONCEPT.1813	MTB magnetised (applied electrically)	L	2F_05.02.Zefiro-Europe.CONCEPT.1656
2F_05.02.Zefiro-Europe.CONCEPT.1814	Brake pipe vented/traction safe PST pressure switch	L	2F_05.02.Zefiro-Europe.CONCEPT.1635
2F_05.02.Zefiro-Europe.CONCEPT.2647	Service brake selector switch position Indirect Brake mode	L	
2F_05.02.Zefiro-Europe.CONCEPT.2649	Brake pipe overcharge button	L	
2F_05.02.Zefiro-Europe.CONCEPT.2650	DBV position released	L	
2F_05.02.Zefiro-Europe.CONCEPT.2651	DBV position braking	L	
2F_05.02.Zefiro-Europe.CONCEPT.2652	DBV position full service brake	L	
2F_05.02.Zefiro-Europe.CONCEPT.2653	DBV position emergency brake position	L	
2F_05.02.Zefiro-Europe.CONCEPT.2654	Urgency brake loop control point 1	L	
2F_05.02.Zefiro-Europe.CONCEPT.2660	Urgency brake loop control point 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.2659	Urgency brake loop control point 3	L	
2F_05.02.Zefiro-Europe.CONCEPT.2658	Urgency brake loop control point 4	L	

2F_05.02.Zefiro-Europe.CONCEPT.2657	Emergency brake loop control point 1	L	
2F_05.02.Zefiro-Europe.CONCEPT.2656	Emergency brake loop control point 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.2655	Emergency brake loop control point 3	L	
2F_05.02.Zefiro-Europe.CONCEPT.2661	Emergency brake loop control point 4	L	
2F_05.02.Zefiro-Europe.CONCEPT.HR52	Transducer bogie x setting out of range TC_BI_#BCU_S_Trans_outrange_bogie_x	SIL 0	

4.1.2.2 *Outputs:*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.2645	Brake applied lamp indication	L	2F_05.02.Zefiro-Europe.CONCEPT.2051
2F_05.02.Zefiro-Europe.CONCEPT.1175	"Passenger emergency brake suspended" lamp inidcation	L	2F_05.02.Zefiro-Europe.CONCEPT.495
2F_05.02.Zefiro-Europe.CONCEPT.1815	TCMS emergency brake relay: energise demand	L	2F_05.02.Zefiro-Europe.CONCEPT.1816
2F_05.02.Zefiro-Europe.CONCEPT.1179	"Parking brake applied" lamp indication	L	2F_05.02.Zefiro-Europe.CONCEPT.1779
2F_05.02.Zefiro-Europe.CONCEPT.1177	"Holding brake applied" lamp indication	L	2F_05.02.Zefiro-Europe.CONCEPT.545
2F_05.02.Zefiro-Europe.CONCEPT.1181	"Sanding active" lamp indication	L	2F_05.02.Zefiro-Europe.CONCEPT.1073
2F_05.02.Zefiro-Europe.CONCEPT.1186	Brake test status lamp (green)	L	2F_05.02.Zefiro-Europe.CONCEPT.1103
2F_05.02.Zefiro-Europe.CONCEPT.2389	Brake test status lamp (red)	L	2F_05.02.Zefiro-Europe.CONCEPT.1103
2F_05.02.Zefiro-Europe.CONCEPT.1555	Parking brake apply demand (EVFS1)	L	2F_05.02.Zefiro-Europe.CONCEPT.1069 2F_05.02.Zefiro-Europe.CONCEPT.1500
2F_05.02.Zefiro-Europe.CONCEPT.1743	Parking brake release demand (EVFS1)	L	2F_05.02.Zefiro-Europe.CONCEPT.1069 2F_05.02.Zefiro-Europe.CONCEPT.1500
2F_05.02.Zefiro-Europe.CONCEPT.2644	Sanding heating distributor pipe DM and T4/5 cars	L	

4.2 Bus Interface

4.2.1 Suppliers to TCMS SW

4.2.1.1 *From Brake Supplier (BCU/WSP/DBV-GW)*

4.2.1.1.1 ***BCU to TCMS***

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1952	BCU start-up ok	L	2F_05.02.Zefiro-Europe.CONCEPT.3553
2F_05.02.Zefiro-Europe.CONCEPT.1139	Parking brake applied	L	2F_05.02.Zefiro-Europe.CONCEPT.1723
2F_05.02.Zefiro-Europe.CONCEPT.2084	Parking brake released	L	2F_05.02.Zefiro-Europe.CONCEPT.2070
2F_05.02.Zefiro-Europe.CONCEPT.1951	DNRA function ok	L	2F_05.02.Zefiro-Europe.CONCEPT.282
2F_05.02.Zefiro-Europe.CONCEPT.1714	DNRA request active	L	2F_05.02.Zefiro-Europe.CONCEPT.490 2F_05.02.Zefiro-Europe.CONCEPT.1045
2F_05.02.Zefiro-Europe.CONCEPT.1950	Brake reservoir pressure available TSA (> 700 kPa)	L	2F_05.02.Zefiro-Europe.CONCEPT.2165
2F_05.02.Zefiro-Europe.CONCEPT.1214	EB loop status (energised)	L	2F_05.02.Zefiro-Europe.CONCEPT.642
2F_05.02.Zefiro-Europe.CONCEPT.1262	Traction train line energised	L	2F_05.02.Zefiro-Europe.CONCEPT.1028 2F_05.02.Zefiro-Europe.CONCEPT.1861
2F_05.02.Zefiro-Europe.CONCEPT.1744	Braking trainline energised	L	2F_05.02.Zefiro-Europe.CONCEPT.1028 2F_05.02.Zefiro-Europe.CONCEPT.1861
2F_05.02.Zefiro-Europe.CONCEPT.1964	Emergency operation trainline energised	L	2F_05.02.Zefiro-Europe.CONCEPT.1028 2F_05.02.Zefiro-Europe.CONCEPT.1861 2F_05.02.Zefiro-Europe.CONCEPT.3609
2F_05.02.Zefiro-Europe.CONCEPT.3228	Dragging brake alert bogie 1	L	2F_05.02.Zefiro-Europe.CONCEPT.512
2F_05.02.Zefiro-Europe.CONCEPT.3229	Dragging brake alert bogie 2	L	2F_05.02.Zefiro-Europe.CONCEPT.3004
2F_05.02.Zefiro-Europe.CONCEPT.1553	Parking brake bogie 1 isolation cock 06.02/01 (RIFS1) isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.762 2F_05.02.Zefiro-Europe.CONCEPT.766
2F_05.02.Zefiro-Europe.CONCEPT.1554	Parking brake bogie 2 isolation cock 06.02/02 (RIFS2) isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1552
2F_05.02.Zefiro-Europe.CONCEPT.1539	Direct brake isolation cock 04.53/01 RIEP isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1524
2F_05.02.Zefiro-Europe.CONCEPT.1633	Brake isolation cock 04.29/01 RIEP-DIST isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1587
2F_05.02.Zefiro-Europe.CONCEPT.1631	Brake frame isolation cock 04.27/01 RICP isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1585
2F_05.02.Zefiro-Europe.CONCEPT.1219	Brake bogie 1 isolation cock 04.38/1 RICF1 isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.120
2F_05.02.Zefiro-Europe.CONCEPT.1626	Brake bogie 2 isolation cock 04.38/2 RICF2 isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.120

2F_05.02.Zefiro-Europe.CONCEPT.3552	EB application pressure switch in emergency position (PAPP)	L	2F_05.02.Zefiro-Europe.CONCEPT.2674
2F_05.02.Zefiro-Europe.CONCEPT.1793	EB application valve isolation cock RIVAPP isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1789 2F_05.02.Zefiro-Europe.CONCEPT.3192
2F_05.02.Zefiro-Europe.CONCEPT.1597	Secondary suspension isolation cock 05.02/01 RISOSP isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1516
2F_05.02.Zefiro-Europe.CONCEPT.1949	DIA EB loop implausible (IP and EB loop status not aligned)	L	2F_05.02.Zefiro-Europe.CONCEPT.2096
2F_05.02.Zefiro-Europe.CONCEPT.3550	DIA DNRA function disturbed (DNRA not available)	L	2F_05.02.Zefiro-Europe.CONCEPT.3549
2F_05.02.Zefiro-Europe.CONCEPT.1993	DIA BCU/WSP internal bus failure (RS232)	L	2F_05.02.Zefiro-Europe.CONCEPT.1905
2F_05.02.Zefiro-Europe.CONCEPT.1115	DIA DEP brake faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.357 2F_05.02.Zefiro-Europe.CONCEPT.1527
2F_05.02.Zefiro-Europe.CONCEPT.3230	DIA Overbrake axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.793
2F_05.02.Zefiro-Europe.CONCEPT.3233	DIA Overbrake axle 2	L	2F_05.02.Zefiro-Europe.CONCEPT.793
2F_05.02.Zefiro-Europe.CONCEPT.3232	DIA Overbrake axle 3	L	2F_05.02.Zefiro-Europe.CONCEPT.793
2F_05.02.Zefiro-Europe.CONCEPT.3231	DIA Overbrake axle 4	L	2F_05.02.Zefiro-Europe.CONCEPT.793
2F_05.02.Zefiro-Europe.CONCEPT.3234	DIA Missing brake axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.792
2F_05.02.Zefiro-Europe.CONCEPT.3237	DIA Missing brake axle 2	L	2F_05.02.Zefiro-Europe.CONCEPT.792
2F_05.02.Zefiro-Europe.CONCEPT.3236	DIA Missing brake axle 3	L	2F_05.02.Zefiro-Europe.CONCEPT.792
2F_05.02.Zefiro-Europe.CONCEPT.3235	DIA Missing brake axle 4	L	2F_05.02.Zefiro-Europe.CONCEPT.792
2F_05.02.Zefiro-Europe.CONCEPT.1116	DIA indirect brake integrity failure	L	2F_05.02.Zefiro-Europe.CONCEPT.162 2F_05.02.Zefiro-Europe.CONCEPT.1080
2F_05.02.Zefiro-Europe.CONCEPT.1112	DIA load transducer faulty TS1	L	2F_05.02.Zefiro-Europe.CONCEPT.1004
2F_05.02.Zefiro-Europe.CONCEPT.1542	DIA load transducer faulty TS2	L	2F_05.02.Zefiro-Europe.CONCEPT.1543
2F_05.02.Zefiro-Europe.CONCEPT.2887	DIA TFS parking brake sensor fault	L	2F_05.02.Zefiro-Europe.CONCEPT.1660 2F_05.02.Zefiro-Europe.CONCEPT.2888
2F_05.02.Zefiro-Europe.CONCEPT.1712	DIA BCU speed sensor faulty axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.1940	DIA BCU speed sensor faulty axle 2	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.1941	DIA BCU speed sensor faulty axle 3	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.1942	DIA BCU speed sensor faulty axle 4	L	2F_05.02.Zefiro-Europe.CONCEPT.662

2F_05.02.Zefiro-Europe.CONCEPT.2894	DIA BCU faulty speed sensor acquistion channel 1	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.2895	DIA BCU faulty speed sensor acquistion channel 2	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.2896	DIA BCU faulty speed sensor acquistion channel 3	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.2897	DIA BCU faulty speed sensor acquistion channel 4	L	2F_05.02.Zefiro-Europe.CONCEPT.662
2F_05.02.Zefiro-Europe.CONCEPT.3238	DIA v > 10 km/h speed output open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3566
2F_05.02.Zefiro-Europe.CONCEPT.3241	DIA v > 10 km/h speed output short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3568
2F_05.02.Zefiro-Europe.CONCEPT.3240	DIA v = 0 km/h speed output open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3562
2F_05.02.Zefiro-Europe.CONCEPT.3239	DIA v = 0 km/h speed output short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3564
2F_05.02.Zefiro-Europe.CONCEPT.1264	DIA DNRA output from BCU open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3555
2F_05.02.Zefiro-Europe.CONCEPT.3226	DIA DNRA output from BCU short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3557
2F_05.02.Zefiro-Europe.CONCEPT.1159	DIA Wheel set 1 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.1677	DIA Wheel set 2 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.1678	DIA Wheel set 3 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.1679	DIA Wheel set 4 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.1126	DIA DEP EV1 valve pressure control open circuit fault	L	2F_05.02.Zefiro-Europe.CONCEPT.1535
2F_05.02.Zefiro-Europe.CONCEPT.1127	DIA DEP EV2 valve pressure control open circuit fault	L	2F_05.02.Zefiro-Europe.CONCEPT.1535
2F_05.02.Zefiro-Europe.CONCEPT.1128	DIA DEP EV1 valve pressure control short circuit fault	L	2F_05.02.Zefiro-Europe.CONCEPT.982
2F_05.02.Zefiro-Europe.CONCEPT.1129	DIA DEP EV2 valve pressure control short circuit fault	L	2F_05.02.Zefiro-Europe.CONCEPT.1537
2F_05.02.Zefiro-Europe.CONCEPT.3561	DIA TSA - pressure transducer faulty (Short/Open)	L	2F_05.02.Zefiro-Europe.CONCEPT.3559
2F_05.02.Zefiro-Europe.CONCEPT.1132	DIA DEP pilote pressure sensor 04.28 (Tpil) faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.983
2F_05.02.Zefiro-Europe.CONCEPT.2369	DIA Indirect pilot pressure sensor TDIS faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.2277
2F_05.02.Zefiro-Europe.CONCEPT.2540	DIA Emergency brake integrity failure	L	2F_05.02.Zefiro-Europe.CONCEPT.3320
2F_05.02.Zefiro-Europe.CONCEPT.3250	DIA dump valve axle 1 active for more than 10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658

2F_05.02.Zefiro-Europe.CONCEPT.3253	DIA dump valve axle 2 active for more than 10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658
2F_05.02.Zefiro-Europe.CONCEPT.3252	DIA dump valve axle 3 active for more than 10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658
2F_05.02.Zefiro-Europe.CONCEPT.3251	DIA dump valve axle 4 active for more than 10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658
2F_05.02.Zefiro-Europe.CONCEPT.3254	DIA BCU dump valve axle 1: short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.3257	DIA BCU dump valve axle 2: short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.3256	DIA BCU dump valve axle 3: short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.3255	DIA BCU dump valve axle 4: short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.3258	DIA BCU dump valve axle 1: open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.3261	DIA BCU dump valve axle 2: open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.3260	DIA BCU dump valve axle 3: open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.3259	DIA BCU dump valve axle 4: open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.3262	DIA dump valve axle 1 security circuit failure (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3265	DIA dump valve axle 2 security circuit failure (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3264	DIA dump valve axle 3 security circuit failure (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3263	DIA dump valve axle 4 security circuit failure (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3266	DIA pressure incoherence DV1 (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3269	DIA pressure incoherence DV2 (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3268	DIA pressure incoherence DV3 (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3267	DIA pressure incoherence DV4 (BCU WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3270	DIA ED11_BCU_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3273	DIA ED12_BCU_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3272	DIA ED21_BCU_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3271	DIA ED22_BCU_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766

2F_05.02.Zefiro-Europe.CONCEPT.3274	DIA ED31_BCU_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3277	DIA ED32_BCU_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3276	DIA ED41_BCU_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3275	DIA ED42_BCU_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3278	DIA ED11_BCU_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3279	DIA ED12_BCU_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3280	DIA ED21_BCU_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3281	DIA ED22_BCU_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3282	DIA ED31_BCU_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3283	DIA ED32_BCU_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3284	DIA ED41_BCU_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3285	DIA ED42_BCU_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3872	DIA Missing second stage braking BCU	L	2F_05.02.Zefiro-Europe.CONCEPT.3860 2F_05.02.Zefiro-Europe.CONCEPT.3864
2F_05.02.Zefiro-Europe.CONCEPT.3873	DIA Undue second stage braking BCU	L	2F_05.02.Zefiro-Europe.CONCEPT.3865 2F_05.02.Zefiro-Europe.CONCEPT.3866
2F_05.02.Zefiro-Europe.CONCEPT.3874	DIA Missing third stage braking BCU	L	2F_05.02.Zefiro-Europe.CONCEPT.3867 2F_05.02.Zefiro-Europe.CONCEPT.3868
2F_05.02.Zefiro-Europe.CONCEPT.3875	DIA Undue third stage braking BCU	L	2F_05.02.Zefiro-Europe.CONCEPT.3868 2F_05.02.Zefiro-Europe.CONCEPT.3869
2F_05.02.Zefiro-Europe.CONCEPT.3929	DIA Missing Fourth Stage braking	L	2F_05.02.Zefiro-Europe.CONCEPT.3927
2F_05.02.Zefiro-Europe.CONCEPT.3876	DIA Triple stage control faulty BCU	L	2F_05.02.Zefiro-Europe.CONCEPT.3870 2F_05.02.Zefiro-Europe.CONCEPT.3871
2F_05.02.Zefiro-Europe.CONCEPT.1259	BCU minor fault	L	2F_05.02.Zefiro-Europe.CONCEPT.3793
2F_05.02.Zefiro-Europe.CONCEPT.1828	DIA - WSP function disturbed	L	2F_05.02.Zefiro-Europe.CONCEPT.3546
2F_05.02.Zefiro-Europe.CONCEPT.2371	Ready for brake test	L	2F_05.02.Zefiro-Europe.CONCEPT.665 2F_05.02.Zefiro-Europe.CONCEPT.666
2F_05.02.Zefiro-Europe.CONCEPT.2373	Brake test step 1 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.2514 2F_05.02.Zefiro-Europe.CONCEPT.3720
2F_05.02.Zefiro-Europe.CONCEPT.2374	Brake test step 1.1 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.2531

2F_05.02.Zefiro-Europe.CONCEPT.2375	Brake test step 4 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.2790
2F_05.02.Zefiro-Europe.CONCEPT.2387	Brake test step 4.1 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.2844
2F_05.02.Zefiro-Europe.CONCEPT.2386	Brake test step 5 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.2576
2F_05.02.Zefiro-Europe.CONCEPT.2388	Brake test step 5.1 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.3678
2F_05.02.Zefiro-Europe.CONCEPT.3792	Brake test step 5.2 passed	L	2F_05.02.Zefiro-Europe.CONCEPT.2583
2F_05.02.Zefiro-Europe.CONCEPT.3810	WSP test passed	L	2F_05.02.Zefiro-Europe.CONCEPT.1881
2F_05.02.Zefiro-Europe.CONCEPT.3812	WSP test not passed	L	2F_05.02.Zefiro-Europe.CONCEPT.1881
2F_05.02.Zefiro-Europe.CONCEPT.3242	ED bit 1 active axle 1 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3249	ED bit 2 active axle 1 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3248	ED bit 1 active axle 2 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3247	ED bit 2 active axle 2 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3246	ED bit 1 active axle 3 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3245	ED bit 2 active axle 3 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3244	ED bit 1 active axle 4 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3243	ED bit 2 active axle 4 WSP redundancy	L	2F_05.02.Zefiro-Europe.CONCEPT.3545
2F_05.02.Zefiro-Europe.CONCEPT.3539	BCU is controlling WSP function axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1869
2F_05.02.Zefiro-Europe.CONCEPT.3540	BCU is controlling WSP function axle 2	L	2F_05.02.Zefiro-Europe.CONCEPT.1869
2F_05.02.Zefiro-Europe.CONCEPT.3541	BCU is controlling WSP function axle 3	L	2F_05.02.Zefiro-Europe.CONCEPT.1869
2F_05.02.Zefiro-Europe.CONCEPT.3542	BCU is controlling WSP function axle 4	L	2F_05.02.Zefiro-Europe.CONCEPT.1869
2F_05.02.Zefiro-Europe.CONCEPT.3535	Redundant WSP activity axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.3222
2F_05.02.Zefiro-Europe.CONCEPT.3536	Redundant WSP activity axle 2	L	2F_05.02.Zefiro-Europe.CONCEPT.3222
2F_05.02.Zefiro-Europe.CONCEPT.3537	Redundant WSP activity axle 3	L	2F_05.02.Zefiro-Europe.CONCEPT.3222
2F_05.02.Zefiro-Europe.CONCEPT.3538	Redundant WSP activity axle 4	L	2F_05.02.Zefiro-Europe.CONCEPT.3222

2F_05.02.Zefiro-Europe.CONCEPT.HR71	**Heavy Slide** 2 = No heavy slide 1 = Heavy slide 0 = not valid/ not in charge to this control Unit TC_BI_#BCU_S_HeavySlide_X		
2F_05.02.Zefiro-Europe.CONCEPT.3543	Redundant speed zero (digital output)	L	2F_05.02.Zefiro-Europe.CONCEPT.1867
2F_05.02.Zefiro-Europe.CONCEPT.3544	Redundant speed > 10 km/h (digital output)	L	2F_05.02.Zefiro-Europe.CONCEPT.1873
2F_05.02.Zefiro-Europe.CONCEPT.1544	Brake pipe pressure TCG (10 = 1kPa)	A	2F_05.02.Zefiro-Europe.CONCEPT.533 2F_05.02.Zefiro-Europe.CONCEPT.602 2F_05.02.Zefiro-Europe.CONCEPT.985
2F_05.02.Zefiro-Europe.CONCEPT.1638	Indirect brake pilote pressure TDIS (10 = 1kPa)	A	2F_05.02.Zefiro-Europe.CONCEPT.1639
2F_05.02.Zefiro-Europe.CONCEPT.2368	DEP brake pilot pressure Tpil (10 = 1kPa)	A	2F_05.02.Zefiro-Europe.CONCEPT.983
2F_05.02.Zefiro-Europe.CONCEPT.1110	Load pressure bogie 1 (TS1) 10=1 kPa	A	2F_05.02.Zefiro-Europe.CONCEPT.536
2F_05.02.Zefiro-Europe.CONCEPT.1838	Load pressure bogie 2 (TS2) 10=1 kPa	A	2F_05.02.Zefiro-Europe.CONCEPT.536
2F_05.02.Zefiro-Europe.CONCEPT.3924	Auxiliary brake reservoir pressure (TSA)	A	2F_05.02.Zefiro-Europe.CONCEPT.3925
2F_05.02.Zefiro-Europe.CONCEPT.1943	BCU speed axle 1	A	2F_05.02.Zefiro-Europe.CONCEPT.1958
2F_05.02.Zefiro-Europe.CONCEPT.1944	BCU speed axle 2	A	2F_05.02.Zefiro-Europe.CONCEPT.1958
2F_05.02.Zefiro-Europe.CONCEPT.1945	BCU speed axle 3	A	2F_05.02.Zefiro-Europe.CONCEPT.1958
2F_05.02.Zefiro-Europe.CONCEPT.1946	BCU speed axle 4	A	2F_05.02.Zefiro-Europe.CONCEPT.1958
2F_05.02.Zefiro-Europe.CONCEPT.HR72	Reference speed		
2F_05.02.Zefiro-Europe.CONCEPT.3539	**WSP function in charge to BCU** 1 = Axle protected by BCU for wheel slide 0 = Axle not protected by BCU for wheel slide TC_BI_#BCU_S_axleprotwsp_X		
	DNRA in charge to BCU 1 = Axle protected by BCU for DNRA 0 = Axle not protected by BCU for DNRA TC_BI_#WSP_S_axleprotDNRA_X		

2F_05.02.Zefiro-Europe.CONCEPT.HR73	**Dump Valves command status:** 2 = hold 1 = vent 0 = fill TC_BI_#BCU_S_DV_X		
2F_05.02.Zefiro-Europe.CONCEPT.HR74	**BCU transducers check (per bogie)** 2 = Check OK 1 = Check NOK 0 = not valid/not already done TC_BI_#BCU_S_Trans_outranb_1 Reset: Power cycle (reset) BCU OR new brake test	SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR75	**BCU pressure switch check (per bogie)** 2 = Check OK 1 = Check NOK 0 = not valid/ not already done **TC_BI_#BCU_S_PCFbtcheckb_X** Reset: Power cycle (reset) BCU OR new brake test	SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR76	**TACU_BSS_axle 1 (only Motor coach)** 1 = TACU speed sensor on axle 1 out TC_BI_#BCU_TACU_WSS_axle 1		
2F_05.02.Zefiro-Europe.CONCEPT.HR77	**TACU_BSS_axle 2 (only Motor coach)** 1 = TACU speed sensor on axle 2 out TC_BI_#BCU_TACU_WSS_axle 2		
2F_05.02.Zefiro-Europe.CONCEPT.HR78	**TACU_BSS_axle 3 (only Motor coach)** 1 = TACU speed sensor on axle 3 out TC_BI_#BCU_TACU_WSS_axle 3		
2F_05.02.Zefiro-Europe.CONCEPT.HR79	**TACU_BSS_axle 4 (only Motor coach)** 1 = TACU speed sensor on axle 4 out TC_BI_#BCU_TACU_WSS_axle 4		
2F_05.02.Zefiro-Europe.CONCEPT.HR80	~~Max gradient to hold the train with holding brake (%)~~		
2F_05.02.Zefiro-Europe.CONCEPT.HR81	~~Max performance available in emergency (m/sec*sec)~~		
2F_05.02.Zefiro-Europe.CONCEPT.HR82	~~Max service brake performance available in service braking (m/sec*sec)~~		
2F_05.02.Zefiro-Europe.CONCEPT.3932	WSP reference speed BCU	A	2F_05.02.Zefiro-Europe.CONCEPT.3934
2F_05.02.Zefiro-Europe.CONCEPT.3227	BCU life sign	A	2F_05.02.Zefiro-Europe.CONCEPT.3789
2F_05.02.Zefiro-Europe.CONCEPT.HR46	F_SS_INCOH_X	Event SIL 0 Diagnostic SIL 2	

2F_05.02.Zefiro-Europe.CONCEPT.HR47	F_SS_DISTURBED_X	Event SIL 0 Diagnostic SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR46	Counter of F_SS_INCOH_X: C_SS_ INCOH	SIL 0	
2F_05.02.Zefiro-Europe.CONCEPT.HR117	BCU_X_STATUS (U8) Bit 0 = coasting Bit 1 = braking Bit 2 = traction Bit 3 = emergency braking Bit 4 -7 not valid	Functionality in SIL 2 Signal in SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR120	BCU_X_ dragging 1 = BCU X dragging for brake status problem 0 = not in error	Functionality in SIL 0 Signal in SIL 0	

4.2.1.1.2 *WSP-CU to TCMS*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1935	WSP-CU start-up ok	L	2F_05.02.Zefiro-Europe.CONCEPT.1877
2F_05.02.Zefiro-Europe.CONCEPT.1986	WSP test passed	L	2F_05.02.Zefiro-Europe.CONCEPT.1881
2F_05.02.Zefiro-Europe.CONCEPT.3811	WSP test not passed	L	2F_05.02.Zefiro-Europe.CONCEPT.1881
2F_05.02.Zefiro-Europe.CONCEPT.1987	ED bit 1 axle 1/2/3/4 active	L	2F_05.02.Zefiro-Europe.CONCEPT.1884
2F_05.02.Zefiro-Europe.CONCEPT.1988	ED bit 2 axle 1/2/3/4 active	L	2F_05.02.Zefiro-Europe.CONCEPT.1884
2F_05.02.Zefiro-Europe.CONCEPT.3301	WSP minor fault	L	2F_05.02.Zefiro-Europe.CONCEPT.3794
2F_05.02.Zefiro-Europe.CONCEPT.1238	Sanding is active	L	2F_05.02.Zefiro-Europe.CONCEPT.1073 2F_05.02.Zefiro-Europe.CONCEPT.1728
2F_05.02.Zefiro-Europe.CONCEPT.1772	Sanding isolation cock (RICS) isolated	L	2F_05.02.Zefiro-Europe.CONCEPT.1770
2F_05.02.Zefiro-Europe.CONCEPT.1239	Sanding level low in sandbox wheel 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1768
2F_05.02.Zefiro-Europe.CONCEPT.1854	Sanding level low in sandbox wheel 2	L	2F_05.02.Zefiro-Europe.CONCEPT.1768
2F_05.02.Zefiro-Europe.CONCEPT.1133	DIA - WSP function disturbed	L	2F_05.02.Zefiro-Europe.CONCEPT.345
~~2F_05.02.Zefiro-Europe.CONCEPT.HR83~~	~~**TACU_WSS_axle 1 (only Motor coach)**~~ ~~1 = TACU speed sensor on axle 1 out~~ ~~TC_BI_#WSP_TACU_WSS_axle 1~~		
~~2F_05.02.Zefiro-Europe.CONCEPT.HR84~~	~~**TACU_WSS_axle 2 (only Motor coach)**~~ ~~1 = TACU speed sensor on axle 2 out~~		

	~~TC_BI_#WSP_TACU_WSS_axle 1~~		
~~2F_05.02.Zefiro-Europe.CONCEPT.HR85~~	~~**TACU_WSS_axle 3 (only Motor coach)**~~ ~~1 = TACU speed sensor on axle 3 out~~ ~~TC_BI_#WSP_TACU_WSS_axle 1~~		
~~2F_05.02.Zefiro-Europe.CONCEPT.HR86~~	~~**TACU_WSS_axle 4 (only Motor coach)**~~ ~~1 = TACU speed sensor on axle 4 out~~ ~~TC_BI_#WSP_TACU_WSS_axle 1~~		
2F_05.02.Zefiro-Europe.CONCEPT.HR87	**Dump Valves command status:** 2 = hold 1 = vent 0 = fill TC_BI_#WSP_S_DV_X_		
2F_05.02.Zefiro-Europe.CONCEPT.HR88	**Diameter correction achieved** 1 = Diameter correction achieved 0 = Not updated TC_BI_#WSP_S_Diamcorrect_st Reset to 0: CPU power cycle (reset)		
2F_05.02.Zefiro-Europe.CONCEPT.HR89	~~**Diameter correction end**~~ ~~1 = Diameter correction end~~ ~~0 = not valid~~ ~~TC_BI_#WSP_S_Diamcorrect_end~~		
2F_05.02.Zefiro-Europe.CONCEPT.HR90	**DNRA in charge to WSP** 1 = Axle protected by WSPCU for DNRA 0 = Axle not protected by WSPCU for DNRA TC_BI_#WSP_S_axleprotDNRA_X		
2F_05.02.Zefiro-Europe.CONCEPT.3304	DNRA request active (redundancy)	L	2F_05.02.Zefiro-Europe.CONCEPT.2001
2F_05.02.Zefiro-Europe.CONCEPT.HR39	**Axle protected by WSP function in charge to WSPCU (WSP Control Unit)** 1 = Axle protected by WSPCU for wheel slide 0 = Axle not protected by WSPCU for wheel slide **TC_BI_#WSP_S_axleprotwsp_x**	Diagnosticated in SIL 2 Event in SIL 0	
2F_05.02.Zefiro-Europe.CONCEPT.1143	WSP activity on axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.484 2F_05.02.Zefiro-Europe.CONCEPT.743
2F_05.02.Zefiro-Europe.CONCEPT.1664	WSP activity on axle 2	L	2F_05.02.Zefiro-Europe.CONCEPT.484
2F_05.02.Zefiro-Europe.CONCEPT.1661	WSP activity on axle 3	L	2F_05.02.Zefiro-Europe.CONCEPT.484
2F_05.02.Zefiro-Europe.CONCEPT.1665	WSP activity on axle 4	L	2F_05.02.Zefiro-Europe.CONCEPT.484

2F_05.02.Zefiro-Europe.CONCEPT.1261	Brake cylinder pressure released	L	2F_05.02.Zefiro-Europe.CONCEPT.515
2F_05.02.Zefiro-Europe.CONCEPT.HR91	**Heavy Slide** 2 = No heavy slide 1 = Heavy slide 0 = not valid/ not in charge to this control unit TC_BI_#WSP_S_HeavySlide_X	Diagnosticated in SIL 2 Event in SIL 0	
~~2F_05.02.Zefiro-Europe.CONCEPT.HR111~~	~~TCMS shall set an event in case of WSP heavy slide signal received (2F_05.02.Zefiro-Europe.CONCEPT.HR91) directed for maintenance~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~
~~2F_05.02.Zefiro-Europe.CONCEPT.HR115~~	~~TCMS shall set an event for maintenance in case of 2F_05.02.Zefiro-Europe.CONCEPT.HR114~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~
~~2F_05.02.Zefiro-Europe.CONCEPT.HR116~~	~~TCMS shall set an event for maintenance in case of 2F_05.02.Zefiro-Europe.CONCEPT.HR113~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~
2F_05.02.Zefiro-Europe.CONCEPT.3297	Speed zero (digital output set)	L	2F_05.02.Zefiro-Europe.CONCEPT.1867
2F_05.02.Zefiro-Europe.CONCEPT.3299	Speed v > 10 km/h (digital output set)	L	2F_05.02.Zefiro-Europe.CONCEPT.1873
2F_05.02.Zefiro-Europe.CONCEPT.1938	DIA BCU/WSP internal bus failure (RS232)	L	2F_05.02.Zefiro-Europe.CONCEPT.1905
2F_05.02.Zefiro-Europe.CONCEPT.3551	DIA EB loop implausible (IP and EB loop status not aligned)	L	2F_05.02.Zefiro-Europe.CONCEPT.2096
2F_05.02.Zefiro-Europe.CONCEPT.2363	DIA Missing second stage braking	L	2F_05.02.Zefiro-Europe.CONCEPT.2271
2F_05.02.Zefiro-Europe.CONCEPT.2364	DIA Undue second stage braking	L	2F_05.02.Zefiro-Europe.CONCEPT.2274
2F_05.02.Zefiro-Europe.CONCEPT.3288	DIA Missing third stage braking	L	2F_05.02.Zefiro-Europe.CONCEPT.2274
2F_05.02.Zefiro-Europe.CONCEPT.3287	DIA Undue third stage braking	L	2F_05.02.Zefiro-Europe.CONCEPT.2274
2F_05.02.Zefiro-Europe.CONCEPT.3918	DIA Missing fourth stage braking	L	2F_05.02.Zefiro-Europe.CONCEPT.3913
2F_05.02.Zefiro-Europe.CONCEPT.3289	DIA Triple Stage DIA1 HW2 failure	L	2F_05.02.Zefiro-Europe.CONCEPT.3581
2F_05.02.Zefiro-Europe.CONCEPT.3290	DIA Triple Stage DIA2 HW2 failure	L	2F_05.02.Zefiro-Europe.CONCEPT.3583
2F_05.02.Zefiro-Europe.CONCEPT.3579	DIA Triple stage valve EVTS1 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3575
2F_05.02.Zefiro-Europe.CONCEPT.3580	DIA Triple stage valve EVTS2 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3577
2F_05.02.Zefiro-Europe.CONCEPT.2365	DIA Triple stage control faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.2170
2F_05.02.Zefiro-Europe.CONCEPT.2002	DIA Wheel set 1 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660

2F_05.02.Zefiro-Europe.CONCEPT.2003	DIA Wheel set 2 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.2004	DIA Wheel set 3 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.2005	DIA Wheel set 4 detected as non-rotating	L	2F_05.02.Zefiro-Europe.CONCEPT.660
2F_05.02.Zefiro-Europe.CONCEPT.1684	DIA WSP-CU faulty supply speed sensor 1 (open/short circuit)	L	2F_05.02.Zefiro-Europe.CONCEPT.794
2F_05.02.Zefiro-Europe.CONCEPT.1685	DIA WSP-CU faulty supply speed sensor 2 (open/short circuit)	L	2F_05.02.Zefiro-Europe.CONCEPT.794
2F_05.02.Zefiro-Europe.CONCEPT.1686	DIA WSP-CU faulty supply speed sensor 3 (open/short circuit)	L	2F_05.02.Zefiro-Europe.CONCEPT.794
2F_05.02.Zefiro-Europe.CONCEPT.1687	DIA WSP-CU faulty supply speed sensor 4 (open/short circuit)	L	2F_05.02.Zefiro-Europe.CONCEPT.794
2F_05.02.Zefiro-Europe.CONCEPT.1688	DIA Faulty WSP-CU speed sensor 1 acquisition channel	L	2F_05.02.Zefiro-Europe.CONCEPT.795
2F_05.02.Zefiro-Europe.CONCEPT.1689	DIA Faulty WSP-CU speed sensor 2 acquisition channel	L	2F_05.02.Zefiro-Europe.CONCEPT.795
2F_05.02.Zefiro-Europe.CONCEPT.1690	DIA Faulty WSP-CU speed sensor 3 acquisition channel	L	2F_05.02.Zefiro-Europe.CONCEPT.795
2F_05.02.Zefiro-Europe.CONCEPT.1691	DIA Faulty WSP-CU speed sensor 4 acquisition channel	L	2F_05.02.Zefiro-Europe.CONCEPT.795
2F_05.02.Zefiro-Europe.CONCEPT.1233	DIA TCF pressure sensor axle 1 faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.1622
2F_05.02.Zefiro-Europe.CONCEPT.1623	DIA TCF pressure sensor axle 2 faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.1622
2F_05.02.Zefiro-Europe.CONCEPT.1625	DIA TCF pressure sensor axle 3 faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.1622
2F_05.02.Zefiro-Europe.CONCEPT.1624	DIA TCF pressure sensor axle 4 faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.1622
2F_05.02.Zefiro-Europe.CONCEPT.1992	DIA WSP pressure incoherence on DV1 (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3293	DIA WSP pressure incoherence on DV2 (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3292	DIA WSP pressure incoherence on DV3 (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.3291	DIA WSP pressure incoherence on DV4 (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.1908
2F_05.02.Zefiro-Europe.CONCEPT.1151	DIA WSP dump valve axle 1 active for >10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658
2F_05.02.Zefiro-Europe.CONCEPT.1673	DIA WSP dump valve axle 2 active for >10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658
2F_05.02.Zefiro-Europe.CONCEPT.1674	DIA WSP dump valve axle 3 active for >10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658
2F_05.02.Zefiro-Europe.CONCEPT.1675	DIA WSP dump valve axle 4 active for >10s security circuit activated	L	2F_05.02.Zefiro-Europe.CONCEPT.658

2F_05.02.Zefiro-Europe.CONCEPT.1157	DIA WSP dump valve axle 1 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.1231	DIA WSP dump valve axle 2 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.1232	DIA WSP dump valve axle 3 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659 2F_05.02.Zefiro-Europe.CONCEPT.1039
2F_05.02.Zefiro-Europe.CONCEPT.1676	DIA WSP dump valve axle 4 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.659
2F_05.02.Zefiro-Europe.CONCEPT.1847	DIA WSP dump valve axle 1 open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.1848	DIA WSP dump valve axle 2 open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.1849	DIA WSP dump valve axle 3 open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1852 2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.1850	DIA WSP dump valve axle 4 open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1904
2F_05.02.Zefiro-Europe.CONCEPT.1857	DIA WSP dump valve axle 1 security circuit failure (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3295	DIA WSP dump valve axle 2 security circuit failure (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3296	DIA WSP dump valve axle 3 security circuit failure (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3294	DIA WSP dump valve axle 4 security circuit failure (WSP test)	L	2F_05.02.Zefiro-Europe.CONCEPT.3223
2F_05.02.Zefiro-Europe.CONCEPT.3302	DIA DNRA output from WSP-CU open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3555
2F_05.02.Zefiro-Europe.CONCEPT.3303	DIA DNRA output from WSP-CU short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3557
2F_05.02.Zefiro-Europe.CONCEPT.1235	DIA ED11_WSP_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.1255	DIA ED12_WSP_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3597	DIA ED21_WSP_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3598	DIA ED22_WSP_B1 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3599	DIA ED31_WSP_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3600	DIA ED32_WSP_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3601	DIA ED41_WSP_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.3602	DIA ED42_WSP_B2 open circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1766
2F_05.02.Zefiro-Europe.CONCEPT.1234	DIA ED11_WSP_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765

2F_05.02.Zefiro-Europe.CONCEPT.1767	DIA ED12_WSP_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3603	DIA ED21_WSP_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3604	DIA ED22_WSP_B1 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3605	DIA ED31_WSP_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3606	DIA ED32_WSP_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3607	DIA ED41_WSP_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.3608	DIA ED42_WSP_B2 short circuit (DM and M cars)	L	2F_05.02.Zefiro-Europe.CONCEPT.1765
2F_05.02.Zefiro-Europe.CONCEPT.1205	DIA Sanding valve 10.04 (EVCS) short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1773
2F_05.02.Zefiro-Europe.CONCEPT.1206	DIA Sanding valve 10.04 (EVCS) open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.1773
2F_05.02.Zefiro-Europe.CONCEPT.1855	DIA Sanding command wheel 1 open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3589
2F_05.02.Zefiro-Europe.CONCEPT.1856	DIA Sanding command wheel 1 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3589
2F_05.02.Zefiro-Europe.CONCEPT.3587	DIA Sanding command wheel 2 open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3592
2F_05.02.Zefiro-Europe.CONCEPT.3588	DIA Sanding command wheel 2 short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3592
2F_05.02.Zefiro-Europe.CONCEPT.3570	DIA v > 10 km/h speed output open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3566
2F_05.02.Zefiro-Europe.CONCEPT.3571	DIA v > 10 km/h speed output short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3568
2F_05.02.Zefiro-Europe.CONCEPT.3573	DIA zero speed output open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3562
2F_05.02.Zefiro-Europe.CONCEPT.3574	DIA zero speed output short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3564
2F_05.02.Zefiro-Europe.CONCEPT.1666	WSP-CU speed axle 1	A	2F_05.02.Zefiro-Europe.CONCEPT.1662
2F_05.02.Zefiro-Europe.CONCEPT.1667	WSP-CU speed axle 2	A	2F_05.02.Zefiro-Europe.CONCEPT.1662
2F_05.02.Zefiro-Europe.CONCEPT.1668	WSP-CU speed axle 3	A	2F_05.02.Zefiro-Europe.CONCEPT.1662
2F_05.02.Zefiro-Europe.CONCEPT.1669	WSP-CU speed axle 4	A	2F_05.02.Zefiro-Europe.CONCEPT.1662
2F_05.02.Zefiro-Europe.CONCEPT.3933	WSP reference speed WSP-CU	A	2F_05.02.Zefiro-Europe.CONCEPT.3934
2F_05.02.Zefiro-Europe.CONCEPT.1980	Brake cylinder pressure (TCF) axle 1	A	2F_05.02.Zefiro-Europe.CONCEPT.1866

2F_05.02.Zefiro-Europe.CONCEPT.1981	Brake cylinder pressure (TCF) axle 2	A	2F_05.02.Zefiro-Europe.CONCEPT.1866
2F_05.02.Zefiro-Europe.CONCEPT.1983	Brake cylinder pressure (TCF) axle 3	A	2F_05.02.Zefiro-Europe.CONCEPT.1866
2F_05.02.Zefiro-Europe.CONCEPT.1982	Brake cylinder pressure (TCF) axle 4	A	2F_05.02.Zefiro-Europe.CONCEPT.1866
2F_05.02.Zefiro-Europe.CONCEPT.1939	Watchdog counter (life sign WSP-CU) to TCMS	A	2F_05.02.Zefiro-Europe.CONCEPT.1878
2F_05.02.Zefiro-Europe.CONCEPT.HR45	WSP status in automatic drive mode: TC_BI_#WSP_S_StatusAuto	Event SIL 0 Diagnostic SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR46	F_SS_INCOH_X	Event SIL 0 Diagnostic SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR47	F_SS_DISTURBED_X	Event SIL 0 Diagnostic SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR46	Counter of F_SS_INCOH_X: C_SS_ INCOH	Event SIL 0 Diagnostic SIL 2	
2F_05.02.Zefiro-Europe.CONCEPT.HR47	The specific counter refreshed C_SS_DISTURBED_X The refreshed acceleration value A_SS_DISTURBED_X The maximum acceleration achieved AMAX_SS_DISTURBED_X	SIL 0	
2F_05.02.Zefiro-Europe.CONCEPT.HR118	WSP_X_STATUS (U8) Bit 0 = coasting Bit 1 = braking Bit 2 = traction Bit 3 = emergency braking Bit 4 -7 not valid	Functionality in SIL 2 Signal in SIL 2	

4.2.1.1.3 *DBV-GW to TCMS*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1272	Brake pipe assimilation active	L	2F_05.02.Zefiro-Europe.CONCEPT.3348
2F_05.02.Zefiro-Europe.CONCEPT.3316	Overcharge active	L	2F_05.02.Zefiro-Europe.CONCEPT.3345
2F_05.02.Zefiro-Europe.CONCEPT.1645	Isolation cock RIPCG cock in isolated position	L	2F_05.02.Zefiro-Europe.CONCEPT.1021 2F_05.02.Zefiro-Europe.CONCEPT.1640 2F_05.02.Zefiro-Europe.CONCEPT.1788
2F_05.02.Zefiro-Europe.CONCEPT.1644	Cabin panel isolation cock in isolated position (RIM)	L	2F_05.02.Zefiro-Europe.CONCEPT.2132
2F_05.02.Zefiro-Europe.CONCEPT.1634	RIBK cock in position EP brake (0=IP, 1=EP)	L	2F_05.02.Zefiro-Europe.CONCEPT.1590
2F_05.02.Zefiro-Europe.CONCEPT.1268	DBV-GW minor fault	L	2F_05.02.Zefiro-Europe.CONCEPT.3362
2F_05.02.Zefiro-Europe.CONCEPT.1830	DBV-GW major fault (IEP brake pipe control faulty)	L	2F_05.02.Zefiro-Europe.CONCEPT.2262
2F_05.02.Zefiro-Europe.CONCEPT.1270	BP leakage detected (Train separation)	L	2F_05.02.Zefiro-Europe.CONCEPT.2246
2F_05.02.Zefiro-Europe.CONCEPT.1273	IEP brake mode active	L	2F_05.02.Zefiro-Europe.CONCEPT.357 2F_05.02.Zefiro-Europe.CONCEPT.1024 2F_05.02.Zefiro-Europe.CONCEPT.2429

2F_05.02.Zefiro-Europe.CONCEPT.2343	DBV filling position FS	L	2F_05.02.Zefiro-Europe.CONCEPT.1027 2F_05.02.Zefiro-Europe.CONCEPT.2266 2F_05.02.Zefiro-Europe.CONCEPT.2440 2F_05.02.Zefiro-Europe.CONCEPT.2442
2F_05.02.Zefiro-Europe.CONCEPT.2344	DBV release position RP	L	2F_05.02.Zefiro-Europe.CONCEPT.2137
2F_05.02.Zefiro-Europe.CONCEPT.2345	DBV last release position 1A	L	2F_05.02.Zefiro-Europe.CONCEPT.2137
2F_05.02.Zefiro-Europe.CONCEPT.2348	DBV last apply position 1B	L	2F_05.02.Zefiro-Europe.CONCEPT.2137
2F_05.02.Zefiro-Europe.CONCEPT.2346	DBV maximum service brake position MB	L	2F_05.02.Zefiro-Europe.CONCEPT.2137
2F_05.02.Zefiro-Europe.CONCEPT.2347	DBV emergency brake position EB	L	2F_05.02.Zefiro-Europe.CONCEPT.2137
2F_05.02.Zefiro-Europe.CONCEPT.2370	Ready for brake test	L	2F_05.02.Zefiro-Europe.CONCEPT.665
2F_05.02.Zefiro-Europe.CONCEPT.2352	DIA Isolation valve VISO is not opening on order	L	2F_05.02.Zefiro-Europe.CONCEPT.2227
2F_05.02.Zefiro-Europe.CONCEPT.3305	DIA Isolation valve VISO is opening without order	L	2F_05.02.Zefiro-Europe.CONCEPT.2228
2F_05.02.Zefiro-Europe.CONCEPT.2356	DIA Neutral valve EVNEU is not opening on order	L	2F_05.02.Zefiro-Europe.CONCEPT.2231
2F_05.02.Zefiro-Europe.CONCEPT.3306	DIA Neutral valve EVNEU is opening without order	L	2F_05.02.Zefiro-Europe.CONCEPT.2233
2F_05.02.Zefiro-Europe.CONCEPT.1166	DIA MR pipe sensor for brake pipe filling function (TMP) fault	L	2F_05.02.Zefiro-Europe.CONCEPT.986 2F_05.02.Zefiro-Europe.CONCEPT.1078
2F_05.02.Zefiro-Europe.CONCEPT.1167	DIA Brake pipe pilot pressure sensor faulty (TRE DM car)	L	2F_05.02.Zefiro-Europe.CONCEPT.987 2F_05.02.Zefiro-Europe.CONCEPT.1079
2F_05.02.Zefiro-Europe.CONCEPT.1171	DIA Brake pipe pressure sensor 03.07 (TBP) faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.1083
2F_05.02.Zefiro-Europe.CONCEPT.1251	DIA Quick release valve EVR open / short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3357 2F_05.02.Zefiro-Europe.CONCEPT.3358
2F_05.02.Zefiro-Europe.CONCEPT.1252	DIA Neutral valve EVNEU open / short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3338 2F_05.02.Zefiro-Europe.CONCEPT.3339 2F_05.02.Zefiro-Europe.CONCEPT.3340 2F_05.02.Zefiro-Europe.CONCEPT.3341
2F_05.02.Zefiro-Europe.CONCEPT.1253	DIA BP exhaust valve EVPCG open circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3366
2F_05.02.Zefiro-Europe.CONCEPT.3666	DIA BP exhaust valve EVPCG short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.3368
2F_05.02.Zefiro-Europe.CONCEPT.1267	DBV or IP brake used in DEP mode	L	2F_05.02.Zefiro-Europe.CONCEPT.364
2F_05.02.Zefiro-Europe.CONCEPT.1141	DIA DBV GW Minitrol CAN communication faulty	L	2F_05.02.Zefiro-Europe.CONCEPT.348 2F_05.02.Zefiro-Europe.CONCEPT.1099
2F_05.02.Zefiro-Europe.CONCEPT.1240	DIA DBV PWM fault	L	2F_05.02.Zefiro-Europe.CONCEPT.348
2F_05.02.Zefiro-Europe.CONCEPT.1169	DIA Brake pipe filling valve (EVS) in DM car short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.783

2F_05.02.Zefiro-Europe.CONCEPT.1170	DIA Brake pipe filling/venting valve open circuit (EVS/EVF)	L	2F_05.02.Zefiro-Europe.CONCEPT.784
2F_05.02.Zefiro-Europe.CONCEPT.3309	DIA Brake pipe venting valve EVF short circuit	L	2F_05.02.Zefiro-Europe.CONCEPT.2280
2F_05.02.Zefiro-Europe.CONCEPT.3308	DIA RE (relay valve) target pressure not achieved	L	2F_05.02.Zefiro-Europe.CONCEPT.3364
2F_05.02.Zefiro-Europe.CONCEPT.1246	DIA EVR valve quick release not opening on order	L	2F_05.02.Zefiro-Europe.CONCEPT.3352
2F_05.02.Zefiro-Europe.CONCEPT.3307	DIA EVR valve quick release opening without order	L	2F_05.02.Zefiro-Europe.CONCEPT.3355
2F_05.02.Zefiro-Europe.CONCEPT.3661	DIA Overcharge command permanently active	L	2F_05.02.Zefiro-Europe.CONCEPT.3659
2F_05.02.Zefiro-Europe.CONCEPT.2342	Brake pipe pressure TBP minitrol	A	2F_05.02.Zefiro-Europe.CONCEPT.602
2F_05.02.Zefiro-Europe.CONCEPT.3312	Brake pipe pre-control pressure TRE	A	2F_05.02.Zefiro-Europe.CONCEPT.3662
2F_05.02.Zefiro-Europe.CONCEPT.3314	Main pipe pressure TMP	A	2F_05.02.Zefiro-Europe.CONCEPT.3663
2F_05.02.Zefiro-Europe.CONCEPT.3315	DBV-GW life sign	A	2F_05.02.Zefiro-Europe.CONCEPT.3664

4.2.1.2 *From Propulsion*

4.2.1.2.1 ***From PCU1 Bogie 1***

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1602	Available ED brake effort MCM 1	A	2F_05.02.Zefiro-Europe.CONCEPT.371 2F_05.02.Zefiro-Europe.CONCEPT.384
2F_05.02.Zefiro-Europe.CONCEPT.2391	Available ED brake effort MCM 2	A	2F_05.02.Zefiro-Europe.CONCEPT.371
2F_05.02.Zefiro-Europe.CONCEPT.1604	Achieved ED brake effort in kN MCM 1	A	
2F_05.02.Zefiro-Europe.CONCEPT.2407	Achieved ED brake effort in kN MCM 2	A	
2F_05.02.Zefiro-Europe.CONCEPT.1990	ED brake slide control bit 1 axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1989
2F_05.02.Zefiro-Europe.CONCEPT.1991	ED brake slide control bit 2 axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1989
2F_05.02.Zefiro-Europe.CONCEPT.2405	ED brake slide control bit 1 axle 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.2406	ED brake slide control bit 2 axle 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.HR92	Axle 1 speed	U16	
2F_05.02.Zefiro-Europe.CONCEPT.HR93	Axle 2 speed	U16	

4.2.1.2.2 *From PCU2 Bogie 2*

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.2409	Available ED brake effort MCM 1	A	2F_05.02.Zefiro-Europe.CONCEPT.371 2F_05.02.Zefiro-Europe.CONCEPT.384
2F_05.02.Zefiro-Europe.CONCEPT.2410	Available ED brake effort MCM 2	A	2F_05.02.Zefiro-Europe.CONCEPT.371
2F_05.02.Zefiro-Europe.CONCEPT.2411	Achieved ED brake effort in kN MCM 1	A	
2F_05.02.Zefiro-Europe.CONCEPT.2412	Achieved ED brake effort in kN MCM 2	A	
2F_05.02.Zefiro-Europe.CONCEPT.2413	ED brake slide control bit 1 axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1989
2F_05.02.Zefiro-Europe.CONCEPT.2414	ED brake slide control bit 2 axle 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1989
2F_05.02.Zefiro-Europe.CONCEPT.2415	ED brake slide control bit 1 axle 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.2416	ED brake slide control bit 2 axle 2	L	
2F_05.02.Zefiro-Europe.CONCEPT.HR103	Axle 3 speed	U16	
2F_05.02.Zefiro-Europe.CONCEPT.HR104	Axle 4 speed	U16	

4.2.2 TCMS SW to Suppliers

4.2.2.1 *To Brake Control Units (BCU/WSP-CU/DBV-GW)*

4.2.2.1.1 To BCU

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.3840	Urgency brake demand		2F_05.02.Zefiro-Europe.CONCEPT.2056
2F_05.02.Zefiro-Europe.CONCEPT.2394	Dynamic brake active	L	2F_05.02.Zefiro-Europe.CONCEPT.1963
2F_05.02.Zefiro-Europe.CONCEPT.2395	Traction activated	L	2F_05.02.Zefiro-Europe.CONCEPT.3839
2F_05.02.Zefiro-Europe.CONCEPT.2442	Urgency brake active	L	2F_05.02.Zefiro-Europe.CONCEPT.2056
2F_05.02.Zefiro-Europe.CONCEPT.2443	Emergency brake active	L	2F_05.02.Zefiro-Europe.CONCEPT.3318
2F_05.02.Zefiro-Europe.CONCEPT.1919	Parking brake apply demand (monitoring only)	L	2F_05.02.Zefiro-Europe.CONCEPT.1069
2F_05.02.Zefiro-Europe.CONCEPT.1920	Parking brake release demand (monitoring only)	L	2F_05.02.Zefiro-Europe.CONCEPT.1069
2F_05.02.Zefiro-Europe.CONCEPT.2399	Train brake test active	L	2F_05.02.Zefiro-Europe.CONCEPT.1088

2F_05.02.Zefiro-Europe.CONCEPT.2383	Brake test step 1	L	2F_05.02.Zefiro-Europe.CONCEPT.2513 2F_05.02.Zefiro-Europe.CONCEPT.2514
2F_05.02.Zefiro-Europe.CONCEPT.3795	Brake test step 1.1	L	2F_05.02.Zefiro-Europe.CONCEPT.2528 2F_05.02.Zefiro-Europe.CONCEPT.2531
2F_05.02.Zefiro-Europe.CONCEPT.2385	Brake test step 4	L	2F_05.02.Zefiro-Europe.CONCEPT.2789 2F_05.02.Zefiro-Europe.CONCEPT.2790
2F_05.02.Zefiro-Europe.CONCEPT.3796	Brake test step 4.1	L	2F_05.02.Zefiro-Europe.CONCEPT.2796 2F_05.02.Zefiro-Europe.CONCEPT.2844
2F_05.02.Zefiro-Europe.CONCEPT.3797	Brake test step 5	L	2F_05.02.Zefiro-Europe.CONCEPT.2576 2F_05.02.Zefiro-Europe.CONCEPT.2578
2F_05.02.Zefiro-Europe.CONCEPT.3799	Brake test step 5.1	L	2F_05.02.Zefiro-Europe.CONCEPT.3677 2F_05.02.Zefiro-Europe.CONCEPT.3678
2F_05.02.Zefiro-Europe.CONCEPT.3798	Brake test step 5.2	L	2F_05.02.Zefiro-Europe.CONCEPT.2581 2F_05.02.Zefiro-Europe.CONCEPT.2583
2F_05.02.Zefiro-Europe.CONCEPT.3803	Brake test DNRA	L	2F_05.02.Zefiro-Europe.CONCEPT.3514
2F_05.02.Zefiro-Europe.CONCEPT.3804	WSP test active	L	2F_05.02.Zefiro-Europe.CONCEPT.1881
2F_05.02.Zefiro-Europe.CONCEPT.3806	WSP test ED bit 1	L	2F_05.02.Zefiro-Europe.CONCEPT.1885
2F_05.02.Zefiro-Europe.CONCEPT.3805	WSP test ED bit 2	L	2F_05.02.Zefiro-Europe.CONCEPT.1888
2F_05.02.Zefiro-Europe.CONCEPT.2400	Friction brake demand (kN)	A	2F_05.02.Zefiro-Europe.CONCEPT.617
2F_05.02.Zefiro-Europe.CONCEPT.2402	Watchdog counter CCU-O	A	
2F_05.02.Zefiro-Europe.CONCEPT.HR94	**ED braking** 1 = braking with ED PCU1_XF_TrBrAchX		
2F_05.02.Zefiro-Europe.CONCEPT.HR95	**Traction status** 1 = traction 0 = not traction TC_BO_#BCU_S_TrAct		
2F_05.02.Zefiro-Europe.CONCEPT.HR96	**PCF1 status** 1 = digital input signal is 1 0 = digital input signal is 0 TC_BO_1BCU_PCF1		
2F_05.02.Zefiro-Europe.CONCEPT.HR97	**PCF2 status** 1 = digital input signal is 1 0 = digital input signal is 0 TC_BO_1BCU_PCF2		
2F_05.02.Zefiro-Europe.CONCEPT.HR98	**Coupled side (only DM)** 1 = Coupled Cab 0 = Not Coupled Cab TC_BO_#BCU_S_ThisCabCoup		

2F_05.02.Zefiro-Europe.CONCEPT.HR99	**End of Train (only DM)** Boolean 1 = End of Train 0 = Not End of Train TC_BO_#BCU_S_ThisCabEoT		
2F_05.02.Zefiro-Europe.CONCEPT.HR100	**Cab active (only DM)** 1 = Cab Active 0 = Cab Not Active TC_BO_#BCU_S_ThisCabAct		
2F_05.02.Zefiro-Europe.CONCEPT.HR101	**At least one cab active in the train consist** 1 = At Least one cab is active on train (one at least in all units in the composition) 0 = No cabs active on train TC_BO_#BCU_S_OneCab	N/A	
2F_05.02.Zefiro-Europe.CONCEPT.HR102	**Parking status** 1 = Train is in parking status 0 = No parking status TC_BO_#BCU_S_Parking	N/A	
2F_05.02.Zefiro-Europe.CONCEPT.HR44	**Automatic Drive** 1 = automatic drive 0 = not automatic drive TC_BO_#BCU_S_AutoDrive	N/A	

4.2.2.1.2 To WSP-CU

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1215	Dynamic brake active	L	2F_05.02.Zefiro-Europe.CONCEPT.1963
2F_05.02.Zefiro-Europe.CONCEPT.1217	Traction activated	L	2F_05.02.Zefiro-Europe.CONCEPT.3839
2F_05.02.Zefiro-Europe.CONCEPT.3808	Urgency brake active	L	2F_05.02.Zefiro-Europe.CONCEPT.2056
2F_05.02.Zefiro-Europe.CONCEPT.1925	Emergency brake active	L	2F_05.02.Zefiro-Europe.CONCEPT.3318
2F_05.02.Zefiro-Europe.CONCEPT.1734	Sanding device enable (Driving direction DM1)	L	2F_05.02.Zefiro-Europe.CONCEPT.742 2F_05.02.Zefiro-Europe.CONCEPT.1733
2F_05.02.Zefiro-Europe.CONCEPT.1211	Automatic sanding enable	L	2F_05.02.Zefiro-Europe.CONCEPT.1501
2F_05.02.Zefiro-Europe.CONCEPT.1216	Sanding command	L	2F_05.02.Zefiro-Europe.CONCEPT.315 2F_05.02.Zefiro-Europe.CONCEPT.1728
2F_05.02.Zefiro-Europe.CONCEPT.1212	Train brake test active	L	2F_05.02.Zefiro-Europe.CONCEPT.1088
2F_05.02.Zefiro-Europe.CONCEPT.1213	WSP test active	L	2F_05.02.Zefiro-Europe.CONCEPT.1881
2F_05.02.Zefiro-Europe.CONCEPT.1921	WSP test bit1 ED	L	2F_05.02.Zefiro-Europe.CONCEPT.1885

2F_05.02.Zefiro-Europe.CONCEPT.1922	WSP test bit 2 ED	L	2F_05.02.Zefiro-Europe.CONCEPT.1885
2F_05.02.Zefiro-Europe.CONCEPT.3807	Brake test DNRA	L	2F_05.02.Zefiro-Europe.CONCEPT.3514
2F_05.02.Zefiro-Europe.CONCEPT.1549	OPTIONAL: Enable magnetic track brake	L	2F_05.02.Zefiro-Europe.CONCEPT.755
2F_05.02.Zefiro-Europe.CONCEPT.1550	OPTIONAL: Activate magnetic track brake (manual test activation)	L	2F_05.02.Zefiro-Europe.CONCEPT.1368
2F_05.02.Zefiro-Europe.CONCEPT.1108	Main pipe pressure	A	
2F_05.02.Zefiro-Europe.CONCEPT.2026	Average brake pipe pressure	A	2F_05.02.Zefiro-Europe.CONCEPT.2027
2F_05.02.Zefiro-Europe.CONCEPT.1924	Watchdog counter CCU-O	A	
2F_05.02.Zefiro-Europe.CONCEPT.HR44	**Automatic Drive** 1 = automatic drive 0 = not automatic drive TC_BO_#WSP_S_AutoDrive	N/A	
2F_05.02.Zefiro-Europe.CONCEPT.HR102	**Parking status** 1 = Train is in parking status 0 = No parking status TC_BO_#WSP_S_Parking	N/A	

4.2.2.1.3 TCMS to DBV-GW

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.2351	Local cab active	L	2F_05.02.Zefiro-Europe.CONCEPT.2129
2F_05.02.Zefiro-Europe.CONCEPT.2360	Isolation valve command active	L	2F_05.02.Zefiro-Europe.CONCEPT.3688
2F_05.02.Zefiro-Europe.CONCEPT.2362	OPTIONAL: Master controller brake request (0-100%)	A	2F_05.02.Zefiro-Europe.CONCEPT.2266
2F_05.02.Zefiro-Europe.CONCEPT.2642	Train brake test active	L	2F_05.02.Zefiro-Europe.CONCEPT.1088
2F_05.02.Zefiro-Europe.CONCEPT.2382	DEP active	L	2F_05.02.Zefiro-Europe.CONCEPT.2148
2F_05.02.Zefiro-Europe.CONCEPT.2384	Brake test step 3 BP continuity test	L	2F_05.02.Zefiro-Europe.CONCEPT.3391
2F_05.02.Zefiro-Europe.CONCEPT.3945	Coupled composition active	L	2F_05.02.Zefiro-Europe.CONCEPT.3943 2F_05.02.Zefiro-Europe.CONCEPT.3944
2F_05.02.Zefiro-Europe.CONCEPT.2643	Watchdog counter CCU-O	A	

4.2.2.1.4 To Propulsion (PCU)

Nr	Functional name	Type	Linked to requirement
2F_05.02.Zefiro-Europe.CONCEPT.1605	Traction/ED brake effort demand	A	2F_05.02.Zefiro-Europe.CONCEPT.183

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[4] SFRS ATP (Train Protection): FA020025863 rev00
[5] SFRS Conventional Train Control: FA020023107 rev 00[6] SFRS Drive control: FA020022675 rev 00[7] Prestazioni di Treno ED040019912 rev. 1
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[16] COMMISSION REGULATION (EU) No 1302/2014 of 18 November 2014 concerning a technical specification for interoperability relating to the ‘rolling stock - locomotives and passenger rolling stock’ subsystem of the rail system in the European Union as amended by Commission Implementing Regulation (EU) 2018/868 of 13 June 2018, Commission Implementing Regulation (EU) 2019/776 of 16 May 2019 and Commission Implementing Regulation (EU) 2020/387 of 9 March 2020